

Stochastic and Thermal Exactness

Diophantine certificates for exact events
in uniformly convergent systems

Positive stochastic erasure, exact value iteration of stochastic games, thermal Hamiltonians with undecidable confinement boundaries, positive mixing realizations of polynomials, and coercive Green operators

AI-assisted research report prepared for Vladimir Reshetnikov from six manuscripts

(ProveIt's incoming reports: batch 78, manuscripts 02, 10 and 03; batch 79, manuscripts 12, 10 and 14), October 2, 2026

Abstract

Six manuscripts share one pattern: a numerical approximation converges uniformly, at a rate known in advance, while an *exact* event of the same system is undecidable.

Part I (source 02) compiles integer matrix mortality into strictly positive rational column-stochastic matrices with no common invariant complex line, each entry within any prescribed band around $1/n$ and every product contracting total variation by η^T . A product has rank one exactly when the source projection is mortal, so exact erasure is undecidable for at most eight positive 7×7 matrices and many-one complete for a nine-state family with a variable alphabet. Explicit horizon- T natural quartics have one witness per erasing labelled word.

Part II (source 10) builds a fixed turn-based stochastic game with discount $1/2$, reward $1/4$, dyadic transition rows of full support and the known value vector $\frac{1}{2}\mathbf{1}$. Value iteration converges at rate 2^{-n} , yet exact attainment of $\frac{1}{2}\mathbf{1}$ from a dyadic terminal payoff is computably enumerable complete. Each horizon has an orthant-nonnegative quadratic certificate with exactly one complete natural witness.

Part III (source 03) turns a natural arithmetic circuit into a diagonal Hamiltonian $H = (1 + \sum_i n_i)Q$ of degree at most five whose zero states biject with the witnesses and whose positive spectrum is effectively confined. Compact resolvent and finite thermal trace are equivalent to finite witness multiplicity and are undecidable on a padded universal family; a uniformly computable critical fugacity t_* satisfies $t_* < 1$ exactly on halting inputs.

Part IV (source 11) realizes every nonzero rational polynomial P with $r(P) + 1$ states and commuting, strictly positive doubly stochastic matrices; on one fixed alphabet, exact arrival of one coordinate at its equilibrium value is computably enumerable complete.

Part V (sources 12 and 13, one theorem by two routes) gives uniformly coercive operators $\alpha I - A$ whose Green data have finite support and are rational exactly on halting inputs.

The report is AI-assisted and unrefereed; nothing in it is formalized in Lean or Rocq, and priority is not certified for any Part.

Status. Proofs and executable finite checks; not kernel-checked. Section 1 states the common spine, the notation, the relation to neighbouring reports and to the repository's reviews of these manuscripts, and what is not claimed; Appendix F gives the provenance.

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1 Scope, common spine and conventions

[write] This section was written when the three manuscripts of batch 78 were merged, and extended on the same day when batch 79 added Parts IV and V. Parts I–V print the six sources; text there without a [write] marker is the source’s own.

1.1 The five Parts

This report is built from six AI-assisted research manuscripts; Appendix F gives the full provenance. Three were delivered with batch 78 of ProveIt’s incoming reports (manuscripts 02, 10 and 03). They are called *source 02*, *source 10* and *source 03* after their batch-78 manuscript numbers, which are also the file prefixes of their shipped programs and data. None cites another, they were written from three different snapshots of the repository on the same day, and they share no lemma.

Three more were delivered with batch 79 (manuscripts 12, 10 and 14). They are called after the next free file prefixes of this report, in the order of their Parts: *source 11* is batch-79 manuscript 12 (Part IV), *source 12* is batch-79 manuscript 10 (Part V, base) and *source 13* is batch-79 manuscript 14 (Part V, second route). Source 10 (batch-78 manuscript 10, Part II) and batch-79 manuscript 10 (source 12) are different manuscripts. Sources 12 and 13 prove one theorem by two independent routes from one repository snapshot; source 11 re-uses Part I’s lift without naming it, knowing Part I’s source only through the research programme’s review [23].

- **Part I (source 02), *Exact erasure without a common invariant line*.** A finite alphabet of rational column-stochastic matrices is driven by an external schedule; a word *erases* when its product has rank one, so that the output distribution no longer depends on the input distribution. Two invertible tensor guards lift integer mortality into strictly positive stochastic matrices with no common invariant complex line and the exact rank identity $\text{rank } S_w = 1 + 2 \text{rank } A_{\pi(w)}$ (Theorem 5.1). Rank-one existence is undecidable for at most eight positive 7×7 matrices (Corollary 5.3) and many-one complete for nine states with a variable alphabet (Corollary 6.3). Under independent full-support switching, existence of an erasing word, almost-sure finite erasure and finite expected erasure time coincide (Theorem 9.1). Horizon- T natural quartics have exactly one witness per erasing labelled word (Theorems 10.1 and 10.2).
- **Part II (source 10), *Exact convergence as computation*.** A counter program is compiled through homogeneous dyadic min/max circuits into a genuine two-player turn-based stochastic game with discount $1/2$, running reward $1/4$, at most two actions per state and every transition probability at least $1/(2N)$. The Bellman operator is a $1/2$ -contraction with the known fixed point $\frac{1}{2}\mathbf{1}$, and exact finite attainment of that fixed point from a dyadic terminal payoff is computably enumerable complete for a suitable fixed game (Theorem 17.1). Point-mass selectors and chance-only games give decidable boundaries, and chance-only scalar equality is Skolem-equivalent (Section 23). Each horizon has an orthant-nonnegative quadratic with exactly one complete natural witness (Theorem 24.2).
- **Part III (source 03), *Thermal arithmetic*.** A natural arithmetic circuit becomes a diagonal Hamiltonian $H = (1 + \sum_i n_i)Q$ on occupation modes, of joint degree at most five, whose zero-energy basis states are in bijection with the witnesses while every positive energy is at least $1 + \sum_i n_i$ (Theorem 32.1). Compact resolvent and finite thermal trace hold exactly when the witnesses are finitely many (Theorem 34.1), and both are undecidable on a padded

universal family (Theorem 34.6). A uniformly computable critical fugacity satisfies $t_*(c) < 1$ exactly on halting inputs (Theorem 36.1).

- **Part IV (source 11), *Mixing does not remove arithmetic*.** The translation space of a nonzero rational polynomial P in k variables, of dimension $r(P)$, is lifted to $r(P) + 1$ states with commuting, invertible, strictly positive doubly stochastic rational matrices in any band $((1 - \theta)/s, (1 + \theta)/s)$, and one accepting coordinate equals $1/s + \varepsilon q^{|n|} P(n)$ after the count vector n (Theorem 45.1, Corollary 45.2). The state count is optimal for the exact series (Theorem 45.3) and at most $\binom{k+3}{2}$ for quartics, sharply (Theorem 46.1). A fixed alphabet with the input loaded along a rational quadratic curve or a line segment of initial distributions has a computably enumerable complete exact equilibrium-hit problem (Theorems 47.2 and 47.4); polynomial degree equals joint nilpotence depth (Theorem 48.2); separated targets are decidable (Theorem 49.1).
- **Part V (sources 12 and 13), *Coercive Green operators*.** A branch-history lift turns a counter machine into a computable path forest. For every integer $\alpha \geq 3$ the operator $L_\alpha = \alpha I - A$ is uniformly coercive with a polynomial-time approximable inverse, yet finite support of $L_\alpha^{-1} e_s$, rationality of the Green value $g_\alpha(s)$ and a unique normalized integer certificate with continuant coefficients are each equivalent to halting (Theorems 58.2 and 59.1, Corollary 59.3); the rational values form a Pell curve and each specified level is decidable (Theorems 60.1 and 60.2). Source 13 proves the same theorem by its own route on a connected graph of maximum degree three with a dipole source (Theorem 73.2), shows membership over every finite-prime localization decidable (Corollary 75.4) and four nonzeros per row optimal (Theorem 76.1), and gives a horizon-indexed natural quadratic with one complete witness (Theorem 77.2).

Each Part keeps its own definitions, letters and numbering of statements. Parts I and II are both about stochastic operators with full support (stochastic games extend stochastic matrices by minima and maxima); Part III shares no lemma with either, and it is placed here because of the spine of Section 1.2, not because of a shared construction. Part IV re-uses Part I's lift and is the counterpart of its commuting theorem (Section 42.4); Part V shares no lemma with Parts I–IV and joins them through the spine, its killed random walk (Theorem 65.2) being its stochastic reading.

1.2 The common spine: uniform approximation, undecidable exactness

In each Part an approximation statement holds with a computable rate that does not depend on the instance, and an exact statement about the same object is undecidable. Table 1 lines them up.

Three further features recur. First, every Part proves that no total computable function of the input bounds the time or the witness height of a positive instance (Theorem 11.2, Corollary 27.1, Theorem 37.1, Corollary 49.2, Theorem 61.2, Corollary 75.6); this is the bounded-search argument of `cdc:bd:prop:nobound` in [22] and `pqc:qm:cor:nobound` in [21], applied to new predicates. Second, a fixed positive tolerance restores decidability or a finite cutoff in Parts I, II and IV ($\eta^T \leq \varepsilon$; Proposition 28.1; Theorem 49.1), and in Part V so do a specified rational level and a finite set of primes (Theorem 60.2, Corollary 75.4); in Parts III and V no computable positive separation from the boundary exists (Corollary 36.3, Theorem 61.2). Third, Parts I–III each state an MRDP consequence of fixed arity and degree at most four obtained by circuit quadratization (Proposition 11.1; Section 27; Corollary 32.3 with Theorem 32.1); none of them controls the multiplicity of the original MRDP witnesses. Part IV uses MRDP the other way round, as the input front end of its fixed alphabets

Part	Uniform approximation	Exact predicate and its status	Bounded or static certificate
I (02)	$\text{TV}(S_w p, S_w q) \leq \eta^T \text{TV}(p, q)$ for every word of length T (Proposition 4.2)	some product has rank one: undecidable on the seven-state class, c.e.-complete on the nine-state class (Corollaries 5.3, 6.3)	horizon- T quartic, $T((n-1)^2 + k)$ natural witnesses, one per erasing labelled word (Theorem 10.2)
II (10)	$\ \Phi^n(v) - \frac{1}{2}\mathbf{1}\ _\infty \leq$ $2^{-n} \ v - \frac{1}{2}\mathbf{1}\ _\infty$, and $4^{-n}/2$ on encoded inputs (Section 21)	$\exists n \Phi^n(v) = \frac{1}{2}\mathbf{1}$: c.e.-complete for a fixed game (Theorem 17.1)	horizon- T quadratic, $(N+2b)T$ natural witnesses, one complete witness (Theorem 24.2)
III (03)	excited trace, regulated partition function and t_* uniformly computable (Theorems 33.3, 35.1, 36.1)	$t_*(c) < 1$: c.e.-complete; compact resolvent, finite thermal trace: co-c.e.-complete on a padded family (Theorems 36.1, 34.6)	static polynomial of degree at most five with a fixed number of modes, one zero state per witness (Theorem 32.1)
IV (11)	$\text{TV}(pM_w, u_s) \leq \theta^L$ for every word of length L (Theorem 45.1)	$\exists n p(t)M^n \mathbf{e}_1 = 1/s$: c.e.-complete on a fixed alphabet (Theorems 47.2, 47.4)	no horizon polynomial: the count vector itself, one canonical word per natural root (Section 50)
V (12, 13)	Neumann tail $(2/\alpha)^{K+1}/(\alpha-2)$; polynomial-time intervals for $g_\alpha(s)$ (Theorems 57.1, 61.1)	finite support, rational Green value, rational resistance: c.e.-complete (Corollary 59.3; Theorems 73.2, 74.1)	supplied-support slice or horizon- T quadratic, one witness (Theorems 63.1, 77.2)

Table 1: [write] The common spine of the five Parts.

(Lemma 47.1), and also keeps the original multiplicity; Part V states no fixed-arity MRDP corollary, its finite certificates having a supplied support or a supplied horizon.

The slogan itself is not new in the collection: Parts V and VII of [21] already prove contraction-versus-exactness results, and sources 02 and 10 say so themselves (Sections 2 and 17); of sources 11–13 only source 13 cites one of those Parts’ manuscripts (source 17 of [21], as an antecedent). What each Part adds is stated in Section 1.5.

1.3 Notation across the Parts

No symbol was renamed and no normalization was changed. Each Part keeps its source’s letters, and a letter means what its own Part says it means. Table 2 lists the letters that Parts I–III use differently; Table 8, printed at the opening of Part IV, does the same for Parts IV and V (readings after the list below), and Table 10, at the opening of Part V, compares the two sources of Part V notion by notion. In Parts IV and V only macro spellings were unified: the inner-product and norm macros of sources 11–13 print with this report’s definitions (delimiter sizes only), and source 13’s `\cref` references are written out.

Table 2: [write] Letters used differently in the three Parts.

Letter	Part I (source 02)	Part II (source 10)	Part III (source 03)
n, N	n : state dimension; N : nilpotent shift, also N_T (erasing words of length T) and the input bit length in $b(N)$	N : number of game states; n : iteration count	N_i : number operators; N : index of the example H_N ; $n \in \mathbb{N}^m$: occupation tuple
K	common lift scale, $K \geq 2$	layer normalizer $K = K_L$, a power of two	K : halting set; K : compact remainder operator
D	common column sum $D = nK$	common probability denominator	$D = (B + 1)^m$: tuples in a truncation box
M	largest entry of $ \Delta_i $	pipeline period $M = L + 1$	number of zero states $\#Z$
T	word length or horizon; $T_i = p_i S_i$ labelled transitions	horizon	T_q : trace-class remainder
H	H_n : zero-mass hyperplane; H_i : integer restriction matrices	first halting time	the Hamiltonian
Q	PCP connector $Q = c\ell$; Q_T : full quartic	Q_T : complete quadratic	compiled quartic penalty
S, s	S_i : stochastic matrices; s_t : selector residuals	S : game operator; s : clock register	$s(n) = \sum_i n_i$: total occupation
E	coordinate map E	E_T : endpoint sum	energy level
L	guard L ; length of an erasing word	final circuit layer	minimum zero-state height
R	guard R	power of two in the Skolem lift	excited sum $R(q, t)$
P	cyclic permutation (Section 7)	integer transition numerators $P_{ia,j}$; chance-only matrix	source polynomial; P_0 : zero projection
η	contraction band	reset weight	requested width in the proof of Theorem 36.1
m, k	m source matrices, k letters	m control states, k counters	m modes, k source witnesses
q, t	q : probability of a block; t : time	t : time step	q : Boltzmann parameter; t : fugacity

Watch for these readings:

- **Erasure.** In Part I a word *erases* when its stochastic product has rank one: every input distribution is sent to one common output distribution, which need not be uniform (a *uniform* erasing word gives J_n). In Part II the *erasing map* F_{era} and the *erasing halt convention* send a halted counter configuration, and then the whole simulated pipeline vector, to the zero vector; the exact event observed there is attainment of the fixed point $\frac{1}{2}\mathbf{1}$, not rank one of a product. The two notions are unrelated constructions that happen to share a word.

- **Uniform.** Part I: a uniform erasing word (target J_n), uniform switching (equal letter probabilities), and the uniform-input quartic (matrix entries as parameters). Part II: uniformly mixing (every row of full support) and the common uniform reset. Part III: uniformly computable (one algorithm for every instance).
- **Contraction and confinement.** Part I contracts total variation by η per letter; Part II contracts the supremum norm by $\delta = 1/2$ and the span by $\delta(1 - \eta)$. Part III has no dynamics at all: its *confinement* is the spectral inequality $H > 0 \Rightarrow H \geq 1 + s$, and its diagonal evolution only changes phases (Section 37).
- **Certificate.** The certificates of Parts I and II are horizon-indexed: their arity grows with T . The Hamiltonian of Part III has a fixed number of modes for a fixed circuit. None of the three is a fixed-arity single-fold representation of a computably enumerable complete set.
- **Fugacity, thermal.** In Part III these are mathematical parameters of a positive trace; no physical reservoir, laboratory implementation or physical decision procedure is assumed (source 03’s own caveat).

Further readings to watch in Parts IV and V:

- **Stochastic convention.** Part I uses column-stochastic matrices acting on column distributions; Part IV uses row-stochastic (in fact doubly stochastic) matrices with distributions as rows. Part IV’s “no word erases” (Proposition 45.5) is Part I’s erasure: for doubly stochastic products rank one means $M_w = J_s$ (compare Corollary 4.3).
- **Exact hit.** Part IV’s predicate is one coordinate equal to its equilibrium value $1/s$, while the whole distribution stays nonuniform; it is not erasure and not attainment of a fixed point as in Part II.
- **Green value and resistance.** Part V.A’s $g_\alpha(s)$ is the one-terminal root value; Part V.B’s $\mathcal{R}_x$ is the two-terminal resistance, $2g_L$ on halting inputs.
- **Certificate.** Part IV’s witness is the count vector itself, of fixed arity with unbounded entries. Part V’s linear certificate is an unbounded finite-support array (a quasi-Diophantine witness); its ordinary quadratics have a supplied support (Theorem 63.1) or a supplied horizon (Theorem 77.2). None of them is a fixed-arity single-fold representation of a computably enumerable complete set.
- **Well-conditioned.** In Part V the condition number bound $(\alpha+2)/(\alpha-2)$ (or $(m+3)/(m-3)$) and the closeness of the normalized operator to the identity are analytic facts; finite precision still decides nothing about exact rationality or support.

1.4 Relation to neighbouring reports and to the repository’s reviews

This report sits in the collection’s `hilbert-tenth-problem` category beside four earlier reports and the signal-machine report written in the same batch. Its closest neighbour is the report on probabilistic, quantum and continuous computation [21], cited here by label names (it is a separate document):

- *Part II of [21] (quantum mortality).* Theorem `pqc:qm:thm:47` imports the undecidability of mortality for six integer 3×3 matrices and packs it into seven rational 4×4 Kraus operators, so that a zero-probability outcome word exists exactly when the source is mortal.

Corollary 5.3 runs the same pipeline with a different exact predicate (rank one of a positive stochastic product) and Theorem 11.2 is the argument of `ppc:qm:cor:nobound`. Proposition `ppc:qm:prop:nonnegative` decides mortality for nonnegative matrices; Part I shows that this positivity boundary does not transfer to rank one (note after Corollary 5.3), which bears on Question `ppc:q:signed`.

- *Part V of [21]*. Source 13’s witness-faithful quartic energy (`ppc:ql:thm:compiler`) is a bounded Boolean energy landscape over $\mathbb{R}$ and expressly treats no Gibbs distribution; Part III is the unbounded natural counterpart with a thermal trace. Source 17’s reversible chains have a uniform spectral gap and an undecidable exact stationary equality, the same slogan as Part II for a different object.
- *Part VII of [21]*. Source 12 proves r.e.-complete exact extinction for a 2-adic 1/2-contraction (`ppc:ad:cor:complete`); source 14 a real piecewise-linear 1/2-contraction whose point-to-point reachability is decidable (`ppc:sc:thm:main` (iii)); source 15 the unsquared complementarity device and the degree-two unique-witness trace polynomial (`ppc:pm:lem:clamp`, `ppc:pm:thm:trace`). Part II is complementary to all three: a real, order-preserving 1/2-contraction whose exact fixed-point attainment is c.e.-complete because its clock is erased at halting (note after Section 17’s comparison), and its quadratic specializes source 15’s device to min/max gates. Part II also answers Part VII’s unlabelled question “Structural subclasses” negatively in part (note after the table of Section 23).
- *Parts VIII and IX of [21]*. Corollary `ppc:tu:cor:fixedquartic` and its remark (a fixed MRDP quartic whose witness multiplicity is uncontrolled) are the observation that Part III’s spectral classification makes exact. Theorem `ppc:iq:thm:geometric` (geometric stopping clocks with undecidable records) is related only to Part I’s zero-one law.

[write] *Batch 79 (Parts IV and V)*. In Part V of [21], source 13 (*Probabilities from Well-Conditioned Quartics*) is a bounded polynomial energy landscape, unrelated in construction to Part V’s coercive operators, and source 17 (*Arithmetic of Rapidly Mixing Reversible Computation*) is cited by source 13 here as an antecedent (Section 70); its uniformly gapped reversible chains bear on Proposition 48.4 and on source 12’s question of Section 68.11. Part VII of [21] has the same slogan as Parts IV and V for contractions. Part IV bears on Questions `ppc:q:signed` and `ppc:q:promise` of [21]: a positive commuting doubly stochastic class keeps a computably enumerable complete exact hit, while finite search settles every target off the critical value.

The report on canonical Diophantine certificates [22] supplies four results that the Parts reprove, or use without citation, and that are printed here as pointers or marked second routes, never as new: the unique-extension quadratization `cdc:wf:lem:quadratization` (and `cdc:cs:lem:quartic`), which is the core of Theorem 32.1 and of the fixed-arity quartic corollaries of Parts I and II; the absence of a computable witness-height bound `cdc:bd:prop:nobound`; the finite-fold substrate equivalence `cdc:cs:thm:finitefold`, the template of Theorem 34.4; and the positivity-restricted degree classification `cdc:of:thm:classification`, which shows that radial isolation below degree four is impossible for a universal family (note in Section 39). Parts IV and V add pointers to `cdc:wf:lem:quadratization` (Lemma 47.1), to `cdc:pt:prop:unipotent` (the binomial step of Theorem 48.2), to `cdc:bd:prop:nobound` (Corollary 49.2, Theorem 61.2, Corollary 75.6) and to the quadratic bounded counter execution at `cdc:eq:bigM` (Theorem 77.2). The reports *group-theoretic-substrates* and *signal-machine-collision-certificates* have no overlap with this one. *liveness-beyond-halting* [41] has none with Parts I–III; source 13’s horizon quadratic (Theorem 77.2) is a variant of its degree-two counter-machine compilers `lbh:qd:thm:compiler` and `lbh:ql:thm:compiler`, printed

with a pointer. Batch-79 manuscript 01, which sources 12 and 13 cite as *Linear Quasi-Diophantine Universality*, was placed in the report *polynomial-witness-histories* [40]; its regime is the opposite one (a dense nonclosed range instead of a bounded inverse).

The Hilbert’s-tenth-problem research programme (read-only for this report) had already reviewed the three batch-78 manuscripts and two of the three batch-79 manuscripts before they were placed here, and reviewed the third soon after; this report does not present its placement as a first review:

- [23] reviewed sources 02 and 03 (commit 44b28c395). It accepts source 02 within its stated domains, replays its suite and all eight certificates and adds independent checks; it reports no defect. It accepts source 03’s mathematics for natural arithmetic DAGs and finds that the delivered low-level `Compiler/Atom` interface accepts invalid descriptors; it supplies the tested patch `thermal_exact_dag_guards.patch`. The shipped compiler here is the unpatched original (note in Section 38.5).
- [24] reviewed source 10 (commit 126028588). It accepts the mathematics and finds two implementation defects: the independent checker does not bind its scaled initial vector to the declared input, and an explicitly supplied integer scale in the optional counter loader is divided in floating point; it supplies the patches `exact_convergence_input_binding.patch` and `exact_convergence_integer_scale.patch`. The shipped programs here are the unpatched originals (notes in Section 26 and Appendix D).
- The note `matrix_pcp_trace.md` [25] uses the same classical radix-three encoding of Post correspondence tiles as Lemma 6.1, with a leading sentinel instead of Part I’s fourth coordinate and idempotent connector.
- [write] *Batch 79*. [42] (commit 899391bde, made before the placement commit bbaf322e5) reviewed sources 12 and 13. It accepts both within their presentation and domain assumptions, reran their suites (110,522 and 38,524 checks) and all five example exports, added independent checks, and found no defect in source 12. In source 13’s reference API it found an export alias that can erase the live input binding and a prime-deduplication order that accepts an inexact scalar, and it supplies the tested patch `well_conditioned_certificate_snapshots.patch`; the shipped program is the unpatched original (note in Section 79). Source 11 was inventoried by the programme ([43], commit b927b2290) and reviewed after placement [44] (commit 5b633fcf9): it passes, with no defect and no repair (opening of Part IV). The programme has also authenticated the 64 files placed by bbaf322e5 against the six delivered archives (`review_placement_bbaf322e5.md` in the same directory) and has emitted complete paid evaluators of source 13’s horizon quadratic [45] (note in Section 77).

The formal project. The report sits in the collection, not in `Computability/HilbertTenthProblem`, and placement beside a Lean/Rocq development confers no formal status. The three sources import MRDP only as a classical theorem; source 10 names the project’s interface, the theorem `Diophantine.mrdp` with its variants `Diophantine.mrdp_iff` and `Diophantine.mrdp_dioph_iff` in `Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean` (unchanged from source 10’s pin to this write). The project has formalized none of this report’s statements: it has no `stochastic-matrix`, `stochastic-game`, `Bellman-operator` or `occupation-number Hamiltonian` module. Sources 11–13 also use MRDP only as a classical theorem; source 13 names `Diophantine.mrdp`, `Diophantine.mrdp_iff` and `Diophantine.mrdp_dioph_iff` and the theorem `re_tm0` of `Computability/HilbertTenthProblem/Lean/Diophantine/Paper1980/TuringPartrec100.lean`, both files unchanged from its pin 44983ed7e to this write. The project has formalized none of the statements of Parts IV and V either: it has no `Markov-chain realization`, `Hankel-rank`,

coercive-operator, Green-function or resistor-network module.

1.5 Status and what is not claimed

The report claims conventional mathematical proofs, by its sources, of the results summarized in Section 1.1. What each source presents as its own contribution is narrower: source 02 the two-guard no-common-line compiler, its higher-dimensional subspace exclusion, the exact-forgetting interpretation and the counted certificate interface; source 10 the stochastic-game realization under the simultaneous restrictions of Theorem 17.1 with its complete arithmetic interface; source 03 the radial factor $1 + \sum_i n_i$ with its effective spectral decomposition, the multiplicity classification and the critical-fugacity construction; source 11 the explicit optimal-size combination of positive realization, input-geometry trade-offs, reverse translation and reproducible interface; source 12 the combined explicit coercive normal form with a canonical integer witness and its interfaces; source 13 the connected coercive compiler, the exact Lucas charge law, the finite-prime bound and the row-sparsity threshold.

Every limitation stated by a source is kept in its Part. Collected here, the report does **not** claim:

- historical priority for any Part (all six sources say their searches were targeted, and sources 11–13 credit commuting probabilistic-automaton undecidability, Hankel-rank minimality, stochastic embeddings, inverse systems, history recording, inverse decay and the network-walk correspondence to the literature), or any refereeing, proof-assistant verification, or rebuild of the repository’s Lean or Rocq developments; the finite checks (50,582, 26,077 and 7,936 assertions in Parts I–III; 110,522 and 38,524 checks in Part V; Part IV’s 1,310 count vectors, 1,424 words and 269 nilpotence products) illustrate and do not prove;
- an improvement of the repository’s 75-operation complete-certificate or 87-operation universal-polynomial records; no Part instantiates a numerical universal polynomial, alphabet, game or network;
- a finite-fold or single-fold Diophantine representation of computably enumerable sets; the bounded certificates of Parts I and II have horizon-dependent arity, and the fixed-arity MRDP quartics leave the original witness multiplicity uncontrolled; Part IV keeps the multiplicity of the MRDP representation it starts from, and Part V’s witnesses are unbounded finite-support arrays whose ordinary quadratics have a supplied support or horizon;
- *Part I*: irreducibility (a common invariant hyperplane remains); one fixed universal stochastic alphabet with an ordinary input loader (the seven-state result is a family of alphabets parameterized by the input); many-one completeness of the eight-letter bound (only of the nine-state variable-alphabet family); anything about undecidable output entropy of hidden Markov models; optimality of the dimension, alphabet or certificate counts. Its headline answer to a MathOverflow question rests on an external discussion that is not archived in the repository;
- *Part II*: a numerical universal game (the shipped 128-state games are one-counter examples); hardness of ordinary value iteration from zero or of solving finite discounted games; a resolution of the Skolem problem or of the one-player Bellman reachability frontier; robustness to perturbed tables or floating point; novelty of the chance-only Skolem specialization (Vahanwala);
- *Part III*: a resolution of the finite-fold problem; physical hypercomputation or autonomous

step-by-step dynamics; a numerical value of the mode count of the universal family; that t_* is a noncomputable real (it is computable; its exact endpoint is undecidable); geometric locality from four-mode support; robustness to coefficient errors;

- *Part IV*: a printed numerical universal alphabet (the shipped examples are illustrative, and the universal theorems import MRDP); a lower bound for representing only the zero set of P , or for noncommuting or differently encoded competitors (Theorem 45.3 concerns the exact centered series); undecidability of a supplied finite word’s acceptance (a finite rational computation); a decision or undecidability result for single cubics (Corollary 48.3 is a reduction); simulation of a Turing machine by an autonomous finite chain; any procedure using physical noise, sampling or finite-precision measurement; an arbitrary preassigned q near 1 at no cost;
- *Part V*: a new ordinary universal integer-polynomial bound or a fixed-arity ordinary compiler for the sparse witness; a finite physical halting oracle (the networks are countable and finitely presented, and finite-precision measurement is no exact test); polynomial-time neighbour exploration in source 13’s connected graph (only computability is claimed); a decision theorem for arbitrary equations over $\mathbb{Z}[S^{-1}]$ (Corollary 75.4 is family-specific and concerns the unit-source solution); a uniform algorithm deciding the graph type in Theorem 76.1; validity in positive characteristic (Theorem 66.2); universality of restricted reversible two-counter models; optimality of the quadratic counts; individual novelty of undecidable rationality of computable reals, reversible simulation or path continued fractions.

Part I

Exact erasure without a common invariant line (source 02)

[write] Part I prints batch-78 manuscript 02, *Exact Erasure without a Common Invariant Line: Positive stochastic computation, tensor guards, and Diophantine certificates*, “Research report prepared for Vladimir Reshetnikov; Mathematical development and implementation: ChatGPT”, October 2, 2026. Its labels carry the prefix `ste:ee:`. Its abstract and status statement follow; its two appendices are Appendices A and B.

Abstract (source 02). We give an explicit reduction from integer matrix mortality to exact finite-time forgetting by strictly positive rational stochastic matrices. Two tensor-factor guards remove every common invariant complex line without creating spurious zero products. From m matrices of dimension d , the construction produces $m+2$ matrices of dimension $2d+1$, with every entry arbitrarily close to the uniform stochastic matrix and a prescribed uniform total-variation contraction factor. A product has rank one precisely when its source-letter projection is mortal. Consequently, rank-one product existence is undecidable already for at most eight positive rational 7×7 column-stochastic matrices having no common invariant complex line. Clearing one common denominator gives the same conclusion for strictly positive integer matrices. This answers the specific no-common-invariant-line restriction raised in a June 2026 MathOverflow discussion; it does not establish irreducibility.

A separate, fully printed four-dimensional correspondence-problem compiler gives a many-one complete nine-state stochastic family with a variable-size alphabet. A higher tensor construction excludes all invariant subspaces of dimensions $1, \dots, r-1$. For independent full-support switching, existence of an erasing word is equivalent both to almost-sure finite exact forgetting and to

finite expected forgetting time, despite deterministic exponential approximate forgetting on every instance. Explicit mass-compressed horizon- T natural-domain quartics have $T((n-1)^2 + k)$ witnesses and $T((n-1)^2 + 1) + (n-1)^2$ squared residuals, with exactly one witness per accepted labelled word. We also prove a sharp $n-1$ erasure horizon when the induced matrices on the zero-mass hyperplane commute, and exhibit arbitrarily small non-erasing perturbations preserving all common invariant subspaces. The package includes exact-arithmetic code, independent certificate replay, and 50,582 passing finite checks.

Status (source 02). Full mathematical proofs are supplied below. The new constructions are not proof-assistant verified or independently refereed. Established mortality, PCP, and DPRM results are cited as dependencies. The literature search is targeted, not a certification of global priority. No improvement to ProveIt’s fixed universal polynomial operation count is claimed.

2 Research question, contributions, and provenance

A stochastic matrix usually suggests information loss rather than universal computation. If every entry is positive and close to $1/n$, every step loses a large fraction of the distinction between two initial distributions. It is therefore tempting to expect that determining whether all distinction is ever lost exactly should be straightforward. The distinction between *small* and *zero*, however, changes the problem completely.

This report studies a finite alphabet $S_1, \dots, S_k$ of rational stochastic matrices. A word is an externally selected schedule. The question is whether some finite schedule makes the output distribution independent of the input distribution. Equivalently, does a matrix product have rank one? The control alphabet and rational coefficients are finite data; the word length is unbounded.

2.1 A concrete question that the construction answers

In a MathOverflow question posted on June 2, 2026, Victor Miller asked about the rank-one analogue of matrix mortality. Benjamin Steinberg observed that $A \mapsto A \oplus [1]$ transfers mortality to rank-one product existence. The subsequent discussion isolated the restriction that the generators have no common invariant line, and suggested an affine modification without a complete argument for that restriction [4]. We prove the missing restricted statement, with positivity and strong stochastic contraction added. ([write] The MathOverflow question and its June 3 discussion are external sources, not archived in the repository; that the construction answers their restriction is this source’s claim, not checked at the write.)

The answer has a necessary qualification. Every column-stochastic matrix preserves the hyperplane $\mathbf{1}^\top x = 0$. Our construction has no common invariant *line*, even over $\mathbb{C}$, but it is not an irreducible matrix family. It answers the stated line restriction, not a stronger irreducibility requirement that may arise in a particular application. It also does not settle any general question about computing hidden-Markov entropy.

2.2 Main results and their logical dependencies

The principal construction, Theorem 5.1, is elementary once a matrix mortality instance is supplied. Its central identity is

$$\text{rank } S_w = 1 + 2 \text{rank } A_{\pi(w)}, \quad (1)$$

where π deletes the two guard letters. The guards are invertible on their private tensor factor, so inserting them anywhere cannot fabricate a mortal source computation. One guard removes possible shared eigenvectors in a transverse direction; an affine displacement in the other removes a common stationary line.

For the small numerical undecidability bounds, we import only the established mortality bounds of Cassaigne, Halava, Harju, and Nicolas [1]. Their conventions allow oracle Turing reductions, so we do not silently promote every numerical bound in their paper to a many-one completeness statement. Instead, Section 6 prints a separate PCP reduction. That route gives many-one completeness in dimension nine, with alphabet size part of the input. The classical halting-to-PCP dependency is standard and has also been treated by formal reduction proofs [2].

The Diophantine contribution is explicit at a fixed horizon. Its witness bijection is proved directly, rather than inferred from DPRM. The latter theorem is used only for the separate fixed-arity, unbounded representation in Section 11. Its use does not preserve a unique witness per unbounded computation.

2.3 Relationship to the inspected ProveIt material

The repository was inspected at the pinned commit

928ea97017a25ebe56d240c84f27d2275d818c75.

The relevant sources are its Hilbert-tenth-problem README, the October 2 substrate review, and the README of the combined probabilistic/quantum/continuous report [5, 6, 7]. These already contain extensive work on exact observations, contractions, matrix substrates, and bounded certificates. We do not present the generic claim “computably enumerable implies Diophantine” or “contraction can coexist with undecidable exactness” as new.

The proposed contribution is more specific: the two-guard no-invariant-line compiler, its higher-dimensional invariant-subspace exclusion, its exact stochastic forgetting interpretation, and the ensuing explicitly counted certificate interface. The targeted inspection did not amount to rereading every manuscript in the repository, and the public-source search is not an exhaustive priority audit.

The October 2 review emphasizes the difference between an external-horizon polynomial and one fixed polynomial accepting arbitrarily long computations [6]. We retain that distinction throughout. The review’s stated 87-operation fixed universal polynomial is not improved by any count in this report. State dimension, residual count, arithmetic-circuit size, and number of unbounded Diophantine witnesses are different resources.

[write] After the pin, the research programme reviewed this manuscript, byte for byte the same package [23]: it accepts the results within their stated domains and quantifiers, replays the suite and all eight certificates, adds independent tensor-rank, natural selector/prefix and signed-selector checks, and reports no defect. The collection’s report [21] has a parallel pipeline in its Part II (Theorem `pqc:qm:thm:47`: six integer 3×3 mortality matrices compiled into seven rational 4×4 Kraus operators, zero-probability word $\Leftrightarrow$ mortal); Corollary 5.3 below is its positive stochastic counterpart with a different exact predicate. Neither that report nor the research notes contain the no-common-line compiler, the idempotent PCP connector, the stochastic zero-one law or the compressed quartic. Section 1.4 gives the comparisons.

3 Exact forgetting and the zero-mass representation

All stochastic matrices in this article are *column-stochastic*. Probability distributions are columns, so $p \mapsto Sp$. We set $\mathbb{N} = \{0, 1, 2, \dots\}$. Matrices called positive have every entry strictly greater than zero. Integer matrix mortality allows negative entries in the source matrices.

For a word $w = i_1 \cdots i_T$ in chronological order, define

$$S_w = S_{i_T} \cdots S_{i_1}, \quad S_\epsilon = I.$$

The same convention applies to every matrix alphabet. A nonempty word is required for mortality and erasure.

Definition 3.1. An *erasing word* for $S_1, \dots, S_k$ is a nonempty word w for which $S_w p$ is the same distribution for every initial distribution p . A *uniform erasing word* satisfies $S_w = J_n$, where

$$u_n = \frac{1}{n} \mathbf{1}_n, \quad J_n = u_n \mathbf{1}_n^\top.$$

Lemma 3.2 (Equivalent erasure tests). *For a nonnegative column-stochastic $n \times n$ matrix S , the following are equivalent: S erases all initial distributions; all its columns are equal; $\text{rank } S = 1$; and S vanishes on*

$$H_n = \{x \in \mathbb{R}^n : \mathbf{1}_n^\top x = 0\}.$$

If S is also row-stochastic, these conditions are equivalent to $S = J_n$.

Proof. The images of the point-mass distributions are precisely the columns of S . Thus erasure is equivalent to equal columns. Equal columns have rank one, not zero, since their sums are one. Conversely, rank one makes the columns scalar multiples of a fixed nonzero vector; their equal nonzero column sums force all these multiples to be equal. The vectors $e_j - e_1$, $j > 1$, span H_n , giving the hyperplane condition. Finally, a matrix with equal columns has the form $p \mathbf{1}^\top$; its row sums are np_a , which are all one precisely when $p = u_n$. $\square$

Lemma 3.3 (Erasure persists). *If a word w is erasing, every word containing w as a contiguous block is erasing. Appending further letters after erasure can change the common output distribution, but cannot restore dependence on the initial distribution.*

Proof. Multiplication on either side cannot increase matrix rank. Every product remains column-stochastic and so has rank at least one. A product containing the rank-one factor S_w therefore has rank exactly one. $\square$

To separate total mass from information, let $h = n - 1$ and introduce

$$E = \begin{pmatrix} I_h \\ -\mathbf{1}_h^\top \end{pmatrix}, \quad F = [I_h \ 0] - \frac{1}{n} \mathbf{1}_h \mathbf{1}_n^\top, \quad F_0 = nF. \quad (2)$$

These are explicit rational coordinate maps, not existentially chosen bases.

Lemma 3.4 (Coordinates). *One has*

$$FE = I_h, \quad F u_n = 0, \quad \mathbf{1}_n^\top E = 0, \quad EF = I_n - J_n.$$

Every $x \in \mathbb{Q}^n$ has the unique decomposition

$$x = Ey + su_n, \quad y = Fx, \quad s = \mathbf{1}_n^\top x.$$

For any rational column-stochastic S , its matrix in these coordinates is

$$\begin{pmatrix} C & b \\ 0 & 1 \end{pmatrix}, \quad C = FSE, \quad b = FSu_n, \quad (3)$$

and $\text{rank } S = 1 + \text{rank } C$.

Proof. The displayed identities follow by multiplying the matrices in (2). They prove existence and uniqueness of the decomposition. Column-stochasticity preserves s , while applying F to $S(Ey + su_n)$ gives $Cy + sb$. To compute the rank, subtract from each of the first h output rows its last-column entry times the last row. This leaves the block diagonal matrix $\text{diag}(C, 1)$ and hence $\text{rank } 1 + \text{rank } C$. $\square$

The same algebra holds over $\mathbb{C}$. We will use the complexified hyperplane and coordinates when excluding invariant lines. The distinction between a common invariant subspace and a common eigenvector is important: a one-dimensional invariant subspace is generated by a common eigenvector, but its eigenvalue may depend on the generator.

4 An explicit positive affine lift

Let (C_i, b_i) , $1 \leq i \leq k$, be finitely many integer affine data, with $C_i \in \mathbb{Z}^{h \times h}$ and $b_i \in \mathbb{Z}^h$. Fix a rational number $0 < \eta < 1$, chosen independently of the input coefficients. Define

$$\Delta_i = EC_i F_0 + nEb_i \mathbf{1}_n^\top, \quad (4)$$

$$M = \max_{i,a,b} |(\Delta_i)_{ab}|, \quad K = \max\{2, 1 + \lceil M/\eta \rceil\}, \quad D = nK, \quad (5)$$

$$B_i = K \mathbf{1}_n \mathbf{1}_n^\top + \Delta_i, \quad S_i = B_i/D. \quad (6)$$

Here B_i and D are integers. The same K and D are used for every generator.

Theorem 4.1 (Positive affine lift). *The matrices in (6) are positive rational column-stochastic and satisfy the strict entrywise bounds*

$$\frac{1-\eta}{n} < (S_i)_{ab} < \frac{1+\eta}{n}. \quad (7)$$

In the coordinates of Lemma 3.4,

$$S_i \sim \begin{pmatrix} K^{-1}C_i & K^{-1}b_i \\ 0 & 1 \end{pmatrix}. \quad (8)$$

For every word w of length T ,

$$\text{rank } S_w = 1 + \text{rank } C_w. \quad (9)$$

All entries of B_i are strictly positive integers and all its column sums equal D .

Proof. Since $K > M/\eta$, each entry of $\Delta_i/(nK)$ has absolute value less than η/n . Adding J_n proves (7) and positivity. The column sums of Δ_i vanish because $\mathbf{1}^\top E = 0$, so the column sums of S_i are one and those of B_i are D . Using $F_0 = nF$, rewrite the lift as

$$S_i = J_n + K^{-1}EC_iF + K^{-1}Eb_i\mathbf{1}_n^\top.$$

On $Ey + su_n$ this gives

$$S_i(Ey + su_n) = E(K^{-1}C_iy + K^{-1}b_is) + su_n,$$

which proves (8). Multiplying these upper block-triangular matrices gives a top-left block $K^{-T}C_w$. Its translation block need not vanish, but Lemma 3.4 shows that it does not affect the rank. Since $K \neq 0$, (9) follows. $\square$

Proposition 4.2 (Uniform approximate forgetting). *For all probability distributions p, q and every word w of length T ,*

$$\text{TV}(S_w p, S_w q) \leq \eta^T \text{TV}(p, q), \quad \text{TV}(p, q) = \frac{1}{2} \|p - q\|_1. \quad (10)$$

Proof. The lower bound in (7) allows the decomposition

$$S_i = (1 - \eta)J_n + \eta Q_i,$$

where Q_i is nonnegative and column-stochastic. Such a matrix is nonexpansive in ℓ^1 : for any real vector z ,

$$\|Q_i z\|_1 \leq \sum_a \sum_b (Q_i)_{ab} |z_b| = \|z\|_1.$$

For $z = p - q$ one has $J_n z = 0$, giving a factor η in one step. Iteration proves the result. $\square$

In particular, for a fixed accuracy $\varepsilon > 0$, a horizon with $\eta^T \leq \varepsilon$ works for every input instance and every schedule. This is a bound on the difference between two trajectories driven by the same schedule. When translations are present, it is *not* a claim that arbitrary switching converges to one common stationary distribution.

Corollary 4.3 (Doubly stochastic special case). *For $A_1, \dots, A_m \in \mathbb{Z}^{d \times d}$, apply the lift with $h = d$, $C_i = A_i$, and $b_i = 0$. Then S_i is doubly stochastic, and*

$$S_w = J_{d+1} + K^{-T}EA_wF. \quad (11)$$

Thus $A_w = 0$ if and only if S_w has rank one, if and only if $S_w = J_{d+1}$.

Proof. The zero translations imply $S_i u_n = u_n$, which is row-stochasticity. The block product has zero translation and top-left block $K^{-T}A_w$. Conjugating it back gives (11); the equivalences follow from $FE = I_d$. $\square$

[write] *Batch 79, 2 October 2026.* Part IV applies this doubly stochastic lift, transposed to row distributions and with a rational scale, to the commuting unipotent translation operators of a polynomial (Theorem 45.1); there the exact event is one coordinate at its equilibrium value, not rank one (Section 42.4).

The baseline lift alone is a direct transport of mortality, not the main no-line result: its generators share the stationary line $\mathbb{C}u_n$.

5 Two guards eliminate every common invariant line

The obstacle in the baseline is structural. Merely adding an affine term need not remove eigenvectors already shared by the original matrices inside H_n . We therefore reserve a tensor factor for two invertible guards.

Set

$$R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \quad (12)$$

Given $A_1, \dots, A_m \in \mathbb{Z}^{d \times d}$, take $h = 2d$, $n = 2d + 1$, and use the following $m + 2$ affine data:

$$(C_i, b_i) = (A_i \otimes I_2, 0), \quad 1 \leq i \leq m, \quad (13)$$

$$(C_R, b_R) = (I_d \otimes R, 0), \quad (14)$$

$$(C_L, b_L) = (I_d \otimes L, e_1). \quad (15)$$

The vector e_1 in (15) is the first standard basis vector of $\mathbb{Z}^{2d}$. Apply the common-scale positive affine lift to all these pairs.

For a word w in the augmented alphabet, let $\pi(w)$ be its source-letter subword and $\gamma(w)$ its guard-letter subword. Both retain chronological order. An empty source subword has product I_d .

Theorem 5.1 (Line-free mortality compiler). *The construction (13)–(15) is effective and has the following properties.*

- (i) *All $m + 2$ generators are positive rational column-stochastic matrices of size $2d + 1$, satisfying (7) and (10) for the prescribed η .*
- (ii) *The generators have no common invariant one-dimensional subspace of $\mathbb{C}^{2d+1}$.*
- (iii) *Every word w satisfies*

$$C_w = A_{\pi(w)} \otimes G_{\gamma(w)}, \quad \text{rank } S_w = 1 + 2 \text{rank } A_{\pi(w)}, \quad (16)$$

where $G_{\gamma(w)}$ is a product of R, L and is invertible.

- (iv) *The source family is mortal if and only if the target family has an erasing word, if and only if it has a uniform erasing word.*

The same rank-one existence equivalence and no-common-line property hold for the positive integer numerators $B_1, \dots, B_{m+2}$.

Proof. The first assertion is Theorem 4.1 and Proposition 4.2.

For the tensor identity, each source linear part acts only on the first tensor factor and each guard only on the second. Thus source factors commute with guard factors. Moving source factors past guards, without changing the order within either subword, gives the first identity in (16). Both R and L have determinant one. Therefore $G_{\gamma(w)}$ is invertible, including $G_\epsilon = I_2$. Multiplying on the second factor by its inverse shows

$$\text{rank}(A_{\pi(w)} \otimes G_{\gamma(w)}) = \text{rank}(A_{\pi(w)} \otimes I_2) = 2 \text{rank } A_{\pi(w)}.$$

The affine rank formula now proves (16).

We next rule out a common invariant complex line ℓ . There are two cases. If $\ell \subseteq H_n^{\mathbb{C}}$, use the E coordinates. A nonzero vector spanning it would be an eigenvector of both $I_d \otimes R$ and $I_d \otimes L$; scaling both by K^{-1} does not affect their eigenvectors. The only eigenvalue of each unipotent guard is one. Their eigenspaces are respectively

$$\mathbb{C}^d \otimes \text{span}(e_1) \quad \text{and} \quad \mathbb{C}^d \otimes \text{span}(e_2),$$

whose intersection is zero. This is impossible.

If $\ell \not\subseteq H_n^{\mathbb{C}}$, choose $v \in \ell$ with $\mathbf{1}^T v = 1$. Every column-stochastic generator acts on v by its line eigenvalue, and preservation of total mass forces that eigenvalue to be one. Write $v = Ey + u_n$. The unshifted R guard gives

$$y = K^{-1}(I_d \otimes R)y.$$

The matrix on the right has only eigenvalue $K^{-1} \neq 1$, since $K \geq 2$. Consequently $y = 0$ and $v = u_n$. But the shifted L guard sends

$$u_n \mapsto u_n + K^{-1}Ee_1 \neq u_n,$$

contradicting invariance. This proves (ii) over $\mathbb{C}$, not just over $\mathbb{Q}$.

If $A_v = 0$ for a nonempty source word v , use the same word without guards. Its affine translations are zero, so its lifted product is exactly J_n . Conversely, if S_w has rank one, (16) gives $A_{\pi(w)} = 0$. The projection cannot be empty, since $A_e = I_d \neq 0$. This establishes (iv).

Finally, each $B_i = DS_i$ with a common nonzero scalar D . Scaling a generator by a nonzero scalar preserves all its invariant subspaces, and $B_w = D^{|w|}S_w$ preserves every product rank. The integer conclusion follows. $\square$

Remark 5.2 (Why the guard arguments must both be present). The two unipotent guards without an affine displacement would still fix u_n . An affine displacement without a protected tensor factor could leave an invariant line inside the information hyperplane. Our two cases address these independent obstructions. Invertibility of the guards is equally essential: a singular guard could destroy source information and introduce false erasing words.

5.1 Small numerical consequences

Cassaigne et al. prove undecidability of mortality for the four parameter pairs $(d, m) = (3, 6), (5, 4), (9, 3), (15, 2)$ [1]. Applying the two explicit lifts gives Table 3. The entries are sufficient bounds transported from that paper, not a claim to list an optimal or exhaustive current frontier. ([write] Part II of [21] imports the same six-matrix 3×3 bound from Neary [20, Corollary 12], through binary tag systems and the Post correspondence problem for five pairs of words. Both attributions are kept; this report does not decide which publication first proved the $(3, 6)$ bound.)

Source mortality (d, m)	Doubly stochastic (n, k)	No common line (n, k)
(3, 6)	(4, 6)	(7, 8)
(5, 4)	(6, 4)	(11, 6)
(9, 3)	(10, 3)	(19, 5)
(15, 2)	(16, 2)	(31, 4)

Table 3: Undecidable exact-erasure families. Generator counts are upper bounds. All target entries may be confined to any prescribed band $((1 - \eta)/n, (1 + \eta)/n)$ with rational $0 < \eta < 1$.

Corollary 5.3. *For any fixed rational $0 < \eta < 1$, it is undecidable whether a finite product of at most eight positive rational 7×7 column-stochastic matrices has rank one, even under both promises that the entrywise band (7) holds and that the generators have no common invariant complex line. The corresponding rank-one problem for positive integer matrices with one common positive column sum is also undecidable.*

Proof. Apply Theorem 5.1 to the (3,6) mortality family. An algorithm on the promised target class would decide every source instance, since the compiler always meets the promises and preserves the answer. $\square$

This is a family of finite alphabets parameterized by the source input. It is not a claim that the displayed three-state example below, or one fixed numerical eight-generator alphabet, simulates every program. Rational precision is unbounded across input instances; fixing all numerator and denominator bounds would leave only finitely many instances.

[write] *Relation to the nonnegative-mortality boundary.* Part II of [21] proves that mortality is decidable for entrywise nonnegative matrices (Proposition `pqc:qm:prop:nonnegative`) and asks, in Question `pqc:q:signed`, for further decidable classes, among them families with a common invariant cone. The generators of Corollary 5.3 are strictly positive and share the invariant cone of nonnegative vectors, yet their rank-one problem is undecidable: the positivity boundary for the zero predicate does not transfer to rank one, because rank one in dimension n is mortality of the signed induced matrices in dimension $n - 1$ (Proposition 8.1).

6 A direct correspondence-problem compiler

To separate completeness from the numerical mortality bounds, we now give a standalone many-one reduction. The starting problem is PCP: for a finite list of pairs (u_i, v_i) , does some nonempty index word satisfy

$$u_{i_1} \cdots u_{i_T} = v_{i_1} \cdots v_{i_T}?$$

We use its classical computably enumerable completeness. The halting-to-PCP reduction is an external dependency, not re-proved here; see the formal treatment of PCP-related reductions by Forster, Heiter, and Smolka [2]. Restriction to a two-letter alphabet does not lose generality: encode a finite alphabet injectively by equal-length binary codewords, which preserves equality of all concatenations.

6.1 A four-dimensional integer front end

Use letters a, b as digits 1, 2 in base three, read most significant digit first. Write $\text{val}(s)$ for the resulting nonnegative integer, including $\text{val}(\epsilon) = 0$. This encoding is injective, and

$$\text{val}(st) = 3^{|t|} \text{val}(s) + \text{val}(t). \quad (17)$$

Define

$$M_i = \begin{pmatrix} 3^{|u_i|} & 0 & \text{val}(u_i) & 0 \\ 0 & 3^{|v_i|} & \text{val}(v_i) & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad c = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad \ell = \begin{pmatrix} 1 & -1 & 0 & 1 \end{pmatrix}, \quad Q = c\ell. \quad (18)$$

The letter q denotes the extra generator Q . It is deliberately idempotent: $\ell c = 1$ and $Q^2 = Q \neq 0$.

Lemma 6.1 (PCP block identity). *For a nonempty source word $v = i_1 \cdots i_T$, let $U_v = u_{i_1} \cdots u_{i_T}$ and $V_v = v_{i_1} \cdots v_{i_T}$. Then*

$$M_v = \begin{pmatrix} 3^{|U_v|} & 0 & \text{val}(U_v) & 0 \\ 0 & 3^{|V_v|} & \text{val}(V_v) & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \ell M_v c = \text{val}(U_v) - \text{val}(V_v).$$

The corresponding scalar for the empty word is $\ell I_4 c = 1$, not zero.

Proof. The product convention is chronological. Multiplying a later tile on the left updates the first translation by $3^{|u_i|} \text{val}(U) + \text{val}(u_i) = \text{val}(Uu_i)$, and similarly for the second. Induction proves the matrix formula. The scalar follows by direct multiplication. For the empty word use $M_\epsilon = I_4$; the block formula with a zero final diagonal entry is asserted only for a nonempty word. $\square$

Theorem 6.2 (Full-word PCP reduction). *The integer alphabet $M_1, \dots, M_m, Q$ is mortal if and only if the PCP instance has a solution. More precisely, write any word containing $s \geq 1$ connector letters as*

$$w = v_0 q v_1 q \cdots q v_s,$$

where each v_j is a possibly empty source word. Its product is zero if and only if at least one internal block v_j , $1 \leq j < s$, is a nonempty PCP solution. Words without a connector never have zero product.

Proof. By $Q = c\ell$ and the chronological convention,

$$M_w = (M_{v_s} c) \left(\prod_{j=1}^{s-1} \ell M_{v_j} c \right) (\ell M_{v_0}). \quad (19)$$

Both endpoint vectors are nonzero. For a nonempty v_s , the third coordinate of $M_{v_s} c$ is one; for an empty block it is the nonzero vector c . For nonempty v_0 , the first coordinate of ℓM_{v_0} is $3^{|U_{v_0}|} > 0$; for an empty block it is $\ell \neq 0$. Thus the outer product in (19) is nonzero, and vanishing occurs exactly when an internal scalar is zero. Lemma 6.1 identifies those zero scalars with nonempty PCP matches; an empty internal block contributes one. Without Q , the third diagonal entry of every nonempty product is one, excluding mortality.

Conversely, a PCP match v gives the zero word qvq . The entire map from tiles to matrices is an explicit computable map, so this is a many-one reduction. $\square$

The fourth coordinate has a specific purpose. A more naive connector based on a scalar comparison that vanishes for the empty word would have $Q^2 = 0$, making every instance mortal. Here the extra coordinate is killed by the first genuine tile but makes the empty-block scalar one. This is a boundary condition, not a cosmetic padding coordinate.

[write] The three-dimensional radix-three tiles are the classical encoding that the research note [25] also uses, there with a leading sentinel digit so that the empty word has a nonzero code. The fourth coordinate and the idempotent connector $Q = c\ell$, $\ell c = 1$, are this source's own.

Corollary 6.3 (A many-one complete stochastic substrate). *For any fixed rational $0 < \eta < 1$, exact erasure is computably enumerable complete under many-one reductions for positive rational 9×9 column-stochastic families with a variable-size alphabet, even with no common invariant complex line and with the entrywise band (7).*

Proof. A PCP instance with m tiles maps by Theorem 6.2 to $m + 1$ integer 4×4 matrices, then by Theorem 5.1 to $m + 3$ stochastic 9×9 matrices. Both maps are many-one and preserve the exact answer. Erasure is computably enumerable: enumerate all finite words and multiply their rational matrices exactly. This proves completeness on the promised class. Equivalently, one may regard the source tile list as the input parameter of the compiled family, so promise validation is not part of the reduction. $\square$

This gives a precise sense in which the substrate supports universal finite computation. A halting instance can be transformed into a finite rational gate alphabet such that a finite exact-forgetting schedule exists exactly when the computation halts. It does not supply an autonomous universal stochastic machine with one fixed gate alphabet and a separate ordinary input loader.

7 Excluding all small common invariant subspaces

The tensor guard method is not limited to lines. Let $r \geq 2$, and let P be the cyclic permutation matrix on $\mathbb{C}^r$, while

$$Z_r = \text{diag}(1, 2, \dots, r).$$

For an integer mortality instance of dimension d , use $h = rd$, $n = rd + 1$, and

$$(A_i \otimes I_r, 0), \quad (I_d \otimes P, 0), \quad (I_d \otimes Z_r, e_1). \quad (20)$$

Apply the same common-scale lift. The name Z_r here denotes a diagonal matrix, not an integer witness domain.

Lemma 7.1 (The guard algebra). *The complex unital algebra generated by P and Z_r is $\text{Mat}_r(\mathbb{C})$. Every subspace of $\mathbb{C}^d \otimes \mathbb{C}^r$ invariant under $I_d \otimes P$ and $I_d \otimes Z_r$ has the form $W \otimes \mathbb{C}^r$ for some $W \leq \mathbb{C}^d$.*

Proof. Since the diagonal entries of Z_r are distinct, Lagrange interpolation produces the coordinate projections:

$$E_{jj} = \prod_{a \neq j} \frac{Z_r - aI}{j - a}.$$

For each i, j choose a power P^t taking e_j to e_i . Then $E_{ii}P^tE_{jj} = E_{ij}$, so the generated algebra contains every matrix unit.

A subspace invariant under the two generators is invariant under their products and complex linear combinations, hence under all $I_d \otimes E_{ij}$. Define $W = \{v : v \otimes e_1 \text{ belongs to the subspace}\}$. Applying $I_d \otimes E_{1j}$ to $\sum_j v_j \otimes e_j$ shows that every coefficient v_j belongs to W . Conversely, applying $I_d \otimes E_{j1}$ shows that $W \otimes \mathbb{C}^r$ is contained in the subspace. This proves equality. $\square$

Theorem 7.2 (Invariant-subspace and product-rank gaps). *For the lift of (20), every common invariant complex subspace has dimension congruent to 0 or 1 modulo r . There is no common invariant line and, consequently, no nonzero common invariant subspace of any dimension $1, \dots, r-1$. Every product satisfies*

$$\text{rank } S_w = 1 + r \text{rank } A_{\pi(w)}. \quad (21)$$

In particular, a product is either rank one or has rank at least $r+1$. Mortality, erasure, and existence of a uniform erasing word remain equivalent.

Proof. Both P and Z_r are invertible. The source/guard tensor factorization therefore proves (21) exactly as before. Zero translations on source-only words give the uniform-target equivalence.

Let U be a common invariant subspace. Its intersection with H_n^C is invariant under the linear guard restrictions and has dimension divisible by r , by Lemma 7.1. If $U \subseteq H_n^C$, this is its full dimension. Otherwise, restriction of the nonzero mass functional to U has a one-dimensional image, so $\dim U = \dim(U \cap H_n^C) + 1$. This proves the congruence statement.

A common invariant line cannot be inside H_n^C because its dimension would be a positive multiple of r . Outside H_n^C normalize a spanning vector to mass one. The unshifted P guard forces its transverse part y to satisfy $y = K^{-1}(I_d \otimes P)y$. Every eigenvalue of P has modulus one, whereas $K > 1$, so $y = 0$. The translated Z_r guard does not fix u_n , again a contradiction. Excluding dimension one and using the congruence statement excludes all positive dimensions smaller than r . $\square$

The rank gap makes the exactness phenomenon sharper: products cannot drift through ranks $2, \dots, r$ on their way to erasure. Their entries can nevertheless approach a rank-one matrix arbitrarily closely without ever becoming rank one. The common hyperplane of dimension rd remains, so the theorem must not be reported as an irreducible-semigroup result.

For example, applying this construction to the $d = 3$, $m \leq 6$ mortality bound gives undecidable erasure for at most eight positive stochastic matrices in dimension $3r + 1$, with no invariant subspace of dimensions $1, \dots, r - 1$. Applying it to the explicit PCP front end gives many-one completeness in dimension $4r + 1$ with variable alphabet size.

8 Decidable boundaries and a sharp commuting horizon

8.1 An exact dimension correspondence

Proposition 8.1. *For $n \geq 2$, rank-one product existence for rational column-stochastic $n \times n$ matrices reduces effectively, without changing the number of generators, to mortality for integer $(n - 1) \times (n - 1)$ matrices. Conversely, integer mortality in dimension $n - 1$ reduces to erasure for positive rational doubly stochastic matrices of dimension n , in any prescribed band (7).*

Proof. For the first direction, compute $C_i = FS_iE$ using the explicit rational coordinates of Lemma 3.4. The restriction of S_w to H_n has coordinate matrix C_w , so erasure is equivalent to $C_w = 0$. Clear the rational denominators in each C_i , or use one common positive denominator. Every product is merely multiplied by a nonzero scalar, preserving zero products. The converse is Corollary 4.3. $\square$

This correspondence is useful independently of particular known dimension bounds. For example, the three-state positive doubly stochastic erasure problem has exactly the corresponding two-dimensional mortality difficulty. We do not infer a current open/closed classification of two-dimensional mortality from a dated source; the reduction equivalence itself is the assertion here.

8.2 Commuting information dynamics

The affine stochastic generators need not commute in the next theorem. It is enough that their restrictions to the zero-mass hyperplane commute.

Theorem 8.2 (Sharp commuting horizon). *Let $S_1, \dots, S_k$ be rational column-stochastic $n \times n$ matrices, $n \geq 2$. Suppose that the induced $(n-1)$ -dimensional matrices $C_i = FS_iE$ commute pairwise. If an erasing word exists, one of length at most $n-1$ exists. A finite exact rank-reduction algorithm finds such a word or certifies that no erasing word exists. The bound $n-1$ is sharp, even for a single positive doubly stochastic generator.*

Proof. Set $h = n-1$ and start with $P = I_h$, $V = \text{im } P = \mathbb{Q}^h$. Since P is a product of commuting C_i , every C_i leaves V invariant. If some generator satisfies $\text{rank}(C_i P) < \text{rank } P$, replace P by $C_i P$, append that letter, and continue. Each such step strictly decreases the nonnegative integer $\dim V$, so at most h steps are possible.

If $P = 0$, the chosen word is erasing. Otherwise suppose that no generator lowers the rank. Every $C_i : V \rightarrow V$ then has full rank on V and is invertible there. All products remain invertible on this nonzero space, so none can be the zero matrix on $\mathbb{Q}^h$. Thus failure to find a further rank-lowering generator is a correct no-erasure certificate. Conversely, if any zero product exists it annihilates V , which is impossible while all $C_i|_V$ are invertible. Hence the algorithm cannot stop at a nonzero V in a mortal instance.

For sharpness, take the $h \times h$ nilpotent shift N with ones immediately above the diagonal. It has $N^h = 0$ and $N^t \neq 0$ for $0 \leq t < h$. The single-generator positive doubly stochastic lift of N therefore first erases at time $h = n-1$. $\square$

The algorithm uses rational matrix multiplication and Gaussian rank tests; it has no unbounded word search. The proof does not assume that a rank reduction already makes a matrix zero, and it allows arbitrary choices among available rank-lowering generators. The common-invariant-image property provided by commutation is what makes these greedy choices safe.

For one generator, erasure is exactly nilpotency of C , and $C^{n-1} = 0$ is a complete test. For $n = 2$, every induced C_i is a scalar, so erasure occurs if and only if at least one generator already has rank one. These are genuine finite boundaries, not numerical heuristics for the undecidable cases.

[write] *Batch 79, 2 October 2026.* Commutation decides erasure, not a single-coordinate hit. Part IV builds commuting, invertible, strictly positive doubly stochastic alphabets, on which no word erases (Proposition 45.5), and on which “some word brings the accepting coordinate exactly to $1/s$ ” is computably enumerable complete for a fixed alphabet (Theorems 47.2 and 47.4).

9 Random schedules, exact forgetting, and a hidden-state interpretation

Now choose each letter independently with probabilities $p_1, \dots, p_k > 0$, $\sum_i p_i = 1$. Define the exact-forgetting time

$$\tau = \inf\{T \geq 1 : \text{rank}(S_{I_T} \cdots S_{I_1}) = 1\},$$

with $\inf \emptyset = \infty$. The event concerns exact equality, not a numerical distance threshold.

Theorem 9.1 (A zero–one law for finite exact forgetting). *For a finite column-stochastic alphabet under independent full-support switching, the following conditions are equivalent:*

- (i) *an erasing word exists;*
- (ii) $\mathbb{P}(\tau < \infty) > 0$;

(iii) $\mathbb{P}(\tau < \infty) = 1$;

(iv) $\mathbb{E}\tau < \infty$.

If v is an erasing word of length L and $q = \prod_{j=1}^L p_{v_j} > 0$, then

$$\mathbb{P}(\tau > t) \leq (1 - q)^{\lfloor t/L \rfloor}, \quad \mathbb{E}\tau \leq L/q. \quad (22)$$

For uniform letter probabilities, $q = k^{-L}$.

Proof. If no erasing word exists, no finite prefix can have rank one, so $\tau = \infty$ for every schedule. Suppose instead that an erasing word v of length L exists. Partition the random schedule into successive disjoint blocks of length L . Each block equals v with probability q , independently of the others. The first matching block occurs after a geometric number G of blocks, with $\mathbb{E}G = 1/q$. By Lemma 3.3, regardless of the preceding prefix, that block makes the entire prefix product rank one. Hence $\tau \leq LG$, which proves (22) and finite expectation. Finite expectation implies almost-sure finiteness, and almost-sure finiteness implies positive probability. These implications establish all equivalences. $\square$

Corollary 9.2. *On the seven-state promised class of Corollary 5.3, both almost-sure finite exact forgetting and finiteness of its expected time are undecidable. They are many-one complete on the variable-alphabet nine-state family of Corollary 6.3. Nevertheless, the deterministic approximation bound (10) holds for every schedule on every instance.*

Proof. Theorem 9.1 identifies each of these random-schedule predicates with existence of a finite erasing word. The reductions and contraction estimates already proved therefore apply unchanged. $\square$

There is no contradiction with the higher logical complexity of many other almost-sure termination questions. Here a single finite erasing block is an absorbing certificate that recurs in independent full-support sampling. That special structure collapses the almost-sure predicate to an existential finite-word predicate. Removing independence, full support, or the absorbing rank-one property changes the argument.

[write] Part IX of [21] (Theorem `pqc:iq:thm:geometric`) also combines a geometric stopping clock with an undecidable exact event, for completed records of rational quantum instruments; the statements are different and neither implies the other.

9.1 A labelled hidden-state model with trivial output statistics

The stochastic interpretation can be made into a finite labelled transition model. For label i , use the nonnegative transition matrix

$$T_i = p_i S_i.$$

Since $\sum_i T_i$ is column-stochastic, these matrices specify joint probabilities of the next hidden state and the emitted label. Conditional on any current hidden state, label i has probability p_i , because $\mathbf{1}^\top T_i = p_i \mathbf{1}^\top$. Thus the emitted labels are independent with the prescribed common distribution.

After observing a word $w = i_1 \cdots i_T$, its probability is $\prod_t p_{i_t}$, independent of the initial hidden distribution. The normalized hidden-state update is exactly S_w . Consequently, exact independence of the posterior from the initial hidden distribution is precisely rank-one product existence.

The output entropy rate in this particular construction is simply $-\sum_i p_i \log p_i$. It is not an undecidable quantity. This observation is important: even when output statistics are completely elementary, deciding whether filtering ever forgets its initial distribution *exactly* can be undecidable. The result addresses exact filtering, not the entropy question that motivated the MathOverflow discussion.

9.2 Finite-horizon probabilities

Under uniform switching, let N_T be the number of labelled words of length T with rank-one product. By persistence,

$$\mathbb{P}(\tau \leq T) = N_T/k^T. \quad (23)$$

This is an exact rational number computable by finite enumeration. The next section realizes N_T as the number of natural solutions of an explicit quartic. Duplicate numerical generators, when present, remain distinct labels; this matches the probability model and the certificate count.

10 Explicit natural-domain Diophantine certificates

Fix a numerical input alphabet $B_1, \dots, B_k \in \mathbb{Z}_{>0}^{n \times n}$, each with column sum $D > 0$, and set $S_i = B_i/D$. In this section $T \geq 1$ is an externally supplied horizon. A polynomial is produced for that horizon; its arity is not fixed as T varies.

10.1 A full-prefix certificate with no minors

Introduce natural selector variables $e_{t,i}$, $1 \leq t \leq T$, $1 \leq i \leq k$, and natural prefix matrices $X_t \in \mathbb{N}^{n \times n}$. The matrix $X_0 = I_n$ is constant. Use residuals

$$s_t = \sum_{i=1}^k e_{t,i} - 1, \quad (24)$$

$$r_{t,ab} = (X_t)_{ab} - \sum_{i=1}^k e_{t,i} \sum_{c=1}^n (B_i)_{ac} (X_{t-1})_{cb}, \quad (25)$$

$$q_{ab} = (X_T)_{ab} - (X_T)_{a1}, \quad 1 \leq a \leq n, \quad 2 \leq b \leq n. \quad (26)$$

Define the integer polynomial

$$\mathcal{Q}_T = \sum_{t=1}^T s_t^2 + \sum_{t=1}^T \sum_{a,b=1}^n r_{t,ab}^2 + \sum_{a=1}^n \sum_{b=2}^n q_{ab}^2. \quad (27)$$

Theorem 10.1 (Word-bijective full certificate). *The polynomial $\mathcal{Q}_T$ has total degree at most four in its natural witnesses. Its natural zero set is in bijection with the labelled erasing words of length T . It has exactly*

$$T(n^2 + k) \quad \text{witness variables,} \quad T(n^2 + 1) + n(n - 1) \quad \text{squared residuals.} \quad (28)$$

Proof. A sum of integer squares vanishes exactly when every residual vanishes. Natural selectors with sum one have exactly one entry equal to one and every other entry zero. No separate Boolean equation is needed. Let i_t be that unique selected letter. The prefix equations then force, successively,

$$X_t = B_{i_t} \cdots B_{i_1}.$$

All these entries are natural, and each column sum equals D^t . Thus the terminal equations say precisely that the stochastic matrix X_T/D^T has equal columns, which is equivalent to erasure by Lemma 3.2.

Conversely, each erasing word supplies those one-hot selectors and exact prefix products, and all residuals vanish. The selectors and the recurrence determine every witness uniquely. Different labelled words have different selector tuples, even when some numerical generators coincide. This proves the bijection. The prefix residuals have degree at most two because B_i are fixed numerical coefficients, and all other residuals are linear. The displayed counts simply count selectors, matrix entries, and equations. $\square$

The terminal test uses equal columns rather than 2×2 minors. It is sound because the common, nonzero column sum has already been established by the retained recurrence. If the target is specifically J_n , replace (26) by

$$(X_T)_{ab} - (X_T)_{11} = 0 \quad ((a, b) \neq (1, 1)).$$

This uses $n^2 - 1$ terminal residuals. It is unnecessary in the doubly stochastic case, but it distinguishes a general rank-one product from the uniform matrix in the line-free construction.

10.2 A smaller certificate in information coordinates

Mass conservation allows a substantial further reduction. Put $h = n - 1$ and compute the integer restriction matrices

$$(H_i)_{ac} = (B_i)_{ac} - (B_i)_{an}, \quad 1 \leq a, c \leq h. \quad (29)$$

These may have negative entries. They are the coordinates of the action of B_i on the zero-mass hyperplane: $B_i E = E H_i$. For a prefix product X_t ,

$$(H_{i_t} \cdots H_{i_1})_{ab} = (X_t)_{ab} - (X_t)_{an}. \quad (30)$$

Each entry on the right lies between $-D^t$ and D^t , since both nonnegative entries belong to columns of sum D^t .

Introduce natural matrices $Y_t \in \mathbb{N}^{h \times h}$ and use the fixed shifts

$$p_t = D^t \quad (0 \leq t \leq T), \quad Y_0 = I_h + \mathbf{1}_h \mathbf{1}_h^T.$$

The intended value is

$$(Y_t)_{ab} = (H_{i_t} \cdots H_{i_1})_{ab} + p_t.$$

Retain the selector residuals s_t and define

$$\rho_{t,ab} = (Y_t)_{ab} - p_t - \sum_{i=1}^k e_{t,i} \sum_{c=1}^h (H_i)_{ac} ((Y_{t-1})_{cb} - p_{t-1}), \quad (31)$$

$$\theta_{ab} = (Y_T)_{ab} - p_T. \quad (32)$$

Let

$$\mathcal{I}_T = \sum_t s_t^2 + \sum_{t,a,b} \rho_{t,ab}^2 + \sum_{a,b} \theta_{ab}^2. \quad (33)$$

Theorem 10.2 (Mass-compressed natural quartic). *The natural zeros of $\mathcal{I}_T$ are in bijection with the labelled erasing words of length T . Every residual has degree at most two, and the polynomial has degree at most four. Its exact counts are*

$$\boxed{T((n-1)^2 + k) \text{ witnesses,} \quad T((n-1)^2 + 1) + (n-1)^2 \text{ residuals.}} \quad (34)$$

There is a canonical bijection between the natural zero sets of $\mathcal{Q}_T$ and $\mathcal{I}_T$, preserving the selected labelled word.

Proof. The selectors again determine a unique word. Write $Z_t = Y_t - p_t \mathbf{1}_h \mathbf{1}_h^\top$. Equation (31) becomes $Z_t = H_{i_t} Z_{t-1}$, with $Z_0 = I_h$. Hence every Y_t is forced uniquely. The intended values are natural because (30) bounds each entry of Z_t below by $-D^t$. Thus no additional range witnesses or nonnegativity slack equations are needed.

The terminal residuals are zero exactly when $Z_T = 0$, which means that the full product vanishes on the zero-mass hyperplane. Lemma 3.2 identifies this with rank-one product existence. This proves both directions and the word-bijection. The counts and degree bounds follow directly from the formulas.

For the zero-set bijection, keep the selectors and map a full prefix certificate to the compressed one by

$$(Y_t)_{ab} = (X_t)_{ab} - (X_t)_{an} + D^t.$$

On valid zeros these values are natural. Conversely, the selectors in a compressed zero uniquely reconstruct the full prefix products. Both directions preserve the word and are inverse on the zero sets. $\square$

This is a bijection on the valid natural zero sets, not an identity between the two polynomials away from zero. The map from arbitrary full witness assignments need not preserve nonnegativity; only its zero-set behavior is asserted. The distinction matches the source-audit discipline emphasized in ProveIt’s October 2 review [6].

For the seven-state, eight-letter construction, the full certificate uses $57T$ witnesses and $50T + 42$ residuals. The compressed certificate uses

$$\boxed{44T \text{ natural witnesses and } 37T + 36 \text{ residuals.}} \quad (35)$$

The improvement is $13T$ witnesses and $13T + 6$ residuals. No assertion about an optimal arithmetic circuit or a smallest possible polynomial follows from these ledgers.

The constants $p_t = D^t$ are computed for the supplied horizon. Their bit lengths are $O(t \log D)$, and printing all of them has a horizon-dependent cost. They are not an uncharged fixed-arity exponentiation primitive. The next subsection makes the uniform-input version polynomial without treating those powers as free input constants.

10.3 When the numerical matrices become polynomial parameters

A degree statement must specify whether matrix entries are coefficients or variables. In (25), treating $(B_i)_{ac}$ as polynomial input parameters turns a term $e_{t,i}(B_i)_{ac}(X_{t-1})_{cb}$ into a cubic. It would therefore be incorrect to reuse the full-prefix quartic degree claim unchanged in that setting.

Here is a fully uniform compressed alternative for fixed n, k, T . Let the positive entries of B_i and the common column sum D be input parameters, subject to the stated stochastic-numerator

promise. Introduce natural variables $z_t, W_t \in \mathbb{N}^{h \times h}$, and $Y_t \in \mathbb{N}^{h \times h}$ at each step, in addition to the selectors. Set $z_0 = 1$ and the same constant Y_0 as before. Use the residuals

$$z_t - Dz_{t-1} = 0, \quad (36)$$

$$(W_t)_{ac} - \sum_i e_{t,i}((B_i)_{ac} + D - (B_i)_{an}) = 0, \quad (37)$$

$$(Y_t)_{ab} - z_t - \sum_c ((W_t)_{ac} - D)((Y_{t-1})_{cb} - z_{t-1}) = 0, \quad (38)$$

along with the selector equations and terminal equations $(Y_T)_{ab} - z_T = 0$. Square and sum all these residuals.

Proposition 10.3 (Uniform-input quartic). *On the positive, common-column-sum input domain, the resulting polynomial has total degree at most four, including its matrix and denominator input parameters. It has exactly*

$$T(k + 2(n-1)^2 + 1) \quad \text{natural witnesses} \quad (39)$$

and

$$T(2(n-1)^2 + 2) + (n-1)^2 \quad \text{squared residuals.} \quad (40)$$

Its natural zeros are again in bijection with the labelled erasing words of length T .

Proof. Every displayed residual is quadratic or linear in all inputs and witnesses. One-hot selectors force $z_t = D^t$ and

$$(W_t)_{ac} = (H_{i_t})_{ac} + D.$$

Since $0 \leq (B_i)_{ab} \leq D$, these selected values are natural. The Y recurrence then reduces exactly to (31), so the preceding proof supplies uniqueness and naturalness of the remaining witnesses. There are $k + 2h^2 + 1$ variables and $2h^2 + 2$ residuals per step, and h^2 terminal residuals, yielding the counts. $\square$

The input promises themselves are decidable finite equalities and inequalities. They are not counted as witnesses in (39). To make a polynomial whose accepted inputs also enforce these promises, include their finite arithmetic conditions in the input-domain representation; one should then charge those additional conditions separately. This proposition supplies the uniform schema and its proof. The shipped numerical certificate evaluators implement Theorems 10.1 and 10.2, not an expanded symbolic version of Proposition 10.3.

10.4 Exact counts of solutions and limits of uniqueness

The two numerical quartics have at most k^T natural zeros, exactly one per erasing labelled word. Combining the bijection with (23) gives

$$\mathbb{P}(\tau \leq T) = \frac{\#\{x \in \mathbb{N}^{T((n-1)^2+k)} : \mathcal{I}_T(x) = 0\}}{k^T}. \quad (41)$$

This identity is about a finite solution set for one externally fixed horizon. It is not a finite-fold Diophantine representation of arbitrary computably enumerable sets. A different polynomial is constructed for every T , and an unbounded union over those polynomials is not automatically one fixed-arity finite-fold polynomial.

11 Fixed unbounded Diophantine representations and impossibility of bounds

11.1 What DPRM supplies, and what it does not

Fix an effective encoding $e \in \mathbb{N}$ of finite rational stochastic alphabets. Validity of the finite rational data, positivity, and common-column-sum claims is decidable. The set of valid codes possessing an erasing word is computably enumerable by exact word enumeration. The Davis–Putnam–Robinson–Matiyasevich (DPRM) theorem identifies computably enumerable sets with Diophantine sets [3]. Hence there is an integer polynomial P and a fixed number of natural witnesses such that

$$e \text{ admits exact erasure} \iff \exists x_1, \dots, x_s \in \mathbb{N} : P(e, x_1, \dots, x_s) = 0. \quad (42)$$

The same conclusion holds for the compiled mortality or PCP input predicates.

Proposition 11.1 (Fixed-arity quartic consequence). *The unbounded erasure predicate has a fixed-arity integer-polynomial representation of total degree at most four over natural witnesses.*

Proof. Start from (42). Split $P = P_+ - P_-$, where P_+ and P_- have nonnegative integer coefficients. Evaluate these two polynomials by finite addition/multiplication circuits over natural numbers. Introduce one natural wire variable for each gate output and impose the appropriate equation $z - u - v = 0$, $z - uv = 0$, or $z - c = 0$. Identify the two final outputs by a linear equation. Squaring and summing all gate and final-output equations gives total degree at most four, including e as a parameter. For fixed original witnesses, all new circuit-wire values are unique and natural. Conversely, any zero of the sum of squares reproduces the two evaluations and satisfies $P = 0$. The circuit and its arity are fixed because P is fixed. $\square$

This is an existence-and-compilation theorem, not a printed small universal polynomial. It leaves the number and multiplicity of the original DPRM witnesses uncontrolled. The explicit horizon-dependent polynomials of the previous section are independent constructions with better traceability, not substitutes for this fixed-arity argument. None of these observations resolves a general finite-fold representation problem.

[write] The circuit quadratization in this proof is the unique-extension quadratization `cdc:wf:lem:quadratization` of [22], and the quartic lowering `pgc:lem:quarticlower` of [21]; it is printed as the source’s own route, without a novelty claim. Part II proves the same corollary for its own predicate (Section 27).

11.2 No uniform computable exactness horizon

Let the size of an input be the bit length of its finite binary encoding. The following statement applies to either the fixed seven-state undecidable family or the nine-state many-one complete family. ([write] “The fixed seven-state undecidable family” means the seven-state, at-most-eight-letter promised class of Corollary 5.3: a class of input-dependent alphabets of fixed dimension, not one fixed alphabet, as the paragraph after that corollary explains.)

Theorem 11.2 (No computable yes-instance horizon). *There is no total computable function $b : \mathbb{N} \rightarrow \mathbb{N}$ such that every erasing input of bit length N has an erasing word of length at most $b(N)$. The conclusion remains true at every fixed rational contraction factor $0 < \eta < 1$ used above.*

Proof. Such a bound would decide erasure: compute $b(N)$ and enumerate all words up to that length, multiplying exactly and checking equal columns. If a word erases, answer yes; otherwise the assumed bound permits a no answer. Applied to the compiler output, this would decide source mortality or PCP, contradicting the proved reduction and its external undecidability dependency. $\square$

By contrast, approximate forgetting has the uniform computable horizon $\eta^T \leq \varepsilon$, independently of the matrix coefficients. The undecidability concerns exact zero information, not the task of reaching any fixed positive tolerance.

[write] The proof of Theorem 11.2 is the bounded-search argument of `cdc:bd:prop:nobound` in [22] and `pqc:qm:cor:nobound` in [21], applied to exact erasure.

Corollary 11.3 (No computable uniform finite-expectation bound). *Under uniform independent switching, there is no total computable bound in the input bit length for $\mathbb{E}\tau$ on all inputs where that expectation is finite. Nor is there a computable function $b(N)$ guaranteeing $\mathbb{P}(\tau \leq b(N)) \geq 1/2$ on every such input.*

Proof. Let $L_{\min}$ be the shortest erasing-word length. Every finite value of τ is at least $L_{\min}$, so a finite upper bound for $\mathbb{E}\tau$ bounds $L_{\min}$. A bound with positive probability of erasure by a given horizon also implies existence of a word by that horizon. Either would contradict Theorem 11.2 after taking an integer ceiling when necessary. $\square$

For any fixed arity representation such as (42), an analogous argument excludes a computable bound on all necessary natural witness coordinates in terms of the input size. A complete computably enumerable family of *non-erasure* certificates is also impossible on the undecidable class: parallel enumeration of yes and no certificates would be a decision algorithm. These are uniform impossibility statements, not obstacles to finding useful certificates or tight bounds for particular instances.

12 Exact erasure is not robust under perturbation

The preceding hardness holds arbitrarily close to the uniform matrix. That is a statement about the location of the input family, not robustness of individual yes instances. Indeed, erasure is destroyed by arbitrarily small perturbations that preserve the invariant-subspace restrictions.

Theorem 12.1 (Subspace-preserving non-erasing perturbations). *Let $S_1, \dots, S_k$ be positive rational column-stochastic matrices of dimension $n \geq 2$. For arbitrarily small positive rational δ , all matrices*

$$S_i^{(\delta)} = (1 - \delta)S_i + \delta I_n \quad (43)$$

are invertible. They remain positive and column-stochastic, have exactly the same common invariant complex subspaces as the original family, and have no erasing word. If the original family contracts total variation by at most η , the perturbed family contracts it by at most $\eta_\delta = (1 - \delta)\eta + \delta < 1$ for $0 < \delta < 1$.

Proof. For each i , $f_i(\delta) = \det((1 - \delta)S_i + \delta I_n)$ is a polynomial and $f_i(1) = 1$. It therefore has only finitely many roots. Outside the finite union of these root sets, all perturbed generators are invertible. Every interval $(0, \varepsilon)$ contains a positive rational outside that finite set. A finite product of invertible matrices is invertible and, since $n \geq 2$, cannot have rank one.

Positivity and column-stochasticity follow immediately from the convex combination for $0 < \delta < 1$. For every subspace U , invariance under S_i implies invariance under $S_i^{(\delta)}$. Conversely,

$$S_i = \frac{S_i^{(\delta)} - \delta I_n}{1 - \delta}$$

shows the reverse implication whenever $\delta \neq 1$. Applying this to every generator proves equality of the common invariant-subspace families. Finally, for $z = p - q$,

$$\|S_i^{(\delta)} z\|_1 \leq (1 - \delta)\eta\|z\|_1 + \delta\|z\|_1,$$

which proves the contraction claim. $\square$

For a prescribed open entrywise band, first choose a compiled family strictly inside that band. Its positive distance from the finitely many band endpoints allows sufficiently small perturbations to remain inside it. Thus even within the line-free, strongly contracting promised class, non-erasing instances occur arbitrarily close to every constructed erasing instance.

This theorem also explains why floating-point rank tests cannot certify exact erasure. A determinant or singular value that is numerically tiny can be nonzero for a nearby rational input with a completely different exact answer. The shipped checks therefore use integer arithmetic and exact rational Gaussian elimination, with explicit rejection of floating-point and Boolean values in integer witness fields.

13 Worked examples and executable verification

13.1 A completely printed three-state example

Take the scalar source alphabet $A_0 = [0]$, tensor degree two, and $\eta = 1/2$. The compiler gives $K = 13$, $D = 39$, and

$$B_0 = \begin{pmatrix} 13 & 13 & 13 \\ 13 & 13 & 13 \\ 13 & 13 & 13 \end{pmatrix}, \quad B_R = \begin{pmatrix} 14 & 14 & 11 \\ 12 & 15 & 12 \\ 13 & 10 & 16 \end{pmatrix}, \quad B_L = \begin{pmatrix} 18 & 15 & 15 \\ 14 & 14 & 11 \\ 7 & 10 & 13 \end{pmatrix}. \quad (44)$$

Every column sums to 39, every entry divided by 39 lies between $1/6$ and $1/2$, and the family has no common invariant complex line by the guard proof. The source generator is already $S_0 = J_3$; this is a readable sanity-check instance, not a universal numerical machine.

The chronological word $(0, L)$ has numerator product

$$B_L B_0 = \begin{pmatrix} 624 & 624 & 624 \\ 507 & 507 & 507 \\ 390 & 390 & 390 \end{pmatrix}$$

and denominator $39^2 = 1521$. Thus its stochastic product is

$$\begin{pmatrix} 16/39 \\ 13/39 \\ 10/39 \end{pmatrix} \mathbf{1}_3^\top,$$

which is rank one but not J_3 . This explicitly demonstrates why the line-free construction requires a distinction between erasure and a specified uniform output, even though existence of the two kinds of word is equivalent.

The full horizon-two certificate has 24 natural witnesses and 26 residuals. Its compressed version has 14 witnesses and 14 residuals. For the compressed certificate, the restriction matrices are

$$H_0 = 0, \quad H_R = 3R, \quad H_L = 3L.$$

After the source letter, the information restriction is zero and remains zero. Hence the two Y matrices are respectively $39\mathbf{1}_2\mathbf{1}_2^\top$ and $152\mathbf{1}_2\mathbf{1}_2^\top$. These are the exact values in the delivered JSON certificate, not floating-point approximations.

Under uniform random switching, erasure occurs precisely when letter 0 first appears. Therefore

$$N_T = 3^T - 2^T, \quad \mathbb{P}(\tau \leq T) = 1 - (2/3)^T, \quad \mathbb{E}\tau = 3.$$

The enumerated values for $T = 1, \dots, 5$ are 1, 5, 19, 65, 211, matching the certificate counts. This instance also permits direct checks of the zero-one law and the difference between a uniform erasing product and an arbitrary rank-one product.

13.2 A seven-state, eight-letter numerical packet

The delivered seven-state instance has source dimension three and six labelled source generators:

$$N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad I_3, \quad -I_3, \quad \text{diag}(2, 1, 1), \quad I_3 + E_{12}, \quad I_3 + E_{23}.$$

The two tensor guards give eight target letters. With $\eta = 1/10$, the compiler selects $K = 131$ and $D = 917$. All 392 integer entries of the eight numerator matrices are printed in `examples/seven_state_instance.json`. ([write] Shipped as `data/02-exact-erasure-seven_state_instance.json`; every `examples/*.json` file of source 02 is shipped as `data/02-exact-erasure-*.json`.)

The chronological word (N, N, N) erases because $N^3 = 0$, while neither N nor N^2 is zero. The numerator product is

$$B_N^3 = 110156459 \mathbf{1}_7 \mathbf{1}_7^\top, \quad D^3 = 771095213 = 7 \cdot 110156459.$$

The full certificate has 171 witnesses and 192 residuals. The compressed certificate has 132 witnesses and 147 residuals. Both independent evaluations return polynomial value exactly zero. A separate uniform-target full certificate checks that this source-only word produces precisely J_7 .

This finite example illustrates the format and counts of the undecidable class. It does not prove undecidability by enumeration; that conclusion rests on the all-input reduction and the cited mortality theorem.

13.3 End-to-end PCP example

Use the two tiles

$$(u_0, v_0) = (a, ab), \quad (u_1, v_1) = (ba, a).$$

The index word $(0, 1)$ is a match because both concatenations are aba . The four-dimensional front end has three generators, including connector $q = 2$. The chronological word $(q, 0, 1, q)$ is zero.

After the two-guard lift, this becomes an erasing word in a nine-state, five-letter positive stochastic instance.

Both that numerical instance and its certificates are included. The full certificate has 344 natural witnesses and 400 residuals; the compressed one has 276 natural witnesses and 324 residuals. Every residual evaluates to zero in both. The example exercises the complete chain

PCP match $\rightarrow$ integer zero product $\rightarrow$ positive stochastic erasure $\rightarrow$ natural quartic zero.

The general proof also handles arbitrary connector placement, adjacent connectors, and empty component strings, rather than checking only this successful shape.

13.4 Scope and results of the executable checks

The shipped suite is deterministic. It uses Python’s exact integers and `fractions.Fraction`; no external Python package is required. Rational matrix rank is computed by exact Gaussian elimination. The finite random source sample uses seed 20261002, recorded in the result file.

The completed run performed **50,582 passing assertions**. Its principal categories are shown in Table 4; the full category ledger is in `verification/check_results.json`. ([write] Shipped as `data/02-exact-erasure-check_results.json`; the suite was rerun at the write on a copy with the delivered layout and reproduced it apart from the recorded Python version.) Counts denote assertions, not distinct mathematical instances or independently proved theorems.

Check category	Assertions
Mortality versus erasure for enumerated words	10,188
Exact tensor product-rank formula	10,188
Full word coordinate/block identity	10,188
Independent full-prefix quartic replay	3,448
Independent compressed quartic replay	3,448
Full and compressed size ledgers, combined	6,896
Exact total-variation contraction inequalities	3,601
PCP arbitrary connector-word reduction	1,638
All remaining algebraic, domain, mutation, and example checks	987
Total	50,582

Table 4: Finite exact checks run for this report. The two numerical certificate formats are checked by a module that imports no generator code.

The suite includes scalar source alphabets with coefficients from -2 to 2 , eight seeded two-dimensional source examples, the seven-state packet, a higher tensor example, one-generator nilpotent examples attaining the commuting bound, and PCP examples with empty words and adjacent connectors. It also changes individual witness entries, changes selectors, and rejects values of type `float`, `bool`, and `str` in natural-number fields.

The independent checker evaluates the defining residuals directly, without calling the generator. It is independent at the implementation-module level, not an independent human review. The all-instance reduction, the nonexistence of common complex lines, the infinite-horizon probability theorem, and the DPRM dependency are established mathematically in the text, not by these finite experiments. The uniform-parameter schema has a hand proof but is not one of the implemented certificate formats.

14 Integration and formalization plan for ProveIt

The present package is self-contained and does not modify the repository. It is suited to a separate research-report directory, with a link from the substrate review rather than a change to the fixed universal-operation record. ([write] This report is that directory. A link from the substrate review is for the owner of the research programme to add; the review of this manuscript [23] already exists.)

A useful formalization order follows the mathematical dependency graph. First formalize the explicit maps E, F , the affine block decomposition, and the rank-one/equal-column equivalence over $\mathbb{Q}$. Next establish the tensor product identity and the two common-line obstructions over $\mathbb{C}$. The mortality undecidability theorem can initially be an explicitly declared external interface, rather than an unproved statement hidden inside an implementation. The PCP front end offers a separate path to a fully concrete reduction chain.

The certificate layer is especially suitable for an early machine-checked milestone. It needs only finite sums, natural one-hot selectors, a matrix-product recurrence, and the equivalence between a zero sum of squares and vanishing of each residual. The compressed form additionally needs the column-sum invariant, the difference formula (30), and the offset bounds. This proves an exact zero-set bijection, not merely soundness of a successful example.

For source integration, four interfaces should remain visibly separate:

1. the computable mortality/PCP-to-stochastic compiler, including its common scale and all numerical input coefficients;
2. the exact finite-word rank and invariant-subspace theorems;
3. the bounded polynomial family and its literal witness/residual counts;
4. the unbounded DPRM polynomial existence theorem and its unspecified original witness multiplicities.

No existing Lean or Rocq file in ProveIt was compiled during this task, and no new theorem here is presented as proof-assistant verified. The accompanying `PROOF_STATUS.md` states these boundaries explicitly. ([write] Shipped as `02-exact-erasure-PROOF_STATUS.md`, beside `02-exact-erasure-SOURCE_AUDIT.md`.)

15 Further research questions and concrete next targets

The following are research directions exposed by the present proofs. Except for the precisely cited MathOverflow restriction already answered, this list does not claim that every formulation is a previously catalogued open problem.

15.1 Reduce the tensor overhead without losing the line guarantee

The present no-line compiler uses dimension $2d + 1$ and two guard letters. Can one construct a uniformly valid compiler in dimension $d + 2$, or with one additional letter, while retaining positivity and a word-projection soundness theorem? The obstacle is not positivity, which the affine lift handles uniformly; it is removing all source invariant lines without allowing guards to create false zero products. A useful first goal is to characterize the affine guard systems whose linear products cannot lower the rank of a protected source word.

15.2 Irreducibility beyond a missing invariant line

Our stochastic models necessarily have a common invariant hyperplane. Thus no construction confined to normalized column-stochastic matrices can make the whole linear action irreducible. A meaningful stronger target is either irreducibility on the information hyperplane, or a positive integer matrix construction without a common invariant hyperplane after normalization is abandoned. The first target is not achieved here either: source invariant subspaces can survive as $W \otimes \mathbb{C}^r$. A reduction must explain how to remove those spaces while still decoding mortality.

15.3 Low-dimensional classifications via the exact correspondence

Proposition 8.1 translates state-dimension questions directly into mortality questions. It can be used to transport new decidability results, bounded-alphabet results, or structural restrictions in either direction. The three-state case is especially focused: its transverse dimension is two. Any claimed classification should consult current primary sources rather than inherit the status labels of a 2014 survey table.

15.4 Regularly constrained and non-independent scheduling

For a finite-state controller, only a regular language of schedules may be allowed. The unrestricted mortality compiler does not by itself preserve such constraints when guard letters are erased. One needs a product construction that tracks both source computation and controller state. On the probabilistic side, identify conditions under which a reachable erasing block recurs almost surely. Full-support independent switching is sufficient here, but zero probabilities, transient controller states, and forbidden blocks can invalidate the zero-one argument.

15.5 Sharper canonical certificates and fully charged arithmetic

The compressed quartic reduces natural witnesses by using the invariant mass rather than storing it. Can additional structure of the tensor restrictions reduce the h^2T information entries, while retaining a bijection for every labelled word? One possibility is to certify the source projection and guard product separately; another is to exploit the rank gap without recording a full matrix at every step. Every proposal should count input parsing, selectors, scale arithmetic, coefficient generation, and reconstruction obligations. An improvement to $44T$ witnesses is a bounded-compiler result, not automatically an improvement to the repository's 87-operation universal construction.

15.6 Fixed-alphabet universality with an ordinary input interface

The nine-state many-one complete family encodes the problem in the gate alphabet. A different objective is one fixed numerical stochastic alphabet with a separate ordinary integer or finite-word input representation. Such a construction needs a concrete universal correspondence or matrix system and a fully paid input loader. It would make the relationship to the fixed universal Diophantine polynomials in ProveIt substantially tighter.

[write] *Batch 79, 2 October 2026: answered for the exact-hit predicate.* Part IV gives one fixed commuting alphabet of strictly positive rational doubly stochastic matrices whose input is loaded only into the initial distribution, along a rational quadratic curve (Theorem 47.2) or a rational line segment (Theorem 47.4), with a computably enumerable complete exact equilibrium-hit problem; a

variant with a fixed initial distribution reads the input as a unary prefix (Section 47). This answers the question for that predicate at the level of existence: the alphabet comes from MRDP and is not printed numerically (Part IV’s own question of Section 53.1). For rank-one erasure the question stays open.

15.7 Structural non-erasure certificates

The commuting case supplies a complete finite certificate: a nonzero invariant image on which all generators are invertible. Can useful larger decidable classes be characterized by analogous invariant-space certificates? Candidate restrictions include simultaneously triangularizable transverse actions, prescribed rank patterns, or semigroups with a bounded descending-image chain. A proposed algorithm must account for the possibility that noncommuting invertible-on-current-image maps carry that image elsewhere; the commuting proof cannot simply be reused without that invariant-image property.

[write] *Batch 79*. Part IV’s alphabets lie in the simplest certificate class: every generator is invertible, so no word erases (Proposition 45.5).

15.8 Precision-sensitive versus robust observation predicates

The perturbation theorem shows that exact erasure is destroyed by generic small invertibilizing perturbations, even when the invariant-subspace restrictions are unchanged. A quantitative alternative is to study coefficient uncertainty sets together with a fixed tolerance on output dependence. Which robust predicates admit computable finite certificates, and how do their costs scale as the tolerance tends to zero? The universal bound $\eta^T \leq \varepsilon$ solves one natural approximate-forgetting question; more delicate targets could depend on a specified output distribution, a restricted schedule language, or nonuniform contraction rates.

[write] *Batch 79, 2 October 2026*. Part IV’s Theorem 49.1 settles one case for its alphabets: a rational target at positive distance from equilibrium is decidable with the explicit horizon $\theta^H < |c - 1/s|$, while targets at distance zero, strict cutpoints included (155), can stay computably enumerable complete.

15.9 A formal proof of the guard compiler and the projection bijection

A focused formal milestone is Theorem 5.1 together with Theorem 10.2, keeping mortality undecidability as a separately identified dependency. A second milestone is the complete PCP chain and its connector boundary condition. These targets are concrete enough to test the repository’s arithmetic and matrix APIs, and they would turn the present hand-checked compiler into a reusable trusted frontend.

16 Conclusion

Exact stochastic erasure supports computational hardness even when every gate is positive, every step uniformly contracts total variation, and no common invariant complex line exists. The mechanism is explicit: preserve the source computation on one tensor factor, use invertible guards on the other, and lift the resulting affine dynamics into the interior of the stochastic simplex.

The strongest precise line-restriction result in this report is the $(d, m) \mapsto (2d + 1, m + 2)$ compiler. It supplies the promised seven-state numerical undecidability bound, while the independent four-dimensional PCP front end supplies a nine-state many-one complete family. The probabilistic interpretation then identifies exact finite forgetting, almost-sure finite forgetting, and finite expected forgetting time without changing the underlying finite-word predicate.

The Diophantine layer is equally explicit at a bounded horizon. A zero-set bijection reduces the seven-state certificate from $57T$ to $44T$ natural witnesses by storing only shifted information coordinates. The fixed-arity unbounded representation remains a separate DPRM consequence. Together, these results give a concrete new computational frontend and a carefully delimited research contribution, rather than a claim that a generic invocation of DPRM or contraction has solved every arithmetic-compression problem.

Part II

Exact convergence as computation (source 10)

[write] Part II prints batch-78 manuscript 10, *Exact Convergence as Computation: Diophantine Certificates for Uniformly Mixing Stochastic Games*, subtitled *Fixed-game universality, canonical quadratic witnesses, and the distinction between scalar and vector reachability*, under the kicker “Diophantine computational substrates”, “Prepared for Vladimir Reshetnikov; Mathematical development and exact-arithmetic implementation: ChatGPT”, October 2, 2026. Its labels carry the prefix `ste:ec:`. Its research-status statement and abstract follow; its two appendices are Appendices C and D.

Research status (source 10). The central results below are supported by complete mathematical arguments and a delivered, executable compiler. The accompanying standard-library Python suite passes 26,077 exact-arithmetic checks. These checks are not a proof-assistant verification, and the manuscript is not peer reviewed. The literature and repository comparison is targeted rather than exhaustive; priority for the precise constrained simulation theorem is not established. No resolution of the general Skolem problem, no single-fold MRDP theorem, and no improved fixed universal-polynomial operation bound is claimed.

Abstract (source 10). We construct a finite, turn-based, two-player stochastic game whose exact value iteration is computationally universal despite unusually strong regularity. The discount is $1/2$, every running reward is $1/4$, every transition probability is dyadic, every available action assigns probability at least $1/(2N)$ to each of the N states, and the value vector is the known constant $\frac{1}{2}\mathbf{1}$. Nevertheless, for a suitable fixed game, deciding whether an input dyadic terminal-payoff vector ever reaches that value vector in finitely many Bellman iterations is computably enumerable complete.

The proof gives a compiler from counter programs to homogeneous linear/min/max circuits and then to genuine stochastic-game transition tables. Paired positive and negative channels remove signed coefficients; a correctly initialized cyclic evaluation pipeline removes nested gates; and a common uniform reset provides full support without destroying the simulated computation. An erasing halt convention turns computational halting into finite attainment of the fixed point. A zero-reward variant gives complete coordinate-equality and finite-consensus problems.

For each externally specified horizon T , the complete computation admits an orthant-nonnegative quadratic Diophantine certificate with exactly one natural witness tuple when the prescribed endpoint condition holds. With N game states and b binary-decision states, it uses $(N +$

2b) T witnesses. Two boundary results clarify the source of hardness: deterministic min/max-selector games have a finite abstraction, whereas chance-only fixed-point attainment is a kernel-stabilization problem. Scalar equality for chance-only games, in contrast, is Skolem-equivalent; an explicit dyadic, full-support specialization is included and related to the existing ergodic Markov-chain literature. Complete example polynomials, exact tests, and an independent serialized-certificate checker accompany the article.

17 Research question, contribution, and source boundary

A computational substrate need not resemble a tape machine. Here the substrate is a familiar numerical recursion: take a vector, form several convex averages of its coordinates, choose a minimum or maximum at each coordinate, and repeat. All coefficients are rational. A discount makes the recursion converge rapidly. The question is whether such benign-looking iteration can still contain an unbounded exact computation.

The substantive question is not whether a computably enumerable relation has some Diophantine representation. MRDP already answers that question. Instead we ask for a reduction that respects a restrictive native interface: nonnegative stochastic coefficients, at most two actions per state, common positive transition support, a fixed discount, a trivial limiting value, and a complete, witness-controlled arithmetic description of bounded runs.

17.1 The central theorem

Let a state be owned by a minimizing player, a maximizing player, or have just one action. Write $p_{ia} \in \mathbb{Q}^N$ for the stochastic row of action a at state i . Put

$$S(v)_i = \text{opt}_{a \in A_i} p_{ia} v, \quad \Phi(v) = \frac{1}{4} \mathbf{1} + \frac{1}{2} S(v). \quad (45)$$

Here opt is the owner's minimum or maximum; for a one-action state there is no choice. The vector v is a terminal-payoff table, not a probability distribution over the states.

Theorem 17.1 (Fixed-game universality with a known fixed point). *There are a fixed positive integer N and a fixed finite game of the form (45) such that:*

- (i) *each state has one or two actions; every $p_{ia,j}$ is dyadic and $p_{ia,j} \geq 1/(2N)$;*
- (ii) *Φ maps $[0, 1]^N$ into itself, is a $1/2$ -contraction in the supremum norm, and has the unique fixed point $v_* = \frac{1}{2} \mathbf{1}$;*
- (iii) *the set of dyadic $v \in [0, 1]^N$ for which $\Phi^n(v) = v_*$ for some $n \geq 0$ is computably enumerable complete.*

The reduction is effective. More generally, the construction compiles every finite counter program into such a game, with an explicitly computed terminal-payoff encoding of each initial configuration.

The hard input is the initialization. The transition table, rewards, discount, and target in the fixed-game assertion do not depend on that input. A numerical transition table for a specifically chosen universal counter machine is not printed or shipped. Fixed-game existence follows by applying the fully specified compiler to a fixed universal counter machine. The shipped numerical tables are worked examples, not universal-machine records.

17.2 Why the source comparison matters

The repository contains an MRDP interface stating the existence of one ordinary integer polynomial and one finite witness dimension for an entire computably enumerable set [15]. Its computational-substrate reports already discuss contracting dynamics and quadratic neural certificates [7]. Consequently, neither the observation that contraction can coexist with computation nor the generic use of quadratic complementarity is claimed here as a discovery. The contribution is the *stochastic-game realization under the simultaneous restrictions in Theorem 17.1*, together with its complete arithmetic interface.

[write] *Comparison with the collection.* Within this collection a 2-adic 1/2-contraction already has an r.e.-complete exact extinction (Corollary `pqc:ad:cor:complete` in Part VII of [21], source 12 there), while a real four-dimensional piecewise-linear 1/2-contraction has decidable point-to-point reachability because its radial clock is divided by B at every step and never vanishes (`pqc:sc:thm:main` (iii), source 14 there). The present construction is the real, order-preserving counterpart of the former: its live clock $s' = \sum_e e$ of (54) is erased at halting (Lemma 20.2), so the whole vector reaches the fixed point. It therefore shows that source 14’s decidable point reachability comes from that compiler’s non-erasable clock, not from real homogeneous contractions as such; nothing printed there is contradicted, and the picture agrees with the comparison section “Two routes to one phenomenon” of Part VII of [21]. Part V of [21] (source 17 there) has the same slogan for a different object: a uniform spectral gap with an undecidable exact stationary equality.

A particularly close external comparison is the January 26, 2026 revision of Varonka and Watanabe [10]. It proves point-reachability results for one-player min- or max-Bellman operators, including arbitrary dimension when source and target are componentwise comparable, and full decidability in dimension two. Our construction uses both owners, a larger fixed dimension, and initial errors of both signs. It therefore does not settle their remaining one-player problem or contradict their positive results. It does show why monotonicity and rapid fixed-point convergence alone cannot justify unrestricted extensions to two-player operators.

The chance-only scalar-equality boundary is related to established work, not a new undecidability assertion. Vahanwala proves Skolem and positivity completeness for ergodic Markov chains with a dimension-efficient reduction [11]. We supply a less dimension-efficient specialization with dyadic probabilities and a prescribed common reset. The general Skolem problem remains unresolved in the July 2026 work of Luca, Ouaknine, and Worrell, whose general decidability implication is conditional [12]. We do not resolve that problem.

17.3 What is and is not computed

There is no difficulty in obtaining the infinite-horizon game value in (45): every play receives the same running reward $1/4$ at every step, and

$$\sum_{t \geq 0} 2^{-t} \frac{1}{4} = \frac{1}{2}.$$

Every strategy profile therefore has infinite-horizon payoff $1/2$. The hard property is *finite exact attainment from an input terminal payoff table*. The usual initialization $v = \mathbf{0}$ is not the input family used in the reduction. In particular, this manuscript does not say that solving ordinary finite discounted games is undecidable.

Value iteration at horizon n is the finite-horizon game value with terminal payoff v and the indicated running reward. Its computation by backward induction is immediate: after choosing

an action, average the remaining horizon's values, then optimize according to the current state's owner. Thus the construction uses actual game semantics, not an arbitrary piecewise-affine map merely renamed a Bellman operator. Stochastic games and discounted values originate in Shapley's framework [8]; approximation stopping criteria, such as those studied in [13], address a different question from finite exact attainment.

18 Conventions and the three observation problems

We take $\mathbb{N} = \{0, 1, 2, \dots\}$. A dyadic rational is an element of $\mathbb{Z}[1/2]$. All finite rational objects have an effective exact encoding. “Computationally enumerable complete” means complete under total computable many-one reductions; no polynomial-time reduction bound is asserted.

We distinguish three predicates for an exact trajectory v_n :

$$\text{scalar equality: } (v_n)_r = (v_n)_s, \quad (46)$$

$$\text{consensus: } v_n \in \mathbb{R}\mathbf{1}, \quad (47)$$

$$\text{fixed-point attainment: } v_n = v_*. \quad (48)$$

Each may be imposed at a specified $n = T$, or existentially over all n . These are different decision problems. In particular, a scalar zero in a linear recurrence is not the same as a whole vector becoming zero.

For a vector x , set

$$\|x\|_\infty = \max_i |x_i|, \quad \text{span}(x) = \max_i x_i - \min_i x_i.$$

The latter is a seminorm, since it vanishes on constant vectors. All stochastic rows in this article act on column vectors of payoffs from the left. The state of an exact computation is the entire payoff vector. It is not the state visited by a single random sample path.

A *bounded certificate* has an externally supplied natural horizon T . Its arity is allowed to depend on T . A *fixed-arity Diophantine representation* uses one fixed polynomial and a fixed number of existential variables for all inputs and all possible running times. Confusing these notions would invalidate the final MRDP discussion.

19 A homogeneous counter-machine compiler

19.1 Source instruction set

A program has m control states and $k \geq 1$ natural counters. Its instructions are

$$\text{INC}(i, q), \quad \text{DECJZ}(i, q_+, q_0), \quad \text{HALT}.$$

An increment increases counter i and jumps to q . A decrement instruction jumps to q_0 without changing counters when counter i is zero; otherwise it decreases that counter and jumps to q_+ . There is one halt state h . The ordinary operational convention makes it absorbing.

This instruction set supports universal computation with finitely many counters, indeed with two. The distinction between counter instruction sets matters: the present zero-test/decrement with independently specified targets contains the undecidable CM2 instruction set described and

analyzed by Dudenhefner [9]. A fixed universal program can be obtained from a universal interpreter in this standard counter model. Universality of this source model is the sole external computability ingredient in the hardness proof; the stochastic and arithmetic translations are proved here.

19.2 Configuration coordinates and exact zero tests

Encode a configuration $(q, n_1, \dots, n_k)$ at scale $s > 0$ as

$$x = (s, p_1, \dots, p_m, c_1, \dots, c_k), \quad p_j = s \mathbf{1}_{j=q}, \quad c_i = s 2^{-n_i}. \quad (49)$$

The dimension is $d = 1 + m + k$. Every coordinate lies in $[0, s]$. The homogeneous gate

$$z_i = \max(0, 2c_i - s) \quad (50)$$

returns s when $n_i = 0$ and 0 otherwise. Indeed, a positive counter has $c_i \leq s/2$. No discontinuous test has been used.

For a nonbranching instruction at state j , its enable value is p_j . For a decrement/zero instruction testing counter i , its two enable values are

$$e_{j,0} = \min(p_j, z_i), \quad e_{j,+} = \min(p_j, s - z_i). \quad (51)$$

On encoded configurations each enable value is 0 or s , and precisely one is s when the active control state is included among the branches.

Associate to each branch e its target state $q(e)$ and multipliers $\alpha_{e,i} \in \{1/2, 1, 2\}$. An increment of counter i uses $1/2$, a positive decrement uses 2 , and an unchanged counter uses 1 . Put

$$p'_j = \sum_{e:q(e)=j} e, \quad (52)$$

$$c'_i = \sum_e \min(2e, \alpha_{e,i} c_i). \quad (53)$$

An inactive summand vanishes. An active summand equals its required counter value: in the decrement case $2c_i \leq s$, and otherwise the value is also at most s . The cap $2e = 2s$ is therefore harmless.

19.3 Two halt conventions

We use two related maps. The *absorbing map* F_{abs} includes a halt branch targeting h with all multipliers 1, and sets $s' = s$. It exactly implements the ordinary absorbing machine.

The *erasing map* F_{era} omits the halt branch and sets

$$s' = \sum_e e. \quad (54)$$

On a nonhalting configuration it performs the same machine step at the same scale. On a halting configuration it outputs the entire zero vector. Both maps take $\mathbf{0}$ to $\mathbf{0}$.

Lemma 19.1 (Counter simulation). *The two maps are continuous, positively homogeneous, dyadic circuits using only linear, minimum, and maximum gates. The absorbing map sends (49) to the encoding of its successor. If the source machine first reaches its halt state after H steps, then $F_{\text{era}}^{H+1}(x) = \mathbf{0}$, whereas $F_{\text{era}}^j(x)$ is a nonzero encoded configuration for $0 \leq j \leq H$. If the machine never halts, all these iterates remain nonzero and their clock coordinate remains s .*

Proof. Equation (50) gives the stated zero test. Exactly one branch is enabled at a nonhalting configuration, so (52) writes a one-hot successor state and (53) applies exactly the required multiplier to each counter coordinate. For the erasing map all branches have enable value zero at the halt configuration, including every counter summand, and (54) vanishes. Homogeneity and continuity hold gate by gate; the constant zero is a homogeneous linear gate. Induction proves all trajectory claims. $\square$

Using unrestricted fan-in for linear sums, the displayed circuit can be organized with depth at most seven: linear zero-test preparation, maximum, branch preparation, positive-branch selection, enable scaling, gated counter terms, and final sums. This is a bound for this particular source construction, not for arbitrary circuits or for binary fan-in arithmetic. The stochastic implementation permits a finite distribution with more than two successor states, so such sums do not violate the two-action restriction.

20 From signed circuits to genuine stochastic games

20.1 Layering and normalization

Consider any positively homogeneous dyadic circuit $F : \mathbb{R}^d \rightarrow \mathbb{R}^d$ whose gates are finite linear combinations and binary minima or maxima. Insert copy gates so every edge enters the immediately following layer. Pad *all outputs* to one common final layer L , chosen at least as deep as every gate retained in the circuit. The latter qualification handles unused gates and the initially halted edge case. Inputs are layer zero.

Choose powers of two $C_\ell \geq 1$ so that every linear gate $g = \sum_j a_j g_j$ at layer ℓ satisfies $\sum_j |a_j| \leq C_\ell$. Define

$$K_0 = 1, \quad K_\ell = C_1 \cdots C_\ell, \quad K = K_L, \quad M = L + 1. \quad (55)$$

For input x , every gate value at layer ℓ has absolute value at most $K_\ell \|x\|_\infty$. This follows by induction for linear gates and by $|\min(a, b)|, |\max(a, b)| \leq \max(|a|, |b|)$ for the other gates.

20.2 Paired channels and a baseline state

Create a baseline state o , with its only transition back to itself. For every circuit input and gate g , create two states g^+, g^- . They will hold a value and its negative. On a centered initialization the baseline value is zero. A retained gate at layer ℓ is intended to hold g/K_ℓ in its positive channel and $-g/K_\ell$ in its negative channel.

For a linear gate $g = \sum_j a_j g_j$ at layer ℓ , give g^+ a single action with probabilities

$$\frac{|a_j|}{C_\ell} \quad \text{to } g_j^+ \text{ if } a_j \geq 0, \quad \text{to } g_j^- \text{ if } a_j < 0, \quad (56)$$

and send the remaining probability to o . For g^- swap the signs of the selected channels. Repeated destinations are combined. Nonnegativity of the remaining probability is exactly the normalization inequality.

For $g = \min(g_1, g_2)$, make g^+ a MIN state. Its two actions assign probability $1/C_\ell$ to g_1^+ and g_2^+ , respectively, and the remaining probability to o . Make g^- a MAX state with the analogous negative-channel actions. For a maximum gate reverse the two owners. Thus the negative channel

really is the negative output, including at ties. All rows are stochastic and dyadic; every state has at most two actions.

Finally, each input-register channel has a single action copying its corresponding output channel at layer L . This creates a cyclic pipeline. Denote its undiscounted, zero-reward operator by S_0 .

If there are G retained gates in addition to the d inputs, the number of states before padding is

$$N_0 = 1 + 2(d + G). \quad (57)$$

Pad to a power of two $N \geq N_0$, with every extra state copying o . Padding will keep a subsequent uniform reset dyadic. If the circuit has $g_{\min \max}$ min/max gates, then the game has exactly $b = 2g_{\min \max}$ binary-decision states. No optimality claim is attached to these counts.

20.3 The invariant subspace

Let $V \subseteq \mathbb{R}^N$ consist of vectors whose baseline and padding coordinates vanish and whose paired channels are negatives of each other. The construction gives

$$S_0(V) \subseteq V, \quad \frac{1}{N} \sum_i w_i = 0 \quad (w \in V). \quad (58)$$

The first assertion follows separately for linear gates, min/max pairs, copy registers, and the baseline. The second is cancellation of the pairs. In particular, signed arithmetic is represented inside a genuine order-preserving stochastic operator. The order on the full space does not prohibit this: distinct generic vectors of V are incomparable in the componentwise order.

20.4 Initialization is part of the compiler

Given $x \in [-1, 1]^d$, evaluate the finite circuit once on x . Initialize its input channels to $\pm x_j$, each layer- ℓ gate to its paired values $\pm g(x)/K_\ell$, and the baseline and padding coordinates to zero. Call the resulting vector $w(x) \in V \cap [-1, 1]^N$.

This is a finite, computable loader. It computes only the circuit's one-step gate values. It does not inspect whether the source program will halt. Zero-filling the internal gates instead would be unsound: the first feedback step would copy zero outputs into all registers and could falsely signal a halted or erased computation.

Lemma 20.1 (Every microstep is accounted for). *Let $r_n \in \mathbb{R}^d$ be the positive input-register coordinates of $S_0^n(w(x))$. Then*

$$r_n = K^{-q_n} F^{q_n}(x), \quad q_n = \left\lceil \frac{n}{M} \right\rceil, \quad (59)$$

where $q_0 = 0$. In particular, $r_0 = x$ and $r_1 = \dots = r_M = F(x)/K$.

Proof. Each layer computes its gates from the preceding layer with an additional factor $1/C_\ell$. Following an input-register vector through all L layers and then the feedback edge therefore gives, for every $n \geq 0$,

$$r_{n+M} = F(r_n)/K. \quad (60)$$

At time zero each layer was already evaluated consistently on x . For $0 \leq t \leq \ell$, the layer- ℓ values at time t are still the normalized gate values evaluated on x : prove this by induction on ℓ , using the

preceding layer at time $t - 1$ for $t > 0$. Thus the final layer has its initial output for times $0, \dots, L$, and the feedback registers equal $F(x)/K$ for times $1, \dots, M$. Positive homogeneity now iterates (60). For each residue class modulo M , its initialized member has exactly the exponent specified in (59); every subsequent member increases that exponent by one. $\square$

Lemma 20.2 (Erasure drains the whole pipeline). *Use $F = F_{\text{era}}$. If the source first reaches its halt state after H steps, then*

$$S_0^n(w(x)) = \mathbf{0} \quad \text{for all } n \geq (H + 1)M. \quad (61)$$

If the source never halts, $S_0^n(w(x)) \neq \mathbf{0}$ for every n .

Proof. In the halting case $F^{H+1}(x) = \mathbf{0}$ and all its later iterates vanish. Lemma 20.1 makes all registers zero from time $HM + 1$ onward. After at most L more steps all internal gates have received only zero data. Every gate is homogeneous and hence maps zero inputs to zero. The time bound is $HM + 1 + L = (H + 1)M$. In the nonhalting case the clock register is $K^{-q_n}s > 0$ at every finite time, so the whole vector cannot vanish. $\square$

The extra L steps cannot in general be omitted. Registers can have become zero while stale, unused, or not-yet-overwritten internal gates remain nonzero. The full-vector theorem uses the drain argument, not just the register simulation.

21 Uniform mixing preserves exact computation

21.1 A common-reset lemma

For any zero-reward stochastic min/max operator S , any rational $0 < \eta < 1$, and $\bar{v} = N^{-1} \sum_j v_j$, define

$$\hat{S}(v) = (1 - \eta)S(v) + \eta\bar{v}\mathbf{1}. \quad (62)$$

It is itself a stochastic-game operator: replace every action row by

$$\hat{p}_{ia} = (1 - \eta)p_{ia} + \frac{\eta}{N}\mathbf{1}^\top. \quad (63)$$

The common term may be moved outside either a minimum or a maximum. Every new transition probability is at least η/N .

Lemma 21.1 (Exact preservation of all equality times). *Let $B = \delta\hat{S}$ with $0 < \delta < 1$. For every initial vector v there are scalars c_n such that*

$$B^n(v) = [\delta(1 - \eta)]^n S^n(v) + c_n \mathbf{1}. \quad (64)$$

Consequently, every pairwise equality or strict order between coordinates of $S^n(v)$ is preserved at exactly the same n by $B^n(v)$.

Proof. Stochastic min/max operators are positively homogeneous and additively homogeneous:

$$S(av + c\mathbf{1}) = aS(v) + c\mathbf{1} \quad (a \geq 0).$$

Put $a_n = [\delta(1 - \eta)]^n$. The assertion holds at $n = 0$ with $c_0 = 0$. Substitution into (62) gives the induction step with

$$c_{n+1} = \delta c_n + \delta \eta a_n \overline{S^n(v)}.$$

Subtracting two coordinates removes c_n , and a_n is strictly positive at every finite n . $\square$

This lemma does not assume that an input is a valid encoded computation. It is an identity for every real vector. In particular, the reset is not an approximation justified only on a finite test set.

21.2 Contraction estimates

Lemma 21.2 (Supremum and span contractions). *For any common reward r and $\Phi = r\mathbf{1} + \delta\hat{S}$,*

$$\|\Phi(v) - \Phi(u)\|_\infty \leq \delta\|v - u\|_\infty, \quad (65)$$

$$\text{span}(\Phi(v) - \Phi(u)) \leq \delta(1 - \eta) \text{span}(v - u). \quad (66)$$

Proof. If $m\mathbf{1} \leq v - u \leq M\mathbf{1}$, monotonicity and additive homogeneity give $m\mathbf{1} \leq S(v) - S(u) \leq M\mathbf{1}$. This proves its supremum nonexpansiveness and bounds the span of the difference by $M - m$. The same supremum argument applies to $\hat{S}$, which is a stochastic operator. For the span, the reset contribution $\eta(\bar{v} - \bar{u})\mathbf{1}$ is constant and therefore disappears. Multiplication by δ completes both inequalities. $\square$

On the invariant space V of (58), the mean vanishes at every iterate, so a stronger identity is available:

$$(\delta\hat{S})^n(w) = [\delta(1 - \eta)]^n S_0^n(w), \quad w \in V. \quad (67)$$

Now choose $\delta = \eta = 1/2$, define

$$v(x) = \frac{w(x) + \mathbf{1}}{2}, \quad \Phi(v) = \frac{1}{4}\mathbf{1} + \frac{1}{2}\hat{S}(v), \quad (68)$$

and use additive homogeneity. One obtains

$$\Phi^n(v(x)) = \frac{1}{2}\mathbf{1} + \frac{4^{-n}}{2} S_0^n(w(x)). \quad (69)$$

Every initial coordinate belongs to $[0, 1]$, all coefficients are dyadic, and all reset rows have entries at least $1/(2N)$. On these encoded inputs, the error is bounded by $4^{-n}/2$ in the supremum norm. For arbitrary $v \in [0, 1]^N$, the uniform bound is $2^{-n}\|v - \frac{1}{2}\mathbf{1}\|_\infty$.

22 Proof of fixed-game universality and its variants

Proof of Theorem 17.1. Fix a universal counter program and apply the erasing compiler. Choose the power-of-two state padding and the reset and reward in (68). All syntactic assertions about the resulting fixed game follow from the explicit row construction. Since a stochastic operator fixes constant vectors, $\Phi(\frac{1}{2}\mathbf{1}) = \frac{1}{2}\mathbf{1}$. The supremum contraction proves uniqueness. The cube is preserved because each convex average and its minimum or maximum lies in $[0, 1]$, whence every output of Φ lies in $[1/4, 3/4]$.

Encode an input of the fixed universal program by the dyadic vector $v(x)$. Equations (69) and (61) show

$$\exists n \Phi^n(v(x)) = \frac{1}{2}\mathbf{1} \iff \exists n S_0^n(w(x)) = \mathbf{0} \iff \text{the source computation halts.}$$

All factors discarded in these equivalences are strictly positive at finite times. The map from a source input to the full payoff vector is computable. Thus the fixed game's attainment set is c.e.-hard. It is c.e. because rational iteration and equality are decidable at every finite time, so successive times can be searched. The same reasoning applies uniformly when the source program is supplied to the compiler. $\square$

22.1 Uniformly stopping reachability games

The operator Φ also has a reachability interpretation. From every nonterminal state, every action goes to a success terminal with probability $1/4$, to a failure terminal with probability $1/4$, and otherwise continues according to its row $\hat{p}_{ia}$ with total probability $1/2$. There are now $N + 2$ states. Both terminals are absorbing.

The continuation probability for n steps is exactly 2^{-n} under every strategy profile, and the expected number of transitions before stopping is exactly two. Success probability is exactly $1/2$ from every nonterminal state. With a supplied boundary payoff table on nonterminal states, backward reachability iteration is exactly Φ .

Corollary 22.1 (Rapid stopping does not imply decidable exact attainment). *There is a fixed, uniformly stopping, finite turn-based reachability game with success-probability vector $\frac{1}{2}\mathbf{1}$ for which finite exact attainment of that vector from an input dyadic boundary table is c.e.-complete. The full-support lower bound applies to the conditional transition kernel on the nonterminal states; it does not apply to the absorbing terminal rows.*

Universality here belongs to the arithmetic of the iterates. A typical random path is short, and we do not assert that it physically executes an arbitrarily long counter computation.

22.2 Zero running rewards and scalar observations

For the absorbing source map, let $B = \frac{1}{2}\hat{S}$ have zero running rewards. The shifted encoding satisfies

$$B^n(v(x)) = \frac{2^{-n}}{2}\mathbf{1} + \frac{4^{-n}}{2}S_0^n(w(x)). \quad (70)$$

Let r be the positive halt-state register, and s the positive clock register. For initial scale one, put $q_n = \lceil n/M \rceil$. Before halting, their difference is

$$(B^n(v(x)))_r - (B^n(v(x)))_s = -\frac{4^{-n}}{2K^{q_n}}; \quad (71)$$

after the simulated state is halted, the difference is zero. If the source's first halting time is $H \geq 1$, the first equality time is

$$n_{\text{eq}} = (H - 1)M + 1. \quad (72)$$

For an initially halted source it is zero. The separate observation “halt-register value greater than baseline value” is true exactly at simulated halted configurations, giving c.e.-complete strict-inequality reachability as well.

For the erasing source map, consensus in (70) is equivalent to $S_0^n(w(x)) = \mathbf{0}$: the latter vector belongs to V , and the only constant vector in V is zero. Thus finite consensus is c.e.-complete in a suitable fixed zero-reward game, although its unique fixed point is zero and all initial payoffs in the reduction are nonnegative.

22.3 One-sided errors are easy

Proposition 22.2. *For the full-support game in Theorem 17.1, if $v \leq v_*$ or $v \geq v_*$ componentwise, finite attainment occurs if and only if $v = v_*$ already.*

Proof. Suppose $v - v_* \leq 0$ and some coordinate is strictly negative. Every full-support action average of $v - v_*$ is then strictly negative, so every coordinate of $\Phi(v) - v_*$ is strictly negative. This remains true at every later iterate. The nonnegative case is identical with signs reversed. $\square$

The encoded hard initializations straddle the fixed point through their paired positive and negative errors. This is essential, not a dispensable technicality.

23 Two decidable boundaries and one Skolem boundary

The combination of averaging and mixed min/max control is important. Each of two natural restrictions has a direct decision procedure for full fixed-point attainment.

23.1 No averaging: finite selector dynamics

Suppose every action row is a point mass. Then S chooses one of the current coordinates at each state, possibly after a minimum or maximum. If v has $m_0 \leq N$ distinct coordinate values, every vector $S^n(v)$ has all coordinates in this fixed finite set. There are at most m_0^N such vectors.

Proposition 23.1 (Deterministic-selector decidability). *Scalar equality, consensus, and attainment of the uniform fixed point are decidable for zero-reward or common-reward discounted games with point-mass action rows. At most m_0^N distinct undiscounted vectors need be visited before a repetition proves that no new observation will occur.*

Proof. For scalar equality and consensus, positive discount factors and common additive rewards do not affect the observations. Iterate the finite selector map and test each vector until it repeats. For the uniform fixed point $v_* = r/(1 - \delta)\mathbf{1}$, translate to the error $v - v_*$. The error at time n is $\delta^n S^n(v - v_*)$, so fixed-point attainment is zero-vector reachability in the same finite abstraction. $\square$

23.2 No choices: whole-vector zeros are easy

Suppose each state has only one action, so $S(v) = Pv$ for one rational stochastic matrix. For a common reward and its uniform fixed point,

$$\Phi^n(v) - v_* = \delta^n P^n(v - v_*).$$

Proposition 23.2 (Kernel-stabilization test). *Finite fixed-point attainment for a chance-only N -state game is equivalent to $P^N(v - v_*) = \mathbf{0}$ and is therefore decidable by exact linear algebra.*

Proof. The kernels $\ker P^j$ form an increasing chain. Their dimensions can strictly increase at most N times. If two consecutive kernels agree, all subsequent kernels agree, by taking inverse images under P . Hence $\bigcup_{j \geq 0} \ker P^j = \ker P^N$. $\square$

This conclusion does *not* solve the scalar zero problem. A single coordinate may vanish while the whole orbit vector stays nonzero.

23.3 Scalar equality: a dyadic, full-support Skolem specialization

Consider the integer-matrix form of the Skolem problem:

$$\text{given } A \in \mathbb{Z}^{d \times d}, x \in \mathbb{Z}^d, \quad \exists n \geq 0 \ e_1^\top A^n x = 0 ? \quad (73)$$

Every integer linear recurrence is put into this form by its companion matrix. Conversely, Cayley–Hamilton gives a recurrence for every such matrix sequence.

Choose powers of two

$$C \geq \max \left(1, \max_i \sum_j |A_{ij}| \right), \quad R \geq \max(1, \|x\|_\infty).$$

Use a baseline and d paired channels. For the positive channel i^+ , put mass $|A_{ij}|/C$ on j^+ or j^- according to the sign of A_{ij} ; put the remaining mass on the baseline. Swap channels for i^- . Pad to a power of two $N \geq 2d + 1$ and add a half-uniform reset. Initialize centered channels to $\pm x_i/R$, then shift to $[0, 1]$. There are no min/max decisions.

For either zero rewards or a common reward, let $r = 1^+$ and let o be the baseline. With discount $1/2$, the difference is

$$v_{n,r} - v_{n,o} = \frac{e_1^\top A^n x}{2R(4C)^n}. \quad (74)$$

Every coefficient is dyadic and every transition has probability at least $1/(2N)$. The formula follows by induction on the paired channels before reset and by Lemma 21.1 after reset.

Proposition 23.3 (Scalar-equality equivalence). *For the class of chance-only dyadic games with discount $1/2$ and a half-uniform reset, scalar-equality reachability is computably many-one equivalent to the Skolem problem. The game is part of the input in this statement; no fixed chance-only game is asserted c.e.-complete.*

Proof. Equation (74) gives one reduction. In the reverse direction, cancel common rewards and the positive discount. Equality of two coordinates is the zero of $(e_r - e_s)^\top P^n v$. Clear the denominators of P and v : with DP integral and av integral, the integer sequence $(e_r - e_s)^\top (DP)^n (av)$ has exactly the same zeros. Cayley–Hamilton supplies its integer recurrence effectively. $\square$

This is an explicit specialization of an established hardness phenomenon. It is not a claim that Skolem-hardness of ergodic chains was previously unknown, nor an improvement over the smaller dimension achieved in [11]. The specialization is useful because its probability and reset constraints match the nonlinear compiler exactly.

Substrate	Scalar equality	Uniform fixed-point attainment
Point-mass rows; MIN and MAX allowed	Decidable by finite selector states	Decidable by the same finite abstraction
Chance only; rational stochastic rows	Skolem-equivalent, even with the stated dyadic reset	Decidable by kernel stabilization
MIN, MAX, and stochastic rows	c.e.-complete for a suitable fixed game	c.e.-complete for a suitable fixed game

The table is not a classification of every one-player subclass. In particular, allowing stochastic rows and only MAX states, or only MIN states, is a separate frontier.

[write] *Bears on a question of [21]*. Part VII of [21] asks (Question “Structural subclasses”) which restrictions force half-space reachability to be decidable: coordinatewise monotonicity, nonnegative weights, boundedly many activation patterns, or a genuine Bellman-operator structure. Theorem 17.1 and Section 22 answer this negatively in part: the operators here are coordinatewise monotone, have nonnegative (stochastic) coefficients and a genuine two-player Bellman structure, and still the strict-inequality observation “halt-register value greater than baseline value”, scalar equality and exact fixed-point attainment are c.e.-complete for suitable fixed games. The one-player case (only MIN or only MAX states) stays open (Section 29.2).

24 Canonical quadratic Diophantine certificates

We now compile the native game recursion directly, rather than importing an MRDP polynomial. Every existential variable below ranges over $\mathbb{N}$. The nonnegative domain is essential.

24.1 Integer scaling of zero-reward runs

Let every transition probability be $P_{ia,j}/D$, with $P_{ia,j} \in \mathbb{N}$ and $\sum_j P_{ia,j} = D$. Let the discount be $1/2$ and $v_0 = A/d_0$, where $A \in \mathbb{N}^N$ and $d_0 \geq 1$. Define

$$X_t = d_0(2D)^t v_t. \quad (75)$$

For zero rewards the recurrence becomes

$$X_{t+1,i} = \text{opt}_{a \in A_i} L_{ia}(X_t), \quad L_{ia}(X) = \sum_j P_{ia,j} X_j, \quad X_0 = A. \quad (76)$$

In particular, every X_t is integral and nonnegative. The exponentially large denominator is not an uncharged existential exponentiation gadget: T is external, and (76) itself has no exponentiation in its transition constraints.

24.2 A min/max gate with unique gaps

For two nonnegative integers L_0, L_1 , a claimed output Z , and natural gaps u, v , define

$$q_{\min}(L_0, L_1, Z; u, v) = (L_0 - Z - u)^2 + (L_1 - Z - v)^2 + uv, \quad (77)$$

$$q_{\max}(L_0, L_1, Z; u, v) = (Z - L_0 - u)^2 + (Z - L_1 - v)^2 + uv. \quad (78)$$

Lemma 24.1 (Canonical complementarity). *On natural inputs, (77) vanishes exactly when $Z = \min(L_0, L_1)$ and $(u, v) = (L_0 - Z, L_1 - Z)$. Formula (78) has the analogous statement for the maximum. These polynomials are nonnegative on the entire real nonnegative orthant.*

Proof. Every displayed summand is nonnegative on that orthant. A zero therefore forces both linear residuals and uv to vanish. In the minimum case, $L_0 = Z + u$ and $L_1 = Z + v$, so Z is a lower bound; since $uv = 0$, it equals at least one of the two values. The gaps are then forced. The maximum case is identical with inequalities reversed. Conversely, the asserted output and gaps make every term zero. $\square$

When the two actions tie, both gaps are zero. There is no selector bit whose value can be chosen in two ways. Also, uv is *not squared*. Squaring it would unnecessarily raise the degree from two to four. Its sign is already controlled by the domain. This use of complementarity is consistent with the repository's prior quadratic neural-certificate work [7]; the contribution here is its full game-level specialization and resource ledger. ([write] The repository work meant is source 15 of Part VII of [21]: Lemma `pqc:pm:lem:clamp`, an unsquared complementarity certificate on $\mathbb{N}$ of degree two, and Theorem `pqc:pm:thm:trace`, one degree-two polynomial with canonical scales and a unique natural zero; compare also `pqc:sc:lem:relucert`. Lemma 24.1 specializes that device to minimum and maximum gates; the game-level gate types and counts are this source's.)

For a one-action state use the single square $(Z - L_0)^2$.

24.3 The complete polynomial

For each step $t = 0, \dots, T - 1$ introduce the next state vector X_{t+1} , and two gaps for each binary-decision state. Sum the state terms (77), (78), or the one-action square. Add an endpoint sum E_T . The resulting *one polynomial* is

$$Q_T(A; W) = \sum_{t=0}^{T-1} \sum_{i=1}^N q_{t,i} + E_T. \quad (79)$$

For scalar equality use $E_T = (X_{T,r} - X_{T,s})^2$. For consensus use $E_T = \sum_{i=2}^N (X_{T,i} - X_{T,1})^2$. To test a specified integer endpoint vector Y_T , use $E_T = \sum_i (X_{T,i} - Y_{T,i})^2$.

Theorem 24.2 (Unique natural witnesses at every fixed horizon). *For fixed game data and an externally specified T , the polynomial (79) has integer coefficients and degree at most two, and is nonnegative on the real nonnegative orthant. For each supplied initial vector, its natural zero set is empty when the actual endpoint does not meet the observation, and consists of exactly one complete witness tuple when the observation holds.*

With b binary-decision states and e endpoint squares, it has exactly

$$\boxed{(N + 2b)T \text{ natural witnesses, } (N + b)T + e \text{ squared linear forms, } bT \text{ products.}} \quad (80)$$

Here $e = 1$ for a scalar pair, $e = N - 1$ for consensus, and $e = N$ for a specified full endpoint. These are counts of the stated construction, not minimality results.

Proof. There are N next-state coordinates and $2b$ gaps per step. A one-action state contributes one squared linear form; a binary state contributes two squares and one complementarity product. This gives the ledger.

All residuals are linear in the variables and all products have degree two. Nonnegativity prevents cancellation between any two terms. At a zero, starting from the supplied X_0 , Lemma 24.1 and the one-action constraints force every X_1 coordinate and every first-step gap. Induction forces the whole trajectory and all gaps. The endpoint squares then hold exactly when the actual endpoint satisfies the observation. Conversely that trajectory and those forced gaps make all terms vanish whenever the observation holds. The case $T = 0$ has no existential variables and reduces to the endpoint test on X_0 ; the unique tuple in the true case is the empty tuple. $\square$

This proves uniqueness of the *entire* witness tuple, not just of a projected trajectory. It does not assert that a first-hit witness is unique when time is also allowed to vary.

24.4 The fixed-point theorem's nonzero reward

For $\Phi(v) = \frac{1}{4}\mathbf{1} + \frac{1}{2}S(v)$ use the slightly enlarged scaling

$$X_t = 4d_0(2D)^t v_t, \quad X_0 = 4A. \quad (81)$$

Then

$$X_{t+1,i} = \text{opt}_a \left(\sum_j P_{ia,j} X_{t,j} + d_0(2D)^{t+1} \right). \quad (82)$$

For fixed T , every power appearing here is a known integer coefficient. The expression is linear in X_t and d_0 . To test the actual fixed point $\frac{1}{2}\mathbf{1}$, use

$$E_T = \sum_{i=1}^N \left(X_{T,i} - 2d_0(2D)^T \right)^2. \quad (83)$$

Thus the full fixed-point-attainment certificate is still quadratic, including in its displayed input parameters A, d_0 . Its count is $(N + 2b)T$ witnesses, $(N + b)T + N$ squares, and bT products. The external input conditions are $d_0 \geq 1$ and $0 \leq A_i \leq d_0$ when the payoff vector is required to lie in the unit cube.

For completeness, the implementation permits any rational discount $\delta = a/b_0 \in (0, 1)$ and any nonnegative common rational reward r . Let D clear the transition probabilities, let q be a rational constant point target when one is prescribed, and choose

$$m = \text{lcm}(\text{den}(r), \text{den}(q)), \quad \lambda = md_0,$$

with target denominator one when there is no point target. Then

$$X_t = \lambda(b_0 D)^t v_t, \quad X_0 = mA,$$

uses the integer recurrence

$$X_{t+1,i} = \text{opt}_a \left(a \sum_j P_{ia,j} X_{t,j} + \lambda(b_0 D)^{t+1} r \right).$$

Every forcing term is an emitted integer; no unnamed relation remains in the exported polynomial. The endpoint $X_{T,i} = \lambda(b_0 D)^T q$ is also integral. For the main theorem this rule gives exactly (81)–(83).

24.5 Size and precision

For a zero-reward run, $\max_i X_{t,i} \leq D^t \max_i A_i$ and each gap is at most $D^{t+1} \max_i A_i$. The bit length of every witness is therefore $O(\text{bits}(A) + T \log(D + 1))$. For the common-reward construction and cube inputs, $X_t \leq 4d_0(2D)^t \mathbf{1}$, giving the analogous linear-in- T bit-length estimate. The number of witnesses is also linear in T .

These bounds concern a horizon supplied in unary or otherwise charged by its numerical value. They are not polynomial bounds in $\log T$. The large dense reset rows can make the unexpanded polynomial sizeable: the count of variables alone does not count the coefficient storage or the work needed to evaluate all dense linear forms.

25 Worked examples and complete emitted objects

25.1 A two-state polynomial that can be checked by hand

Consider the zero-reward game with discount $1/2$ and effective action rows

$$\begin{aligned}\text{MIN state:} & \quad (1/2, 1/2), (1/4, 3/4), \\ \text{MAX state:} & \quad (3/4, 1/4), (1/2, 1/2).\end{aligned}$$

It is obtained from a half-uniform reset on an averaging/copy game. Start at $v_0 = (1, 0)$. The discounted trajectory begins

$$v_1 = (1/8, 3/8), \quad v_2 = (1/8, 1/8).$$

Here $D = 4$, and $X_t = 8^t v_t$ gives $X_1 = (1, 3)$ and $X_2 = (8, 8)$. Use trajectory variables $(x, y) = X_1$, $(z, w) = X_2$, and gaps a, b, c, d, e, f, g, h . The complete scalar-equality polynomial is

$$\begin{aligned}Q = & (2 - x - a)^2 + (1 - x - b)^2 + ab \\ & + (y - 3 - c)^2 + (y - 2 - d)^2 + cd \\ & + (2x + 2y - z - e)^2 + (x + 3y - z - f)^2 + ef \\ & + (w - 3x - y - g)^2 + (w - 2x - 2y - h)^2 + gh \\ & + (z - w)^2.\end{aligned}\tag{84}$$

Its unique natural zero is

$$(x, y, z, w) = (1, 3, 8, 8), \quad (a, b, c, d, e, f, g, h) = (1, 0, 0, 1, 0, 2, 2, 0).$$

The expanded polynomial has 45 nonzero monomials. Both the complete factored polynomial and its expansion are delivered. The machine-readable variable ordering interleaves each time slice's coordinates and gaps; its label array specifies that ordering explicitly.

This small game is not universal. Its purpose is to expose the actual Diophantine construction, including its degree, natural domain, tie behavior, and full witness count.

25.2 A compiled countdown program

The source has one counter and two control states:

State	Instruction
0	If the counter is zero, go to 1; otherwise decrement it and stay at 0.
1	Halt.

From state zero with counter two, halting is reached after three source steps. The complete compiler exports two versions.

Quantity	Absorbing/scalar version	Erasing/fixed-point version
Padded game states N	128	128
Binary-decision states b	12	10
Layered nodes, including inputs	52	41
Pipeline period M	8	8
Normalization K	128	64
Common reward	0	1/4
Initial payoff denominator	512	256
First observed event	equality at 17	full fixed point at 30
Natural witnesses at that horizon	2,584	4,440
Squared linear forms	2,381	4,268
Complementarity products	204	300

Both use discount $1/2$, a half-uniform reset, and dyadic full-support rows. The first equality time is $2M + 1 = 17$, as (72) predicts. In the erasing version, Lemma 20.2 gives the safe full-drain bound $(H + 1)M = 32$; the actual full fixed point is attained earlier, at time 30. The test checks all coordinates at every preceding time, not only the two observed registers.

The numerical tables above describe this illustrative one-counter program. They are not a claim of universality with 128 states. The game exports include every base action row and the explicit common-reset fraction. The certificate exports have the reset already expanded into all integer linear forms; they do not leave transition semantics as an oracle.

26 Reproducibility, checks, and verification limits

The archive has one standard-library implementation, a test/export driver, and a separate checker that does not import the generator. Python 3.13.5 was used; the code uses language features available in Python 3.10 and later. No floating-point approximations, numerical optimization, or random search for integer roots are used. The reproducible pseudorandom seed is 20261002; it controls only finite test cases, not theorem selection or proof validity.

The code exports all linear forms and all complementary products. Thus `countdown_certificate.json` and `fixedpoint_certificate.json` each specify one complete polynomial in factored form, a full variable list, a complete natural assignment, and a resource ledger. Factored representation avoids a gratuitous expansion of the dense reset terms. The 12-witness example additionally includes its fully expanded sparse polynomial.

[write] The two large certificates are *not stored* in this report: `countdown_certificate.json` (11,921,969 bytes) and `fixedpoint_certificate.json` (21,813,796 bytes) are regenerated exactly by the shipped programs, and the README of this report says how and gives their SHA-256 hashes. They also survive in the original archive of the arrival commit. The other delivered artifacts are shipped as `data/10-exact-convergence-*.json`; the delivered `artifacts/test_run.txt`, a byte copy of `verification.json`, is not.

To regenerate the examples and tests:

```
python code/run_checks.py
python code/check_certificate.py artifacts/fixedpoint_certificate.json \
    --game artifacts/fixedpoint_game.json
```

[write] These commands use the delivered layout: `code/run_checks.py` imports `bellman_diophantine` and `check_certificate` by their delivered names and rewrites every file of `artifacts/`. In this report the programs are `code/10-exact-convergence-*.py`; the README gives a recipe that runs on a copy.

The recorded run contains 26,077 assertions. Selected categories are:

Check category	Assertions
Exhaustive bounded local min/max certificate equivalences	17,150
Raw counter-map successor comparisons	640
Layered counter-map successor comparisons	640
Positive homogeneity comparisons	640
Common-reset coordinate-difference identities	1,920
Original-pipeline checks at every microstep	248
Erasing-pipeline checks at every microstep	100
Chance-only Skolem-lift exact identities	225
Expanded versus factored small-polynomial evaluations	100
Each countdown witness coordinate incremented individually	2,584
Independent complete-export checks	3

The full category list, including ties, input rejection, time-zero cases, nonzero rewards, rational targets, and initially halted pipelines, is in `artifacts/verification.json`. ([write] Shipped as `data/10-exact-convergence-verification.json`. The suite was rerun at the write on a copy with the delivered layout: 26,077 assertions, and all eight regenerated exports equal the delivered files apart from line endings.)

For the 2,584 coordinate-mutation checks, the exact change in the polynomial under a unit increment of one coordinate is computed from its squared-linear coefficients and the paired gap values; the test does not rerun a dense polynomial thousands of times. Separate small-instance tests directly evaluate mutated assignments. The distinction is recorded to avoid inflating what the test suite independently establishes.

The independent checker validates natural domains, variable indices, counts, and the integer value of the complete exported polynomial. When a game export is supplied, it reconstructs the reset rows independently, checks the input payoff vector, and recomputes every scaled trajectory coordinate and every gap. This catches serialization or exporter mistakes that a generator-only self-check could miss.

[write] *A defect in the delivered checker.* The research programme’s review of this manuscript [24] found that the checker compares the declared input with the game’s initialization but replays the trajectory from a separate scaled initial vector that it does not bind to that input. Changing only the declared input numerators of a genuine one-step certificate, and supplying a game with the same changed initial payoffs, still yields polynomial value zero and a successful independent game check although the actual endpoint fails the observation. The integer polynomial itself remains an honest certificate for its own input; the false assertion is the checker’s binding. The review’s patch `exact_convergence_input_binding.patch` binds the input vector, denominator, witness scale, reward, discount, endpoint mode and game, and leaves every valid delivered example accepted; the review also reconstructs every coefficient of the three delivered polynomials from their games and inputs. The checker shipped here, `code/10-exact-convergence-check_certificate.py`, is the unpatched original.

None of this proves an infinite theorem by testing. The inductive proofs in Sections 19–24 carry that obligation. No Lean, Rocq, or Isabelle formalization of the new game compiler was produced or run. The repository’s MRDP interface is an inspected source, not a formal verification of these additional constructions.

27 From bounded quadratics to a fixed Diophantine relation

For the fixed game of Theorem 17.1, encode each valid rational payoff vector by a natural number. Its finite-attainment set E is c.e. and c.e.-complete. MRDP, in precisely the fixed-arity form exposed by the repository [15], yields

$$\exists k \exists P \in \mathbb{Z}[n, y_1, \dots, y_k] \forall n \in \mathbb{N} : \quad n \in E \iff \exists y \in \mathbb{N}^k P(n, y) = 0. \quad (85)$$

The ordinary polynomial P and its witness dimension are fixed before the input. We do not print its coefficients or claim an operation count for it. A constructive formal route to MRDP through arithmetic computation is also described in the published Coq development [14].

The bounded quadratics do not themselves give a fixed quadratic polynomial for unbounded time: Q_T has $(N + 2b)T$ witnesses, and its time-dependent coefficients in the reward version also depend on T . Writing “ $\exists T, W \ Q_T = 0$ ” does not turn a varying-arity family into one first-order polynomial equation.

At each fixed horizon the transition graph is described by finitely many linear equalities and inequalities, one finite collection for each action comparison pattern. It is consequently Presburger-definable on its integer parameters. The quadratic complementarity package compresses this finite union into one equation over the orthant. The computational difficulty arises from unbounded iteration, not from a hidden claim that one fixed quadratic polynomial represents every enumerable set.

27.1 A quartic existence corollary, with the correct caveat

One can convert any fixed integer polynomial in (85) into a fixed sum of squares of quadratic residuals. Separate its positive and negative parts, evaluate each by additions and multiplications on natural registers, and impose equality of their outputs. A multiplication gate has residual $z - xy$ and an addition gate residual $z - x - y$. Squaring and summing produces a fixed nonnegative polynomial of degree at most four with the same projected zero set.

The added circuit registers are uniquely determined by the original witness y . The original MRDP witnesses are not thereby made unique. Thus neither the single-fold nor the finite-fold Diophantine representation problem is settled. The horizon-indexed uniqueness in Theorem 24.2 is deliberately a different assertion.

[write] The circuit argument of this subsection is the unique-extension quadratization `cdc:wf:lem:quadratization` of [22] and `pqc:lem:quarticlower` of [21]; Part I proves the same corollary for exact erasure (Proposition 11.1). No novelty is claimed for it.

27.2 No computable global search cutoff

Corollary 27.1. *For the fixed universal game there is no total computable bound, as a function of the length of the input payoff encoding, on the first exact attainment time of all inputs that eventually*

attain the fixed point. Likewise, for any fixed polynomial representation of its attainment set, there is no total computable input-length bound that guarantees a solution with all witness coordinates below that bound whenever a solution exists.

Proof. Either bound would turn semidecision into decision: compute the bound and search the finite time interval, or the finite witness box. That would decide a c.e.-complete set. $\square$

[write] This is the bounded-search argument of `cdc:bd:prop:nobound` in [22]; for contractions in the collection it appears as `pqc:sc:cor:gap` and `pqc:ad:cor:nobound` in Part VII of [21], and in Part I as Theorem 11.2.

The repository README read during this audit reports 75 arithmetic operations for its complete certificate and 87 for one universal polynomial [5]. Nothing here changes either figure. Our counted objects are native game tables and horizon-dependent quadratic certificates, not a competing numerically instantiated universal Diophantine polynomial.

28 Precision, robustness, and why approximation is not a decision

The main operator’s fixed point is explicit and its approximation error is uniformly bounded. On compiled initializations the stronger estimate

$$\|\Phi^n(v(x)) - v_*\|_\infty \leq 4^{-n}/2$$

holds whether or not the source computation halts. Therefore any prescribed absolute accuracy is obtained after a computably bounded number of steps, while exact finite attainment remains undecidable.

For the absorbing scalar construction, a nonzero prehalting register gap has the exact value in (71). If $K = 2^\kappa$, its magnitude is

$$2^{-(1+2n+\kappa\lceil n/M \rceil)}.$$

For a known finite horizon, its bit requirement is explicit. Two separately approximated coordinates with absolute error less than one quarter of this gap can distinguish it from zero, provided a corresponding promise about the two alternatives is available. There is no contradiction with the lack of a computable bound on how far one must look for a future exact zero. Precision requirements and time requirements must not be conflated.

Proposition 28.1 (A fixed positive threshold restores a finite cutoff). *For a zero-reward discounted game on $[0, 1]^N$ and a supplied rational $\varepsilon > 0$, the property $\exists n (v_{n,r} - v_{n,s} \geq \varepsilon)$ is decidable.*

Proof. Every coordinate of v_n lies in $[0, \delta^n]$, so the difference is at most δ^n . Once $\delta^n < \varepsilon$, no later time can qualify. Compute such a cutoff and check the finite prefix exactly. $\square$

[write] The same elementary argument decides threshold reachability for source 14’s map in Part VII of [21] (Propositions `pqc:sc:prop:away` and `pqc:sc:prop:threshold`).

A positive absolute tolerance does not solve the zero-threshold predicate: it changes the predicate. Strict inequalities with threshold zero can still encode halting, as Section 22 shows. Similarly, full support is not arbitrary robustness against perturbing the transition table. The theorem proves preservation under one very specific, exactly known common reset. It does not say that all nearby

games have the same halting language, or that rounded floating-point implementations retain universality.

Finite-precision implementations have finitely many representable states under a specified rounding and overflow policy. Their exact reachability can in principle be decided by finite exploration. That fact is compatible with the exact-rational result, because the required precision is not bounded uniformly over all running times.

29 Research agenda and concrete next targets

29.1 Instantiate and minimize a fixed universal game

The most direct next deliverable is to choose an explicit published universal counter program, enter its entire instruction table, and export the resulting game and proof certificate. The output should include the natural-input loader, not merely a claim about a universal circuit. Minimize the number of states, binary decisions, stochastic successors, normalization factors, and the maximum common denominator separately. These objectives need not agree. The present depth-seven source compiler makes the microstep period bounded by eight, but it is not a size-optimal construction.

A particularly concrete optimization is to share delay copies, normalized linear forms, and zero tests while preserving the two-channel invariant. Any change must be checked against the actual feedback schedule and full pipeline drain, not only against one-step circuit equivalence.

29.2 Remove one player’s genuine choices

Does an analogous fixed-game finite-attainment theorem hold when every nontrivial decision state belongs to MAX, or all belong to MIN? The present proof does not remove the opposite owner: the negative channel of a minimum is a maximum. Any attempted one-player translation must replace this exact dependency, rather than silently switching objectives. This question is close to the remaining Bellman-operator reachability frontier in [10], but their exact hypotheses and observation must be matched before claiming a resolution.

Even a restricted subclass with either a proof of undecidability or a uniform decision procedure would clarify which portion of the two-player compiler carries the essential nonlinearity.

29.3 Prescribed randomness and sparse supports

Can all chance computations be routed through one shared randomizing state, or through a constant-size family of fair-coin gadgets? Can one retain a constant number of stochastic successors per action before the common reset? A global reset and sparse support cannot both describe the same expanded row when N grows, so the cost model must distinguish a native “reset to uniform” instruction from its explicitly expanded probability table.

The current theorem uses an exact common reset. It would be valuable to characterize all reset kernels that preserve the simulated quotient modulo $\mathbb{R}\mathbf{1}$, and all nonuniform resets that still admit an invariant paired-channel subspace. This is a precise algebraic question about the reset’s action on V , not a claim of unrestricted noise robustness.

29.4 One-dimensional observations versus whole-vector attainment

The scalar/whole-vector distinction is already decisive for chance-only systems: Skolem-equivalence on one side, kernel stabilization on the other. Investigate the minimum number of observed coordinates required for hardness in one-player stochastic systems. Related targets include a specified affine subspace, a consensus line, a nonzero point, and a fixed point. They should be studied as distinct problems rather than collapsed under the word “reachability.”

For the two-player construction, can full fixed-point attainment be reduced to equality of one designated pair with no additional input promises? The present reductions give completeness of each problem separately; they do not furnish a global equivalence on every arbitrary initialization.

29.5 Witness-economical native certificates

The quadratic compiler spends two natural gaps per binary comparison. Determine the minimum arity of a single-fold orthant-nonnegative quadratic representation of the binary min and max graphs under a fixed interface. One gap can encode distance from one action but loses the sign of the other difference; naive elimination risks increasing degree or introducing multiple witnesses. A useful result would be a proved lower bound under clearly stated interface restrictions, or an explicit smaller compiler.

Time-dependent reward terms are emitted as integer coefficients. A shared canonical clock could trade coefficient size for additional variables. Compare the two representations using bit complexity and literal evaluator costs, not only total polynomial degree.

29.6 Integrate with the repository’s fixed-arity MRDP route

A formal interface should expose exact game-step evaluation, its computability, and the semidecision procedure for the chosen observation. The repository’s MRDP theorem can then supply the abstract fixed polynomial. Producing a fully expanded native-to-MRDP compiler with audited operation counts is a separate engineering and mathematical project.

Track three properties independently: preservation of the accepted input set, preservation of projected trajectories, and preservation of complete witness tuples. The present bounded compiler proves all three at fixed T . An unbounded history compression must not inherit the strongest property without a separate proof.

29.7 Formalize the proof-critical interfaces

The highest-value formalization targets are the counter encoding, layer-normalization invariant, every-microstep pipeline lemma, erasing pipeline drain, common-reset identity, and the zero-set uniqueness theorem. These are short enough to be individually inspectable and strong enough to expose most plausible compiler mistakes.

A proof-carrying exporter could attach a machine-checkable derivation of each stochastic row and each quadratic residual. The independent checker in this archive is a useful testing boundary, but it does not replace a kernel-checked theorem connecting exported coefficients to the original counter semantics.

30 Conclusion

The main result isolates a computationally universal exact iteration inside a tightly constrained stochastic-game interface. Its transition rows are positive and dyadic; its discount is fixed; its limiting value is obvious; and its error decays geometrically. None of these facts supplies a decision procedure for finite exact attainment from arbitrary rational initial values.

The construction’s nontrivial obligations are explicit: signed values are realized by paired channels, nested computation is realized by a cyclic pipeline, initialization accounts for every phase, and halting erases the entire pipeline rather than only its visible registers. Uniform resetting preserves the computation in a precise invariant subspace.

At fixed time, the same substrate has a particularly economical arithmetic semantics: one orthant-nonnegative quadratic polynomial with a unique complete natural witness whenever its endpoint condition holds. At unbounded time, MRDP supplies a fixed Diophantine relation, but not the bounded family’s uniqueness. The chance-only and selector-only boundaries explain why both the observation and the computational primitives must be specified carefully. These distinctions are the foundation for further native compilers and formal developments, rather than shortcuts around them.

Part III

Thermal arithmetic (source 03)

[write] Part III prints batch-78 manuscript 03, *Thermal Arithmetic: Diophantine Witnesses, Effective Excited Spectra, and Undecidable Confinement Boundaries*, under the kicker “ProveIt research continuation”, “Research report prepared for Vladimir Reshetnikov”, October 2, 2026 (the manuscript names no assistant). Its PDF metadata gave the shorter subtitle *Diophantine Witnesses and Undecidable Confinement Boundaries*; the printed subtitle is used here. Its labels carry the prefix `ste:ta:`. Its abstract and status statement follow; its appendix is [Appendix E](#).

Abstract (source 03). An existential Diophantine representation controls whether witnesses exist, but generally not how many there are or how far away they lie. We construct a witness-faithful spectral normal form that separates these difficulties. A natural arithmetic circuit with k witness inputs and g materialized gates gives a diagonal, nonnegative, integer-valued Hamiltonian on $m = k + g + 1$ occupation modes, of joint polynomial degree at most five, with an explicit decomposition into positive commuting terms supported on at most four modes. Its zero-energy basis states are in bijection with the original witnesses, while every positive-energy state satisfies $E \geq 1 + \sum_i n_i$.

This radial isolation makes the positive-energy resolvent effectively compact and the excited thermal trace uniformly computable, even when the zero-energy space is infinite. Compact resolvent and finite unconfined thermal trace are equivalent to finite witness multiplicity. A free-witness padding construction gives unconditional undecidability of both properties on a fixed universal family; requiring compactness on every input instead is exactly a finite-fold representation condition.

A second construction produces a uniformly computable critical fugacity $t_*(c) \in (0, 1]$ for each encoded computation. The exact assertion $t_*(c) < 1$ is equivalent to halting, although every regulated partition function and every finite-precision approximation to $t_*(c)$ is computable. We prove explicit tail estimates, quantitative witness-height bounds, and the absence of uniform

computational cutoffs. An exact Python compiler and 7,936 passing regression assertions accompany the proofs.

Status (source 03). The proofs below establish the stated normal-form and boundary results from standard Diophantine representability. Their priority in the literature is not established. This report does not resolve the finite-fold conjecture, claim physical hypercomputation, provide a numerical universal polynomial, or claim proof-assistant verification.

31 Research question, scope, and relationship to ProveIt

31.1 Why another computational substrate?

The central question of this report is not simply whether a Turing-complete model has a Diophantine halting predicate. Once a model has an effective finite observation semantics, standard representability already supplies such a predicate. The more informative question is whether an arithmetic representation can separate three independent obstacles:

existence of witnesses, multiplicity of witnesses, height of witnesses.

We use a finite number of occupation-number modes, each with infinitely many basis states. The substrate is a static polynomial energy landscape, not a step-by-step simulator. Its zero-energy questions represent computation; its positive-energy states have effective, uniform confinement. In particular, infinitely many witnesses can coexist with an entirely computable excited-state thermal trace.

The repository’s October 2 review of computational substrates distinguishes horizon-dependent certificates from one fixed universal polynomial, and existence-preserving compilation from witness-preserving compilation. It also identifies uncontrolled multiplicity in a fixed-quartic MRDP route [6]. ([write] The fixed-quartic route meant is Corollary `pqc:tu:cor:fixedquartic` of [21], Part VIII, and the remark after it, which the cited review discusses.) Here multiplicity is no longer an optional implementation detail: it is exactly the obstruction to compact resolvent and to a finite unconfined thermal trace. The present construction builds on that distinction rather than proposing another uncharged bounded-history encoding.

31.2 Source snapshot and limits of the audit

The repository was inspected at commit

6914ccca6685baf53b7a35f25efc89366c76ba74.

The relevant audit file has blob identifier

8bd07142c8b4298fb636e47b40d2d84e993a40ce.

The repository’s reported 87-operation fixed universal polynomial remains a separate arithmetic-circuit result [5]. This report neither improves that bound nor counts an existential MRDP polynomial as a free primitive in a numerical operation ledger. The source inspection was not a rebuild of the Lean or Rocq developments, and the proofs below do not assume that every research manuscript in the repository has been formally verified.

31.3 Established foundations and the proposed contribution

We use the standard fact that every computably enumerable relation over natural numbers has an existential polynomial representation, and that one fixed polynomial can represent a universal such relation [16]. A natural arithmetic-circuit expansion then replaces a high-degree polynomial equation by quadratic gate equations with uniquely determined intermediate values. We give that expansion explicitly because its witness multiplicity matters here. ([write] This expansion is the unique-extension quadratization of the collection's report [22], Lemma `cdc:wf:lem:quadratization`, with the definition of witness-faithfulness given there; see also `cdc:cs:lem:quartic` in [22] and, without the uniqueness statement, `pqc:lem:quarticlower` in [21]. In Theorem 32.1 the zero-set bijection and the quartic sum of squares are that lemma; new here are the marker $a - 1$, the radial factor $1 + \sum_i n_i$, the bound $H > 0 \Rightarrow H \geq 1 + \sum_i n_i$, the degree five and the $r(m + 1)$ four-mode ledger.)

The basic substitution of number operators into a squared Diophantine polynomial is not new. Kieu already writes a Hamiltonian of the form $D(N_1, \dots, N_k)^2$ and identifies zero energy with an integer solution [17]. We use only that mathematical observation, not any proposed physical decision procedure. Our additional factor $1 + \sum_i N_i$ removes all nonzero-energy flat directions while leaving the zero set unchanged. The effective spectral decomposition, multiplicity classification, and critical-fugacity construction are the results developed and proved here. ([write] Part V of [21] (source 13 there, Theorem `pqc:ql:thm:compiler`) builds a witness-faithful quartic energy landscape for Boolean circuits over $\mathbb{R}$ and states that it treats no Gibbs distribution; this Part is the unbounded natural counterpart, with a thermal trace.)

Finite-fold and single-fold representation questions are discussed in the primary literature [18, 19]. The equivalences below are reformulations of those representation conditions, not answers to them. In particular, we do not silently replace an arbitrary MRDP representation by one with finitely many witnesses.

31.4 Main conclusions at a glance

Let $Q = \sum_j f_j^2$ be the compiled quartic penalty and $s = \sum_i n_i$. The construction is

$$H = (1 + s)Q. \tag{86}$$

Its essential features are:

- (i) $\ker H$ has one occupation basis vector for each original witness, with no extra auxiliary multiplicity.
- (ii) The positive-energy sector is effectively confined: $H > 0$ implies $H \geq 1 + s$.
- (iii) Compact resolvent and finite thermal trace hold exactly when there are finitely many witnesses.
- (iv) A regulated partition function has a computable threshold location whose exact contact with the boundary $t = 1$ encodes nonhalting.

The novelty claim is deliberately limited: this is a proved package of normal-form, effective-estimation, and boundary results proposed for further research. The literature search and repository audit do not establish that every individual consequence first appears here.

32 Arithmetic conventions and the witness-faithful compiler

32.1 Natural domains and representations

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. A parameter tuple $p \in \mathbb{N}^d$ is fixed when studying an individual operator. Witness coordinates range independently over $\mathbb{N}$. A polynomial $P \in \mathbb{Z}[p, x_1, \dots, x_k]$ represents a relation by

$$A(p) \iff \exists x \in \mathbb{N}^k P(p; x) = 0. \quad (87)$$

A representation is *finite-fold* if each parameter fiber has finitely many witnesses, including zero witnesses outside the relation. It is *single-fold* if each fiber has at most one witness. A representation originally written over positive integers can be moved to our convention by the bijective substitution $x_i = u_i + 1$.

All polynomial degrees stated below are total degrees jointly in parameters and occupation variables unless explicitly qualified. Parameter values can be arbitrarily large, but the polynomial expression and the number of coordinates do not change with their values.

32.2 Natural arithmetic circuits

Write

$$P = P^+ - P^-, \quad P^+, P^- \in \mathbb{N}[p, x].$$

Choose finite directed acyclic circuits for these two nonnegative polynomials using constants in $\mathbb{N}$, parameter and witness inputs, and binary addition and multiplication. Common subexpressions may be shared. Each actual gate introduces one natural coordinate u_j , in a topological order. The corresponding residual is one of

$$u_j - v - w, \quad u_j - vw, \quad (88)$$

where v, w are constants, parameters, original witness coordinates, or outputs of earlier gates. The final residual is the difference between the two output atoms. A final, separate coordinate a has residual $a - 1$.

The purpose of a is not arithmetical expressiveness. It makes the occupation vacuum a positive-energy state, which will ensure that the thermal threshold starts strictly below its target level. It adds exactly one coordinate and one equation, and is uniquely determined at every zero.

Theorem 32.1 (Witness-faithful degree-five compiler). *Suppose the chosen natural circuit has k witness inputs and g materialized gates. There is an effectively constructed integer polynomial $H(p; n)$ with*

$$m = k + g + 1, \quad r = g + 2, \quad (89)$$

occupation coordinates and residual equations, respectively, satisfying the following properties.

For every fixed p , projection onto the first k coordinates is a bijection

$$\{n \in \mathbb{N}^m : H(p; n) = 0\} \longrightarrow \{x \in \mathbb{N}^k : P(p; x) = 0\}.$$

Every occupation value of H is a nonnegative integer, $H(p; 0) \geq 1$, and

$$H(p; n) > 0 \implies H(p; n) \geq 1 + \sum_{i=1}^m n_i. \quad (90)$$

The joint degree of H is at most five. It has a displayed decomposition into $r(m + 1)$ nonnegative commuting terms, each supported on at most four occupation modes.

Proof. Let $f_1, \dots, f_r$ be all gate residuals, the final output difference, and $a - 1$. Put

$$Q(p; n) = \sum_{j=1}^r f_j(p; n)^2, \quad H(p; n) = \left(1 + \sum_{i=1}^m n_i\right) Q(p; n).$$

Since every residual is an integer, $Q = 0$ exactly when all residuals vanish. Fix p and the original witness tuple x . Topological induction through the circuit forces each u_j to be the natural value of its gate. Thus there is exactly one assignment of intermediate coordinates satisfying the gate equations. The marker is then uniquely $a = 1$. At this assignment the remaining output residual is $P(p; x)$. Consequently a source root has exactly one zero-energy lift, and every zero-energy occupation tuple is such a lift.

Each residual has joint degree at most two. Therefore $\deg Q \leq 4$ and $\deg H \leq 5$. Every factor $1 + \sum_i n_i$ is positive on the natural domain, and every nonzero Q is an integer at least one. This proves (90). At the vacuum, the marker residual equals -1 , so $Q(p; 0) \geq 1$.

Finally, expand only the radial factor, not the residual squares:

$$H = \sum_{j=1}^r f_j^2 + \sum_{i=1}^m \sum_{j=1}^r n_i f_j^2. \quad (91)$$

A gate residual involves at most three occupation modes; the output residual involves at most two, and the marker one. Multiplication by n_i adds at most one mode to the support. Each displayed term is nonnegative on the natural domain. As diagonal multiplication operators all these terms commute. There are $r + mr$ terms in this unsimplified display. $\square$

Remark 32.2 (What this ledger does not count). The number $r(m + 1)$ counts displayed positive summands, not additions and multiplications in an optimized evaluation circuit. Formula (86) is usually cheaper to evaluate than fully expanding (91). Constant bit lengths are also not charged by this ledger. Four-mode support does not mean nearest-neighbor geometry, bounded local Hilbert-space dimension, or bounded operator norm. If the degree is measured in creation and annihilation operators rather than number operators, substituting $N_i = a_i^\dagger a_i$ gives a bound of ten instead of five.

Corollary 32.3 (Fixed universal family). *There is a fixed m and one fixed integer polynomial $H(c; n)$ of joint degree at most five such that*

$$c \in \mathsf{K} \iff \ker H_c \neq \{0\},$$

where K is a fixed computably enumerable complete halting set, and every fiber has the properties of Theorem 32.1.

Proof. Choose once and for all a polynomial $U(c; x)$ representing K , as supplied by Diophantine representability. Apply the fixed finite compiler to U . Only c changes from instance to instance. $\square$

No numerical value of m is claimed here: the universal polynomial is not instantiated in the accompanying code. This distinction prevents a low-degree existence theorem from being mistaken for a competitive numerical universal equation.

32.3 A small example with a complete resource count

For $P(p; x, y) = xy - p$, use one gate $u = xy$. The four occupation coordinates are (x, y, u, a) and

$$Q_p = (u - xy)^2 + (u - p)^2 + (a - 1)^2, \quad (92)$$

$$H_p = (1 + x + y + u + a)Q_p. \quad (93)$$

Thus $k = 2$, $g = 1$, $m = 4$, $r = 3$, and there are 15 displayed positive terms. For $p = 6$, the four zero-energy tuples are

$$(1, 6, 6, 1), (2, 3, 6, 1), (3, 2, 6, 1), (6, 1, 6, 1).$$

For $p = 0$, infinitely many tuples occur because $xy = 0$ has infinitely many natural solutions. The circuit has preserved that distinction exactly; it has not attempted to regularize the source multiplicity.

33 The operator and its effectively confined positive spectrum

33.1 Domains are part of the construction

Fix the parameters and suppress them from the notation. Let

$$\mathcal{H}_m = \ell^2(\mathbb{N}^m), \quad N_i |n\rangle = n_i |n\rangle.$$

For the polynomial $H(n)$, define the multiplication operator with maximal domain

$$\mathcal{D}(H) = \left\{ \psi = (\psi_n) : \sum_n H(n)^2 |\psi_n|^2 < \infty \right\}, \quad (H\psi)_n = H(n)\psi_n. \quad (94)$$

Although the formal expression is a polynomial in commuting number operators, (94) specifies the actual closed operator used in all statements.

Lemma 33.1 (Self-adjoint realization). *The operator (94) is nonnegative and self-adjoint. Its restriction to finitely supported vectors is essentially self-adjoint, with closure equal to (94).*

Proof. Symmetry follows from the real diagonal coefficients. If ψ belongs to the adjoint domain of the finite-support restriction, testing against each basis vector forces the adjoint coordinates to be $H(n)\psi_n$. Membership of that vector in ℓ^2 is precisely (94). Thus the adjoint of the restriction is the maximal multiplication operator. The maximal operator is symmetric, and the same coordinate argument identifies its adjoint with itself.

For the core assertion, truncate any $\psi \in \mathcal{D}(H)$ to a finite box. Both the ℓ^2 tail of ψ and that of $H\psi$ tend to zero. Hence finite-support truncations converge in graph norm. Nonnegativity follows from $H(n) \geq 0$ coordinatewise. $\square$

33.2 Energy layers and an exact finite search

For $n \in \mathbb{N}^m$, put $s(n) = \sum_i n_i$. Let

$$\mathcal{Z} = \{n : Q(n) = 0\}, \quad M = \#\mathcal{Z} \in \mathbb{N} \cup \{\infty\}.$$

The radial lower bound is assumed in the next results; consequently they apply both to the compiler and to any directly supplied nonnegative integer polynomial Q used in (86).

Theorem 33.2 (Effective positive-energy layers). *For every integer $E \geq 1$, all basis states with $0 < H(n) \leq E$ lie in the finite simplex $s(n) \leq E - 1$. Their number, with multiplicity, is at most*

$$\binom{E + m - 1}{m}. \quad (95)$$

Every positive eigenvalue is an integer of finite multiplicity, and its exact multiplicity is computable uniformly from the polynomial and the eigenvalue. There is no finite accumulation point of distinct eigenvalues.

Proof. The inequality $H(n) \geq 1 + s(n)$ on nonzero states gives the simplex restriction. The number of natural m -tuples of sum at most $E - 1$ is (95): add a slack coordinate to get a tuple of sum $E - 1$, and count placements of separators. Polynomial evaluation on this finite set decides each positive-energy membership exactly.

The spectrum of a diagonal multiplication operator is the closure of its diagonal values. To see the only potentially nontrivial direction directly, a complex number at positive distance from all those values has a bounded diagonal inverse for $H - \lambda I$; the other direction follows from basis approximate eigenvectors. Here the values are nonnegative integers, so the set of distinct values has no finite accumulation point and is already closed. $\square$

There is also a useful divisor refinement. If $H(n) = E > 0$, then

$$1 + s(n) \mid E, \quad Q(n) = \frac{E}{1 + s(n)}. \quad (96)$$

Thus one may enumerate only shells $s = d - 1$ for divisors $d \mid E$, rather than the entire simplex. This changes the enumeration strategy, not the decidability result.

33.3 Separating the zero projection from a computable compact part

Let P_0 be the diagonal projection onto $\ell^2(\mathcal{Z})$, and let C_B denote projection onto

$$C_B = \{0, 1, \dots, B\}^m.$$

Define K and T_q , for $0 < q < 1$, by their diagonal entries:

$$K(n) = \begin{cases} (1 + H(n))^{-1}, & H(n) > 0, \\ 0, & H(n) = 0, \end{cases} \quad (97)$$

$$T_q(n) = \begin{cases} q^{H(n)}, & H(n) > 0, \\ 0, & H(n) = 0. \end{cases} \quad (98)$$

Theorem 33.3 (Effective compact and trace-class remainders). *The following decompositions hold:*

$$(H + I)^{-1} = P_0 + K, \quad q^H = P_0 + T_q. \quad (99)$$

The operator K is compact, and T_q is trace class. For the finite-rank truncations $K_B = C_B K C_B$ and $T_{q,B} = C_B T_q C_B$,

$$\|K - K_B\| \leq \frac{1}{B + 3}, \quad (100)$$

$$\|T_q - T_{q,B}\|_1 \leq \frac{mq^{B+2}}{(1 - q)^m}. \quad (101)$$

For rational q , these are effective approximation procedures using exact rational finite matrices, uniformly in all polynomial parameters.

Proof. The identities hold on each basis vector. Outside $\mathcal{C}_B$, at least one coordinate is at least $B+1$, so $s(n) \geq B+1$. On positive-energy states this gives $H(n) \geq B+2$. The supremum of the omitted diagonal entries of K is at most $1/(B+3)$, proving (100) and compactness.

For the heat remainder, nonnegative diagonal entries give a trace norm equal to their sum. Since $q^{H(n)} \leq q^{1+s(n)}$ on positive states, a union bound over the coordinate that leaves the box yields

$$\begin{aligned} \sum_{n \notin \mathcal{C}_B, H(n) > 0} q^{H(n)} &\leq q \sum_{i=1}^m \left(\sum_{n_i=B+1}^{\infty} q^{n_i} \right) \prod_{j \neq i} \left(\sum_{n_j=0}^{\infty} q^{n_j} \right) \\ &= \frac{mq^{B+2}}{(1-q)^m}. \end{aligned}$$

The same geometric calculation over the whole space bounds the trace by $q/(1-q)^m$. Exact evaluation of $Q(n)$ tells the finite algorithm which entries are zero and which belong to the remainder. $\square$

Remark 33.4 (Effective strong information is not effective norm information). For each explicitly supplied basis index n , membership in $\mathcal{Z}$ is decidable by evaluating $Q(n)$. The action of P_0 is consequently computable on vectors presented with effective ℓ^2 approximations: project a finite approximation and use $\|P_0\| \leq 1$ to control the tail. This does not supply a uniform finite-rank approximation to P_0 in operator norm. If any zero state lies outside a proposed cutoff, the omitted projection has norm one. The distinction is exactly where an attempted finite spectral decision procedure can fail.

34 Multiplicity, compactness, and thermal finiteness

Theorem 34.1 (Exact multiplicity classification). *For the radially isolated Hamiltonian $H = (1+s)Q$,*

$$\dim \ker H = M.$$

The following are equivalent:

- (i) M is finite;
- (ii) H has compact resolvent;
- (iii) q^H is trace class for some $q \in (0, 1)$;
- (iv) q^H is trace class for every $q \in (0, 1)$.

When $M < \infty$,

$$\mathrm{Tr}(q^H) = M + \mathrm{Tr}(T_q). \tag{102}$$

When $M = \infty$, the essential spectrum is exactly $\{0\}$. When $M < \infty$, the essential spectrum is empty.

Proof. A vector is killed by H exactly when its support lies in $\mathcal{Z}$, proving the dimension statement. By Theorem 33.3, $(H + I)^{-1}$ differs from P_0 by a compact operator. An orthogonal projection is compact exactly when its range is finite dimensional: an infinite orthonormal sequence in its range

has no norm-convergent image subsequence, while a finite-rank projection is compact. This proves (i) $\Leftrightarrow$ (ii).

Similarly, $q^H = P_0 + T_q$ with T_q positive trace class. Its diagonal sum is $M + \text{Tr}(T_q)$, with the usual extended interpretation when $M = \infty$. Thus (i), (iii), and (iv) are equivalent, and (102) follows. The positive eigenvalues are isolated and of finite multiplicity by Theorem 33.2; only the eigenvalue zero can have infinite multiplicity. This proves the essential-spectrum assertions, using the characterization by failure to be an isolated finite-multiplicity eigenvalue. $\square$

Remark 34.2 (Ground states versus zero-energy states). If $M = 0$, the operator still has a ground-state eigenvalue: a nonempty subset of positive integers has a minimum. What fails is specifically the existence of a *zero-energy* state. If $M = 1$, there is a unique zero-energy state up to scalar multiples. These are different statements from uniqueness of a positive-energy ground state in a root-free instance.

34.1 A spectral equivalent of finite-fold representability

Definition 34.3. A polynomial diagonal family is a single integer polynomial $F(p; n)$ in the parameter and finitely many occupation coordinates, nonnegative on all natural tuples, interpreted by the maximal multiplication operator. The number of occupation coordinates is fixed, independent of p .

Theorem 34.4 (Finite-fold/compact-family equivalence). *For a relation $A \subseteq \mathbb{N}^d$, the following conditions are equivalent:*

- (i) *A has a finite-fold natural Diophantine representation.*
- (ii) *There is a polynomial diagonal family F_p with compact resolvent for every p , such that $p \in A$ exactly when $\ker F_p \neq \{0\}$.*

In (ii), whenever (i) holds, one may choose joint degree at most five and the four-mode positive decomposition of Theorem 32.1. The analogous single-fold condition is equivalent to a polynomial diagonal representation with $\dim \ker F_p \in \{0, 1\}$ for every p ; the forward construction can also have compact resolvent on every fiber.

Proof. For (i) $\Rightarrow$ (ii), compile a finite-fold representation. The zero-set bijection preserves finite fiber sizes, and Theorem 34.1 supplies compact resolvent on every fiber.

Conversely, $F(p; n) = 0$ is already a natural polynomial representation of A . Compact resolvent forces every eigenspace, in particular the zero eigenspace, to be finite dimensional. Distinct zero tuples supply orthogonal basis vectors, so there are finitely many such tuples. This is a finite-fold representation. Replacing “finitely many” by “at most one” proves the single-fold assertion in both directions. $\square$

The joint polynomial requirement matters. An arbitrary algorithm that selects a polynomial from p does not itself give one polynomial relation in (p, n) , and invoking representability for that selection may introduce additional witnesses whose multiplicity is uncontrolled. No such substitution occurs in Theorem 34.4.

This equivalence explains why the apparently stronger assertion “there is a universal halting family with compact resolvent on every input” must not be obtained by simply announcing that

circuit auxiliaries are unique. Circuit auxiliaries are unique *over each source witness*; the source fiber may still be infinite.

[write] *Credit and a consequence.* Theorem 34.4 follows the template of the finite-fold substrate equivalence `cdc:cs:thm:finitfold` of [22]: a unique-extension quartic in the forward direction, and the polynomial itself as a representation in the backward direction. What it adds is the replacement of “finite fibre” by “compact resolvent” through Theorem 34.1. Combined with the equivalence `cdc:bd:thm:universal` of [22] it gives the following statement, recorded here as an editorial consequence of the two printed theorems, not as a result of source 03.

Remark 34.5 ([write] All-fibre compactness for universal halting). The following are equivalent: (i) every computably enumerable subset of $\mathbb{N}$ has a finite-fold Diophantine representation over $\mathbb{N}$; (ii) there is one polynomial diagonal family $F(e, x; n)$, in the sense of the definition above, with compact resolvent for every (e, x) and $\ker F_{(e,x)} \neq \{0\}$ exactly when program e halts on input x . When they hold, F can be chosen of joint degree at most five with the four-mode positive decomposition of Theorem 32.1. Indeed, (ii) gives by Theorem 34.4 a finite-fold representation of universal halting, which is statement (b) of `cdc:bd:thm:universal` and hence equivalent to (i), its statement (a); conversely (i) gives (b), and Theorem 34.4 compiles that representation into a family as in (ii). The equivalence decides neither statement.

34.2 An unconditional undecidable compactness family

Theorem 34.6 (Padding transfers halting to loss of compactness). *Let $Q_c(n)$ be the fixed universal compiled penalty in Corollary 32.3. Add one free coordinate $y \in \mathbb{N}$ and define*

$$\tilde{H}_c(n, y) = \left(1 + y + \sum_i n_i\right) Q_c(n). \quad (103)$$

Then

$$\begin{aligned} c \notin \mathbf{K} &\iff \tilde{H}_c \text{ has compact resolvent} \\ &\iff \mathrm{Tr}(q^{\tilde{H}_c}) < \infty \text{ for some/every } 0 < q < 1, \end{aligned}$$

whereas $c \in \mathbf{K}$ gives an infinite-dimensional zero eigenspace and infinite thermal trace for every such q . Thus compactness and thermal finiteness are co-computably-enumerable complete on this fixed padded family; their negations are computably enumerable complete.

Proof. If Q_c has no root, the padded penalty has no root. If it has one root n , every (n, y) is a root, so its root set is infinite. The radial factor includes the new coordinate and hence satisfies the positive-energy lower bound in $m + 1$ modes. Theorem 34.1 gives the asserted analytic alternatives. The exact equivalences with membership and nonmembership in $\mathbf{K}$ give both reductions and the stated enumerability classifications on this family. $\square$

No assumption about finite-fold Diophantine representations occurs in this proof. Also, no claim is made that the unrestricted finite-root problem has exactly this complexity. For a general polynomial, finite root cardinality has the elementary upper-bound description

$$\exists B \forall n (s(n) > B \implies Q(n) \neq 0),$$

but identifying a sharper unrestricted classification would require a separate argument.

35 Regulated partition functions and their analytic boundary

35.1 Two thermal parameters

Fix $0 < q < 1$. It is convenient to regard $q = e^{-\beta}$ as an inverse-temperature parameter in exponential coordinates. The second parameter $0 \leq t < 1$ damps large total occupation independently of the energy:

$$Z(q, t) = \sum_{n \in \mathbb{N}^m} q^{H(n)} t^{s(n)}. \quad (104)$$

Equivalently, this is the trace of the positive diagonal operator $q^H t^{\sum_i N_i}$. We use $0^0 = 1$ in evaluating $t = 0$. The terminology *fugacity* is a mathematical parametrization; no physical particle reservoir or laboratory implementation is assumed.

Separate the zero-state and excited-state contributions:

$$G(t) = \sum_{n \in \mathcal{Z}} t^{s(n)}, \quad (105)$$

$$R(q, t) = \sum_{n \notin \mathcal{Z}} q^{H(n)} t^{s(n)}, \quad (106)$$

$$Z(q, t) = G(t) + R(q, t). \quad (107)$$

For finite M , G is a polynomial recording witness heights. For infinite M , it is an infinite power series with nonnegative integer coefficients. Unique circuit completion makes these coefficients count the source witnesses at their *compiled* total heights; the heights generally differ from those of the original witness coordinates alone.

Theorem 35.1 (Effective partition functions). *For rational q, t in the domains below, the following sums are uniformly computable with exact rational enclosures:*

- (i) $Z(q, t)$ for $0 < q < 1$ and $0 \leq t < 1$;
- (ii) $R(q, t)$ for $0 < q < 1$ and $0 \leq t \leq 1$.

If Z_B and R_B denote sums over the box $\mathcal{C}_B$, then

$$0 \leq Z(q, t) - Z_B(q, t) \leq \frac{mt^{B+1}}{(1-t)^m}, \quad (108)$$

$$0 \leq R(q, t) - R_B(q, t) \leq \frac{mq(qt)^{B+1}}{(1-qt)^m}. \quad (109)$$

In particular,

$$0 \leq R(q, t) \leq \frac{q}{(1-qt)^m} \leq \frac{q}{(1-q)^m}. \quad (110)$$

At $t = 0$, the sums are determined by the single vacuum tuple.

Proof. Since $q^{H(n)} \leq 1$, the full sum is dominated by

$$\sum_{n \in \mathbb{N}^m} t^{s(n)} = (1-t)^{-m}.$$

A union bound over coordinates exceeding B gives (108). For the excited sum, use $q^{H(n)} \leq q^{1+s(n)}$ and apply the same argument with qt in place of t , proving (109) and (110). For rational parameters

every finite term is a rational number computed by integer polynomial evaluation and exponentiation. Increasing B until the displayed rational tail is below a prescribed tolerance gives an effective enclosure. The argument is uniform in the instance parameters, since neither tail uses them. $\square$

This theorem does not assert a fast algorithm close to $t = 1$. In the full trace, the sufficient box size from (108) can be very large. It asserts termination for each rational $t < 1$ and each requested precision. The excited trace, unlike the full trace, remains effectively computable at $t = 1$.

35.2 A sharp analytic dichotomy

Theorem 35.2 (All boundary obstructions lie in the witness series). *For fixed $q \in (0, 1)$, the function $R(q, t)$ is holomorphic in the complex disk $|t| < q^{-1}$. On every smaller closed disk $|t| \leq \rho < q^{-1}$, its box tails satisfy*

$$|R(q, t) - R_B(q, t)| \leq \frac{mq(q\rho)^{B+1}}{(1 - q\rho)^m}. \quad (111)$$

Consequently:

- (i) If $M < \infty$, then $Z(q, t)$ is holomorphic on $|t| < q^{-1}$ and $Z(q, 1) = M + R(q, 1)$.
- (ii) If $M = \infty$, the Taylor series of $Z(q, t)$ at zero has radius of convergence exactly one and $Z(q, t) \rightarrow +\infty$ as $t \uparrow 1$.

Thus a finite-valued holomorphic continuation of $Z(q, t)$ through $t = 1$ exists exactly when $M < \infty$.

Proof. The majorant $q(q\rho)^{s(n)}$ proves locally uniform absolute convergence of the series for R in the larger disk. Each term is a polynomial in t , so the limit is holomorphic; the same majorant gives (111).

If M is finite, G is a polynomial and the first conclusion follows. If M is infinite, monotone convergence along real $t \uparrow 1$ gives $G(t) \uparrow \infty$, since every zero-state term tends to one. The full series converges absolutely for $|t| < 1$ by the geometric bound. It cannot have radius greater than one, because that would give a finite value at $t = 1$, contradicting the divergence just proved. A finite-valued holomorphic continuation in a neighborhood of 1 would similarly be locally bounded on its real approach and is impossible. $\square$

Corollary 35.3 (A promised analytic-continuation problem is undecidable). *For the padded fixed universal family of Theorem 34.6, choose any fixed rational $q \in (0, 1/2)$. The functions*

$$Z_c(q, t) = \sum_{s=0}^{\infty} b_s(c) t^s, \quad b_s(c) = \sum_{s(n,y)=s} q^{\tilde{H}_c(n,y)},$$

have uniformly computable positive rational coefficients and are uniformly computable at rational $0 \leq t < 1$. Nevertheless, deciding whether their radius of convergence is one or at least $q^{-1} > 2$ is undecidable. Equivalently, deciding whether they extend holomorphically through $t = 1$ is co-computably-enumerable complete on this family.

Proof. Each coefficient is a finite exact sum over a simplex shell. A halting input has infinitely many padded zero states and hence radius one; a nonhalting input has no zero states and hence radius at least q^{-1} . Theorem 34.6 gives the completeness statement. $\square$

This corollary has a genuine promise gap in the possible radii. It is not caused by uncertainty in the Taylor coefficients or by an inability to evaluate the function at individual interior points. The missing information is a global boundary property of the entire computable series.

36 A computable critical fugacity with an undecidable endpoint

36.1 An instance-independent thermal separation

The marker coordinate in Theorem 32.1 guarantees $H(0) \geq 1$. For a system of $m \geq 1$ modes, fix

$$q_m = \frac{1}{8(m+1)}, \quad \varepsilon_m = \frac{q_m}{(1-q_m)^m}. \quad (112)$$

Bernoulli's inequality gives the explicit estimate

$$\varepsilon_m \leq \frac{q_m}{1-mq_m} = \frac{1}{7m+8} < \frac{1}{8}. \quad (113)$$

Thus the entire excited contribution, at any $0 \leq t \leq 1$, is less than $1/8$. Let

$$F(t) = Z(q_m, t), \quad c_0 = F(0) = q_m^{H(0)} \leq q_m \leq \frac{1}{16}.$$

All Taylor coefficients of F are positive, since each occupation tuple has a positive weight and each total-occupation shell is nonempty. Therefore F is continuous and strictly increasing on $[0, 1)$.

Theorem 36.1 (The critical fugacity). *For every compiled instance define*

$$t_* = \begin{cases} \text{the unique } t \in (0, 1) \text{ such that } F(t) = 1/2, & \mathcal{Z} \neq \emptyset, \\ 1, & \mathcal{Z} = \emptyset. \end{cases} \quad (114)$$

This definition is well posed. The real number t_ is uniformly computable from the finite polynomial description. For the fixed universal family,*

$$c \in \mathbf{K} \iff t_*(c) < 1, \quad c \notin \mathbf{K} \iff t_*(c) = 1. \quad (115)$$

Hence its strict interior condition is computably enumerable complete, and its exact endpoint condition is co-computably-enumerable complete.

Proof. If there is no zero state, $F = R \leq \varepsilon_m < 1/8$, so there is no crossing. If there is a zero state of total occupation L , then $F(t) \geq t^L$ and consequently $\lim_{t \uparrow 1} F(t) \geq 1$. Since $F(0) \leq 1/16$, continuity gives a crossing of $1/2$; strict monotonicity makes it unique.

It remains to prove computability without assuming that equality of computable reals is decidable. Write $F(t) = \sum_{s \geq 0} b_s t^s$, with $b_s \geq 0$. Termwise differentiation inside the unit disk gives, for $0 < t < 1$,

$$tF'(t) = \sum_{s \geq 1} s b_s t^s \geq F(t) - c_0. \quad (116)$$

In particular, on any interval where $F(t) \geq 3/8$, one has $F'(t) \geq 1/4$.

Given a requested rational width $0 < \eta \leq 1$, maintain an interval $[a, b] \subseteq [0, 1]$ containing t_* , initially $[0, 1]$. At its rational midpoint $u < 1$, compute a rational enclosure $[L, U]$ of $F(u)$ with

width at most $\delta = \eta/16$, using Theorem 35.1. If $U < 1/2$, replace a by u . If $L > 1/2$, replace b by u . These updates preserve the enclosure, including the case $t_* = 1$.

In the remaining case, $1/2 \in [L, U]$ and hence $|F(u) - 1/2| \leq \delta$. Such a value is impossible in a no-root instance, whose entire range lies below $1/8$. Thus a crossing exists. Throughout the interval joining u to the crossing, the function is at least $1/2 - \delta \geq 7/16 > 3/8$. Integrating the derivative lower bound gives

$$|u - t_*| \leq 4 |F(u) - 1/2| \leq 4\delta.$$

Intersecting $[a, b]$ with $[u - 4\delta, u + 4\delta]$ produces a valid interval of width at most $8\delta = \eta/2$. Otherwise ordinary bisection halves the width at every iteration and terminates when it is at most η . Every function evaluation is at a rational point strictly below one and terminates. This proves uniform computability.

Finally, the fixed universal polynomial has a zero state exactly on K . The definition of t_* then gives (115), and with it the completeness assertions. $\square$

Remark 36.2 (The exact boundary is not a noncomputable real). It would be incorrect to describe $t_*(c)$ as a noncomputable number. There is one uniform algorithm that encloses it to any requested precision, even for nonhalting inputs. What is undecidable is the exact assertion that its value equals one. A finite-precision interval very close to the boundary does not distinguish the boundary itself from a sufficiently nearby interior crossing.

36.2 No computable uniform distance from the boundary

Corollary 36.3 (No instance-computable positive separation). *There is no total computable function $d : \mathbb{N} \rightarrow \mathbb{Q}_{>0}$ such that every halting input in the universal family satisfies*

$$1 - t_*(c) \geq d(c).$$

There is likewise no total computable choice of rational $0 \leq t(c) < 1$ guaranteed to satisfy $F_c(t(c)) \geq 3/4$ on every halting input.

Proof. For the first assertion, replace $d(c)$ by $\min(d(c), 1)$ if necessary and compute $t_*(c)$ to error less than $d(c)/4$. A nonhalting input has value one; a halting input is at most $1 - d(c)$. These disjoint alternatives would decide K .

For the second assertion, evaluate $F_c(t(c))$ to error less than $1/8$. On nonhalting inputs its value is below $1/8$, while the promised halting value is at least $3/4$. Again the alternatives are effectively separated and would decide halting. $\square$

36.3 Witness heights determine the crossover scale

Theorem 36.4 (Finite-multiplicity height bounds). *Suppose $1 \leq M < \infty$, and let*

$$L = \min\{s(n) : n \in \mathcal{Z}\} \geq 1.$$

Then the critical fugacity satisfies

$$\frac{\log 2}{L} \leq -\log t_* \leq \frac{\log(M/(1/2 - \varepsilon_m))}{L}. \quad (117)$$

For $M = 1$, more precisely,

$$\frac{1}{2} - \varepsilon_m \leq t_*^L \leq \frac{1}{2}. \quad (118)$$

If the zero set is infinite but all its heights are at least L , one still has the unconditional estimate

$$G(t) \leq \frac{t^{L/2}}{(1 - \sqrt{t})^m}, \quad 0 < t < 1. \quad (119)$$

Proof. At the crossing, $G(t_*) = 1/2 - R(q_m, t_*)$, so

$$1/2 - \varepsilon_m \leq G(t_*) \leq 1/2.$$

A minimum-height zero contributes t_*^L , while every zero contributes at most t_*^L . Hence

$$t_*^L \leq G(t_*) \leq M t_*^L.$$

Combining these inequalities and taking negative logarithms proves (117); with one zero, $G(t) = t^L$ and (118) follows. For the final estimate, use $t^{s(n)} \leq t^{L/2}(\sqrt{t})^{s(n)}$ whenever $s(n) \geq L$, and sum over all natural tuples. $\square$

The theorem identifies a separate height obstruction even in a single-witness example. A witness can be unique and still require a fugacity arbitrarily close to one before its contribution becomes visible. Moreover, the relevant height is that of the compiled tuple. Large multiplication-gate values can make it much larger than the source witness height.

37 What remains undecidable despite the positive-energy gap

37.1 Bottom-of-spectrum separation

Theorem 37.1 (A gap does not provide a search cutoff). *For the fixed universal family, the bottom spectral value satisfies*

$$\inf \text{spec } H_c = 0 \quad \text{or} \quad \inf \text{spec } H_c \geq 1,$$

and distinguishing the alternatives is undecidable. There is no total computable bound $B(c)$ such that a zero state, whenever one exists, always has a representative in $\{0, \dots, B(c)\}^m$. In particular, no total finite-truncation procedure can uniformly approximate the bottom spectral value to error less than $1/4$.

Proof. All energies are nonnegative integers, and energy zero occurs exactly on halting inputs. A computable occupation cutoff with the stated property would permit exact finite search for a zero and thereby decide halting. Any uniform approximation to the bottom spectral value with error less than $1/4$, whether obtained by truncation or otherwise, would distinguish zero from values at least one by comparison with $1/2$. $\square$

This does not conflict with Theorem 33.2. To find all *positive-energy* states below E , one needs only a known finite simplex. A zero state can lie outside every proposed computable instance-dependent cutoff, because multiplication by the radial factor leaves every true zero of Q at energy zero. The two searches have different logical content.

[write] The second sentence of Theorem 37.1 is the proposition `cdc:bd:prop:nobound` of [22] applied to the compiled penalty, whose zero set is that of H_c ; the spectral form “ $\inf \text{spec } H_c \in \{0\} \cup [1, \infty)$, undecidable which” is source 03’s restatement.

37.2 Bounded operator perturbations versus coefficient errors

Proposition 37.2 (A precise perturbative stability statement). *Let V be any bounded self-adjoint operator with $\|V\| \leq \delta < 1/4$. Then*

$$|\inf \text{spec}(H + V) - \inf \text{spec } H| \leq \delta.$$

Thus zero-root and root-free instances retain disjoint bottom-energy ranges, respectively $[-\delta, \delta]$ and $[1 - \delta, \infty)$. Moreover, $H + V$ has compact resolvent exactly when H does.

Proof. For every unit vector in the operator domain, the quadratic form of V lies between $-\delta$ and δ . Taking infima proves the spectral-bottom inequality. Bounded self-adjoint perturbation preserves self-adjointness on $\mathcal{D}(H)$; it can also be seen by a resolvent Neumann series at a sufficiently large imaginary argument. At any common nonreal resolvent point, the resolvent identity expresses the difference of the two resolvents as a product containing either resolvent. If one resolvent is compact, the identity makes the other compact; interchanging the operators proves the converse. $\square$

The proposition is deliberately not a statement of coefficient-noise robustness. For every $\delta > 0$, the coefficient perturbation δN_i has infinite operator norm on the full occupation space. A very small perturbation of polynomial coefficients can become arbitrarily large at high occupation. Exact coefficient specification, unbounded occupation, and exact boundary predicates are substantive features of this representation, not implementation details that a spectral-gap statement removes.

Nor is diagonal evolution a Turing-complete time evolution in the usual operational sense. For an occupation basis vector,

$$e^{-itH}|n\rangle = e^{-itH(n)}|n\rangle :$$

it changes a phase and does not execute a sequence of machine configurations. The universality proved here belongs to static spectral predicates and to the partition-function boundary, not to autonomous simulation or to a physical algorithm for an undecidable problem.

38 Worked examples and exact computational checks

38.1 A single witness that moves toward the boundary

For $N \in \mathbb{N}$, consider the direct two-mode system

$$H_N(x, a) = (1 + x + a)((x - N)^2 + (a - 1)^2). \quad (120)$$

It has exactly one zero, $(N, 1)$, of total height $L = N + 1$. Here $m = 2$, $q_m = 1/24$, and

$$G_N(t) = t^{N+1}.$$

The Hamiltonian in this special example has degree three; the degree-five normal form is a general upper bound, not a requirement that every instance have exact degree five.

Table 5 reports certified crossings obtained by combining the exact ground polynomial with a rational enclosure of the excited trace. Each enclosure has width at most 2^{-30} ; the displayed decimals have absolute error below 2×10^{-9} . The rational endpoints, not just the rounded decimals, are saved in `verification.json`. For example, the enclosure for $N = 1000$ is exactly

N	Zero-state height L	Critical fugacity t_*
0	1	0.458301564
1	2	0.704142627
3	4	0.840894904
10	11	0.938930911
100	101	0.993160652
1000	1001	0.999307785

Table 5: Single-witness crossover values for (120), at $q = 1/24$.

$$\frac{1072998563}{1073741824} \leq t_* \leq \frac{268249641}{268435456}.$$

Every member of this elementary family is decidable directly. Its role is to display the height effect cleanly, not to demonstrate an undecidable individual instance.

38.2 Infinite zero multiplicity with a computable excited trace

Take

$$H(x, a) = (1 + x + a)(a - 1)^2. \quad (121)$$

All $(x, 1)$ are zero states, so

$$G(t) = \frac{t}{1 - t}.$$

For $a \neq 1$, summing the geometric series in x gives the independent formula

$$R(q, t) = \sum_{\substack{a \geq 0 \\ a \neq 1}} \frac{t^a q^{(1+a)(a-1)^2}}{1 - tq^{(a-1)^2}}. \quad (122)$$

At $t = 1$ this excited series is finite by Theorem 35.1, even though the full thermal trace is infinite. The critical fugacity at $q = 1/24$ is enclosed by

$$\frac{337179061}{1073741824} \leq t_* \leq \frac{168589531}{536870912},$$

with midpoint approximately 0.314022472. It is below $1/3$, the crossing of the ground series alone, because the excited contribution is strictly positive.

For an integer $E > 0$, the multiplicity of energy E equals the number of natural $a \neq 1$ satisfying

$$(a - 1)^2 \mid E, \quad \frac{E}{(a - 1)^2} - a - 1 \geq 0.$$

The corresponding x is uniquely the expression on the left of the inequality. This divisor formula was checked independently against finite occupation enumeration for every $1 \leq E \leq 60$.

38.3 An arithmetic infinite fiber: Pell witnesses

A less degenerate infinite zero set comes from

$$H(x, y, a) = (1 + x + y + a)((x^2 - 2y^2 - 1)^2 + (a - 1)^2). \quad (123)$$

Starting with $(x_0, y_0) = (1, 0)$, define

$$x_{j+1} = 3x_j + 4y_j, \quad y_{j+1} = 2x_j + 3y_j.$$

Direct expansion gives

$$(3x + 4y)^2 - 2(2x + 3y)^2 = x^2 - 2y^2.$$

The positive recurrence strictly increases both coordinates after the initial step and therefore produces infinitely many distinct natural solutions. Thus (123) has infinitely many zero-energy states and no compact resolvent, while its excited heat operator has the explicit effective tail bound of Theorem 33.3. The proof needs only this infinite subfamily, not a classification of every Pell solution. The first 32 recurrence steps were checked with exact integers.

38.4 Why the radial factor must contain every mode

The root-free polynomial $Q(x, y) = x^2 + 1$ has an infinitely degenerate eigenvalue one at $(0, y)$ when used as an unweighted energy. Multiplication by $1 + x$ does not help: the omitted y coordinate is still a flat positive-energy direction. Multiplication by $1 + x + y$ does help, because every positive state then has energy at least $1 + x + y$.

The same issue applies to the padding coordinate in Theorem 34.6 and to intermediate circuit variables. Omitting a mode from the radial factor invalidates the uniform positive-energy bounds unless an additional independent argument controls that mode.

38.5 Compiler and verification package

The accompanying `compiler.py` implements natural addition/multiplication circuit construction, commutative gate sharing, exact residual evaluation, canonical witness lifting, rational thermal enclosures, and critical-fugacity enclosure. Sparse polynomial inputs use exponent tuples with parameters first and source witnesses second. Floats and Boolean values are rejected where exact integer inputs are required.

[write] *Editorial note: this holds for `compile_polynomial`; the low-level `Compiler` and `Atom` interface assumes well-formed natural DAG descriptors and does not check them.* The research programme’s review [23] found that the low-level builder accepts a self-referential gate and non-natural constants: `c = Compiler(1); z = c.add(c.variable(0), Atom('mode', 1)); f = c.finish(z, z)` defines the gate $u = x + u$ and gives `f.energy((0,7,1)) == f.energy((0,8,1)) == 0`, two completions over one input; `Atom('const', 0.5)` is accepted and yields energy 0.5; Boolean atoms are accepted. Both behaviours were reproduced on a copy of the shipped compiler at the write. No theorem is affected: the theorems assume a natural DAG whose gate inputs are constants in $\mathbb{N}$, parameters, witnesses or outputs of *earlier* gates (Section 31 and the conventions before Theorem 32.1). The review’s tested patch `thermal_exact_dag_guards.patch` validates atoms and gate descriptors; with it the 7,936-assertion suite still passes and both JSON outputs are unchanged. The shipped `code/03-thermal-arithmetic-compiler.py` is the unpatched original. In this report the package files are shipped as `code/03-thermal-arithmetic-compiler.py`, `code/03-thermal-arithmetic-verify.py`, `code/03-thermal-arithmetic-build.sh`, `data/03-thermal-arithmetic-examples.json`, `data/03-thermal-arithmetic-verification.json`, `data/03-thermal-arithmetic-verification_console.txt`, `data/03-thermal-arithmetic-requirements.txt` and `03-thermal-arithmetic-PROVENANCE.md`; the example below imports the compiler by its delivered name `compiler`.

For example, the product compiler can be invoked as follows:

```
from compiler import compile_polynomial

#  $P(p; x, y) = x*y - p$ ; tuple order is  $(p, x, y)$ .
model = compile_polynomial(
    {(0, 1, 1): 1, (1, 0, 0): -1},
    inputs=2, parameters=1,
)
n = model.canonical((2, 3), (6,))
assert n == (2, 3, 6, 1)
assert model.energy(n, (6,)) == 0
assert (model.modes, model.equations) == (4, 3)
```

The verification suite performs 7,936 assertions. These are reproducible finite and symbolic regression checks, not a substitute for the general proofs and not a proof-assistant formalization. Table 6 groups the actual assertions without claiming statistical independence.

Check group	Assertions
Canonical gate/marker values, signed outputs, penalty identities, source-zero equivalence	1,240
Mutation of a single auxiliary coordinate is rejected	1,120
Exhaustive small supplied-tuple zero-fiber equivalence	2,384
Radial positive-energy lower bounds on supplied tuples	2,384
Symbolic degree, exact resource counts, and mode supports	170
Exact input-type rejection	5
Dimension-uniform thermal bound for $1 \leq m \leq 200$	200
Finite energy shells and independent multiplicities	192
Pell recurrence identity and strict growth	64
Certified partition widths, independent overlap, global bounds	60
Critical enclosures, height bounds, endpoint and algorithm checks	17
Omitted-mode confinement counterexample	100
Total	7,936

Table 6: Executed exact and symbolic checks. Detailed categories and rational outputs appear in `verification.json`.

Six source polynomials are exported in `examples.json`: $x-p$, x^3-p , $xy-p$, x^2+1 , $x^2+xy-py-3$, and the zero polynomial. Their complete compiled polynomial expressions are included. Symbolic expansion independently confirms the degree and support bounds for these examples. Small supplied tuples include noncanonical gate assignments; the suite is not limited to evaluating the intended lifts.

38.6 Certified truncation without enormous denominators

The rational partition routine uses two explicit error budgets. First choose a box so that (108) or (109) is at most half the requested tolerance. Let $D = (B+1)^m$ be the number of tuples in that box. Next choose an integer energy cutoff C such that

$$Dq^C \leq \frac{\text{tolerance}}{2}.$$

Terms with $H(n) \geq C$ may be omitted from the finite rational sum, since their total contribution is at most Dq^C when $0 \leq t \leq 1$. This avoids constructing denominators such as $24^{H(n)}$ for unnecessarily large energies. The output interval is

$$[\text{retained exact sum}, \text{retained exact sum} + \text{box tail} + Dq^C].$$

No omitted term is silently discarded. A configurable resource limit raises an exception rather than claiming a certificate from unfinished work. The optional exact ground-series interface speeds up the examples near $t = 1$; supplying such a formula is an extra, independently justified input, not an assumed ability to solve the general zero-set problem.

39 Further research questions

The following questions are selected to extend the proved results rather than restate the general halting problem.

39.1 Can radial isolation be achieved at degree four?

The direct multiplier converts a quartic sum of gate squares into a quintic. Is there a witness-faithful finite-variable polynomial normal form of degree at most four with both the same zero fibers and an effective bound excluding positive low-energy states at large occupation? The target need not be the exact inequality $H \geq 1 + s$; any computable proper lower bound on the positive sector would preserve much of the analysis. A construction must account for the possibility that fresh auxiliaries create their own low-energy flat directions. A lower bound for a clearly specified class of constructions would also be useful. ([write] Degree at most three is impossible for a universal family: H is nonnegative on the real nonnegative orthant, and an orthant-nonnegative polynomial of degree at most three has a semilinear natural zero set (`cdc:of:thm:classification` and `cdc:of:cor:no-cubic-universal` in [22]; for globally nonnegative coercive polynomials compare `pqc:ql:prop:degree` in [21]), whereas the zero sets of Corollary 32.3 project onto a c.e.-complete set. For universal families the open gap is therefore four versus five.)

39.2 Can four-mode positive terms be reduced to three-mode terms?

The extra mode enters when n_i multiplies a residual square already involving three modes. A useful refinement would lower the maximum support without losing positivity, introducing uncontrolled witness multiplicity, or increasing degree beyond a stated budget. The appropriate initial task is to classify local identities for one multiplication residual and one external radial coordinate. Geometric locality is a separate, harder requirement.

39.3 Instantiate a numerical universal arithmetic family

Take one audited universal polynomial from ProveIt and apply the compiler literally. Record source witness count, shared gate count, coefficient bit lengths, expanded degree, positive-term supports, and the cost of exact energy evaluation. This would turn Corollary 32.3 from a fixed-family existence result into a reproducible numerical normal form. It should not be reported as an improved universal-operation bound unless that operation count is actually lower under the same convention.

39.4 Optimize compiled witness height, not only gate count

Two zero-set-bijective arithmetic circuits can have very different sums of intermediate values. Theorem 36.4 makes this a measurable analytic cost. Define and compare the height overhead of alternative circuits, including product trees, shared powers, redundant checks, and parameter loaders. A circuit with fewer gates might create a much larger critical-fugacity boundary layer. A useful objective is a joint frontier in gate count, maximum intermediate value, and total zero-state occupation.

39.5 Find unconditional all-fiber compact subclasses

Theorem 34.4 identifies the exact representation requirement for all-fiber compactness. Rather than assume it for universal halting, identify interesting restricted computational relations with explicit finite-fold representations and carry out the compact normal form. Preserving known fiber cardinalities under composition is especially important. Any genuinely universal solution would need to confront the finite-fold issue, not merely make the gate completion unique.

39.6 Characterize effective compactness on restricted families

Even where compact resolvent is mathematically guaranteed, a uniform computable modulus for its finite-rank approximation may fail. Which syntactic or arithmetic hypotheses provide such a modulus? An explicit computable witness-height bound is sufficient. Determine whether weaker information, such as an effective bound on zero-state count combined with a controlled arithmetic geometry, also suffices for useful subclasses. The answer should distinguish compactness, computable compactness, and computability of individual matrix entries.

39.7 Classify witness-series singularities

For infinite fibers, $G(t)$ controls all singular behavior at $t = 1$, while R extends to a larger disk. For families with structured witnesses, determine asymptotic expansions of $G(t)$ as $t \uparrow 1$: polynomial poles for free coordinates, logarithmic growth for exponentially sparse families, and possible logarithmic oscillations. The Pell example is a concrete starting point. Any all-orders expansion would directly become an expansion of the full partition function after adding the analytic excited contribution. This offers a connection with the repository's generating-function and transseries work without importing an unproved asymptotic claim into this article.

39.8 What survives non-diagonal bounded couplings?

Proposition 37.2 preserves compactness and a separated bottom-energy test under bounded self-adjoint perturbations. It does not preserve the exact ground projection or the simple partition decomposition. Seek quantitative heat-trace comparison bounds and effective truncations for a precisely specified class of bounded couplings. An important first distinction is whether the perturbation is effectively approximable in norm or merely has computable matrix entries.

39.9 Can a physically motivated cutoff retain a useful conditional theorem?

A fixed maximum occupation makes the space finite dimensional and every zero test decidable. The meaningful question is therefore not whether this reproduces unrestricted halting, but which bounded computational questions it represents with certified precision and resource costs. Compare known runtime or witness-height bounds with occupation limits, coefficient errors, and energy resolution. A valid theorem should state explicitly which promise supplies the missing cutoff.

39.10 Formalize the arithmetic and effective-analysis interface

A modular formalization can begin with the exact zero-fiber bijection, the radial inequality, and finite-box geometric tails. These are independent of a full formalization of unbounded operators. The next layer can treat diagonal operators directly on ℓ^2 , prove the maximal-domain realization, and connect trace and compactness to diagonal sequences. The universal result should depend on a separately audited MRDP interface, and the finite-fold equivalence should remain visibly conditional. The present executable code is a test oracle for this project, not a formal proof artifact.

40 Dependency and claim ledger

Statement	Status and dependencies
Universal Diophantine representation	Established foundation, used from the MRDP/universal-equation literature; no new proof or numerical instance supplied.
Squared number-operator embedding	Established prior construction; explicitly attributed to Kieu. No physical decision claim used.
Witness-faithful compiler	Proved here by topological induction; classical circuit method with an explicit marker and quantitative ledger. [write] The circuit core is <code>cdc:wf:lem:quadratisation</code> of [22] (note in Section 31).
Degree-five radial isolation and four-mode positive display	Proved here by integer positivity and support counting.
Positive spectral cutoffs and effective remainders	Proved here from the radial inequality, geometric sums, and diagonal operator arguments.
Compactness and thermal multiplicity classification	Proved here; exactly preserves the source root count.
All-fiber compact universal representation	Not asserted unconditionally. The exact finite-fold equivalence is proved.
Padded compactness and analytic-radius undecidability	Unconditional consequence of fixed universal representability and free-witness padding.
Uniformly computable critical fugacity	Proved here using certified interior evaluations and a positive derivative bound.
Undecidable exact endpoint and absent cutoffs	Proved by reductions to halting. The critical real itself is computable.
Numerical and symbolic checks	7,936 executed assertions; evidence about the implementation and selected examples, not proof-assistant verification.

Statement	Status and dependencies
Priority or breakthrough status	Not established by this targeted search. The article supplies explicit proved results and a research program, not a certified historical first.

41 Conclusion

A small structural modification of an arithmetic penalty yields a useful separation: difficult zero states are left exactly where the Diophantine witnesses put them, while every positive state is forced into an effective radial energy bound. This separates unknown witness multiplicity from an otherwise computable excited spectrum and gives exact correspondences between finite-fold arithmetic, compact resolvent, and finite thermal trace.

The boundary construction adds a second distinction. A family of partition functions can be uniformly computable throughout its regulated domain, and its threshold location can itself be a uniformly computable real, while the exact endpoint predicate remains complete for nonhalting. No inaccessible approximation oracle is hidden in the proof; the obstruction is the impossibility of a uniform positive separation from the boundary.

The resulting substrate is unconventional but mathematically precise: a fixed finite collection of unbounded occupation modes with polynomial diagonal energies. It is not a physical solution of an undecidable problem. Its value is to expose which parts of a computational representation are existential, which are counting-theoretic, which are controlled by witness height, and which become visible only at an exact analytic boundary.

Part IV

Mixing does not remove arithmetic: exact equilibrium hits of positive commuting alphabets (source 11)

[write] Part IV prints batch-79 manuscript 12, *Mixing Does Not Remove Arithmetic: Optimal Positive Markov Realizations of Diophantine Predicates*, “Research note prepared with ChatGPT; For the ProveIt computational-substrate research program”, 2 October 2026 (PDF metadata author “Research note prepared with ChatGPT”). In this report it is *source 11*, after the file prefix `11-mixing-arithmetic-` of its shipped programs and data; it is unrelated to source 10 of Part II. It pins the repository at `44983ed7e`, arrived with the archives of commit `ef2fc7990` and was placed by commit `bbaf322e5`. Its labels carry the prefix `ste:mx:`. Its Section n is Section $n + 41$ here; its two appendices are Appendices G and H, printed after the provenance appendix so that the letters of Appendices A–F do not change. Its abstract and status box follow. Section 42.4 lists what this Part re-proves of Part I and of [22]; those results are printed with the source’s own proofs and without a novelty claim.

[write] *Review*. The research programme inventoried source 11 ([43], commit `b927b2290`) and reviewed it after placement [44] (commit `5b633fcf9`): it passes on the stated mathematics, the

compiler’s documented interface and both original programs, and no repair is requested. The review read the whole manuscript and code, reproduced all fourteen exports byte for byte, and reconstructed every coefficient of the fourteen exported polynomials from the transition entries in exact rational arithmetic, with further fixtures of its own. It stresses four limits, all consistent with the text: the bound $s \geq r(P) + 1$ is not a lower bound for the zero set alone ($P = x$ and $P = x^2$ have the same natural zeros and need three and four states); the line loader’s root correspondence needs natural values and an integer S (for $t = 0$, the real point $z_0 = 1$, $S = 1/2$ and the signed point $z_0 = -2$, $S = 2$ are spurious zeros of P_t); the line loader is no arithmetic saving (given S , it costs one multiplication and three additions against one and two); and the binary commuting results of [29] are NP-hardness results, not undecidability. The placement checks of batch 79 also reran both programs on a copy and rederived the lift and the printed four-state example independently, without finding an error.

Abstract (source 11). Let $P \in \mathbb{Q}[x_1, \dots, x_k]$ be nonzero, and let $r(P)$ be the dimension of its translation space. We give an effective realization of P by exactly $r(P) + 1$ states, strictly positive rational initial probabilities, one accepting state, and commuting, invertible, strictly positive doubly stochastic transition matrices. For any prescribed rational contraction bound $0 < \theta < 1$, the construction gives

$$\text{acc}_P(n) = \frac{1}{s} + \varepsilon q^{n_1 + \dots + n_k} P(n), \quad s = r(P) + 1, \quad 0 < q < \theta,$$

while every switching sequence contracts total variation by a factor at most θ per letter. The state count is optimal for the exact centered scalar series, not merely for its zero set. A derivative-space argument yields the sharp quartic bound $s \leq \binom{k+3}{2}$.

The construction gives two fixed-alphabet universality refinements. A universal Diophantine predicate can be loaded along a quadratic rational curve of initial probabilities with joint nilpotence depth at most four, or along a single rational line segment with depth at most five. No transition matrix changes with the input. We prove a converse identifying degree-bounded natural Diophantine solvability with exact equilibrium-coordinate reachability in the corresponding nilpotent class. All off-equilibrium rational targets admit explicit finite search bounds, whereas exact equilibrium hits need not. Fourteen rational exports, a symbolic compiler, and an independent **Fraction**-arithmetic checker accompany the proofs.

Status and scope (source 11). The mathematical claims below are proved in the article. The checks are exact finite computations, not proof-assistant certificates. Commuting probabilistic-automaton undecidability, Hankel-rank minimality, inverse-system methods, and quartic Diophantine reductions have substantial prior literature. This note claims a proved, explicit combination and refinement, not verified historical priority or a solution of finite-fold MRDP. It does not instantiate or improve ProveIt’s universal arithmetic-operation bound. The universal input loader uses the classical MRDP theorem; the supplied numerical alphabets are illustrative, not a printed universal alphabet.

[write] Table 8 lists the letters that Parts IV and V use differently, among themselves and against Parts I–III; the readings to watch are listed in Section 1.3.

Table 8: [write] Letters used differently in Parts IV and V,
and against Parts I–III.

Letter	Part IV (source 11)	Part V.A (source 12)	Part V.B (source 13)	Parts I–III
A	commuting unipotent translation matrices A_i	adjacency operator $A = S + S^*$	the grounded operator $A = A_m = mI - J$	Part I: integer mortality matrices
J	averaging matrix J_s	—	adjacency J, J_P	Part I: averaging matrix J_n
E, F	mass-zero embedding of rows, $EF = I_r$, $FE = I_s - J_s$	E : supplied label set; E_X : boundary map; $\mathbb{E}$: expectation; $\mathbb{F}$: finite field	E_x : energy; $\mathbb{F}_p$: finite field	Part I: coordinate maps of columns
q	common transient eigenvalue, $0 < q < \theta$	code name of the scalar c	integral charge; q_t , q_* : states	Part III: Boltzmann parameter
s	number of states, $s = r(P) + 1$	source vertex	s_x source; s_r branch source state; $s_{t,r}$ slack	Part III: total occupation
m	number of MRDP witnesses in (149)	number of branches; a modulus (Section 66)	diagonal parameter	Part I: number of source matrices
L	word length	the operator L_α	run length $L = T + 1$	Part I: guard; Part II: final layer
H	H_s : mass-zero rows; a search horizon	terminal functional $H(u)$	the halting state	Part I: H_n , zero-mass hyperplane
S	$S(z)$: sum of squared gates	forward step S	rail swap S ; finite prime set S	Part II: the game operator
B	— (lower-case b, b_0 , b_1 : a column and two endpoint rows)	history base $\max(2, m + 1)$	history base $R + 1$; B_S : length bound	—
D	—	continuants $D_j(\alpha)$	continuants D_j (the same sequence for $\alpha = m$)	Part I: column sum; Part II: denominator
R	nilpotent parts R_i (156); remainders in the proof of Theorem 45.1	a coefficient ring	number of branches; $\mathcal{R}_x$: resistance; R_S : localized ring	Part I: guard
U	universal polynomial $U(t, x)$	polynomial vector $U \in \mathbb{Z}[X, Y, Z]^Q$	Lucas numbers U_n ; the universal program U	—

θ, ε	band and loading scale	—	—	Part I: band η
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42 The question and the contribution

A matrix can be close to complete averaging and still retain an exact arithmetic signal. Closeness controls its magnitude; it need not decide whether that magnitude is exactly zero. This note makes that distinction quantitative, constructive, and dimension-optimal for a natural polynomial realization problem.

The central research question is the following. Suppose all transition matrices are rational, strictly positive, doubly stochastic, commuting, invertible, and uniformly close to the same averaging matrix. Does this combination make finite exact reachability of the stationary value decidable? The answer is no. More precisely, these constraints can all be imposed on a *fixed* alphabet whose initial distribution varies on a very simple rational curve, with an undecidable exact-hit problem. The proof also shows exactly where the arithmetic lives: in small nilpotent parts of the matrices after their common exponential decay has been removed.

42.1 Results at a glance

The main realization theorem, Theorem 45.1, preserves the entire polynomial value up to a known positive exponential factor. Its matching lower bound is Theorem 45.3. In particular, positivity and arbitrary prescribed mixing speed cost just one state beyond the polynomial’s minimal linear realization. The scalar readout is an actual coordinate probability, not a signed linear functional disguised as acceptance.

Theorem 46.1 translates a derivative-space bound into a sharp state budget for all quartics, including nonhomogeneous ones. Theorem 47.2 then proves universality with a quadratic input curve and fourth-order joint nilpotence. Theorem 47.4 gives a different tradeoff: a straight-line input family, fifth-order joint nilpotence, and a quadratic state bound in the number of Diophantine variables. The latter construction uses a positive-value filter to keep the external input affine rather than quadratic.

The reverse construction in Theorem 48.2 recovers an ordinary polynomial directly from promised matrix data. Thus the degree restriction is not just a feature of one encoding. Finally, Theorem 49.1 separates exact critical-target reachability from effective finite-search problems at positive distance from that target.

42.2 Relationship to existing work

The selected ProveIt sources were reviewed at repository snapshot 44983ed7ebfd (the full hash appears in Appendix B). The Hilbert-tenth-problem README records its separate complete-certificate and universal-polynomial arithmetic bounds as 75 and 87 operations, respectively [5]. Those ledgers are not state counts. No comparison in this article lowers either bound.

[write] Both figures are still those of the project README at this write (Computability/HilbertTenthProblem/README.md: “75 operations for a complete certificate and 87 for one universal polynomial”); the source’s pin is kept as provenance.

The six-substrate review includes an “Exact Erasure” construction using strictly positive stochastic matrices and mortality, and carefully separates bounded certificates from universal interfaces [23]. Our selected route is different: every generator and every finite product is invertible, so full erasure is impossible. The hard event is equality of *one coordinate* with its equilibrium probability. Commutativity, a common uniform stationary law, and minimal translation-module dimension are built into the construction. This is a review of the cited files, not a claim to have audited every report in the repository.

[write] The “Exact Erasure” construction of that review is the manuscript printed as Part I of this report (source 02); the review is [23]. Inside this report the comparison is direct: Theorem 45.1 below uses Part I’s mass-zero embedding and contraction argument (Section 42.4), and the exact events differ. Part I’s predicate is rank one of a product, so that all distinction is erased; here no product erases (Proposition 45.5) and the predicate is one coordinate at its equilibrium value.

Undecidability for commuting probabilistic finite automata is already known. Bell proves strict and non-strict emptiness undecidable even for polynomially ambiguous commutative automata [28]. Bell and Semukhin study exact cutpoint reachability and related small-alphabet restrictions [29]. Our strictly positive automata have exponentially many accepting paths and do *not* preserve polynomial ambiguity. The older conversion of weighted representations to stochastic ones is associated with Turakainen [27]; the present proof instead prints the specific mass-zero embedding needed for our dimension, readout, and mixing claims.

[write] The same embedding, in column convention, is Part I’s positive affine lift (Theorem 4.1, with its doubly stochastic case Corollary 4.3); see Section 42.4.

Hankel rank is the classical measure of minimal weighted realization dimension; Balle and Mohri give the standard theorem and reconstruction perspective [30]. Polynomial derivative spaces also belong to Macaulay inverse-system and apolar-algebra theory [31, 32]. We use elementary proofs of the specialized facts needed here. The quartic Diophantine reduction relies on the classical MRDP theorem; Davis gives a primary account [26]. None of these foundational ingredients is presented as newly discovered.

42.3 What kind of computational substrate is this?

The substrate is a *finite controlled stochastic system with exact acceptance*. A finite word chooses which Markov matrix is applied at each step. The existential question asks whether *some* finite control word has a designated exact output. It is a Diophantine recognizer for an unbounded existential search, not an autonomous finite Markov chain simulating a Turing machine step by step.

For every fixed input and every supplied finite word, its acceptance probability is computable by rational arithmetic. Undecidability concerns the unbounded existence of a suitable word as the input varies. In particular, no claim is made that membership of a supplied word in a fixed rational automaton’s exact cutpoint language is undecidable.

42.4 Second routes and pointers

[write] This subsection was written when source 11 was added to the report. Four arguments of this Part are results already in the collection. They are printed in place with the source’s own proofs, as second routes, and no novelty is claimed for them.

- *The lift.* Theorem 45.1 is the doubly stochastic lift of Corollary 4.3 (Part I) applied to the

commuting unipotent translation operators A_i of (130). Part I works with column-stochastic matrices acting on column distributions and writes $S_w = J_{d+1} + K^{-T} E A_w F$ with an integer scale K ; this Part works with row distributions and writes $M_w = J_s + q^L F A_w E$ in (137) with a rational scale q , so the roles of E and F are transposed. Both contraction proofs split off a multiple of the averaging matrix and use a nonnegative stochastic remainder (Proposition 4.2). New here are the choice of a translation module as the represented space, the readout of one actual coordinate through $E \mathbf{e}_1 = b$, the matching lower bound $s \geq r(P) + 1$ (Theorem 45.3), the quartic and quintic budgets, the fixed-alphabet input curves and the converse.

- *The gate front end.* Lemma 47.1 is `cdc:wf:lem:quadratzation` of [22], the unique-extension quadratzation, which Parts I–III also use.
- *The binomial step.* The expansion (157) in the proof of Theorem 48.2 is the commuting multi-letter form of `cdc:pt:prop:unipotent` of [22].
- *No horizon.* Corollary 49.2 is the bounded-search argument of `cdc:bd:prop:nobound`, which every Part of this report applies (Section 1.2).

Two comparisons with Part I are new. First, the generators here commute, as in Theorem 8.2, where commutation makes *erasure* decidable with horizon $n - 1$. Here erasure never occurs at all (Proposition 45.5), yet the single-coordinate equilibrium hit stays computably enumerable complete on a fixed alphabet (Theorems 47.2 and 47.4): commutation controls the rank of products, not the value of one coordinate. Second, Part I’s research question of Section 15.6 asked for one fixed stochastic alphabet with an ordinary input interface. This Part supplies one for the exact-hit predicate, at MRDP existence level and without a printed numerical alphabet (dated note under that question). Theorem 49.1 turns Part I’s remark that $\eta^T \leq \varepsilon$ works for every instance into a decision procedure for separated targets.

Within the collection, Part V of [21] (source 17 there, *Arithmetic of Rapidly Mixing Reversible Computation*) proves an exact stationary equality undecidable for reversible chains with a uniform spectral gap on a countably infinite state space. Proposition 48.4 concerns finite chains with a single transient eigenvalue; the two statements do not conflict.

43 Conventions and the exact-hit model

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Vectors representing distributions are rows. An $s \times s$ matrix M is row-stochastic when $M \mathbf{1}_s = \mathbf{1}_s$ and all its entries are nonnegative. It is doubly stochastic when also $\mathbf{1}_s^T M = \mathbf{1}_s^T$. Write

$$u_s = \frac{1}{s} \mathbf{1}_s^T, \quad J_s = \frac{1}{s} \mathbf{1}_s \mathbf{1}_s^T, \quad H_s = \{v \in \mathbb{Q}^{1 \times s} : v \mathbf{1}_s = 0\}. \quad (124)$$

Then u_s is stationary for every doubly stochastic matrix. The matrix J_s is complete averaging, and $I_s - J_s$ is the projection onto mass-zero information.

Fix letters $a_1, \dots, a_k$ with transition matrices $M_1, \dots, M_k$. For a word $w = a_{i_1} \cdots a_{i_L}$, set $M_w = M_{i_1} \cdots M_{i_L}$, and set $M_\emptyset = I_s$. The accepting state will always be state 1:

$$\text{acc}_p(w) = p M_w \mathbf{e}_1, \quad \lambda_* = u_s \mathbf{e}_1 = \frac{1}{s}. \quad (125)$$

When the matrices commute, only the count vector $n = (n_1, \dots, n_k)$ matters. We use $M^n = M_1^{n_1} \cdots M_k^{n_k}$ and $|n| = \sum_i n_i$. Every count vector is represented by the canonical word $a_1^{n_1} \cdots a_k^{n_k}$, including the all-zero vector.

The exact equilibrium-hit question is

$$\exists n \in \mathbb{N}^k \quad pM^n \mathbf{e}_1 = \lambda_*. \quad (126)$$

The matrices, initial distribution, and target are exact rational objects. Replacing equality by numerical closeness changes this problem fundamentally.

For two probability rows, total variation is $\text{TV}(p, p') = \frac{1}{2} \|p - p'\|_1$. We will impose the stronger, common-minorization condition

$$\frac{1 - \theta}{s} < (M_i)_{ab} < \frac{1 + \theta}{s} \quad (0 < \theta < 1) \quad (127)$$

for every entry of every matrix. This both gives strict positivity and provides an explicit contraction estimate for arbitrary switching words.

44 Translation spaces, finite differences, and minimal rank

Definition 44.1. For a nonzero polynomial $P \in \mathbb{Q}[x_1, \dots, x_k]$, its translation space and translation rank are

$$\mathcal{V}(P) = \text{span}_{\mathbb{Q}}\{P(x + m) : m \in \mathbb{N}^k\}, \quad r(P) = \dim_{\mathbb{Q}} \mathcal{V}(P).$$

For a collection $\mathcal{P}$ of bounded-degree polynomials, define $\mathcal{V}(\mathcal{P})$ using all translates of all members and write $r(\mathcal{P}) = \dim \mathcal{V}(\mathcal{P})$.

The coefficients may be negative; positivity enters only after the linear representation has been constructed. All spaces in this section are over $\mathbb{Q}$. For a finite supplied family, the bases needed below can be computed by exact Gaussian elimination; the parameter families used later have explicit finite generating sets.

Lemma 44.2. *If P has total degree d , then*

$$\mathcal{V}(P) = \text{span}\{\Delta^\alpha P : |\alpha| \leq d\} = \text{span}\{\partial^\alpha P : |\alpha| \leq d\}, \quad (128)$$

where $\Delta_i f(x) = f(x + \mathbf{e}_i) - f(x)$. The same statement holds for a family. Every nonzero translation space contains the constant polynomial 1.

Proof. Taylor expansion shows that every translate lies in the derivative span. Conversely, on the space of polynomials of degree at most d , differentiation and finite difference satisfy the finite identities

$$1 + \Delta_i = \exp(\partial_i), \quad \partial_i = \log(1 + \Delta_i) = \sum_{j=1}^d \frac{(-1)^{j+1}}{j} \Delta_i^j.$$

These are identities of nilpotent operators, so convergence is irrelevant. Each difference of a polynomial is a linear combination of its nonnegative integer translates, and repeated differences remain such combinations. The logarithm identity therefore puts every derivative in the translation span. Products of the commuting operators give the multivariable assertion. Finally, differentiating a suitable top-degree monomial of a nonzero member produces a nonzero constant. Rescaling gives 1. $\square$

Lemma 44.3. *Evaluation on $\mathbb{N}^k$ is injective on $\mathbb{Q}[x_1, \dots, x_k]$. Consequently the infinite matrix*

$$\mathcal{H}_P(m, n) = P(m + n), \quad m, n \in \mathbb{N}^k, \quad (129)$$

has rank $r(P)$.

Proof. A univariate polynomial vanishing at every natural number is zero. For several variables, regard the polynomial as a polynomial in the last variable; fixing the others at natural values forces every coefficient to vanish there, and induction completes the argument. The rows of $\mathcal{H}_P$ are precisely the evaluations of the translates of P . An injective linear evaluation map preserves their span dimension. $\square$

44.1 The row-oriented linear representation

Choose a basis $f_1, \dots, f_r$ of $V = \mathcal{V}(P)$, or of a joint family space. For each coordinate direction, let T_i be defined by

$$(f_1(x + \mathbf{e}_i), \dots, f_r(x + \mathbf{e}_i)) = (f_1(x), \dots, f_r(x))T_i.$$

Set

$$A_i = T_i^T, \quad b = (f_1(0), \dots, f_r(0))^T, \quad P = \sum_j (a_P)_j f_j, \quad (130)$$

$$a_P A_1^{n_1} \dots A_k^{n_k} b = P(n).$$

The last displayed identity uses a_P as a row. The matrices A_i commute. They are unipotent: every eigenvalue is 1. Indeed, $A_i - I_r$ represents a finite difference operator, up to the harmless transpose convention.

Proposition 44.4. *Let d be the largest degree occurring in the nonzero joint family space. For $N_i = A_i - I_r$, every product N^α with $|\alpha| = d + 1$ is zero, whereas some product with $|\alpha| = d$ is nonzero. In particular, every A_i is invertible and $\det A_i = 1$.*

Proof. Every finite difference lowers total degree by at least one. This proves vanishing after $d + 1$ factors. Choose a member of degree d and a multiindex of a nonzero monomial of that degree. The corresponding d differences give a nonzero constant: the coefficient is multiplied by the product of the factorials of the exponents. Thus a product of length d survives. An identity plus a nilpotent matrix is invertible with finite geometric inverse, and all its eigenvalues are 1. $\square$

This is a *joint* depth statement. It does not say that every individual letter has a Jordan block of size $d + 1$. Mixed products of different nilpotent parts may be needed to detect the full degree.

44.2 An apolar interpretation

Let $D = \mathbb{Q}[\partial_1, \dots, \partial_k]$ act by differentiation. The map $D \rightarrow \mathcal{V}(P)$ sending L to LP has kernel $\text{Ann}(P)$, so

$$r(P) = \dim_{\mathbb{Q}} D / \text{Ann}(P). \quad (131)$$

This quotient is an Artinian local algebra. Its maximal ideal is generated by the images of the derivatives, which are nilpotent. Its socle consists exactly of the elements whose images under $L \mapsto LP$ are constants, hence is one-dimensional. This is the familiar Gorenstein inverse-system setting [31, 32]. Our stochastic dimension will be the length of this algebra plus one mass coordinate. A joint family is a module version of this construction and need not correspond to a one-dimensional socle.

45 A positive doubly stochastic lift with one extra state

The next theorem is the main algebraic construction. It is useful to state it for an already-computed linear representation, because the same transition matrices can then serve many input polynomials.

Theorem 45.1 (Explicit optimal-size lift). *Let $k, r \geq 1$, let $A_1, \dots, A_k \in \mathbb{Q}^{r \times r}$ be commuting unipotent matrices, let $b \in \mathbb{Q}^{r \times 1}$ be nonzero, and $0 < \theta < 1$ rational. Put $s = r + 1$. There are effectively computable commuting rational matrices M_i satisfying (127), a rational $q \in (0, \theta)$, and, for every rational row a , a strictly positive probability row p_a and rational $\varepsilon_a > 0$ such that*

$$p_a M^n \mathbf{e}_1 = \frac{1}{s} + \varepsilon_a q^{|n|} a A^n b \quad (n \in \mathbb{N}^k). \quad (132)$$

Every M_i is doubly stochastic and invertible, with characteristic polynomial $(X - 1)(X - q)^r$. Every switching word of length L satisfies

$$\text{TV}(pM_w, p'M_w) \leq \theta^L \text{TV}(p, p') \quad (133)$$

for all probability rows p, p' . In particular, $\text{TV}(pM_w, u_s) \leq \theta^L$.

Proof. Complete b to an invertible rational $r \times r$ matrix C with first column b . This can be done by adjoining suitable standard basis columns. Define

$$E = \begin{bmatrix} C & -C\mathbf{1}_r \end{bmatrix}, \quad F = E^T (EE^T)^{-1}. \quad (134)$$

The matrix E has full row rank and kernel $\mathbb{Q}\mathbf{1}_s$. Therefore

$$E\mathbf{1}_s = 0, \quad \mathbf{1}_s^T F = 0, \quad EF = I_r, \quad FE = I_s - J_s, \quad E\mathbf{e}_1 = b. \quad (135)$$

For completeness, FE is symmetric and idempotent, with image the orthogonal complement of $\mathbf{1}_s$ and kernel $\mathbb{Q}\mathbf{1}_s$. Hence it is exactly $I_s - J_s$. The inverse in (134) exists because EE^T is positive definite over $\mathbb{R}$; it has rational entries by ordinary matrix inversion.

Write $K_i = FA_iE$ and let

$$K_{\max} = \max_{i,a,b} |(K_i)_{ab}|, \quad q = \frac{\theta}{1 + sK_{\max}}, \quad M_i = J_s + qK_i. \quad (136)$$

Each K_i has zero row sums and zero column sums. Thus each M_i is doubly stochastic. Moreover,

$$|(M_i)_{ab} - 1/s| \leq qK_{\max} < \theta/s,$$

which proves the strict entry bounds. Here $K_{\max} > 0$, because A_i is invertible and $EF = I_r$, so $0 < q < \theta$.

The identities $J_s F = 0$, $EJ_s = 0$, and $EF = I_r$ give

$$M_w = J_s + q^L F A_w E \quad (137)$$

for every nonempty word of length L . The formula also holds for the empty word: its right side is $J_s + FE = I_s$. Commutation of the A_i implies commutation of the M_i .

For a rational row a , put $w = aE$ and choose

$$\varepsilon_a = \frac{1}{s(1 + \|w\|_\infty)}, \quad p_a = u_s + \varepsilon_a w. \quad (138)$$

The entries of w sum to zero, and $|\varepsilon_a w_j| < 1/s$, so p_a is a strictly positive probability row. Now multiply (137) by p_a and $\mathbf{e}_1$. The uniform part gives $1/s$, and the centered part gives

$$\varepsilon_a q^L a E F A_w E \mathbf{e}_1 = \varepsilon_a q^L a A_w b,$$

which is (132). Any other positive rational ε for which $u_s + \varepsilon a E$ is strictly positive is equally valid.

The row map $a \mapsto aE$ is an isomorphism $\mathbb{Q}^{1 \times r} \rightarrow H_s$, and

$$(aE)M_i = q(aA_i)E. \quad (139)$$

Thus M_i acts as 1 on the uniform direction and as qA_i on H_s . This proves its stated characteristic polynomial and $\det M_i = q^r \neq 0$.

Finally, define $R_i = (M_i - (1 - \theta)J_s)/\theta$. It is nonnegative and doubly stochastic. For a mass-zero row v ,

$$vM_i = \theta vR_i, \quad \|vR_i\|_1 \leq \sum_{a,b} |v_a|(R_i)_{ab} = \|v\|_1.$$

Iterating proves (133). Since u_s is stationary and total variation between probability distributions is at most 1, the final assertion follows. $\square$

[write] This is the lift of Corollary 4.3, transposed to rows and applied to unipotent operators (Section 42.4).

Corollary 45.2. *Every nonzero $P \in \mathbb{Q}[x_1, \dots, x_k]$ has a realization satisfying all the conditions of Theorem 45.1 with $s = r(P) + 1$ and*

$$\text{acc}_P(n) - 1/s = \varepsilon q^{|n|} P(n). \quad (140)$$

The same fixed matrices realize every member of a joint translation space, using its own initial row. In particular, $P(n) = 0$ if and only if the accepting coordinate is exactly at equilibrium.

Proof. Use (130) and Theorem 45.1. Lemma 44.2 ensures $b \neq 0$, since evaluation at zero is nonzero on a space containing 1. $\square$

The zero polynomial is a harmless exceptional input: it can be represented by a one-state chain with constant acceptance 1, or loaded with $p = u_s$ in any nonzero compiled module. The reference compiler requires its *joint family* to be nonzero, but allows a zero member to be loaded into that family.

45.1 Why one extra state is necessary

Theorem 45.3 (Exact-series lower bound). *Suppose commuting rational $s \times s$ row-stochastic matrices B_i have a common stationary probability row u , and suppose a probability row p , a fixed column f , and constants $c \neq 0$, $q \neq 0$ satisfy*

$$pB^n f - u f = c q^{|n|} P(n) \quad (n \in \mathbb{N}^k). \quad (141)$$

Then $s \geq r(P) + 1$. For a family realized by the same B_i, u, f, q , with possibly different p and nonzero c , one has $s \geq r(P) + 1$. Thus Corollary 45.2 is dimension-optimal in this exact-series category.

Proof. For every $m \in \mathbb{N}^k$, commutativity gives

$$P(m+n) = c^{-1} q^{-|m|} q^{-|n|} (p-u) B^m B^n f.$$

Every row $(p-u)B^m$ belongs to the mass-zero space, of dimension $s-1$. The linear map sending such a row v to the function $n \mapsto q^{-|n|} v B^n f$ therefore has image dimension at most $s-1$. All translates of P lie in that image. Injectivity of polynomial evaluation now gives $r(P) \leq s-1$. Taking the span over all initial rows proves the family version. $\square$

Remark 45.4. This is not a lower bound for representing *only the predicate* $P(n) = 0$. Different polynomials may have the same natural zero set and very different translation ranks. Nor does it cover arbitrary encodings of the variables, or noncommuting competitors specified only on canonical words. The invariant concerns the precise numerical series (141).

45.2 Exact coordinate hits without any erasure

Proposition 45.5. *In the construction of Theorem 45.1, no finite word sends a nonuniform initial distribution to u_s . Nevertheless, its first coordinate can equal $1/s$ after a finite word, exactly when the encoded polynomial vanishes there.*

Proof. Every finite product is invertible and fixes u_s . If $pM_w = u_s$, then $p = u_s M_w^{-1} = u_s$. The assertion about one coordinate is (140); a vector can be nonzero while a chosen coordinate is zero. $\square$

This is the precise distinction from stochastic mortality. The information vector survives algebraically at every finite time while its norm becomes arbitrarily small.

46 Sharp quartic budgets and a quintic bound

A full monomial basis of degree at most four has $\binom{k+4}{4}$ members. That is unnecessarily large for a single quartic's translation space. The following sharper estimate comes directly from the derivative orders.

Theorem 46.1 (Sharp quartic state bound). *For every nonzero polynomial P of total degree at most four in $k \geq 1$ variables,*

$$r(P) \leq 1 + k + \binom{k+2}{2} = \binom{k+3}{2} - 1. \quad (142)$$

Consequently the positive mixing realization uses at most $\binom{k+3}{2}$ states. This bound is attained, in the exact-series category, by

$$P(x) = (x_1^2 + \cdots + x_k^2)^2. \quad (143)$$

Proof. The derivative span is generated by P itself, its k first derivatives, and its derivatives of order at least two. The last group lies in the space of polynomials of degree at most two, of dimension $\binom{k+2}{2}$. This proves (142) without assuming homogeneity.

For sharpness, put $R = x_1^2 + \cdots + x_k^2$ and $P = R^2$. Its degree-four part spans one dimension. Its first derivatives $4x_i R$ are independent, so the cubic part has dimension k . For $i \neq j$ and for each i ,

$$\partial_i \partial_j P = 8x_i x_j, \quad \partial_i^2 P = 4R + 8x_i^2.$$

The diagonal coefficient matrix is $8I_k + 4\mathbf{1}_k\mathbf{1}_k^T$, which is invertible: its eigenvalues are 8 on the orthogonal complement of $\mathbf{1}_k$ and $8 + 4k$ on that line. Hence the second derivatives span all homogeneous quadratics, of dimension $\binom{k+1}{2}$. The third derivatives include $\partial_i^3 P = 24x_i$, and a fourth derivative gives a nonzero constant. The different homogeneous degrees are independent, giving

$$r(P) = 1 + k + \binom{k+1}{2} + k + 1 = \binom{k+3}{2} - 1.$$

The lower bound in Theorem 45.3 completes sharpness for states. $\square$

The radial example has a very simple zero set. Its role is to saturate a numerical realization bound, not to furnish a hard Diophantine predicate. The derivative dimensions $(1, k, \binom{k+1}{2}, k, 1)$ are consistent with the standard compressed inverse-system picture [32]; it is their translation into an optimal positive stochastic state budget that is used here.

Proposition 46.2 (General split bound). *If P has degree at most $d \geq 1$, then for $1 \leq m \leq d + 1$,*

$$r(P) \leq \binom{k+m-1}{k} + \binom{k+d-m}{k}, \quad (144)$$

where the second term is taken as zero when $m = d + 1$. In particular, a quintic admits a mixing realization with

$$s \leq 2\binom{k+2}{2} + 1 = (k+1)(k+2) + 1. \quad (145)$$

Proof. There are $\binom{k+m-1}{k}$ differential monomials of order less than m . All derivatives of order at least m have degree at most $d - m$, whose ambient polynomial space has dimension $\binom{k+d-m}{k}$. Add the two bounds. For $d = 5$, take $m = 3$ and then add one state. $\square$

No general sharpness assertion for (144) is needed below. The state-minimal construction uses the computed rank, not the upper bound.

47 Fixed alphabets and simple input geometry

The transitions depend on a joint translation module, not on a particular coefficient row. This permits the entire computational input to be moved into the initial distribution. We first record the elementary parameter-loading mechanism, then apply it to a fixed universal Diophantine predicate.

47.1 Polynomial input curves

Suppose $P_t(x) = \sum_{j=0}^h t^j P_j(x)$ and let V contain all translates of the coefficient polynomials P_j . Construct E, F, M_i for V as above. Write

$$a(t)E = w_0 + tw_1 + \cdots + t^h w_h.$$

Choose a positive rational C strictly larger than $\max_\ell \sum_{j=0}^h |(w_j)_\ell|$. Then

$$p(t) = u_s + \frac{\sum_{j=0}^h t^j w_j}{sC(1+t)^h} \quad (t \in \mathbb{N}) \quad (146)$$

is strictly positive and has total mass one. Indeed, $t^j \leq (1+t)^h$ and all w_j have zero coordinate sum. With $\varepsilon(t) = 1/(sC(1+t)^h)$, the master identity becomes

$$p(t)M^n \mathbf{e}_1 - 1/s = \varepsilon(t)q^{|n|}P_t(n). \quad (147)$$

This is an explicit rational map, not a noncomputable encoding into a real constant.

When $h = 2$, let $v = t/(1+t)$. The same curve is a quadratic Bézier curve:

$$p(t) = (1-v)^2 b_0 + 2v(1-v)b_1 + v^2 b_2, \quad (148)$$

where $b_0 = u_s + w_0/(sC)$, $b_1 = u_s + w_1/(2sC)$, and $b_2 = u_s + w_2/(sC)$. All three control points are strictly positive rational probability rows. Thus the curve lies entirely in the interior of the probability simplex.

47.2 A quartic front end with one input-copy equation

We use MRDP in its standard form: for every computably enumerable set $K \subseteq \mathbb{N}$, there is an integer polynomial $U(t, x)$ such that

$$t \in K \iff \exists x \in \mathbb{N}^m U(t, x) = 0. \quad (149)$$

This external theorem is classical [26]. For a universal halting set, U is one fixed polynomial. The following degree reduction is written out to make both the input contract and witness correspondence explicit.

Lemma 47.1 (Natural quadratic gate front end). *From a fixed integer polynomial $U(t, x)$ one can effectively construct natural variables $z = (z_0, \dots, z_{k-1})$, with z_0 a copy of the input, and fixed integer polynomials $q_j(z)$ of degree at most two such that*

$$\exists x U(t, x) = 0 \iff \exists z \in \mathbb{N}^k [z_0 = t \text{ and all } q_j(z) = 0].$$

For each original witness x , the extra gate variables have a unique completion.

Proof. Write $U = U_+ - U_-$ with $U_\pm$ having nonnegative integer coefficients. Evaluate each by a fixed acyclic circuit using natural constants, addition, and multiplication, replacing its input t by z_0 . Give every materialized gate a fresh natural output variable. A gate $v = a + b$ contributes $v - a - b = 0$, and a gate $v = ab$ contributes $v - ab = 0$. Source variables and constants are allowed at the inputs. Finally equate the outputs evaluating U_+ and U_- . Every residual has degree at most two. Topological induction forces each gate value uniquely from z_0 and the original witnesses; all these values are natural. The final equality is exactly $U = 0$. $\square$

[write] This is `cdc:wf:lem:quadratization` of [22] (Section 42.4).

Set

$$S(z) = \sum_j q_j(z)^2, \quad Q_t(z) = (z_0 - t)^2 + S(z) = Q_0(z) - 2tz_0 + t^2. \quad (150)$$

These are integer polynomials. The polynomial S is nonnegative on all real arguments, has degree at most four, and takes natural values on $\mathbb{N}^k$. For each t , the natural roots of Q_t are exactly the complete valid witness tuples for input t .

Theorem 47.2 (Fourth-order universality on a quadratic curve). *For every fixed computably enumerable $K \subseteq \mathbb{N}$ and every prescribed rational $0 < \theta < 1$, one can effectively obtain, from a Diophantine representation of K , a fixed finite commuting alphabet of strictly positive rational doubly stochastic invertible matrices and a rational curve $p(t)$ whose coordinates have degree at most two over $(1+t)^2$, such that*

$$t \in K \iff \exists n \in \mathbb{N}^k \ p(t)M^n \mathbf{e}_1 = 1/s. \quad (151)$$

The matrices satisfy (127); after division by their common transient eigenvalue, their joint nilpotence depth is at most four. The number of states satisfies $s \leq \binom{k+3}{2}$. The initial distributions lie on a quadratic Bézier curve with strictly positive rational control points. For a computably enumerable complete K , this exact-hit problem is computably enumerable complete.

Proof. Apply the joint-module construction to the family Q_t in (150), equivalently to $Q_0, z_0, 1$. Its derivative span is contained in the sum of the span of Q_0 , the span of its k first derivatives, and the full space of degree-at-most-two polynomials. The two extra generators z_0 and 1 , and all their translates, are already in the last ambient space. Thus the quartic rank bound remains valid for this family, even if those two polynomials were not already in $\mathcal{V}(Q_0)$.

Use (146) with $h = 2$. Identity (147) and the positivity of $\varepsilon(t)q^{|n|}$ prove (151). The curve and control-point claims are (148); the depth claim follows from Proposition 44.4. All data are obtained by finite rational operations after a polynomial U is supplied.

The map from t to the finite rational initial row is computable. The right-hand side of (151) is computably enumerable by enumerating all count vectors and performing exact rational tests. If K is complete, (151) is a many-one reduction from that complete set, proving the last assertion. $\square$

47.3 Straight-line universality at fifth order

A square is convenient for forcing the input copy, but it makes the loader quadratic in t . The next construction replaces that square by a sign barrier. It is important that the target $t + 1$ is always strictly positive.

Lemma 47.3 (Positive-value filter). *With S from Lemma 47.1, define*

$$F(z) = (z_0 + 1)(1 - S(z)), \quad P_t(z) = t + 1 - F(z) = t - z_0 + (z_0 + 1)S(z). \quad (152)$$

Then P_t has degree at most five in z , is affine in t , and its natural roots are exactly the roots of Q_t in (150).

Proof. If $S(z) = 0$, then $F(z) = z_0 + 1$, so $P_t(z) = 0$ exactly when $z_0 = t$. If $S(z) \neq 0$, integrality and nonnegativity imply $S(z) \geq 1$, hence $F(z) \leq 0$. It cannot then equal $t + 1 \geq 1$. This proves the full tuple-wise root correspondence. The degree and affine-dependence assertions are immediate from (152). $\square$

Theorem 47.4 (Fifth-order universality on a line segment). *For every computably enumerable complete set K and prescribed rational $0 < \theta < 1$, there are fixed commuting, invertible, strictly positive rational doubly stochastic matrices $M_1, \dots, M_k$ and two fixed strictly positive rational probability rows b_0, b_1 such that*

$$p(t) = \frac{b_0 + tb_1}{1+t}, \quad t \in K \iff \exists n \in \mathbb{N}^k \ p(t)M^n \mathbf{e}_1 = 1/s. \quad (153)$$

The matrices satisfy (127), have only the eigenvalues 1 and $q \in (0, \theta)$, and have normalized joint nilpotence depth at most five. One may take $s \leq 2^{\binom{k+2}{2}} + 1$. No coefficient in the transition alphabet or the two endpoint distributions depends on t .

Proof. Take $P_t = P_0 + t$ from (152). The polynomial P_0 is nonconstant: a constant P_0 could accept at most one natural input, whereas K is complete. It generates a translation space containing 1, by Lemma 44.2. Thus all P_t belong to the same space $V = \mathcal{V}(P_0)$, and the fifth-degree bound of Proposition 46.2 applies without any family enlargement.

In the lift, put $w_0 = a_{P_0}E$ and $w_1 = a_1E$. Choose rational $C > \max(\|w_0\|_\infty, \|w_1\|_\infty)$ and define

$$b_0 = u_s + \frac{w_0}{sC}, \quad b_1 = u_s + \frac{w_1}{sC}.$$

Both are strictly positive probability rows. Their interpolation is

$$\frac{b_0 + tb_1}{1+t} = u_s + \frac{w_0 + tw_1}{sC(1+t)}.$$

The master identity therefore gives

$$p(t)M^n \mathbf{e}_1 - 1/s = \frac{q^{|n|}}{sC(1+t)} P_t(n). \quad (154)$$

Lemma 47.3 proves the equivalence in (153). All matrix, depth, and state assertions follow from the lift and the degree bound. Choosing a complete K gives completeness as in Theorem 47.2. $\square$

The endpoint b_1 , approached as $t \rightarrow \infty$, represents the constant polynomial 1. Its centered acceptance is $q^{|n|}/(sC) > 0$ for every finite word. Thus this endpoint itself has no exact equilibrium hit. The hard inputs are a simple computable sequence of rational points converging to that endpoint, not an exotic or uncomputable initial real.

For fixed construction data, each coordinate of $p(t)$ in Theorem 47.4 has a rational numerator and denominator with bit length $O(\log(t+2))$. The same estimate holds for the quadratic curve. Computing these input rows requires only fixed-degree rational polynomial evaluation. This is an exact symbolic input convention, not a claim that physical preparation or measurement can use unbounded precision at constant cost.

47.4 Strict cutpoints and a fixed initial distribution

For the quartic front end, $Q_t(n)$ is a nonnegative integer. Consequently

$$1 - 2Q_t(n) > 0 \iff Q_t(n) = 0.$$

Loading the polynomial $1 - 2Q_t$ into the same joint module therefore gives

$$t \in K \iff \exists n \, p_{\text{strict}}(t)M^n \mathbf{e}_1 > 1/s. \quad (155)$$

This is an exact strict comparison with no promised positive gap. The argument uses the nonnegativity of Q_t ; it does not assert an analogous root-to-sign conversion for the generally signed quintic P_t without further changes.

There is also a fixed-initial-state formulation. Treat the external t in $Q_t(z)$ as an additional count variable and compile the single quartic $Q(t, z)$. Read the input in unary as a_0^t , and restrict the subsequent existential word to the other letters. Then

$$t \in K \iff \exists n \, pM_0^t M_1^{n_1} \cdots M_k^{n_k} \mathbf{e}_1 = 1/s.$$

The initial distribution and all matrices are fixed, with $s \leq \binom{k+4}{2}$. This pays for an additional letter and an explicit unary input prefix. Allowing the witness suffix to contain a_0 would alter the input and is not part of this protocol.

48 The converse: polynomial degree is nilpotence depth

The forward construction suggests an intrinsic matrix class. Its defining conditions can be checked by rational arithmetic when q and the depth bound are supplied.

Definition 48.1. A rational doubly stochastic alphabet belongs to $\mathcal{C}_d$ if there is a specified rational $q > 0$ such that the matrices

$$R_i = q^{-1}(M_i - J_s) - (I_s - J_s) \quad (156)$$

commute, annihilate the uniform direction on both sides, and satisfy $R^\alpha = 0$ for every $|\alpha| = d + 1$. Equivalently, on H_s , $q^{-1}M_i = I + R_i$ and the nilpotent parts have joint depth at most d . Strict positivity and a prescribed bound (127) may be imposed as additional conditions.

Theorem 48.2 (Effective degree–depth equivalence). *For fixed $d \geq 0$, exact equilibrium-coordinate reachability for rational initial distributions and alphabets in $\mathcal{C}_d$ effectively reduces to solvability over $\mathbb{N}$ of a single rational polynomial of degree at most d . Conversely, every such polynomial reduces to this reachability problem, even with strict positivity, arbitrary prescribed rational mixing bound, commuting invertible matrices, and one accepting state. Matrix dimension and alphabet size are allowed to vary in this equivalence.*

Proof. Put $v = p - u_s$. On the mass-zero subspace, the binomial theorem and joint nilpotence give

$$\begin{aligned} pM^n \mathbf{e}_1 - 1/s &= q^{|n|} v \prod_{i=1}^k (I_s + R_i)^{n_i} \mathbf{e}_1 \\ &= q^{|n|} \sum_{|\alpha| \leq d} (v R_1^{\alpha_1} \cdots R_k^{\alpha_k} \mathbf{e}_1) \prod_{i=1}^k \binom{n_i}{\alpha_i}. \end{aligned} \quad (157)$$

The right-hand sum defines a rational polynomial $g(n)$ of total degree at most d . The term with $\alpha = 0$ uses the identity matrix; the formula is therefore valid at $n = 0$ as well. Since $q > 0$, exact equilibrium is equivalent to $g(n) = 0$. Clearing a common positive denominator gives an integer polynomial with the same natural zero set and no degree increase.

In the other direction, apply the polynomial lift. Proposition 44.4 and (139) put the result in $\mathcal{C}_d$. A nonzero constant polynomial can be realized in two states; a zero polynomial is handled by the trivial equilibrium initialization. All additional restrictions follow from Theorem 45.1. $\square$

The matrix promise is finite and explicit. One checks stochasticity, commutation, and each product of total degree $d + 1$ in (156). There are $\binom{k+d}{d+1}$ such multiindices. Their vanishing implies the vanishing of all longer products because the matrices commute. This is a validity check for the promised class, not a test for reachability itself.

Corollary 48.3. *The variable-dimension equilibrium-hit problem in $\mathcal{C}_4$ is computably enumerable complete. This remains true under the positivity and mixing restrictions above. At depth three the corresponding problem is exactly the natural solvability problem for a single polynomial of degree at most three.*

Proof. The fourth-degree statement follows from the quadratic-gate sum-of-squares reduction and Theorem 48.2. The third-degree statement is the same equivalence with $d = 3$. $\square$

The degree-three equivalence is a precise reduction, not a decision or undecidability claim for that Diophantine class. The article does not settle it. At depth zero the recovered polynomial is constant. At depth one it is linear, so the matrix question reduces to an ordinary natural linear equation; no higher-order matrix behavior is hidden there.

48.1 Why nonnormality is doing real work

Proposition 48.4. *Suppose a doubly stochastic s -state matrix has a simple eigenvalue 1 and all its other eigenvalues equal the same $q \in (0, 1)$. If it is diagonalizable, then*

$$M = J_s + q(I_s - J_s).$$

Consequently, in a commuting family with this common two-point spectrum, diagonalizability of every generator makes the normalized centered scalar signal constant in the count vector.

Proof. The invariant mass-zero subspace contains all nonstationary eigenvectors. The restriction of a diagonalizable operator to this invariant spectral subspace is diagonalizable with its only eigenvalue q , hence equals qI . The action on the uniform direction is 1. This gives the formula. Products then act by $q^{|n|}$ on the mass-zero subspace, leaving only a constant after that factor is divided out. $\square$

A normal matrix is diagonalizable, so nonconstant polynomial signals in this specific class require nonnormality. In particular, all the generators cannot be reversible with respect to the common uniform distribution, because that would make them symmetric. This observation concerns the stated two-point spectrum; it is not a general theorem deciding reachability for reversible chains with several transient eigenvalues.

49 Exact critical targets versus effective robust questions

The common contraction estimate makes the limiting acceptance easy to compute: it is always $\lambda_* = 1/s$. The discontinuity lies in the predicate of *ever attaining* that value, not in a noncomputable limiting real.

Theorem 49.1 (Finite search away from equilibrium). *Suppose every generator satisfies (127). Let p be rational. For a rational target $c \neq 1/s$, choose any integer H satisfying*

$$\theta^H < |c - 1/s|. \tag{158}$$

No word of length at least H has acceptance c . Thus exact reachability of c is decidable by testing the finitely many words of length less than H . If the matrices commute, it suffices to test count vectors with $|n| < H$.

More generally, a rational interval at positive distance from $1/s$ has the same finite-search property. For every rational $\eta > 0$, all sufficiently long words satisfy $|\text{acc}_p(w) - 1/s| < \eta$, irrespective of whether an exact hit ever occurs.

Proof. The contraction estimate gives

$$|\text{acc}_p(w) - 1/s| \leq \text{TV}(pM_w, u_s) \leq \theta^{|w|}.$$

For $|w| \geq H$, this is strictly smaller than the distance to c , or to the specified interval. All shorter products are finite rational computations. For the last assertion, choose H with $\theta^H < \eta$. $\square$

There is no need to use floating-point logarithms to find H . Starting at $H = 0$, multiply a rational accumulator by θ until it is strictly below the rational distance. This terminates and correctly handles equality at the previous step. For example, $\theta = 1/2$ and distance $1/8$ give $H = 4$, not 3. The implementation includes this boundary check.

For commuting k -letter alphabets and $H \geq 1$, the number of count vectors with $|n| < H$ is $\binom{H+k-1}{k}$. This gives a useful explicit search-size bound, although rational numerator and denominator sizes must still be paid for in an implementation.

[write] This is Part I's remark that a horizon with $\eta^T \leq \varepsilon$ works for every instance (after Proposition 4.2), turned into a decision procedure for targets at positive distance. It settles one case of Part I's question of Section 15.8 and bears on Question `pqc:q:promise` of [21].

Corollary 49.2. *For a fixed complete input family in Theorem 47.2 or Theorem 47.4, there is no total computable function $B(t)$ which bounds the length of some exact-hit word whenever one exists.*

Proof. Such a function would decide membership in K : compute $B(t)$ and test all count vectors of total size at most $B(t)$. This would terminate on all inputs and be correct in both the positive and negative cases, contradicting completeness of K . $\square$

[write] This is the bounded-search argument of `cdc:bd:prop:nobound` in [22]; Part I proves the same for erasure (Theorem 11.2).

49.1 Why approximate measurement is not an exact-zero oracle

For an integer polynomial and the basic lift,

$$P(n) \neq 0 \implies |\text{acc}_P(n) - 1/s| \geq \varepsilon q^{|n|}. \quad (159)$$

The lower bound is positive for each supplied tuple, but shrinks exponentially with its length. For a known finite word, rational arithmetic still decides equality exactly. What is absent is a computable universal horizon for the search and a fixed positive observation margin valid at all lengths.

Every symmetric tolerance band around equilibrium is eventually reached by *every* word, so that approximate event cannot retain the Diophantine predicate. Conversely, a rational target separated from equilibrium is settled by finite search. A one-sided interval with equilibrium on its boundary has zero separation and is not covered by that argument. Exact strict cutpoints such as (155) may therefore remain hard. “Robust” must include an actual positive margin; merely replacing equality by a strict inequality does not supply one.

Nothing here proposes an algorithm for the halting problem using physical noise, sampling, thermalization, or finite-precision measurement. Fast mixing hides the signal numerically without deleting its exact algebraic zero set.

50 Witnesses, input cost, and representation boundaries

50.1 Natural roots and canonical words

For a fixed polynomial P , the map

$$n \mapsto a_1^{n_1} \cdots a_k^{n_k}$$

is a bijection between natural root tuples and canonical exact-hit words. No additional existential variable is introduced by the stochastic lift. Among all words, a root tuple of total length L has $L!/(n_1! \cdots n_k!)$ orderings, and all give the same acceptance. Keeping the canonical word removes this combinatorial multiplicity.

The gate front end also preserves witnesses in its stated sense. Over each original root of $U(t, x)$, the input copy and the acyclic gate outputs are unique. Lemma 47.3 leaves the entire tuple unchanged. Neither statement makes the original MRDP representation single-fold or finite-fold. Its pre-existing witness multiplicity is retained, not solved.

50.2 What is fixed, and what is counted

For each chosen universal representation, k, s, q, θ , all matrix entries, the accepting state, and the control points of the initial curve are fixed. The parameter t changes only the rational initial row. The witness count tuple has fixed arity k but unbounded entries. This is why its search can be unbounded without adding a state per simulated time step.

It is not a free compression theorem for arbitrary chronological histories. We begin with an actual Diophantine polynomial, whose representation theorem already carries the computational semantics. The stochastic system then realizes that polynomial's counts. The proof does not infer a unique trace, a halting-time bound, or a smaller arithmetic circuit from the matrix state count.

The reference implementation starts from explicit rational polynomials. It computes their joint derivative module, bases, exact shifts, and the lift. The displayed universal-existence corollaries start one stage earlier, from classical MRDP and a gate normalization proved on paper. The package does not supply numerical matrices for a universal Jones polynomial, nor a counted machine-to-polynomial frontend. This keeps the deliverable's implemented scope separate from its mathematical existence statements.

51 Explicit rational examples

51.1 A four-state exact difference test

Take $P(x, y) = (x - y)^2 - 1$. Its translation space is spanned by $1, x - y, (x - y)^2$, so $r(P) = 3$. For $\theta = 1/2$, the supplied compiler gives $s = 4, q = 1/18, \varepsilon = 1/8$, and

$$p = \frac{1}{8}(1, 3, 2, 2). \quad (160)$$

Its two transition matrices are

$$M_a = \frac{1}{72} \begin{pmatrix} 22 & 17 & 18 & 15 \\ 22 & 21 & 14 & 15 \\ 10 & 17 & 22 & 23 \\ 18 & 17 & 18 & 19 \end{pmatrix}, \quad M_b = \frac{1}{72} \begin{pmatrix} 18 & 17 & 16 & 21 \\ 18 & 21 & 20 & 13 \\ 22 & 17 & 20 & 13 \\ 14 & 17 & 16 & 25 \end{pmatrix}. \quad (161)$$

Every row and every column of each integer numerator sums to 72. Every entry is positive. Direct rational multiplication verifies commutation and gives

$$pM_a^x M_b^y \mathbf{e}_1 = \frac{1}{4} + \frac{(x-y)^2 - 1}{8 \cdot 18^{x+y}}. \quad (162)$$

Thus the accepting coordinate is exactly uniform precisely when $|x - y| = 1$. For $(x, y) = (2, 0)$ the probability is $217/864$, slightly above $1/4$. The empty word gives $1/8$, correctly matching (160).

[write] The values $q = 1/18$ and $\varepsilon = 1/8$ and the matrices (161) belong to the basis chosen by the shipped compiler. Another rational basis of the same three-dimensional translation space gives another valid lift with the same four states; an independent lift computed at placement had $q = 3/56$. The state count and the zero set do not depend on the basis; q , ε and the entries do. The same holds for the values of q and ε quoted in the rest of this section and in Table 9.

Each matrix has characteristic polynomial $(X - 1)(X - 1/18)^3$ and determinant 18^{-3} . The initial row is not uniform, so no word makes the entire distribution uniform. Four states are necessary for the numerical series by Theorem 45.3, even though the predicate $|x - y| = 1$ can be studied by other encodings.

Recompiling the same polynomial with $\theta = 1/100$ keeps four states and gives $q = 1/900$. It changes the transition entries, not the zero set or the translation rank. This provides a concrete check of the arbitrary-mixing-speed claim without invoking a universal representation.

51.2 Multiplication and a maximal quartic rank

For $P(x, y, z) = xy - z$, the derivative space is spanned by $xy - z, x, y, 1$. The compiler therefore uses five states. Its exact signal has $q = 1/14$ and $\varepsilon = 1/10$ when $\theta = 1/2$.

For $P(x, y) = (x^2 + y^2)^2$, the rank is 9 and the number of states is 10, attaining $\binom{2+3}{2} = 10$. The exported lift has $q = 1/390$ and $\varepsilon = 1/20$. Its unique natural root is $(0, 0)$, so its first coordinate is at equilibrium initially, while its full initial distribution remains nonuniform. This is another useful check that coordinate equality does not mean vector equality.

51.3 Quadratic and linear loaders for factorization

The small family

$$Q_t(x, y, z) = (z - t)^2 + (z - xy)^2 \quad (163)$$

has natural roots precisely when $z = t = xy$. Its joint translation rank is 10. The supplied alphabet is fixed at 11 states with $q = 1/80$, for every exported input $t \in \{0, 1, 2, 6, 11\}$. The quadratic loader uses

$$\varepsilon(t) = \frac{1}{44(1+t)^2},$$

with the three control distributions determined by (148). For $t = 6$, the roots in the tested box $\{0, \dots, 6\}^3$ are exactly $(1, 6, 6), (2, 3, 6), (3, 2, 6), (6, 1, 6)$.

Its positive-filter counterpart is

$$P_t(x, y, z) = t - z + (z + 1)(z - xy)^2. \quad (164)$$

It has exactly the same natural roots, now with a straight-line loader. The rank of $\mathcal{V}(P_0)$ is 18, so this example uses 19 states, with $q = 3/1613$ and

$$\varepsilon(t) = \frac{6}{247(1+t)}.$$

The construction uses $C = 13/6$ in Theorem 47.4. The independent checker reconstructs both endpoint distributions from the exports for $t = 0, 1$ and verifies the affine interpolation identity for all five supplied inputs.

These are arithmetic demonstrations, not universal alphabets. They make the input-geometry tradeoff visible on completely specified rational data and exercise root equivalence in a case with several witnesses.

Polynomial or family	Degree	Rank	States	q
$(x - y)^2 - 1$	2	3	4	1/18
$xy - z$	2	4	5	1/14
$(x^2 + y^2)^2$	4	9	10	1/390
Q_t of (163)	4	10	11	1/80
P_t of (164)	5	18	19	3/1613

Table 9: Exact exported dimensions at $\theta = 1/2$. Family rows use one fixed alphabet across all listed inputs. Rank measures numerical realization, not arithmetic difficulty.

52 Implementation, independent checking, and reproducibility

52.1 Compiler contract

The file `code/mixing_compiler.py` is a rational reference implementation using SymPy. It accepts a nonzero joint family of polynomials over $\mathbb{Q}$, a nonempty tuple of distinct variables, and a rational $\theta \in (0, 1)$. It constructs a derivative basis, computes shift matrices by exact coefficient comparison, builds E, F , and emits the matrices from (136).

It rejects floating-point coefficients, undeclared polynomial variables, negative or nonintegral counts, Boolean values in numerical input positions, and polynomials outside the compiled module. Loaded initial scales must be positive and must produce strictly positive distributions. Output matrices and the compiled object are immutable. Monomials are enumerated by graded weak compositions rather than by a full Cartesian grid of exponent bounds. No complexity optimality is claimed for symbolic elimination.

The optional scale argument permits the explicit quadratic and linear input curves used above. Without it, the compiler selects the safe pointwise scale (138). Exported JSON stores every rational as an exact string and includes the input polynomial, basis, shifts, E, F , transitions, initial row, q , θ , and ε .

A minimal use is:

```
from fractions import Fraction
import sympy as sp
from mixing_compiler import compile_family

x, y = sp.symbols("x y")
P = (x-y)**2 - 1
model = compile_family([P], (x, y), Fraction(1, 2))
assert model.states == 4
assert model.acceptance(P, (2, 0)) == sp.Rational(217, 864)
model.export(P, "example.json", "difference test")
```

52.2 What was checked

The shipped execution used Python 3.13.5 and SymPy 1.14.0. The producer `build_examples.py` generates fourteen exports. It symbolically reconstructs the polynomial from the transition matrices by (157) for every export, tests radial quartic ranks for one through four variables, and rejects nineteen deliberately malformed inputs. It writes `verification/compiler_report.json`.

The independent `verify_exports.py` imports neither the compiler nor SymPy. It implements matrix multiplication, powers, and Gaussian elimination with Python’s `fractions.Fraction`. It checks the identities $EF = I$ and $FE = I - J$, reconstructs each transition matrix, checks strict entry margins, row and column sums, commutation, invertibility, and the required nilpotent products. It also computes finite Hankel ranks directly from polynomial values, independently of the derivative-basis code.

Across the fourteen files, the recorded checks cover 1,310 count vectors, 1,424 arbitrary-order words including empty words, and 269 degree-plus-one nilpotent products. They check both scalar identities and full-vector non-erasure, the fixed-matrix property of each input family, the recovered linear and quadratic input curves, and both four-root factorization lists at $t = 6$. All checks pass. The exact counts and per-file results are in `verification/independent_report.json`.

The two test populations overlap; their totals are not a count of independent mathematical theorems. Finite success cannot prove an identity for all inputs. The general proofs in Sections 3–9 ([write] Sections 44–50 here) supply that justification, and the independent arithmetic checks guard against transcription and implementation errors in the delivered examples.

52.3 Reproduction

From the package root, run:

```
python -m pip install -r requirements.txt
python code/build_examples.py
python code/verify_exports.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The independent checker needs only the Python standard library and can be run without installing SymPy when checking the existing JSON files. Building new examples requires SymPy. The article has no external image or font-file dependencies beyond the installed LaTeX packages. No network access is needed after the software dependencies are installed.

[write] These commands use the delivered layout. In this report the three programs are `code/11-mixing-arithmetic-mixing_compiler.py`, `code/11-mixing-arithmetic-build_examples.py` and `code/11-mixing-arithmetic-verify_exports.py`, and the delivered `reproduce.sh` is `code/11-mixing-arithmetic-reproduce.sh`. The fourteen examples, the three files of `verification/`, `provenance.json` and `requirements.txt` are `data/11-mixing-arithmetic-*`. The source’s `article.tex` is printed as this Part; its PDF, its README and its `SHA256SUMS.txt` (26 of 26 verified at placement) are not shipped. `build_examples.py` imports `mixing_compiler` by its delivered name and rewrites `examples/` and `verification/compiler_report.json`, and `verify_exports.py` rewrites `verification/independent_report.json`, both under their parent directory and, on Windows, with CRLF line endings; the README of this report gives a recipe that runs on a copy. At placement both programs ran (about 24 s and 2 min) and reproduced every shipped export and both reports after line-ending normalization.

53 Further research questions and concrete next steps

53.1 Priority A: finish a numerical universal instance

The most direct repository-facing task is to select a frozen universal polynomial or a fully specified arithmetic circuit from the existing H10 pipeline, carry out the natural gate normalization, and compile its *actual* joint translation module. The deliverable should print or export every matrix, the complete input map, its exact rank, and coefficient bit lengths. This would replace the existence-level universal alphabet in this note by a concrete one. It must retain the source polynomial’s full domain contract and cannot infer an arithmetic-operation improvement merely from fewer states.

[write] This instantiation is what the batch-79 answer to Part I’s question of Section 15.6 still lacks (dated note there).

A useful optimization target is the module rank after gate normalization. Different circuits for the same polynomial can give different quartic sums of squares, and hence different derivative spaces. Gate count alone is not the correct objective. An experimentally searchable criterion is the rank of the coefficient matrix of all derivatives through order four, together with the height of the resulting stochastic lift.

53.2 Priority A: line inputs with fourth-order depth

Our quadratic loader uses depth four, while the straight-line loader uses depth five. Can one achieve a universal exact-hit predicate on a fixed rational line segment while keeping depth at most four and comparable state size? The present positive-value filter raises the degree through the factor $z_0 + 1$. A construction avoiding that degree increase would improve a specific input-geometry/depth tradeoff. This note neither proves it impossible nor asserts that this formulation is an established literature-wide open problem.

A related restricted target is an affine input family whose normalized polynomial has the form $A(n) + tB(n)$ with B constant. Understanding which Diophantine predicates admit such degree-bounded representations could separate the cost of loading an input from the cost of enforcing a witness.

53.3 Priority A: mechanically check the algebraic interface

A manageable formalization begins with finite-dimensional rational vector spaces and the identities $EF = I$, $FE = I - J$. The stochastic lift, empty-word case, exact readout, and total-variation contraction can then be proved without formalizing MRDP again. A second stage formalizes the polynomial translation space and its degree filtration. Only after these interfaces are checked is it necessary to connect an existing formal Diophantine representation theorem.

This decomposition separates rational algebra, finite combinatorics, and the external computability theorem. It also gives a precise specification against which the Python producer can emit certificates rather than merely trusted matrices. No Lean or Rocq formalization is included in the present package.

53.4 Priority B: zero-set state complexity

The numerical lower bound $r(P) + 1$ is exact, but can greatly overestimate the cost of representing the predicate $P(n) = 0$. Define a zero-set realization complexity by minimizing over all polynomials with the same natural zero set, and then over admissible exact-cutpoint encodings. Can useful lower bounds be proved for this invariant? Even showing a gap between value complexity and zero-set complexity for structured families would clarify what the rank theorem does and does not measure.

A computational first step is to compare a polynomial, its square, a sum-of-squares presentation, and an equivalent presentation with unique gate variables. Variables and states must be reported separately; adding witness variables can change the optimization problem rather than solve it.

53.5 Priority B: coefficient height and prescribed decay

The construction makes q small enough to enforce positivity. It does not allow an arbitrary preassigned q close to 1 at no cost. What is the optimal tradeoff among state count, rational coefficient height, a requested contraction factor, and the size of the transient eigenvalue? Can a different rational basis of the same translation module dramatically improve the denominators without adding states?

A concrete optimization problem is to minimize $\max_i \|FA_iE\|_{\max}$ over rational choices of the embedding and basis, subject to the fixed coordinate readout. The theorem supplies a feasible point, not an optimum. Large-denominator input distributions may be as important as large matrix denominators in any end-to-end bit-cost study.

53.6 Priority B: third-order and reversible boundaries

Theorem 48.2 identifies the third-order matrix problem with single-cubic natural solvability. Progress on either side transfers to the other. Restrictions such as bounded alphabet size, special sign patterns, or small translation rank may isolate tractable subproblems without deciding the unrestricted class.

The obstruction in Proposition 48.4 disappears when several transient eigenvalues are allowed. It is therefore worth studying commuting reversible alphabets with a common stationary law and multiple positive transient eigenvalues. Their centered signals are exponential polynomials rather than just one exponential times a polynomial. The present proof does not classify their exact-hit problem.

53.7 Priority C: families, partial commutation, and margins

For a parameter family, the joint translation rank can exceed the rank of each individual member. A systematic theory of the minimal fixed alphabet should track this joint invariant, including exceptional parameter values. The affine family $P_0 + t$ is especially economical because 1 already belongs to the translation space; the quadratic family need not have this property for its input-coordinate term.

Partial commutation is another natural extension. One could replace count vectors by traces in a partially commutative monoid and seek a comparable mass-zero realization bound. Any such extension must pay for the retained ordering information rather than silently replace chronological histories by counts.

Finally, the exact/robust boundary suggests a parameterized verification program. Given a

rational target set, a certified distance from equilibrium makes the problem finite. When that distance is zero, sign information and nilpotent coefficients matter. A classification by target geometry and input curve, rather than by mixing speed alone, may yield sharper verification algorithms and more informative impossibility results.

54 Conclusion

A rapidly mixing stochastic system can retain arbitrary Diophantine zero tests in an exponentially small exact signal. The translation space provides the right finite-dimensional carrier, and an explicit mass-zero embedding converts it into a strictly positive, doubly stochastic, commuting, invertible alphabet with a single accepting state. The cost is exactly one state beyond translation rank for the numerical series.

The resulting universal refinements are structurally restrictive: fixed matrices, a common uniform stationary law, a common transient eigenvalue, bounded joint nilpotence depth, and initial inputs lying on a quadratic curve or even a straight line. At the same time, separated targets have effective finite search bounds, and all positive-tolerance neighborhoods of equilibrium are eventually reached automatically. The distinction is exact equality, not slow convergence, complicated spectra, or full-state erasure.

The immediate next mathematical and engineering tasks are explicit universal instantiation, state-versus-height optimization, and formal certification of the lift. The note supplies the proofs and a reproducible rational reference implementation needed to begin those tasks without conflating their remaining costs with the results already established here.

Part V

Coercive Green operators: finite support and rational response in well-conditioned networks (sources 12 and 13)

[write] Part V prints two batch-79 manuscripts that prove one theorem by two routes. They were written independently from the same repository snapshot `44983ed7e` (their 8-gram text overlap is at most 0.009), arrived together with the archives of commit `ef2fc7990` and were placed by commit `bbaf322e5`:

- *source 12* (batch-79 manuscript 10, file prefix `12-coercive-green-`): *Coercive Linear Algebra as a Universal Computational Substrate: Canonical Quasi-Diophantine Certificates, Rational Green Functions, and Uniformly Shunted Resistor Networks*, “Research report prepared for Vladimir Reshetnikov” (no assistant named), 2 October 2026. Labels `ste:cg::`; printed first, as the base (Part V.A, Sections 55–69, Appendices [I](#) and [J](#));
- *source 13* (batch-79 manuscript 14, file prefix `13-well-conditioned-`): *Well-Conditioned Diophantine Computation: Connected Resistor Networks, Compact Support, and a Prime-Localization Barrier*, “Research manuscript prepared for Vladimir Reshetnikov; Mathematical development and reference implementation: ChatGPT” (PDF metadata author “ChatGPT”;

prepared for Vladimir Reshetnikov”), 2 October 2026. Labels `ste:wc::`; printed second, as a second route with its further results (Part V.B, Sections 70–81, Appendices K and L).

The source numbers are this report’s file prefixes. *Source 12 is batch-79 manuscript 10*; it is unrelated to source 10 of Part II (batch-78 manuscript 10).

[write] *The common theorem.* A branch-history lift of counter machines makes every component of the configuration graph a rooted finite path or a rooted ray. The grounded operator, $\alpha I - A$ in source 12 and $mI - J$ in source 13, is uniformly coercive, and its Green column at an empty-history source has finite support, a rational root value and a primitive continuant certificate exactly when the computation halts; the resistance is rational or a fixed quadratic irrational, and no computable function bounds support, charge or time. Statement by statement:

- history lift: Lemma 56.1 and Theorem 56.2 = Lemma 71.1 and Theorem 71.2;
- coercivity and Neumann inverse: Theorem 57.1 = (214), (221) and (223);
- Green column: Lemma 58.1 with Theorem 59.1 = Lemma 72.1;
- unique primitive certificate: Theorem 58.2 = Proposition 72.2;
- resistance: Theorem 65.1 = Theorem 74.1 (one-terminal versus two-terminal source);
- no cutoff: Theorem 61.2 = Corollary 75.6, both the argument of `cdc:bd:prop:nobound` in [22];
- bounded certificates: Theorem 63.1 (a supplied label set) and Theorem 77.2 (a supplied horizon) are different finite interfaces to the same certificate.

Only in source 12: every $\alpha \geq 3$, the Pell classification of the rational values (Theorem 60.1), decidable rational levels (Theorem 60.2), polynomial-time intervals (Theorem 61.1), the polynomial-functional presentation (Theorem 62.1), the killed random walk (Theorem 65.2) and the positive-characteristic obstruction (Theorem 66.2). Only in source 13: the connected network of maximum degree three (Lemma 73.1, Theorem 73.2), which answers source 12’s own research question of Section 68.4; the strongly convex energy (Corollary 74.2); decidable finite-prime localization (Theorem 75.3, Corollary 75.4); the connected row-three threshold (Theorem 76.1); the positivity obstruction (Proposition 76.2); and the electrical layer of the horizon quadratic (Theorem 77.2).

[write] *How the two are printed.* Both are printed in full. Source 12 is the base because it covers every $\alpha \geq 3$ and has more consequences of the common theorem; source 13 follows, and each of its statements shared with source 12 carries a bracketed pointer while its proof is kept as a second route. Source 12’s Section n is Section $n + 54$ here and source 13’s Section n is Section $n + 69$. Both write D_j for the same continuants but with different letters around them; Table 10 lists the notions, and the letters are otherwise left as delivered. Source 13’s `\cref` references are written out (“Lemma ...”), and both sources’ inner-product and norm macros print with this report’s definitions.

[write] *Review.* The research programme reviewed both manuscripts before placement [42] (commit 899391bde). It read every proof, accepts both within their stated presentation and domain assumptions, reran both suites (110,522 and 38,524 checks) and all five example exports, and added independent checks (720 history labels, 36 rational inverse columns, 350 node encodings, 1,400 connected vertices, 16 dipole certificates, 700 finite-prime cases, among others). It found no defect in source 12 and two defects in source 13’s reference API, with a tested patch (note in Section 79); the shipped program is the unpatched original. It also stresses that the exact finite-support witnesses

are unbounded arrays, not fixed tuples of integer unknowns, and that neither report lowers the 87-operation bound.

Table 10: [write] The two sources of Part V, notion by notion.

Notion	Source 12 (Part V.A)	Source 13 (Part V.B)	Tempting false reading
diagonal parameter	$\alpha \geq 3$, operator $L_\alpha = \alpha I - A$	$m \geq 3$ on paths ($A_m^{\text{path}} = mI - J_P$), $m \geq 5$ on the connected graph, odd m for localization	in (225)–(227) source 13 writes $\alpha = \rho^{-1}$ (source 12's λ), not the diagonal
adjacency and operator	A adjacency, L_α the operator, S the forward step	J_P , J adjacency, A_m^{path} , A_m , A the operator, S the rail swap	A is the adjacency in V.A but the operator in V.B; J is not the averaging matrix J_n , J_s of Parts I and IV
continuants	$D_{-1} = 0$, $D_0 = 1$, $D_{j+1} = \alpha D_j - D_{j-1}$	$D_0 = 1$, $D_1 = m$, $D_{j+1} = mD_j - D_{j-1}$; Lucas $U_0 = 0$, $U_1 = 1$, $D_j = U_{j+1}$	none: for $\alpha = m$ the two sequences coincide
run length	$n = T + 1$ configurations	$L = T + 1$	V.B's L is a length, V.A's L_α an operator
root value	$g_\alpha(s)$; $g_n = D_{n-1}/D_n$ on a halting run; ρ_α otherwise	$g_L = D_{L-1}/D_L$; resistance $\mathcal{R}_x = 2g_L$ or $m - \sqrt{m^2 - 4} = 2\rho$	V.B's resistance is two-terminal, twice V.A's one-terminal value
source scalar	c (called q in the code)	charge q ; Q_c in the horizon quadratic	Part IV's q is a transient eigenvalue
branches and history	$m = \Sigma $ branches with tags d_σ ; base $B = \max(2, m + 1)$; histories start at $h = 0$	R branches with tags r ; base $B = R + 1$; histories start at $h = 1$	V.A's m counts branches, V.B's m is the diagonal
counters and tests	two counters a, b , numbered 1 and 2 (0 and 1 in the code); TEST (i, p_+, p_0) lists the positive branch first	d counters numbered from 0; TEST (j, q_0, q_+) lists the zero branch first	the order of the two destinations is reversed
halting	set Halt of halting states; terminal functional $H(u)$	one halting state H	
finite label and prime sets	E supplied labels, W row set	W vertex set of the connected graph; S finite prime set	
the source	vertex s with empty history	s_x ; also s_r (source state of branch r) and $s_{t,r}$ (slack)	

[write] Source 12's abstract, keywords and reading guide follow.

Abstract (source 12). We construct a fixed, finitely described, sparse self-adjoint operator whose inverse is uniformly bounded and efficiently approximable, but whose exact arithmetic behavior represents universal computation. For every integer $\alpha \geq 3$, the operator has the form $L_\alpha = \alpha I - A$, where A is the adjacency operator of a computable disjoint union of finite paths and one-sided infinite rays. Its condition number is at most $(\alpha + 2)/(\alpha - 2)$; after normalization it is within $2/\alpha$ of the identity. Nevertheless, for a computable family of roots s , the conditions that $L_\alpha^{-1}e_s$ has finite support and that its diagonal Green function is rational are both equivalent to halting.

An explicit branch-history lift of two-counter machines gives the graph by fixed guarded affine rules on $Q \times \mathbb{N}^3$. Halting has a unique normalized integer certificate $L_\alpha u = ce_s$, $H(u) = 1$, with u of finite support. If the run has n configurations, then $c = D_n$ and its coefficients are $D_{n-1}, \dots, D_0$, where $D_{-1} = 0$, $D_0 = 1$, and $D_{j+1} = \alpha D_j - D_{j-1}$. The Green function is D_{n-1}/D_n for a halting run and $(\alpha - \sqrt{\alpha^2 - 4})/2$ otherwise. We prove a complete Pell-type classification of the rational output spectrum, a polynomial-time algorithm for each specified rational level, and a polynomial-time approximation algorithm despite undecidable rationality.

The same construction yields a passive resistor-network interpretation and a uniformly killed random walk with bounded mean lifetime. We give a linear polynomial-functional presentation, exact quadratic finite-support slices, and a positive-characteristic obstruction. The accompanying standard-library implementation passes 110,522 exact checks. The contribution is a proved structural normal form and its consequences, not a new ordinary universal integer-polynomial bound, a finite physical halting oracle, or an established claim of literature priority.

Keywords. Diophantine representation; counter machines; coercive operators; finite support; effective resistance; computable real numbers; continued fractions; certificate uniqueness.

Reading guide. The representation theorem is developed in Sections 56–58. The exact observable and its computability boundary are in Sections 59–61. Sections 62 and 63 specify the algebraic language and the ordinary Diophantine interface. A fully numerical example appears in Section 64. Sections 65–68 cover other substrates, limitations, validation, and further research.

55 Question, contribution, and source boundary

55.1 The structural question

A broad Diophantine representation of the halting relation is already provided by the Davis–Putnam–Robinson–Matiyasevich theorem [33]. A new programming-language name, followed only by an appeal to that theorem, does not normally constitute a new result about representation. We instead ask:

Can universal computation have a canonical *linear* finite-support certificate even when its associated infinite-dimensional operator is self-adjoint, strictly positive, uniformly invertible, and arbitrarily well-conditioned? Can an exact arithmetic observable remain undecidable when that observable has a uniform fast approximation algorithm?

The answer to both questions is yes. The construction below separates three issues which are easy to conflate: existence of an analytic solution, finite support of an exact algebraic solution, and decidability of the exact arithmetic type of a computable real number.

The underlying combinatorics is unusually simple: every component is a path. The difficulty is not hidden in enormous vertex degrees, a closing spectral gap, a noncomputable matrix entry, or a variable operator chosen after observing a computation. It lies in whether an effectively presented

path ever ends. A finite table of guarded affine rules generates all rows of the matrix, and a source configuration chooses the component.

55.2 Repository audit and relation to previous work

The public repository was inspected at commit

44983ed7ebfd545de55bfdb50e040c82f3d24295.

The Hilbert-tenth-problem directory contains corrected historical articles, formal material, and a large, separately qualified research corpus [5]. Its README reports established arithmetic counts of 75 operations for a complete certificate and 87 for a universal polynomial. It also distinguishes finite-horizon substrate compilers from fixed-arity, unbounded-time representations. Those distinctions are retained here. Neither number is improved by this article.

[write] Both figures are still those of the project README at this write; the source’s pin is kept as provenance.

An earlier project report, *Linear Quasi-Diophantine Universality* [38], gives a clocked boundary-transport equation with unique binary polynomial certificates over every nonzero commutative coefficient ring. Its Hilbert-space consequence is a dense, nonclosed range and an exact finite-support integrality-gap phenomenon. The present construction does *not* repackage that range-membership result: its operator is onto, its range is closed, its inverse is bounded, and it may be arbitrarily close to the identity after scaling. Its main exact decision problems concern finite support and rationality. Its coefficients are continuants rather than binary digits, and its characteristic-independent validity fails.

[write] *Linear Quasi-Diophantine Universality* is batch-79 manuscript 01 (archive `Linear_Boundary_Transport_Research.zip`, arrival 060e08a07). Batch 79 placed it in the collection’s report *polynomial-witness-histories* [40] (commit 224ca41df), where it is to be printed as a Part with label prefix `pwh:bt::`; that report had not yet been written at this write. Source 13 cites the same manuscript (Section 70).

The classical ingredients are also explicit. Suitable two-counter instruction sets are universal; importantly, changing the available jump instructions can change decidability [9]. Recording enough history to make a computation injective is a classical reversible-computing idea [34]. The connection between resistor networks and random walks is classical [35]. We prove the particular lift, certificates, operator bounds, observable formulas, and computational consequences used here, rather than importing unproved properties of a different machine model.

55.3 Status of the claims

The theorems in this manuscript have mathematical proofs. The executable checks provide additional finite validation, including independent implementations of key identities; they are not a replacement for those proofs. No Lean or Coq formalization of the new construction is supplied. The bibliography and repository comparison establish relevant antecedents, not an exhaustive priority search. In particular, no claim is made that undecidable rationality of computable reals, reversible simulation, or path continued fractions are individually new. The proposed contribution is the combined, explicit, coercive normal form with a canonical integer witness and the precise interfaces proved below.

56 A finite affine history lift

56.1 Machine semantics

Fix a finite state set Q . A configuration of the base machine is $(r, a, b) \in Q \times \mathbb{N}^2$. Every state has exactly one instruction of one of the following forms:

$$\begin{aligned} \text{INC}(1, p) : \quad & (r, a, b) \mapsto (p, a + 1, b), \\ \text{INC}(2, p) : \quad & (r, a, b) \mapsto (p, a, b + 1), \\ \text{TEST}(1, p_+, p_0) : \quad & (r, a, b) \mapsto \begin{cases} (p_+, a - 1, b), & a > 0, \\ (p_0, 0, b), & a = 0, \end{cases} \\ \text{TEST}(2, p_+, p_0) : \quad & (r, a, b) \mapsto \begin{cases} (p_+, a, b - 1), & b > 0, \\ (p_0, a, 0), & b = 0, \end{cases} \\ \text{HALT} : \quad & \text{no successor.} \end{aligned}$$

Destinations are arbitrary states; no fall-through restriction is imposed. The two branches of a test are regarded as distinct branches even when they have the same destination. This is an ordinary deterministic counter model containing the universal instruction patterns discussed in [9]. We do not assert universality for the more restricted reversible two-counter model considered there.

Let Σ be the finite set of all nonhalting branches, and give its members distinct tags $d_\sigma \in \{1, \dots, m\}$, where $m = |\Sigma|$. Put $B = \max(2, m + 1)$. The lifted vertex set is

$$V = Q \times \mathbb{N}^3, \quad v = (r, a, b, h). \quad (165)$$

The extra natural number h records a branch history. If branch σ is enabled at (r, a, b) , define

$$F(r, a, b, h) = (p, a + \delta_1, b + \delta_2, Bh + d_\sigma), \quad (166)$$

where (δ_1, δ_2) is the corresponding counter update. Leave F undefined at halting states. Each individual branch is a fixed affine map restricted by one elementary counter guard.

Lemma 56.1 (Effective injectivity). *The partial map F is injective. Its partial inverse is computable from a vertex by one division with remainder by the fixed base B , one table lookup, and a counter-guard check. Moreover, $h(F(v)) > h(v)$ whenever $F(v)$ is defined.*

Proof. The last base- B digit of the target history is exactly d_σ and therefore determines the branch. The quotient determines the preceding history. For an increment branch, subtract one from the affected target counter and require that counter to be positive. For a decrement branch, add one. For a zero branch, require the affected target counter to be zero and leave it unchanged. Check also that the target state is the branch's destination. These conditions reconstruct at most one predecessor, and a forward application verifies it. Finally, $Bh + d_\sigma > h$ because $B \geq 2$ and $d_\sigma \geq 1$. $\square$

This inverse is defined on *all* natural-number labels, not just on histories known in advance to arise from an input. Labels with inadmissible last digits, mismatched target states, or failing inverse guards simply have no predecessor. Such labels do not create spurious branching.

Theorem 56.2 (Path normal form). *Join v to $F(v)$ whenever the latter is defined. The resulting undirected graph G is a disjoint union of finite paths and one-sided infinite rays. Its degree is at*

most two. Every vertex with $h = 0$ is the initial endpoint of its component. The component of $s = (r, a, b, 0)$ is finite precisely when the base-machine run from (r, a, b) halts. If it halts after T transitions, that component has $n = T + 1$ vertices.

Proof. Determinism gives at most one successor and Lemma 56.1 gives at most one predecessor. A directed cycle would strictly increase h and return to its starting value, which is impossible. Nor can there be an infinite chain of predecessors, since h would be a strictly decreasing sequence of natural numbers. Thus every component has a root and is a finite path or a ray; no bi-infinite component occurs. A history-zero vertex cannot be a target of (166). Its forward path projects, step for step, to the specified base-machine run. The only vertices without a successor are those in halting states. This proves both the equivalence and the vertex count. $\square$

56.2 Fixing the universal graph

Choose once and for all a universal counter program in the stated instruction set. More explicitly, simulate a fixed universal Turing machine by a fixed finite-register machine and use the standard effective two-counter simulation. A computable encoding $x \mapsto s(x)$ maps each program/input pair to an empty-history source of this one counter program. This use of universality is external; the subsequent graph and algebra constructions are explicit for *every* finite instruction table. A numerical smallest universal table is neither required by the theorem nor included in the implementation. Complexity bounds below are measured in the binary size of the *counter configuration supplied*, not in a hypothetically efficient implementation of every possible source-language encoding.

For the fixed universal program, G is a fixed graph. Computing its neighbors uses only the finite table and the vertex label, with polynomial bit complexity. In particular, the graph does not contain matrix entries whose definitions require deciding whether an arbitrary computation halts. Only the unbounded question of which paths have terminal endpoints is hard.

57 A uniformly coercive operator

On the Hilbert space $\ell^2(V)$, with standard basis $(e_v)_{v \in V}$, set

$$S e_v = \begin{cases} e_{F(v)}, & F(v) \text{ exists,} \\ 0, & \text{otherwise,} \end{cases} \quad A = S + S^*, \quad L_\alpha = \alpha I - A, \quad \alpha \in \mathbb{Z}, \alpha \geq 3. \quad (167)$$

Injectivity makes S a partial isometry: distinct nonzero images of basis vectors are distinct basis vectors. Thus $\|S\| \leq 1$, and A is exactly the adjacency operator of the graph in Theorem 56.2.

Theorem 57.1 (Uniform invertibility). *The operator L_α is bounded and self-adjoint, with*

$$(\alpha - 2)I \leq L_\alpha \leq (\alpha + 2)I. \quad (168)$$

It is bijective on $\ell^2(V)$, and

$$\|L_\alpha^{-1}\| \leq \frac{1}{\alpha - 2}, \quad \kappa(L_\alpha) \leq \frac{\alpha + 2}{\alpha - 2}, \quad \|\alpha^{-1}L_\alpha - I\| \leq \frac{2}{\alpha}. \quad (169)$$

All rows contain at most three nonzero entries, drawn from $\{\alpha, -1\}$. The inverse is given by the norm-convergent series

$$L_\alpha^{-1} = \frac{1}{\alpha} \sum_{k=0}^{\infty} (A/\alpha)^k, \quad (170)$$

whose tail after $k = K$ has norm at most

$$\frac{(2/\alpha)^{K+1}}{\alpha - 2}. \quad (171)$$

Proof. The bound $\|A\| \leq \|S\| + \|S^*\| \leq 2$ gives (168). Since $\|A/\alpha\| < 1$, the geometric operator series (170) converges and is a two-sided inverse. Summing its norm bounds gives the inverse bound and (171); the remaining claims follow from $L_\alpha = \alpha I - A$ and the degree bound. $\square$

Proposition 57.2 (Effective inversion on sparse rational data). *For fixed program and fixed α , a finitely supported rational vector z and a precision b determine, in polynomial bit time, a finitely supported rational vector approximating $L_\alpha^{-1}z$ within 2^{-b} in ℓ^2 norm. The complexity is polynomial in b and the full sparse binary description of z .*

Proof. Use the partial sum in (170). Choose K so that the tail bound (171) multiplied by the computable rational upper bound $\|z\|_1$ for $\|z\|_2$ is at most 2^{-b} . For fixed α , one can choose $K = O(b + \text{size}(z) + 1)$. The radius- K neighborhood of a set of r vertices in a path forest contains at most $r(2K + 1)$ vertices. All successive partial products can therefore be stored sparsely on a polynomial-size set. The guarded affine updates increase label bit sizes by at most $O(K)$, and the rational coefficients have polynomial bit size: common input denominators, powers of the fixed α , and walk counts bounded by 2^K suffice. Exact sparse iteration computes the claimed partial sum in polynomial time. If $z = 0$, return zero directly. $\square$

For any prescribed $K_0 > 1$, an integer $\alpha \geq 2(K_0 + 1)/(K_0 - 1)$ guarantees $\kappa(L_\alpha) \leq K_0$. The eventual undecidability result therefore persists as close to condition number one as desired. This statement concerns a fixed choice of α for each target condition bound; it does not take a limit in which an algorithm is permitted to discard the nonzero off-diagonal entries.

Lemma 57.3 (Support of the inverse column). *Every entry of L_α^{-1} is nonnegative. Its (v, w) entry is strictly positive if and only if v, w belong to the same component. Consequently, for an empty-history source s ,*

$$\text{supp}(L_\alpha^{-1}e_s) \text{ is finite} \iff \text{the source computation halts.} \quad (172)$$

Proof. The (v, w) entry of A^k counts length- k walks from w to v and is nonnegative. It is positive for some k exactly when the vertices are connected. Apply (170) and Theorem 56.2. $\square$

The analytic equation $L_\alpha u = e_s$ has a unique ℓ^2 solution for *every* source. Thus (172) is not undecidable analytic solvability, and it is not a range-closure pathology.

58 Canonical normalized integer certificates

Let $R^{(V)}$ denote finite-support arrays with coefficients in a ring R . The terminal functional

$$H(u) = \sum_{v: \text{state}(v) \in \text{Halt}} u_v \quad (173)$$

is well-defined on this algebraic direct sum. It is not asserted to be a bounded functional on the full Hilbert space. The certificate uses only finite support.

Define integers $D_j = D_j(\alpha)$ by

$$D_{-1} = 0, \quad D_0 = 1, \quad D_{j+1} = \alpha D_j - D_{j-1} \quad (j \geq 0). \quad (174)$$

Let $L_{\alpha,n}$ be the $n \times n$ tridiagonal matrix with α on the diagonal and -1 immediately above and below it.

Lemma 58.1 (Continuants and endpoint normalization). *For $n \geq 1$, $\det L_{\alpha,n} = D_n > 0$. Furthermore,*

$$L_{\alpha,n} \begin{pmatrix} D_{n-1} \\ D_{n-2} \\ \vdots \\ D_0 \end{pmatrix} = D_n e_0. \quad (175)$$

For every $n \geq 1$,

$$D_n > (\alpha - 1)D_{n-1} > 0. \quad (176)$$

Proof. Expansion along an endpoint gives the determinant recurrence $\det L_{\alpha,n} = \alpha \det L_{\alpha,n-1} - \det L_{\alpha,n-2}$, with initial values 1 for the empty matrix and α for $n = 1$. For (175), the first row is the defining recurrence for D_n , each interior row vanishes by the same recurrence, and the last row vanishes because $D_1 = \alpha D_0$. When $n = 1$, the single row is simply $\alpha D_0 = D_1$. Finally, $D_1 = \alpha > \alpha - 1$; induction using $D_{n-2} < D_{n-1}$ proves (176). $\square$

Theorem 58.2 (Canonical certificate). *For an empty-history source s , consider*

$$L_\alpha u = c e_s, \quad H(u) = 1, \quad c \in \mathbb{Z}, \quad u \in \mathbb{Z}^{(V)}. \quad (177)$$

This system has a solution if and only if the computation from s halts. When it has a solution, the entire pair (c, u) is unique. If the run is $v_0 = s, v_1, \dots, v_{n-1}$, ending in a halting configuration, then

$$c = D_n, \quad u_{v_j} = D_{n-1-j} \quad (0 \leq j < n), \quad u_v = 0 \text{ off the run}. \quad (178)$$

In particular all occupied coefficients and c are positive integers, and the terminal coefficient is exactly one.

Proof. Every finite-support solution is in $\ell^2(V)$, so invertibility gives $u = c L_\alpha^{-1} e_s$. If $c = 0$, then $u = 0$, contradicting $H(u) = 1$. By Lemma 57.3, a nonzero inverse column has finite support only when the source component is a finite path. Hence any solution implies halting.

Suppose the component has n vertices. Equation (175) gives the finite-support solution (178); there is exactly one halting vertex in that component, and its coefficient is $D_0 = 1$, so it satisfies the normalization. Finally, any other solution is a scalar multiple of the same inverse column. Its terminal coordinate fixes that scalar, proving uniqueness. This argument also covers an already halted source: $n = 1$, $c = \alpha$, and $u = e_s$. $\square$

There is no positivity condition hidden in the quantifiers of (177). Positivity is a consequence of its equations. Nor is the scalar c a freely chosen scaling slack: the terminal normalization fixes it to the path determinant.

Proposition 58.3 (Torsion-free coefficient extension). *Let R be a nonzero commutative unital ring whose additive group is $\mathbb{Z}$ -torsion-free. Replacing $\mathbb{Z}$ by R in (177) preserves existence and uniqueness, with the integer solution (178) interpreted through the canonical map $\mathbb{Z} \rightarrow R$.*

Proof. On a finite path not containing the source, the adjugate identity implies $D_n u = 0$ for a homogeneous solution, hence $u = 0$ by torsion-freeness. On a ray, a finite-support homogeneous solution is zero: at the vertex just beyond its last possibly occupied position, the equation is the negative of that last coefficient, and downward induction finishes. These arguments eliminate all components not containing the source.

If the source component is a ray, the same exterior-row induction first eliminates every positive-index coefficient, then the coefficient at the root, and finally c . The normalization is impossible. If it is a finite path, its sole terminal coefficient is one. Reading the terminal row and then successive rows backwards yields $1, \alpha, D_2, \dots, D_{n-1}$. The root row then gives $c = D_n$. This is both existence and uniqueness. $\square$

In particular the certificate is unchanged over $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ when finite support is retained. The torsion-free hypothesis is stronger than characteristic zero. Section 66 explains why this qualification matters.

59 Exact Green functions and universal rationality

For a source s , define

$$g_\alpha(s) = \langle e_s, L_\alpha^{-1} e_s \rangle, \quad \rho_\alpha = \frac{\alpha - \sqrt{\alpha^2 - 4}}{2}. \quad (179)$$

Then $0 < \rho_\alpha < 1/2$ and $\rho_\alpha^2 - \alpha\rho_\alpha + 1 = 0$.

Theorem 59.1 (Rational-quadratic dichotomy). *For every empty-history source,*

$$g_\alpha(s) = \begin{cases} D_{n-1}/D_n, & \text{if the source run has } n \text{ configurations,} \\ \rho_\alpha, & \text{if the source run never halts.} \end{cases} \quad (180)$$

The first value is rational and strictly smaller than ρ_α ; the second is a quadratic irrational. Thus all source values lie in the fixed quadratic field $\mathbb{Q}(\sqrt{\alpha^2 - 4})$.

Proof. On a finite path, divide (175) by D_n and read the first coordinate. On a ray with coordinates indexed by $j \geq 0$, put $w_j = \rho_\alpha^{j+1}$. This sequence is in ℓ^2 . Its root equation is $\alpha w_0 - w_1 = \alpha\rho_\alpha - \rho_\alpha^2 = 1$, and its remaining rows vanish by the same quadratic identity. Uniqueness of the inverse therefore gives $g_\alpha(s) = w_0 = \rho_\alpha$.

For irrationality, $(\alpha - 1)^2 < \alpha^2 - 4 < \alpha^2$ for integer $\alpha \geq 3$, so the discriminant is not a square. The strict inequality for the finite value follows from Lemma 59.2 below. $\square$

Lemma 59.2 (Exact convergence gap). *Writing $\rho = \rho_\alpha$ and $\lambda = \rho^{-1}$ gives*

$$D_n = \frac{\lambda^{n+1} - \rho^{n+1}}{\lambda - \rho}, \quad (181)$$

$$g_n := \frac{D_{n-1}}{D_n} = \rho \frac{1 - \rho^{2n}}{1 - \rho^{2n+2}}, \quad (182)$$

$$\rho - g_n = \frac{\rho^{2n+1}(1 - \rho^2)}{1 - \rho^{2n+2}} \leq \rho^{2n+1}. \quad (183)$$

The sequence g_n is strictly increasing and converges to ρ .

Proof. The expression in (181) has the initial values and recurrence in (174), since λ, ρ are the two roots of $t^2 - \alpha t + 1$. Division gives (182), and subtraction gives (183). The last denominator is at least $1 - \rho^2$. Alternatively, monotonicity follows from $g_1 = 1/\alpha$ and $g_{n+1} = 1/(\alpha - g_n)$: this increasing map carries every point of $[0, \rho)$ to a strictly larger point below ρ . The displayed gap then gives convergence. $\square$

Corollary 59.3 (Fixed-operator completeness). *For the fixed graph obtained from a universal machine, each of the following sets of encoded empty-history sources is computably enumerable complete under computable many-one reductions:*

$$\{s : g_\alpha(s) \in \mathbb{Q}\}, \quad \{s : g_\alpha(s) < \rho_\alpha\}, \quad \{s : \text{supp}(L_\alpha^{-1}e_s) \text{ is finite}\}.$$

The set $\{s : g_\alpha(s) = \rho_\alpha\}$ is co-computably-enumerable complete. Deciding whether $g_\alpha(s)$ has algebraic degree one or two is therefore impossible in general.

Proof. Each of the first three conditions is exactly halting by Theorems 58.2 and 59.1. Simulating the fixed counter program semidecides them. The fixed universal input encoding reduces the usual halting set to each one. The equality condition is their complement, and the degree statement follows from (180). $\square$

For $\alpha = 3$, the recurrence is especially concrete:

$$D_n = F_{2n+2}, \quad g_n = \frac{F_{2n}}{F_{2n+2}}, \quad \rho_3 = \frac{3 - \sqrt{5}}{2}. \quad (184)$$

Here $F_0 = 0, F_1 = 1$ are the Fibonacci numbers. The subsequence F_{2n+2} has the initial values 1, 3 and satisfies $D_{n+1} = 3D_n - D_{n-1}$, which proves the identity. The first values are $1/3, 3/8, 8/21, 21/55, 55/144, \dots$

60 The rational output spectrum is a Pell curve

The rational values in the preceding theorem are not arbitrary. Their numerators and denominators obey a single quadratic equation, and its positive branch admits an effective descent.

Theorem 60.1 (Complete spectrum classification). *For positive integers $p < q$, the following are equivalent:*

$$q^2 + p^2 - \alpha pq = 1; \quad (q, p) = (D_n, D_{n-1}) \text{ for a unique } n \geq 1. \quad (185)$$

All such pairs are coprime. The index n is recovered by repeatedly replacing

$$(q, p) \mapsto (p, \alpha p - q) \quad (186)$$

until reaching $(1, 0)$. Exactly n replacements occur.

Proof. The transformation $(q, p) \mapsto (\alpha q - p, q)$ preserves $q^2 + p^2 - \alpha pq$. Starting with $(1, 0)$ therefore proves the forward identity for all consecutive continuants. Any common divisor of q, p has its square dividing one, so the pair is coprime.

Conversely, suppose $q > p > 0$ satisfies the equation. For $r = \alpha p - q$, rearrangement gives

$$qr = p^2 - 1. \quad (187)$$

Thus $r \geq 0$ and $r < p$, because $q > p$. The pair (p, r) satisfies the same equation. Descent strictly decreases its positive second coordinate, so it reaches a pair $(q', 0)$; the equation then forces $q' = 1$. Reversing the uniquely specified steps recovers exactly the continuant sequence, proving existence and uniqueness of n . $\square$

This is an ordinary Diophantine description of the *possible output fractions*; it does not by itself bind a fraction to a particular source computation. Dropping that binding would replace a universal-computation question by a decidable recurrence membership problem.

Theorem 60.2 (Decidability of every specified rational level). *For fixed program and fixed $\alpha \geq 3$, the predicate*

$$g_\alpha(s) = r, \quad s \text{ an empty-history source}, \quad r \in \mathbb{Q}, \quad (188)$$

is uniformly decidable. With binary integer input, it is decidable in polynomial time in the bit sizes of s and r .

Proof. Reduce $r = p/q$ to lowest terms with $q > 0$. Reject unless $0 < p < q$ and (185) holds. In the remaining case, run the descent in Theorem 60.1 to obtain its index n . Simulate the source for $n - 1$ transitions. Accept exactly when its first halt occurs at that time, including time zero when $n = 1$. Theorem 59.1 proves correctness: an earlier halt gives a strictly smaller rational level, a later halt gives a larger one, and an infinite run gives the irrational limit.

By (176), $D_n > 2^n$, so $n < \log_2 q$. Euclidean reduction, quadratic arithmetic, and the descent use integers of polynomial bit size. During $O(\log q)$ simulated steps, each counter changes by at most one per step and the history gains only a constant number of bits per step. With a fixed instruction table, this is a polynomial-time computation. $\square$

Consequently there is a uniformly decidable family of level sets whose union, the rationality set, is computably enumerable complete. There is no contradiction: deciding the countable union would require a computable bound on which denominator, or which index n , must be checked.

61 Fast approximation and unbounded exact separation

61.1 A rational interval algorithm

Put $f(t) = 1/(\alpha - t)$. On $[0, 1/2]$, this map is increasing, maps the interval into itself, and has derivative bounded by

$$\sup_{0 \leq t \leq 1/2} f'(t) = \frac{1}{(\alpha - 1/2)^2} \leq \frac{4}{25} < \frac{1}{4}. \quad (189)$$

Its unique fixed point in that interval is ρ_α .

Theorem 61.1 (Uniform polynomial-time approximation). *For fixed program and α , a rational interval containing $g_\alpha(s)$ and of width at most 2^{-b} can be computed in time polynomial in b and the binary size of the source s .*

Proof. Set $n = \max(1, \lceil b/2 \rceil)$ and simulate $n - 1$ transitions. If a halt is reached, compute its exact continuant fraction and return the degenerate interval consisting of that fraction. Otherwise return

$$I_n = [f^n(0), f^n(1/2)]. \quad (190)$$

The lower endpoint is g_n , while the upper endpoint is above ρ . If the run later halts, its number of configurations is greater than n , so its Green value lies between g_n and ρ ; a nonhalting run has value ρ . Therefore the returned interval always contains the answer. By (189),

$$|I_n| \leq \frac{1}{2}(4/25)^n < 2^{-(2n+1)} \leq 2^{-b}.$$

Each endpoint is obtained by n rational fractional-linear updates; its numerator and denominator have $O(n \log \alpha)$ bits. The bit-size and simulation estimates from Theorem 60.2 now give the stated polynomial bound. $\square$

This algorithm does not attempt to recognize a nonhalting run. It uses the same certified interval in every case in which no halt has yet been seen. The conclusion is stronger than mere computability of the real: arbitrarily fine approximation is compatible with the exact undecidability in Corollary 59.3. The separation rests on the absence of a uniform computable gap from the irrational threshold, not on slow numerical convergence.

61.2 Certificate height and the absence of a cutoff

For fixed α ,

$$\log_2 D_n = \Theta(n \log \alpha). \quad (191)$$

This follows either from (181) or from $(\alpha-1)D_{n-1} < D_n < \alpha D_{n-1}$. The denominator of g_n is exactly D_n , because Theorem 60.1 gives coprimality. The unique certificate has n occupied configurations and maximum coefficient D_{n-1} . Along an n -configuration run, history labels use $O(n \log B)$ bits and counter labels use the source bit length plus $O(\log n)$ bits. A literal sparse certificate therefore has size $O(n^2 + n \text{ size}(s))$ for fixed program and α . This is a size bound in terms of actual runtime, not a polynomial bound in source length.

Theorem 61.2 (No computable uniform exact cutoff). *For a fixed universal program, there is no total computable function of source bit length bounding, on all halting sources, any of the following: the halting time; the number of occupied coefficients in the canonical certificate; or its scalar c . There is also no total computable positive rational function $\gamma(m)$ such that*

$$\rho_\alpha - g_\alpha(s) \geq \gamma(m) \quad (192)$$

for every halting source of bit length at most m .

Proof. A halting-time bound would decide halting by simulation. The support size is the halting time plus one, so a support bound would give such a time bound. A bound on $c = D_n$ also bounds n , since $D_n > 2^n$.

Suppose a function γ satisfying (192) existed. Since $\rho < 1/2$, equation (183) gives $\rho - g_n < 2^{-(2n+1)}$. Compute an integer N for which $2^{-(2N+1)} < \gamma(m)$. No halting source of length at most m could have $n \geq N$. This again gives an effective time bound and decides halting, a contradiction. $\square$

[write] This is the bounded-search argument of `cdc:bd:prop:nobound` in [22]. Source 13 proves the same for support and charge and adds the largest prime factor of the charge (Corollary 75.6).

Equivalently, the maximum finite runtime among sources of bounded size has no computable upper bound, and the smallest positive separation from the threshold along those halting sources has no computable positive lower bound. These statements do not require an unqualified claim about which busy-beaver function eventually dominates every computable function under every possible encoding.

62 An explicit polynomial-functional presentation

62.1 The sorts and operations

Identify an array of finite support on V with a vector

$$U = (U_r)_{r \in Q} \in \mathbb{Z}[X, Y, Z]^Q, \quad e_{(r,a,b,h)} \longleftrightarrow e_r X^a Y^b Z^h. \quad (193)$$

The variables X, Y, Z respectively encode the two counters and the history. All polynomials have finite support; their degrees are not bounded in advance.

Define the counter operators

$$M_X P = X P, \quad E_X P = P(0, Y, Z), \quad D_X P = \frac{P - E_X P}{X}, \quad (194)$$

and define M_Y, E_Y, D_Y similarly. The quotient in this notation is an exact polynomial because its numerator is divisible by X . On coefficient arrays, $M_X^* = D_X$, $D_X^* = M_X$, and $E_X^* = E_X$. For each digit $0 \leq d < B$, define

$$J_d P = Z^d P(X, Y, Z^B), \quad (195)$$

$$C_d \left(\sum_{a,b,h} p_{abh} X^a Y^b Z^h \right) = \sum_{a,b,k} p_{ab, Bk+d} X^a Y^b Z^k. \quad (196)$$

Thus J_d appends a base- B digit to each history exponent, and C_d extracts the coefficients in that residue class while dividing the remaining history exponent by B . Directly on monomials, $J_d^* = C_d$. These are coefficient-linear maps. They are not multiplication by fixed polynomials in the ordinary polynomial-ring language.

Let $E_{p,r}$ denote the state matrix unit carrying the r component to the p component. For a branch σ from r to p , set

$$T_\sigma = \begin{cases} M_X \text{ or } M_Y, & \text{increment,} \\ D_X \text{ or } D_Y, & \text{positive decrement branch,} \\ E_X \text{ or } E_Y, & \text{zero branch.} \end{cases}$$

Then the two transport directions are explicitly

$$S = \sum_{\sigma \in \Sigma} E_{p,r} J_{d_\sigma} T_\sigma, \quad S^* = \sum_{\sigma \in \Sigma} E_{r,p} T_\sigma^* C_{d_\sigma}. \quad (197)$$

The counter and history operations commute because they act on different exponents. Formula (197) agrees with the graph definition on every monomial, including labels that are not reachable from any chosen input, and hence on every polynomial by linearity.

Theorem 62.1 (Linear quasi-Diophantine representation). *For a finite program and an empty-history source $s = (r_0, a_0, b_0, 0)$, its halting relation is represented by the following fixed set of linear polynomial-functional identities:*

$$\begin{aligned} (\alpha I - S - S^*)U &= c e_{r_0} X^{a_0} Y^{b_0}, \\ \sum_{r \in \text{Halt}} U_r(1, 1, 1) &= 1, \quad U \in \mathbb{Z}[X, Y, Z]^Q, \quad c \in \mathbb{Z}. \end{aligned} \quad (198)$$

Its witness is unique when it exists. The same assertion holds over every nonzero torsion-free commutative $\mathbb{Z}$ -algebra.

Proof. The identification (193) intertwines every monomial term of (197) with the corresponding graph transport. Evaluation at $(1, 1, 1)$ sums polynomial coefficients, so the second identity is exactly (173). Apply Theorem 58.2 and Proposition 58.3. $\square$

62.2 Making the extra operations explicit

The counter boundary maps can be described by uniquely typed polynomial splits. For example, $P = P_0 + XD$ with $P_0 \in \mathbb{Z}[Y, Z]$ uniquely determines $P_0 = E_X P$ and $D = D_X P$. Likewise every polynomial has a unique expansion

$$P(X, Y, Z) = \sum_{d=0}^{B-1} Z^d P_d(X, Y, Z^B), \quad P_d \in \mathbb{Z}[X, Y, Z], \quad (199)$$

and $P_d = C_d P$. Uniqueness follows by disjointness of the history-exponent residue classes. Thus the functional notation can be replaced by a finite list of linear identities with explicit sort restrictions and the fixed substitution $Z \mapsto Z^B$.

The substitution is part of the language; it has not been eliminated into ordinary integer polynomial arithmetic. Similarly, a source exponent such as a_0 is a monomial input parameter, not a counted constant-size integer loader. The representation is therefore *quasi-Diophantine* in a specified algebraic language. It is not an assertion that an ordinary linear system in a fixed finite tuple of integer unknowns has an undecidable existence problem.

63 Ordinary quadratic slices with complete boundary rows

Let $E \subset V$ be a finite set of supplied configuration labels. A candidate vector is described by coordinates $(u_v)_{v \in E}$ and is set to zero outside E . Define the finite row set

$$W = E \cup N(E) \cup \{s\}, \quad (200)$$

where $N(E)$ contains all graph neighbors of vertices in E . Write $u_w = 0$ when $w \notin E$. Define the ordinary integer polynomial

$$P_{s,E}(c, (u_v)_{v \in E}) = \sum_{w \in W} \left(\alpha u_w - \sum_{\substack{v \in E \\ v \sim w}} u_v - c \mathbf{1}_{w=s} \right)^2 + \left(\sum_{v \in E \cap \text{Halt}} u_v - 1 \right)^2. \quad (201)$$

Here $E \cap \text{Halt}$ denotes the vertices in E whose state is halting. The coefficients and index set of this polynomial are effectively computed from the *supplied* labels and program table.

Theorem 63.1 (Exact finite-slice compiler). *The polynomial $P_{s,E}$ has degree at most two in $|E| + 1$ integer variables. Its displayed sum has at most $3|E| + 2$ residual squares. It has an integer zero if and only if the source halts and its entire history-lifted run is contained in E . In that case the zero is unique: it is (178), with zero coordinates at every extra label.*

Proof. Each residual is affine linear. Since graph degree is at most two, $|W| \leq 3|E| + 1$. An integer sum of squares is zero precisely when every residual is zero. Rows outside W vanish automatically: they have no candidate coefficient, no candidate neighbor, and no source term. Therefore a zero of (201) is exactly a finite-support solution of (177). Theorem 58.2 shows that this solution exists precisely under the stated condition and fixes every coefficient uniquely. $\square$

63.1 Why exterior rows cannot be omitted

Suppose a nonhalting ray is truncated after its first n vertices. The finite tridiagonal system on those vertices alone accepts the continuant column from (175). But the next, omitted vertex has residual -1 , because it is adjacent to the last occupied coefficient $D_0 = 1$. That exterior equation is part of the infinite operator even though its own candidate coefficient is zero. Keeping only rows with labels in E would silently change the problem to an artificially terminated path.

The verifier therefore generates the whole set (200). It also receives the source as an independent argument; a certificate is not allowed to substitute an easier source for the one being verified. These two requirements are important for a source-bound certificate, regardless of how attractive a smaller residual count might appear.

63.2 The fixed-arity boundary

For each fixed E , the problem is just a finite system of integer linear equations, written as a quadratic sum of squares. It is decidable. The unbounded computation enters through existential choice of an arbitrary finite support with arbitrary natural-number labels. Thus the collection of all slices is not one fixed finite-variable quadratic polynomial. Encoding those labels and their unbounded list into a fixed integer tuple is a further arithmetic compilation problem.

MRDP supplies existence of an ordinary Diophantine representation for the resulting computably enumerable source relation [33]. It does not imply that this presentation directly yields a small, unique, or finite-fold ordinary integer witness. A unique polynomial or finite-support witness can have many integer codes or many auxiliary arithmetic certificates. No finite-fold Diophantine conjecture is resolved here.

64 A complete numerical certificate

Consider the two-state program

$$0 : \text{TEST}(1, 0, 1), \quad 1 : \text{HALT}. \quad (202)$$

The positive branch has tag one, the zero branch tag two, and $B = 3$. Start at $(0, 2, 0, 0)$ and take $\alpha = 3$. The complete lifted run is

$$\begin{aligned} v_0 &= (0, 2, 0, 0), & v_1 &= (0, 1, 0, 1), \\ v_2 &= (0, 0, 0, 4), & v_3 &= (1, 0, 0, 14). \end{aligned}$$

There are three transitions and four configurations. The continuants are $D_0 = 1, D_1 = 3, D_2 = 8, D_3 = 21, D_4 = 55$, so the unique certificate is

$$c = 55, \quad (u_{v_0}, u_{v_1}, u_{v_2}, u_{v_3}) = (21, 8, 3, 1), \quad g_3(v_0) = \frac{21}{55}. \quad (203)$$

For direct row verification,

$$\begin{aligned} 3 \cdot 21 - 8 &= 55, & 3 \cdot 8 - 21 - 3 &= 0, \\ 3 \cdot 3 - 8 - 1 &= 0, & 3 \cdot 1 - 3 &= 0. \end{aligned}$$

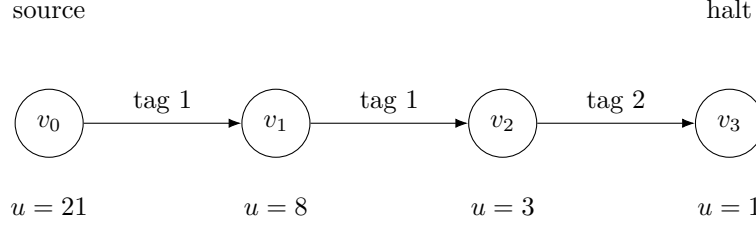


Figure 1: A source-bound history path. The operator uses undirected edges; the arrows display only the computational orientation.

The polynomial vectors are

$$U_0 = 21X^2 + 8XZ + 3Z^4, \quad U_1 = Z^{14}. \quad (204)$$

Formula (197) gives

$$\begin{aligned} (SU)_0 &= 21XZ + 8Z^4, & (SU)_1 &= 3Z^{14}, \\ (S^*U)_0 &= 8X^2 + 3XZ + Z^4, & (S^*U)_1 &= 0. \end{aligned}$$

Hence $(3I - S - S^*)U = (55X^2, 0)$ and $U_1(1, 1, 1) = 1$. This checks both directions of the polynomial transport, not just the forward computation.

For the supplied support $E = \{v_0, v_1, v_2, v_3\}$, the complete ordinary quadratic is

$$\begin{aligned} &(3u_0 - u_1 - c)^2 + (3u_1 - u_0 - u_2)^2 \\ &+ (3u_2 - u_1 - u_3)^2 + (3u_3 - u_2)^2 + (u_3 - 1)^2 = 0. \end{aligned} \quad (205)$$

Its sole integer zero is $(u_0, u_1, u_2, u_3, c) = (21, 8, 3, 1, 55)$. Finally, its output fraction satisfies $55^2 + 21^2 - 3 \cdot 55 \cdot 21 = 1$; the descent is

$$(55, 21) \rightarrow (21, 8) \rightarrow (8, 3) \rightarrow (3, 1) \rightarrow (1, 0),$$

recovering the exact number of configurations.

65 Passive electrical and stochastic substrates

65.1 A countable passive resistor network

Place conductance one on every graph edge. At every vertex v , add a shunt to a common grounded boundary with conductance

$$a_v = \alpha - \deg_G(v) \in \{\alpha, \alpha - 1, \alpha - 2\}. \quad (206)$$

All shunts are strictly positive. They can be realized as a_v parallel unit-conductance resistors, so no irrational component value is needed. The common ground is a Dirichlet boundary; its possible infinite degree does not alter the bounded degree of the non-ground configuration graph.

For a voltage vector w and injected currents j , Kirchhoff's law at v is

$$\sum_{z \sim v} (w_v - w_z) + a_v w_v = j_v, \quad (207)$$

which is precisely $L_\alpha w = j$. The energy identity is

$$\langle w, L_\alpha w \rangle = \sum_{\{v,z\} \in E(G)} |w_v - w_z|^2 + \sum_v a_v |w_v|^2. \quad (208)$$

It holds first for finite-support vectors and then for ℓ^2 vectors by continuity and nonnegative convergence. Since $a_v \geq \alpha - 2$, finite energy implies membership in ℓ^2 ; conversely, bounded degree makes every ℓ^2 vector finite-energy. There is therefore no unspecified boundary choice at infinity in the finite-energy solution.

Theorem 65.1 (Exact resistance represents halting). *In this fixed countable resistor network, inject unit current at an empty-history source s and remove it at ground. The unique finite-energy voltage vector is $L_\alpha^{-1}e_s$, and the source-to-ground effective resistance is $g_\alpha(s)$. Its rationality is equivalent to halting. This remains true with all individual resistors having unit resistance and with condition number as close to one as prescribed.*

Proof. Theorem 57.1 gives the unique solution of (207), and (208) identifies the appropriate finite-energy class. With unit source current, effective resistance is the source voltage, the diagonal Green entry. Apply Theorem 59.1. Parallel unit shunts realize (206) exactly; the condition-number statement is (169). $\square$

The resistance/Green-function interpretation is consistent with the classical framework of [35]; the particular arithmetic classification is established above. This is *static exact-observable universality*. A finite resistor circuit is not being claimed to execute an arbitrary unbounded computation, and a physical measurement is not being claimed to decide exact rationality. The network is countable and finitely specified, and the observable is defined exactly. The exponentially vanishing gap in (183) explains why robust finite-precision detection does not provide a halting oracle.

65.2 A uniformly killed random walk

At each non-ground vertex, move to each graph neighbor with probability $1/\alpha$, and otherwise move to a cemetery state and stop. The substochastic transition operator is

$$P = A/\alpha. \quad (209)$$

Its row sums are at most $2/\alpha < 1$. Let N be the number of visits to non-cemetery vertices before death, counting the initial visit, and let N_s count visits to the starting vertex s .

Theorem 65.2 (Undecidable exact arithmetic with bounded mean lifetime). *The walk dies almost surely. Uniformly in the source,*

$$\mathbb{P}(N > t) \leq (2/\alpha)^t \quad (t \in \mathbb{N}), \quad \mathbb{E}N \leq \frac{\alpha}{\alpha - 2}. \quad (210)$$

Nevertheless,

$$\mathbb{E}_s N_s = \alpha g_\alpha(s) \quad (211)$$

has undecidable rationality for the universal source family. Given any $\varepsilon > 0$, a fixed integer α can make the uniform mean-lifetime bound at most $1 + \varepsilon$ without changing that conclusion.

Proof. At each step, conditional survival probability is at most $2/\alpha$. Induction gives the tail bound, and summing tails gives the expectation bound and almost-sure death. The expected number of visits to the start is the sum of its return probabilities:

$$\mathbb{E}_s N_s = \sum_{k \geq 0} (P^k)_{s,s} = \alpha (L_\alpha^{-1})_{s,s}.$$

The series converges absolutely by the geometric bound. Multiplication by the nonzero integer α preserves rationality, so Corollary 59.3 applies. Finally, $\alpha/(\alpha - 2) = 1 + 2/(\alpha - 2)$ tends to one. $\square$

The undecidable predicate is not almost-sure termination, which is uniformly proved here, nor is it failure of an expected-time bound. It is an exact arithmetic property of a well-behaved expectation. The transient kernel is symmetric with respect to counting measure; no claim of a finite stationary probability distribution on the non-ground state space is needed.

[write] This is the report's spine (Section 1.2) for a substochastic chain: a uniform geometric lifetime bound and a computable expectation whose exact arithmetic type is undecidable. Part V of [21] (source 17 there) has the analogous statement for the stationary laws of reversible chains with a uniform spectral gap.

66 Why positive characteristic breaks the certificate

Analytic coercivity is an ordered-field fact. Its integer consequences extend to torsion-free coefficient rings, but not to every characteristic. The obstruction can be made uniform and completely explicit.

Add a disjoint dummy sector to the program: one state decrements counter one until zero and then enters a halting state. Its empty-history roots supply finite path components of every length $n \geq 2$, and an already halted root supplies length one. This augmentation preserves the universal sector. A second, unused counter supplies arbitrarily many disjoint copies of each dummy path length.

Lemma 66.1 (Singular continuants in every modulus). *For every integer modulus $m \geq 2$ there exists $n \geq 1$ with $D_n \equiv 0 \pmod{m}$.*

Proof. The matrix

$$M = \begin{pmatrix} \alpha & -1 \\ 1 & 0 \end{pmatrix}$$

has determinant one, so it has finite order r in $\text{GL}_2(\mathbb{Z}/m\mathbb{Z})$. Since $M^j(1,0)^\top = (D_j, D_{j-1})^\top$ and $M^{-1}(1,0)^\top = (0, -1)^\top$, the index $n = r - 1$ satisfies the assertion. The order is not one, since the lower-left entry of M is one modulo m , so $n \geq 1$. $\square$

Theorem 66.2 (A complete positive-characteristic obstruction). *For a program containing the dummy sector, and over every nonzero unital commutative ring of positive characteristic, system (177) can have a finite-support solution for a nonhalting source. It therefore fails as a uniform halting representation in positive characteristic.*

Proof. Let the characteristic be $m \geq 2$. Choose a dummy path of length n with $D_n = 0$ in the ring, using Lemma 66.1. On that path place the continuant vector $(D_{n-1}, \dots, D_0)$ and put zero elsewhere. Equation (175) now says $L_\alpha u = 0$, while the terminal coefficient still equals one, so $H(u) = 1$. Set $c = 0$. This pair satisfies (177) for any independently supplied nonhalting source, even one in the universal sector. $\square$

For $\alpha = 3$ modulo two, length two already suffices: $D_2 = 8 = 0$ and $(D_1, D_0) = (3, 1) = (1, 1)$. The length-two matrix becomes $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, whose product with $(1, 1)^\top$ is zero. The terminal coefficient is still one. The archive contains this source-bound false positive and verifies that it fails over the integers.

Adding $c \neq 0$ is not a general repair: the valid normalized certificate for a genuine halting path of singular length has $c = D_n = 0$ in that same ring. Such a restriction would also remove legitimate computations.

Characteristic zero alone does not ensure uniqueness. For example, $R = \mathbb{Z} \times \mathbb{F}_2$ has characteristic zero but has additive torsion. At $\alpha = 3$, place the preceding modulo-two homogeneous vector in each of two disjoint dummy length-two paths, using zero in the $\mathbb{Z}$ component. Their terminal sums add to zero in $\mathbb{F}_2$. Adding this nonzero vector to any valid certificate leaves both equations unchanged. This supplies an explicit reason for using torsion-freeness in Proposition 58.3, rather than merely characteristic zero.

67 Implementation, validation, and integration

67.1 Executable contract

The companion module `code/green_machine.py` is a standard-library Python implementation. It accepts immutable, validated finite instruction tables and exact natural-number configuration labels. Boolean values and floating-point values are not silently accepted as integer inputs. The constructor freezes nested input tables so later caller mutation does not change a compiled program.

Its main interfaces are `step` and `predecessor`; sparse `apply_1`; an independently implemented `polynomial_transport`; bounded `run`; canonical `certificate` and `check_certificate`; the ordinary quadratic-row compiler `slice_rows`; Pell descent and exact rational-level testing; and the rational interval algorithm `green_interval`. A failure to find a halt within a supplied horizon raises a horizon-exhaustion exception. It is never reported as a proof of nonhalting.

The certificate checker evaluates all operator rows induced by the sparse support and all exterior neighbors. It takes the source separately and checks the terminal normalization. It does not trust a stored execution trace, positivity claim, or a purported terminal designation. The finite instruction table determines the semantics.

67.2 What was actually checked

The reproducible suite `code/verify.py` completed with **110,522 exact checks in 43 categories**, using seed 20261002 and Python 3.13.5. Counts are assertion counts, not counts of independently proved theorems. Selected categories are summarized below; the JSON report contains the complete ledger.

Check category	Checks
Exhaustive small rational-spectrum membership	30,300
Successor/predecessor round trips (two directions)	18,792
Graph degree bound	9,396
Independent polynomial transport versus graph transport	3,600
Pell descent and continuant identity (combined)	5,600
Independent exact Gaussian-elimination inverse entries	546
Exhaustive small quadratic zero set	1,056
Source-bound halting certificates	825
Exact irrational enclosure checks on nonhalting increment rays	450
Positive-characteristic false positives	258

The tests cover 30 explicit program tables, 28 generated by the fixed seed; all increment, positive-test, and zero-test branches; history labels that need not be reachable; signed sparse coefficient arrays; and a range of α values and moduli. Independent rational Gaussian elimination is compared with the continuant inverse formula. The finite quadratic interface is checked with extra disconnected labels, exterior-row tests, source substitutions, and coefficient tampering. A small zero set is exhausted over all 1,056 assignments in its declared box. Invalid scalar and constructor inputs are also tested.

These checks do not enumerate all counter programs, prove the analytic statements, or validate a numerical universal-machine table. The general claims rest on the proofs in this article. The implementation gives a replayable artifact for the exact local construction and examples.

[write] In this report the two programs are `code/12-coercive-green-green_machine.py` and `code/12-coercive-green-verify.py`; the examples are `data/12-coercive-green-countdown.json` and `data/12-coercive-green-modular_false_positive.json`; `validation/results.json` is `data/12-coercive-green-results.json`; `SOURCE_AUDIT.md` is `12-coercive-green-SOURCE_AUDIT.md`. Not shipped: the source's PDF and README, and `validation/test_output.txt`, a byte copy of `results.json`. The delivered README notes that the code numbers the two registers 0 and 1 (1 and 2 in the article) and calls the scalar c `q.verify.py` imports `green_machine` by its delivered name and rewrites `examples/` and `validation/results.json` under its parent directory, so it must run on a copy (README of this report). At placement it passed on a copy with 110,522 checks and reproduced both examples after line-ending normalization; the research programme's review [42] replayed it too and found no defect in this package.

67.3 A proposed ProveIt interface

A natural integration target is a separate research package under the Hilbert-tenth-problem computational-substrate materials, not a replacement for the existing universal-polynomial compiler. The logical dependency chain should be kept explicit:

Layer	Obligation
Counter frontend	Exact finite instruction semantics and input binding.
History lift	Injectivity, computable inverse, and strict history growth.
Finite algebra	Path continuants, exterior-row completeness, normalization.
Hilbert analysis	Bounded adjacency, coercivity, inverse positivity.
Observable	Finite-path ratio, ray root, Pell descent, gap formula.
Arithmetic backend	Separate encoding of arbitrary sparse supports, with all auxiliary witnesses and input-loading costs accounted for.

The first three layers can be formalized without infinite-dimensional analysis. One can prove the canonical finite-support representation combinatorially, using the torsion-free argument. The fourth layer is needed for the electrical and Green-function interpretations, and the fifth connects exact values with computation. The final layer is future work; this package does not silently charge its cost to an external uncounted decoder.

68 Further research questions

The following questions concern extensions not proved by the present construction. They are ordered roughly from directly implementable formalization work to more speculative structural goals.

68.1 A certified compiler with a frozen universal frontend

Implement a proof-producing version of the finite branch-table compiler and attach an explicitly listed universal counter program. A useful milestone is a formal theorem that emitted polynomial-functional rows intertwine both graph transport directions for every monomial, including unreachable labels. The current code checks this independently on many examples, but a machine-checked all-input theorem and a frozen universal input-loader would remove the remaining implementation-level universality interface. Optimize measured state count only after this full contract is fixed.

68.2 Eliminating history substitution without losing coercivity

Can the operator be expressed using fewer polynomial coordinates or using only typed polynomial subrings, without $Z \mapsto Z^B$, while retaining a uniform positive lower bound and canonical finite-support witnesses? The history coordinate enforces injectivity and a root for each component. Any replacement must preserve both properties on all labels, not merely on valid executions. A directed clock is sufficient for other representations, but symmetrization can reveal predecessor collisions that were harmless in the directed setting.

68.3 Geometrically local realizations

The present graph is abstractly a path forest, but its effective vertex labels are not a fixed nearest-neighbor lattice embedding. Can the same exact rationality classification be realized by a fixed local rule in a bounded-dimensional grid, with a simple computable source embedding and uniform shunts? Merely drawing a path after simulating its length does not answer this question. A successful construction must make adjacency geometrically local before the halting behavior is known, and must analyze the effect of wire subdivision on the observable and its condition bound.

68.4 A connected network with the same exact classification

The grounded electrical network is connected through its boundary, but the non-ground configuration graph is a forest of separate components. Can one connect those components by a fixed, computable, bounded-degree non-ground structure while preserving a rational-versus-fixed-irrational observable? Generic weak coupling destroys the exact finite-component formula. This calls for exact decoupling identities or algebraically controlled tails, not merely a perturbative estimate.

[write] *Batch 79, 2 October 2026: answered by source 13.* Part V.B connects the components in the way asked for. Two copies of the path forest (rails), one coupling hub per configuration and a backbone ray give a fixed connected computable graph of maximum degree three, and the rail-swap involution together with a dipole source cancels every hub and backbone voltage exactly (Lemma 73.1). The finite-support, rationality and least-charge classification survive, with resistance $2D_{L-1}/D_L$ on halting inputs and $m - \sqrt{m^2 - 4}$ otherwise (Theorems 73.2 and 74.1). The answer uses a signed two-terminal source; Proposition 76.2 shows that a nonnegative finitely supported source cannot have a finitely supported response on any infinite connected graph of this kind.

68.5 Other algebraic thresholds and output degrees

Periodic Jacobi tails have algebraic response functions obtained from fractional-linear fixed-point equations. Which algebraic thresholds can replace ρ_α while retaining a finite alphabet of rational coefficients and a uniform spectral lower bound? Can a similarly explicit construction distinguish more than two prescribed algebraic degrees? The present theorem already gives degree one versus two, but it does not classify attainable higher-degree tails or canonical normalization in those cases.

68.6 Exact off-diagonal and multi-source observables

For vertices on one finite path, inverse entries also have continuant formulas. Determine which source-bound reachability and relative-time relations can be captured by rationality, vanishing, or specified algebraic identities among a fixed number of off-diagonal Green entries. A source pair in different components gives zero automatically; a useful universal relation must control that alternative explicitly. Multi-source versions may expose additional information while retaining positive definiteness.

68.7 Sharp decidable boundaries for effective path families

Find natural restrictions on the finite affine rule table under which rationality of the root response becomes decidable. Candidates include one-counter frontends, eventually periodic symbolic descriptions, or bounded-history variations. The theorem rules out a criterion based only on degree two, coefficient alphabet, self-adjointness, coercivity, or fast approximation. A useful positive criterion must constrain the effective presentation of endpoint reachability itself.

68.8 Characteristic-independent symmetrized certificates

Can one add a finite collection of typed constraints that prevents the modular dummy-component kernels while keeping a useful self-adjoint interpretation? The obstruction in Section 66 shows that the current terminal normalization cannot work over all rings, and simply requiring a nonzero source scalar loses genuine halting paths. A repair would need a different normalization, a changed

coefficient module, or a proof that the extra constraints exclude every unwanted homogeneous component without excluding valid computations.

68.9 Support coding and ordinary witness multiplicity

Translate the canonical sparse witness into a fixed-arity ordinary Diophantine certificate with an explicit operation ledger. Track the fibers of every encoding map: canonical polynomial support does not by itself give a single-fold integer representation. A realistic intermediate goal is a source-bound, unbounded-time ordinary compiler with a proved surjection and fully accounted input conversion. Any reduction in the repository's established arithmetic counts must be measured at that complete interface.

68.10 Robust promise problems and quantitative restrictions

Under what computable restrictions on a source family can one prove a positive separation from ρ_α and thereby turn approximation into an exact classifier? For example, a proved runtime bound gives an explicit gap by (183). Theorem 61.2 excludes a uniform solution for all universal inputs, but leaves meaningful subclasses and promise problems. The challenge is to derive such bounds from structural program properties rather than assume the answer in the promise.

68.11 Uniform killing with additional stochastic constraints

The present walk has geometric killing and a symmetric transient kernel. Can comparable exact-arithmetic universality coexist with a connected, positive-recurrent chain carrying an explicitly computable stationary probability measure and a uniform mixing bound? That question is not settled by the killed construction: a cemetery state changes the invariant measure and makes almost-sure termination trivial. Any extension must separately establish its recurrence and stationary-measure claims.

[write] Part V of [21] (source 17 there) supplies reversible chains with a uniform spectral gap and an undecidable exact stationary equality; whether they can carry this Part's finite-support certificates is addressed by neither source.

68.12 Realizing intermediate computability degrees

Replacing the universal recognizer by a fixed recognizer for another computably enumerable set transfers that set to the source-bound rationality predicate, subject to the chosen effective source encoding. Develop an exact classification of which degrees can be obtained under additional syntactic restrictions on the frontend and under uniform bounds on source-loading complexity. The unrestricted transfer is immediate from the construction; the interesting problem is the tradeoff with a restricted, explicitly specified local rule set.

69 Conclusion

A fixed coercive linear operator can retain the entire halting problem in the exact arithmetic structure of its inverse. There is no ill-posed analytic equation: the inverse exists everywhere, has a known norm bound, and is numerically approximable with a geometric modulus. Yet its source

column has finite support exactly on halting inputs, and its root Green value is rational exactly on those same inputs.

The canonical certificate is particularly concrete. A finite computation of length n produces the continuant column $(D_{n-1}, \dots, D_0)$ and source scalar D_n . The terminal value one fixes the whole witness. At $\alpha = 3$, these are even-index Fibonacci numbers, and the nonhalting resistance is $(3 - \sqrt{5})/2$. The quadratic Pell identity classifies the possible rational outputs, while exact source binding still requires the computation.

The result supplies three linked substrates: a linear polynomial-functional certificate system, a passive uniformly shunted network, and a uniformly killed random walk. Their shared lesson is a precise separation between numerical stability and exact arithmetic decidability. It is this separation, rather than a claim of physical hypercomputation or a new small ordinary universal polynomial, that the construction establishes.

Part V.B: the second route (source 13)

[write] Sections 70–81 print source 13 (batch-79 manuscript 14) in full; its two appendices are Appendices K and L. Its statements shared with source 12 carry bracketed pointers to Part V.A, and their proofs are kept as second routes. Its abstract and keywords follow.

Abstract (source 13). Can an exactly solvable, uniformly well-conditioned linear system still be a universal quasi-Diophantine substrate? We construct a fixed computably presented, connected graph of maximum degree three whose grounded conductance operator $A = 5I - J$ satisfies $2I \leq A \leq 8I$. It has integer coefficients in $\{0, -1, 5\}$ and at most four nonzero entries per row. A computable family of two-terminal sources b_x satisfies

$$U(x) \text{ halts} \iff \exists q \in \mathbb{N}_{>0} \exists z \in \mathbb{Z}^{(W)} : Az = qb_x.$$

Here the superscript denotes finite support, not a fixed witness dimension. Every analytic equation $Au = b_x$ nevertheless has a unique, uniformly computable ℓ^2 solution, with an explicit geometric approximation rate.

The arithmetic certificate has an exact denominator law. A run of T steps has $L = T + 1$ configurations; its least positive integral source charge is D_L , where $D_0 = 1$, $D_1 = 5$, and $D_{j+1} = 5D_j - D_{j-1}$. The primitive finite-support certificate is unique. Effective resistance is rational exactly on halting inputs and otherwise equals $5 - \sqrt{21}$. An elementary rank and valuation argument gives a contrasting decidability theorem: for every finite set of primes S , membership of the solution in $\mathbb{Z}[S^{-1}]^{(W)}$ is decidable, uniformly in S and the input.

We also prove a sharp connected row-sparsity threshold, explain why a positive one-terminal source cannot work, and give a single quadratic polynomial over natural numbers for each external horizon, with a unique complete witness. The package includes full proofs, an exact compiler, a 32-witness worked example, and 38,524 reproducible checks. These are proposed structural contributions, not a claim to solve a named classical open problem, establish historical priority, improve a fixed universal-polynomial operation bound, or settle finite-fold Diophantine representation.

Keywords. Diophantine representation; computable graphs; resistor networks; finite support; positive-definite operators; Lucas sequences; prime localization; counter machines.

70 The research question and the boundary of the contribution

70.1 From existence to structure

The MRDP theorem already implies that every computably enumerable acceptance relation has an ordinary integer-polynomial representation. ProveIt’s `Diophantine.mrdp` and `Diophantine.mrdp_iff` expose precisely such a fixed-polynomial, fixed-finite-witness statement, including input zero [15]. Applying that interface to one more universal language would not, by itself, reveal much about the language.

[write] These three theorems are at lines 33, 42 and 26 of `Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean`, unchanged from the source’s pin to this write.

This manuscript asks more specific questions. Does universality of a linear finite-support certificate require a singular or poorly conditioned analytic operator? Can a connected passive network carry the same arithmetic? How sparse can the operator be? And is its use of unrestricted rational denominators essential?

The answers obtained here are: a uniformly coercive operator suffices; a connected degree-three grounded network suffices; four nonzeros per row are optimal in the fixed connected scalar class specified below; and, for this construction, every finite-prime localization is decidable even though the rational finite-support problem is universal. The prime-localization theorem is not merely a restatement of MRDP: its proof identifies the exact computation-dependent denominator and gives a finite, effective stopping bound.

70.2 Repository and prior-manuscript audit

The repository reference point is the commit

44983ed7ebfd545de55bfdb50e040c82f3d24295.

The inspected Hilbert-tenth-problem README separates formalized results, research reports, complete arithmetic ledgers, and externally bounded substrate compilers [5]. Its stated 87-operation universal polynomial bound is not affected by this work. We have inspected source interfaces, not rebuilt the repository or independently verified all of its research claims.

Two supplied earlier manuscripts are particularly relevant. *Linear Quasi-Diophantine Universality* constructs unique linear polynomial certificates and a bounded operator with dense nonclosed range [38]. The present operator instead has a bounded inverse and full closed range. *Arithmetic of Rapidly Mixing Reversible Computation* separates effective approximation from exact stationary arithmetic in uniformly gapped Markov chains [39]. The present network gives a different invariant, exact finite support, together with an explicit localization theorem and a row-sparsity lower boundary. These earlier research manuscripts are antecedents, not independently established literature results imported as axioms. Every new mathematical assertion needed here is proved below, apart from standard computability and Hilbert-space background.

[write] *Linear Quasi-Diophantine Universality* is batch-79 manuscript 01, placed in the report *polynomial-witness-histories* [40] (note in Section 55). *Arithmetic of Rapidly Mixing Reversible Computation* is source 17 of [21], printed in its Part V. Source 12 of this report, written from the same snapshot on the same day, proves the disconnected form of this manuscript’s main theorem; neither of sources 12 and 13 cites the other.

The history mechanism belongs to the classical reversible-computation tradition [34]. Exponential inverse decay is also a classical phenomenon [36]; our relevant bounds follow directly from a Neumann series. We do not claim novelty for either ingredient. The contribution under examination is the combined compiler and its exact algebraic, connectedness, sparsity, and denominator consequences.

70.3 Conventions and effectivity

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. For a countable set W and a ring R ,

$$R^{(W)} = \{u : W \rightarrow R : \text{supp}(u) \text{ is finite}\}$$

is the algebraic direct sum. On a discrete locally finite graph, finite support is the intended meaning of compact support. An element is represented by a finite list of vertex-coefficient pairs.

A *computable presentation* includes a coding of vertices by natural numbers and an algorithm returning the complete finite neighbor list of each vertex. This is stronger than merely deciding whether a given pair forms an edge. All our coefficients are exact integers. Computable real or Hilbert space vectors are understood through effective rational approximation with an explicit error bound.

A set is Σ_1^0 -complete when it is computably enumerable and every computably enumerable set reduces to it by a computable many-one map. Fix a universal halting set K . Universality statements below concern a single fixed operator and a computable input-to-source map. The operator is not rechosen after inspecting whether the input halts.

71 A finite arithmetic compiler for reversible rooted paths

71.1 The counter-machine interface

A program has a finite state set $Q = \{0, \dots, Q_0 - 1\}$, $d \geq 1$ counters in $\mathbb{N}$, a start state q_* , and exactly one halting state H . Every other state has one instruction of either form

$$\text{INC}(j, q') \quad \text{or} \quad \text{TEST}(j, q_0, q_+).$$

The first increments counter j and changes state to q' . The second goes to q_0 without changing counters when counter j is zero, and otherwise decrements it and goes to q_+ . The halting state has no successor. We count transitions, so an initially halting configuration has halting time zero.

This finite-counter interface supports a fixed universal program when a sufficient fixed number of counters is permitted. One standard construction stores the two halves of a Turing tape as base- β stack integers, with nonzero digits and zero for the empty stack. Fixed-base multiplication, addition of a digit, quotient, and remainder are implemented by decrement loops, finite-control remainders, and a finite supply of scratch counters. The same loops convert an ordinary input counter to a binary input stack. A fixed interpreter then gives a fixed universal program. No two-counter ordinary-input minimality claim is needed. The repository contains a verified finite-Turing-machine interface and counter simulation pipeline relevant to this step [37].

For the mathematical construction we can take any program P . For the completeness results choose one fixed U with

$$x \in K \iff U \text{ halts from } (q_*, x, 0, \dots, 0).$$

The reference implementation compiles arbitrary programs in the stated interface; its small demonstration programs are not advertised as universal.

71.2 Branch tags and a one-coordinate history

Split each test into its zero and positive branches. Number all branches $r = 1, \dots, R$. A branch specifies its source s_r , target t_r , update $\Delta_r \in \{-1, 0, 1\}^d$, and either no guard, a zero guard, or a positive guard. Set $B = R + 1$; in the degenerate $R = 0$ case use $B = 2$.

Let

$$V = Q \times \mathbb{N}^d \times \mathbb{N}_{>0}.$$

A vertex $v = (q, \mathbf{c}, h)$ has the edge

$$(s_r, \mathbf{c}, h) \longrightarrow (t_r, \mathbf{c} + \Delta_r, Bh + r) \quad (212)$$

when branch r is enabled. The natural number h stores the complete branch history as appended base- B digits. Its initial value is 1.

Lemma 71.1 (Exact predecessor recovery). *Every vertex has at most one predecessor and at most one successor. Both can be computed without simulating an unbounded future. Every predecessor has strictly smaller history coordinate.*

Proof. Determinism gives the successor. To recover a predecessor of $(q', \mathbf{c}', h')$, perform Euclidean division $h' = Bh + r$. Reject unless $h \geq 1$ and $1 \leq r \leq R$. The tag determines the only possible branch. Check $q' = t_r$, form $\mathbf{c} = \mathbf{c}' - \Delta_r$, check nonnegativity and its branch guard, and verify (212). There is at most one candidate. Since $B \geq 2$ and $r \geq 1$, a forward edge strictly increases h . $\square$

Theorem 71.2 (Rooted-path compiler). *The underlying undirected graph G_P is a computably presented disjoint union of finite rooted paths and one-sided infinite rooted paths. The component of*

$$s_x = (q_*, (x, 0, \dots, 0), 1)$$

is a path of $L = T + 1$ vertices if the computation first halts after T transitions, and is a ray otherwise.

Proof. Degrees are at most two by Lemma 71.1. Backward iteration must terminate because the positive integer h strictly decreases. There are no directed cycles, and an undirected cycle in a graph with both indegree and outdegree at most one would necessarily be directed. Each component therefore has a root and no branches. Nonhalting states always have a successor, so a finite path ends precisely at a halting state. The vertex s_x has no predecessor because its history is 1. Its forward trajectory is exactly the original computation with its branch history appended. $\square$

[write] [The same as Lemma 56.1 and Theorem 56.2 of source 12, with d counters and histories starting at 1 rather than 0; source 13's proof is kept as a second route.]

A computable bijection $\nu : \mathbb{N} \rightarrow V$ can be obtained by iterating Cantor's pairing function on the counters and $h - 1$, and then adjoining the finite state coordinate. The code gives this bijection and its inverse explicitly. Thus the entire graph is fixed once P is fixed. In particular, one does not decide in advance whether to build a finite path or an infinite one.

72 A path resolvent and its exact integral denominator

Fix an integer $m \geq 3$, and define

$$D_0 = 1, \quad D_1 = m, \quad D_{j+1} = mD_j - D_{j-1} \quad (j \geq 1). \quad (213)$$

Also put $U_0 = 0$, $U_1 = 1$, and $U_{n+1} = mU_n - U_{n-1}$, so that $D_j = U_{j+1}$. These symbols always refer to the displayed fixed value of m .

Let J_P be adjacency on $\ell^2(V)$ and set $A_m^{\text{path}} = mI - J_P$. For a graph of maximum degree two, the inequality $2|ab| \leq |a|^2 + |b|^2$ on each edge gives

$$(m-2)I \leq A_m^{\text{path}} \leq (m+2)I. \quad (214)$$

This proves boundedness, self-adjointness, and a bounded inverse.

Lemma 72.1 (Finite and infinite Green columns). *Let the source be the root of a component. If that component has L vertices numbered $0, \dots, L-1$, the unique solution of $A_m^{\text{path}}v = \delta_0$ is*

$$v_i = \frac{D_{L-i-1}}{D_L} \quad (0 \leq i < L), \quad (215)$$

and is zero on all other components. If the component is a ray, then

$$v_i = \rho^{i+1}, \quad \rho = \frac{m - \sqrt{m^2 - 4}}{2} \in (0, 1). \quad (216)$$

Proof. The finite equations are $mv_0 - v_1 = 1$, followed by $mv_i - v_{i-1} - v_{i+1} = 0$, with omitted exterior values zero. Substituting (215) verifies the first equation by $mD_{L-1} - D_{L-2} = D_L$, the interior ones by the recurrence, and the last by $D_1 = mD_0$. The case $L = 1$ reads $v_0 = 1/m$ separately. The positive recurrence implies all numerators and denominators are positive. For a ray, $m\rho - \rho^2 = 1$ and $m\rho^{i+1} - \rho^i - \rho^{i+2} = 0$; the sequence is in ℓ^2 . Invertibility gives uniqueness in both cases and zero on other components. $\square$

[write] [The same as Lemma 58.1 with Theorem 59.1 of source 12, with m for α and L for n ; source 13's proof is kept as a second route.]

Proposition 72.2 (Complete integral fiber). *For a root s and $q \in \mathbb{N}_{>0}$, the equation $A_m^{\text{path}}z = q\delta_s$ has a finite-support integral solution exactly when the root component is finite, of length L , and $D_L \mid q$. Every such pair is*

$$q = kD_L, \quad z_i = kD_{L-i-1}, \quad k \in \mathbb{N}_{>0}. \quad (217)$$

The pair with greatest common divisor one across q and all nonzero coordinates is unique and corresponds to $k = 1$.

Proof. Every finite-support solution is in ℓ^2 , so Lemma 72.1 determines it. A ray solution has infinitely many nonzero entries. On a finite path the last entry is q/D_L , which proves necessity of divisibility; the other entries then give sufficiency. The last coordinate in the primitive pair is 1, and the common divisor of the scaled pair is exactly k . $\square$

[write] [The same as Theorem 58.2 of source 12, which fixes the scale by the terminal normalization $H(u) = 1$ instead of listing the whole fibre; source 13's proof is kept as a second route.]

The determinant of the L -vertex tridiagonal matrix is D_L , by expansion along an end row. Also

$$\gcd(D_j, D_{j-1}) = 1 \quad (218)$$

by the Euclidean algorithm applied to (213). Thus the root fraction D_{L-1}/D_L is already reduced. At $m = 3$, the even Fibonacci subsequence satisfies the same recurrence, and

$$D_L = F_{2L+2}, \quad q_{\min} = F_{2T+4}.$$

This Fibonacci specialization concerns the disconnected path operator; our main connected integer network will use $m = 5$.

73 Connecting every component without losing its computation

73.1 The two-rail construction

For every $v = \nu(n)$ make two vertices v^+ and v^- . Copy each edge of G_P within each rail. Add two further vertices c_n, d_n and edges

$$\{v^+, c_n\}, \quad \{v^-, c_n\}, \quad \{c_n, d_n\}, \quad \{d_n, d_{n+1}\} \quad (n \geq 0). \quad (219)$$

Call the resulting graph $\mathcal{G}_P$, with vertex set W and adjacency J . It is connected because the d_n form a backbone ray and every rail vertex is attached to that ray. Each rail vertex has degree at most three, each c_n has degree three, and each d_n has degree two or three.

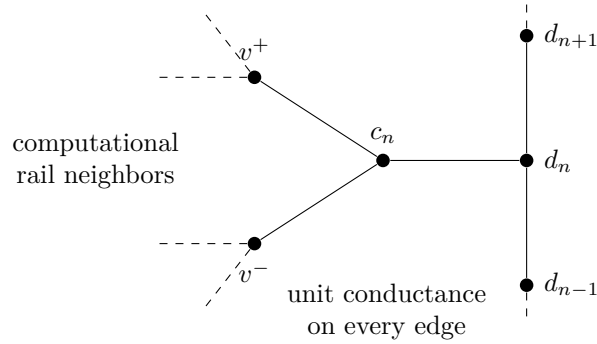


Figure 2: One attachment cell, with $v = \nu(n)$. The dashed rail edges represent at most two computational neighbors, not a prescribed geometric embedding. At $n = 0$ the lower backbone edge is omitted. Ground shunts are not shown.

Let S swap v^+ with v^- and fix every c_n, d_n . This involution is a graph automorphism. Fix now $m \geq 5$, and define

$$A_m = mI - J, \quad b_x = \delta_{s_x^+} - \delta_{s_x^-}. \quad (220)$$

(The localization theorem will additionally use odd m .) The same edge estimate as before gives

$$(m - 3)I \leq A_m \leq (m + 3)I. \quad (221)$$

Lemma 73.1 (Antisymmetric decoupling). *The unique solution $u = A_m^{-1}b_x$ satisfies*

$$u(v^-) = -u(v^+), \quad u(c_n) = u(d_n) = 0.$$

Its restriction to the positive rail solves $A_m^{\text{path}}u_+ = \delta_{s_x}$.

Proof. The involution commutes with A_m and sends b_x to $-b_x$. Consequently both Su and $-u$ solve $Aw = -b_x$, so uniqueness gives $Su = -u$. At fixed vertices this forces zero. The equation at v^+ is therefore the original path equation; at c_n the opposite rail values cancel, and at d_n all values are zero. Conversely, lifting a path solution with opposite signs on the rails and zero on the backbone solves every equation, including all attachment equations. $\square$

Theorem 73.2 (Connected, coercive quasi-Diophantine universality). *There is a fixed computably presented connected graph of maximum degree three, with conductance operator $A = 5I - J$, and a computable family of sources b_x having exactly two nonzero entries $+1, -1$, such that*

$$x \in K \iff A^{-1}b_x \in \mathbb{Q}^{(W)} \iff \exists q \in \mathbb{N}_{>0} \exists z \in \mathbb{Z}^{(W)} : Az = qb_x. \quad (222)$$

The operator has at most four nonzeros per row, entries in $\{0, -1, 5\}$, and $2I \leq A \leq 8I$. If x halts after T transitions, let $L = T + 1$. The least possible charge is D_L , the nonzero support has exactly $2L$ vertices, and the complete integral fiber consists of the positive multiples of a unique primitive certificate.

Proof. Apply the construction to the fixed universal program U . Theorem 71.2, Lemma 73.1 and Proposition 72.2 prove every stated equivalence, the support size, and the complete fiber description. Rational finite support is equivalent to clearing finitely many denominators. The graph and its complete neighbor algorithm were specified independently of the input. The bounds and coefficient count follow from maximum degree three. $\square$

[write] Together with Lemma 73.1, this answers source 12's research question of Section 68.4 (dated note there).

The displayed quantifier over $\mathbb{Z}^{(W)}$ is genuinely algebraic but is not an ordinary finite-dimensional Diophantine quantifier. Checking a proposed finite-support witness is effective: evaluate $Az - qb_x$ on the union of its support, the neighbors of that support, and the source vertices. All other rows vanish automatically. This also directly proves the computable-enumerability side of the assertion.

73.2 A passive circuit, not an analog oracle

Give every graph edge conductance 1. At a vertex w , add a conductance $m - \deg(w)$ to a common zero-potential ground. Kirchhoff's nodal equation is

$$\sum_{v \sim w} (u_w - u_v) + (m - \deg(w))u_w = b_w,$$

which is exactly $A_m u = b$. At $m = 5$ the shunts belong to the finite set $\{2, 3, 4\}$. The degree-three claim refers to the connected graph before adding the common ground reference; including that ground as a vertex would give it infinite degree.

The signal is a two-terminal current injection and withdrawal. No active or negative conductance is used. What is computationally universal is the *exact finite-support or arithmetic question* for an infinite computably specified network. Neither a finite physical device nor finite precision measurement is asserted to decide a halting problem.

73.3 Arbitrarily close to the identity

Corollary 73.3. *For arbitrarily large fixed odd m , the same construction satisfies*

$$\|I - A_m/m\| \leq \frac{3}{m}, \quad \kappa_2(A_m) \leq \frac{m+3}{m-3},$$

while retaining (222) with the corresponding D_L . Thus the normalized operators can be arbitrarily close to the identity, and the condition-number upper bound can be arbitrarily close to one.

Proof. Use $\|J\| \leq 3$ and (221). The path resolvent and its finite-support dichotomy hold for each fixed $m \geq 5$. $\square$

74 Fast analytic approximation and undecidable exact arithmetic

74.1 The inverse is uniformly computable

For every $m \geq 5$, the operator-norm Neumann expansion gives

$$A_m^{-1} = \frac{1}{m} \sum_{k=0}^{\infty} (J/m)^k, \quad \left\| A_m^{-1} - \frac{1}{m} \sum_{k=0}^N (J/m)^k \right\| \leq \frac{(3/m)^{N+1}}{m-3}. \quad (223)$$

In (223), the norm bars denote the ordinary operator norm; the bound is just the tail of a geometric series. For a finite rational source, each partial sum has finite rational support and is computed from a finite neighborhood. Multiplying the right side by the source norm bounds the vector approximation error. For b_x that norm is $\sqrt{2}$.

This proves considerably more than existence of an analytic solution: there is one uniform algorithm producing arbitrarily accurate rational finite-support approximations for every input, including nonhalting inputs. The neighbor exploration cost is not asserted to be polynomial. The distinction is not computable inversion versus noncomputable inversion; it is computable approximation versus an undecidable exact support test.

[write] [Compare Theorem 57.1 and Proposition 57.2 of source 12, where the path-forest neighbourhoods also give a polynomial bit-time bound.]

74.2 The exact resistance dichotomy

Write

$$g_L = \frac{D_{L-1}}{D_L}, \quad \mathcal{R}_x = \langle b_x, A_m^{-1} b_x \rangle.$$

The latter is the voltage difference under a unit two-terminal current, or effective resistance in the grounded network.

Theorem 74.1 (Rational versus fixed quadratic resistance). *For the fixed universal network and $m \geq 5$,*

$$\mathcal{R}_x = \begin{cases} 2D_{L-1}/D_L, & \text{if } U(x) \text{ halts with } L = T + 1, \\ m - \sqrt{m^2 - 4}, & \text{if } U(x) \text{ does not halt.} \end{cases} \quad (224)$$

Consequently rationality of $\mathcal{R}_x$ is Σ_1^0 -complete on this computable family; equality to the displayed irrational constant is Π_1^0 -complete. At $m = 5$ the nonhalting value is $5 - \sqrt{21}$.

Proof. The source pairs with the two opposite root values, so the answer is twice the root value in Lemma 72.1. The finite values are rational. For integer $m \geq 3$, the integer $m^2 - 4$ lies strictly between $(m-1)^2$ and m^2 , hence is not a square. This makes the nonhalting value irrational. The assertions of completeness now follow by equality of the corresponding input sets with K and its complement. No claim is made about rationality tests for arbitrary computable real numbers outside this family. $\square$

[write] [The same as Theorem 65.1 of source 12 for a dipole source: the value is twice the root value g_L , and the nonhalting value $m - \sqrt{m^2 - 4}$ is 2ρ . The closed forms (225)–(227) below are (181)–(183) of source 12; here α denotes ρ^{-1} , source 12’s λ , not a diagonal parameter. Source 13’s proof is kept as a second route.]

For precise quantitative separation, let $\alpha = \rho^{-1}$ and $\delta = \sqrt{m^2 - 4} = \alpha - \rho$. Solving the recurrence gives

$$D_j = \frac{\alpha^{j+1} - \rho^{j+1}}{\delta}, \quad (225)$$

$$g_L = \rho \frac{1 - \rho^{2L}}{1 - \rho^{2L+2}}, \quad (226)$$

$$\rho - g_L = \frac{\delta \rho^{2L+2}}{1 - \rho^{2L+2}} > 0. \quad (227)$$

The last expression strictly decreases with L , proving that the rational values g_L strictly increase to ρ . If the first $N \geq 1$ configuration vertices have been inspected and the run has not yet halted, its eventual root value lies between g_N and ρ . If it has halted, the exact fraction is known. Equation (227) is an exponential error bound for this scalar procedure. Thus a number of simulated transitions linear in the requested number of accuracy bits suffices, with a constant depending only on fixed m . This does not give a bound on when the procedure will discover exact rationality.

74.3 A uniformly strongly convex energy

Define on $\ell^2(W; \mathbb{R})$

$$E_x(u) = \frac{1}{2} \langle u, A_m u \rangle - \langle b_x, u \rangle. \quad (228)$$

The unique minimizer is $u_x = A_m^{-1} b_x$, and completion of the square shows

$$E_x(u) - E_x(u_x) = \frac{1}{2} \langle u - u_x, A_m(u - u_x) \rangle. \quad (229)$$

For $m = 5$ this yields

$$\|u - u_x\|^2 \leq E_x(u) - E_x(u_x) \leq 4 \|u - u_x\|^2.$$

Corollary 74.2 (Undecidable attainment on an algebraic domain). *The restriction of E_x to $\mathbb{Q}^{(W)}$ has an infimum equal to its Hilbert space minimum. That infimum is attained exactly when $x \in K$. The same attainment equivalence holds on $\mathbb{R}^{(W)}$. Moreover,*

$$\min_{\ell^2} E_x = -\frac{1}{2} \mathcal{R}_x$$

is rational exactly on halting inputs.

Proof. Finite rational arrays are dense in ℓ^2 , and the energy is continuous. Attainment on either subdomain forces equality with the unique Hilbert space minimizer by (229). That minimizer has finite support exactly in the finite-path case, where it is rational. The formula for the minimum follows by substituting $A_m u_x = b_x$. $\square$

Here coercivity, strict convexity, invertibility, and an explicit analytic error bound all hold. The algebraic finite-support domain is not closed in the Hilbert topology. It is this domain restriction, rather than a nonclosed operator range, that carries the obstruction.

75 A finite-prime localization restores decidability

75.1 The fixed-amplitude question

For a finite set of primes S , put

$$R_S = \mathbb{Z}[S^{-1}] = \{a/b \in \mathbb{Q} : \text{all prime divisors of } b \text{ belong to } S\}.$$

We now ask whether the *unit-source* solution $A_m^{-1} b_x$ belongs to $R_S^{(W)}$. The free charge q in (222) must not be silently reintroduced: allowing arbitrary q already makes integral certificates universal. Equivalently, the localized question permits a scaled integral certificate only when q itself has no prime factors outside S .

By the final coordinate $1/D_L$ in (215),

$$A_m^{-1} b_x \in R_S^{(W)} \iff [U(x) \text{ halts at length } L \text{ and } D_L \text{ has no prime factor outside } S]. \quad (230)$$

To make this a decision procedure we need an effective upper bound on such L . We give an elementary proof rather than invoke a primitive-divisor theorem or a general effective S -unit theorem.

75.2 Ranks of apparition

Throughout this section $m \geq 3$ is odd. For a prime p , let

$$z_p = \min\{n \geq 1 : p \mid U_n\}. \quad (231)$$

Lemma 75.1 (Effective rank and exact divisibility). *The integer z_p exists, can be found by a finite modular recurrence, and satisfies $1 \leq z_p \leq p^2 - 1$. For every $n \geq 1$,*

$$p \mid U_n \iff z_p \mid n.$$

Proof. The recurrence map $(a, b) \mapsto (b, mb - a)$ is invertible over $\mathbb{F}_p$. Its orbit from $(0, 1)$ stays among the $p^2 - 1$ nonzero pairs and returns to $(0, 1)$; at a return time its first coordinate is zero. This proves existence and the search bound.

For the divisibility assertion use

$$M = \begin{pmatrix} m & -1 \\ 1 & 0 \end{pmatrix}, \quad M^n = U_n M - U_{n-1} I \quad (n \geq 1). \quad (232)$$

The identity follows by induction, or from $M^2 = mM - I$. Modulo p , the matrix M^n is scalar exactly when $U_n = 0$, because its lower-left entry is U_n . The integers $n \in \mathbb{Z}$ for which the power of the invertible matrix M is scalar form a subgroup of $\mathbb{Z}$. Its least positive element is z_p , proving the claim. $\square$

75.3 A valuation-lifting proof with the prime two included

The recurrence and $U_1 = 1$ imply

$$\gcd(U_n, U_{n-1}) = 1 \quad (n \geq 2). \quad (233)$$

For $n = 1$ the corresponding assertion with $U_0 = 0$ is also true.

Lemma 75.2 (Exact valuation on rank multiples). *For odd $m \geq 3$, every prime p , and every $k \geq 1$,*

$$v_p(U_{z_p k}) = v_p(U_{z_p}) + v_p(k). \quad (234)$$

Proof. Write $z = z_p$, $a = -U_{z-1}$, $b = U_z$, and $v = v_p(b) \geq 1$. Equation (233) gives $p \nmid a$. From $M^z = aI + bM$ and the binomial theorem, taking lower-left entries yields

$$U_{zk} = \sum_{j=1}^k \binom{k}{j} a^{k-j} b^j U_j. \quad (235)$$

The $j = 1$ term has valuation $v + v_p(k)$. The identity $j \binom{k}{j} = k \binom{k-1}{j-1}$ gives

$$v_p \binom{k}{j} \geq v_p(k) - v_p(j).$$

Consequently, for $j \geq 2$, the j th term of (235) has valuation at least

$$v + v_p(k) + (j-1)v - v_p(j).$$

For odd p , $(j-1)v - v_p(j) \geq 1$: indeed $v \geq 1$ and $v_p(j) \leq j-2$ for $j \geq 2$. All these terms therefore have strictly larger valuation than the first.

It remains to check $p = 2$ rather than import an odd-prime argument unchanged. Since m is odd, U_n modulo two has the repeating pattern 0, 1, 1, so $z_2 = 3$. Moreover $U_3 = m^2 - 1$ is divisible by eight. Thus $v \geq 3$; in particular $v \geq 2$, which makes $(j-1)v - v_2(j) \geq 1$ for every $j \geq 2$. Again the first term is the unique term of lowest valuation. The valuation of the sum is its valuation, proving (234) for every prime. $\square$

No estimate here is asymptotic. The small-rank values, including the exceptional behavior at two, enter an exact computable constant.

Theorem 75.3 (An explicit finite-prime length bound). *For a finite prime set S define*

$$c_p = \max\{0, v_p(U_{z_p}) - v_p(z_p)\}, \quad C_S = \prod_{p \in S} p^{c_p}, \quad (236)$$

with the empty product equal to 1. Let B_S be the first integer $L \geq 1$ satisfying

$$(m-1)^L > C_S(L+1). \quad (237)$$

Then B_S is effectively computable. If D_L has no prime factor outside S , then $L < B_S$.

Proof. By Lemmas 75.1 and 75.2, for every $n \geq 1$ and prime p ,

$$v_p(U_n) \leq c_p + v_p(n). \quad (238)$$

When $z_p \nmid n$ the left side is zero. Otherwise the exact valuation is $v_p(U_{z_p}) + v_p(n) - v_p(z_p)$, proving the inequality. If U_n has all its prime factors in S , it follows that

$$U_n \leq C_S \prod_{p \in S} p^{v_p(n)} \leq C_S n. \quad (239)$$

On the other hand $U_1 = 1$, $U_2 = m$, and the positive increasing recurrence implies $U_{n+1} \geq (m-1)U_n$, so

$$U_n \geq (m-1)^{n-1} \quad (n \geq 1). \quad (240)$$

Put $n = L + 1$. Smoothness forces $(m-1)^L \leq C_S(L+1)$. The ratio $(m-1)^L/(L+1)$ is strictly increasing for $L \geq 1$, because its successive ratio is $(m-1)(L+1)/(L+2) > 1$; it tends to infinity. The first failure can be found by integer arithmetic, and every later length fails as well. All constants in (236) are computable by the finite rank search and exact integer valuations. $\square$

Corollary 75.4 (Uniform localized decidability). *For each fixed odd $m \geq 5$, the predicate*

$$(x, S) \mapsto [A_m^{-1}b_x \in \mathbb{Z}[S^{-1}]^{(W)}]$$

is decidable for finite prime sets S . More generally the same algorithm works uniformly for any finite counter program supplied to the compiler.

Proof. Compute B_S . If $B_S \leq 1$, reject. Otherwise simulate through transition number $B_S - 2$, including the initial configuration. If a halt is found at length $L < B_S$, compute D_L and repeatedly divide out primes in S . Accept exactly when the remaining integer is 1. If no halt has occurred, reject. Equation (230) and Theorem 75.3 prove both directions, including inputs which eventually halt after this finite bound. $\square$

The theorem is family-specific. It is not a decision theorem for arbitrary Diophantine equations over $\mathbb{Z}[S^{-1}]$, nor a claim that those rings cannot support other universal representations. In fact our own unrestricted charge equation $Az = qb_x$ remains universal with integral z .

Example 75.5 (A small explicit bound). At $m = 3$ and $S = \{2, 3\}$, the ranks are $z_2 = 3$ and $z_3 = 2$. The valuations give $c_2 = 3$, $c_3 = 1$, and $C_S = 24$. The first L with $2^L > 24(L+1)$ is $L = 8$, so every admissible length is below eight. Checking those seven values gives precisely $L = 1, 2, 5$, with denominators 3, 8, 144. This example illustrates the disconnected Fibonacci case; the same algorithm, with $m = 5$, applies to the connected network.

75.4 Noncomputable certificate resources

Corollary 75.6. *For a fixed universal program, none of the following resources on its halting inputs has a total computable upper bound as a function of input: the finite-support size, the least charge D_L , or the largest prime factor of D_L .*

Proof. A support bound gives a halting-time bound because support has size $2L$. A charge bound gives a length bound from $(m-1)^L \leq D_L$. Either would decide halting by bounded simulation.

For the last assertion, suppose $f(x)$ bounds every prime factor of D_L whenever x halts. Compute the finite set of primes at most $f(x)$ and apply Corollary 75.4. Under the assumed bound it accepts every halting input and no nonhalting input, again deciding K . $\square$

[write] [The support and charge clauses are the same as Theorem 61.2 of source 12; all three clauses are the bounded-search argument of `cdc:bd:prop:nobound` in [22]. The prime-factor clause is new with source 13.]

Thus the decidable finite-prime fragments form an effectively indexed cover of the rational finite-support halting relation, but no computable input-dependent prime cutoff captures all its positive instances.

76 Sharp sparsity boundaries and the necessity of a dipole

76.1 Connected scalar operators with at most three entries per row

We make the lower-bound class explicit. An operator has a computably presented countable vertex set, rational entries, a nonzero diagonal, symmetric off-diagonal support, and complete finite neighbor lists. The support graph records nonzero off-diagonal entries. In particular, positive-definite scalar conductance matrices belong to this class.

Theorem 76.1 (A fixed connected row-three decision boundary). *For every fixed connected operator in the preceding class with at most three nonzero entries per row, finite-support rational solvability of $Au = b$ is decidable as a function of the supplied finite rational source b . Consequently no such fixed connected operator can realize (222). Four entries per row are sufficient by Theorem 73.2, and hence are optimal in this specified class.*

Proof. The nonzero diagonal leaves at most two off-diagonal neighbors. A connected simple graph of maximum degree two is a finite path, a finite cycle, a one-sided ray, or a two-sided infinite path. A single vertex is included in the finite case.

For a finite graph, solve its finite rational system. For a two-sided path, let K_b be the finite interval spanned by the source support. If a finite-support solution had a nonzero coordinate beyond that interval, choose the outermost such coordinate on that side. At the next vertex, the source is zero, the solution value and the value farther out are zero, but the inward neighbor is nonzero. That row reads a nonzero edge coefficient times a nonzero solution value equal to zero, a contradiction. Thus every finite-support solution is contained in K_b .

For a ray, include its endpoint together with the source support when forming K_b . The same outermost-coordinate argument beyond the far end bounds every finite-support solution by this interval. For a zero source, the zero solution is immediately sufficient; otherwise these constructions apply directly.

The needed finite paths can be found by exhaustive search in the computable connected neighbor graph. Restrict unknowns to K_b , set exterior values to zero, and include equations at K_b and all its neighbors. This is a finite rational linear system, decidable by Gaussian elimination. Including the neighbor rows is essential for sufficiency. $\square$

There is a quantifier qualification in this theorem. The decision algorithm exists for *each fixed* connected graph. In the ray case it may hardcode that graph's unique endpoint. We do not assert a uniform algorithm which, from an arbitrary graph description, decides whether it is a ray or a two-sided path, or finds an endpoint when one exists. This nonuniformity in the fixed operator does not weaken the claimed exclusion of a single fixed universal row-three operator.

If symmetry and positive definiteness are additionally assumed, analytic uniqueness is automatic; the finite-support decision proof does not need that stronger hypothesis. The lower bound does

not concern block-valued vertices, hidden auxiliary sorts, or a representation in which the operator itself is allowed to vary with the input.

76.2 The disconnected threshold is three

Without connectedness, two nonzeros per row and a nonzero diagonal leave maximum off-diagonal degree one. Components have one or two vertices. For a fixed rational positive-definite operator, every finite rational source therefore has a finite rational solution obtained by solving finitely many blocks. The disconnected path construction in Section 72 uses three nonzeros per row and already realizes the universal finite-support problem. Thus connectedness increases the sharp scalar threshold from three to four under these conventions.

76.3 Why a single positive source cannot work

Proposition 76.2 (Positivity obstruction). *Let G be an infinite connected graph of maximum degree Δ , and $m > \Delta$. Then every coordinate of $(mI - J_G)^{-1}\delta_s$ is strictly positive. In particular it never has finite support.*

Proof. The Neumann series has nonnegative entries. For any vertex v , connectivity provides a finite walk from s to v , of some length k , so the (v, s) entry of J_G^k is positive. The corresponding positive term occurs in the inverse series. $\square$

The same argument works for a nonzero finitely supported nonnegative source. The two-terminal dipole is not cosmetic: its opposite signs let an exact graph symmetry cancel all backbone voltages while preserving a nontrivial computation in the antisymmetric sector. The network is passive, but its imposed current source is signed.

77 One quadratic polynomial at every external horizon

The linear module equation is the unbounded-support representation. This section supplies a separate, completely finite arithmetic interface, with all configuration and circuit data paid for. Its arity depends on a supplied horizon T . The construction is useful for exact checking and formalization; it is not the main source of universality or a new fixed-arity quadratic representation of all halting computations.

77.1 All variables and all linear residuals

Fix a program P , dimension d , branch count R , and number Z of test instructions. Input coordinates $x_0, \dots, x_{d-1}$ are parameters, not witnesses. For $0 \leq t \leq T$, introduce the state q_t , counters $c_{t,j}$, and history h_t . For $0 \leq t < T$ and $1 \leq r \leq R$, introduce a natural selector $b_{t,r}$ and routed counters $a_{t,r,j}$. For every positive test branch also introduce a natural slack $s_{t,r}$. Finally introduce $u_0, \dots, u_T$ and a charge Q_c . All these unknowns range over $\mathbb{N}$.

The initial and terminal residuals are

$$q_0 - q_*, \quad h_0 - 1, \quad c_{0,j} - x_j \ (0 \leq j < d), \quad q_T - H. \quad (241)$$

At each $t < T$, include the following linear residuals:

$$\sum_r b_{t,r} - 1, \quad (242)$$

$$q_t - \sum_r s_r b_{t,r}, \quad q_{t+1} - \sum_r t_r b_{t,r}, \quad (243)$$

$$c_{t,j} - \sum_r a_{t,r,j} \quad (0 \leq j < d), \quad (244)$$

$$c_{t+1,j} - c_{t,j} - \sum_r \Delta_{r,j} b_{t,r} \quad (0 \leq j < d), \quad (245)$$

$$h_{t+1} - B h_t - \sum_r r b_{t,r}. \quad (246)$$

Here s_r in (243) denotes the fixed *source state* of branch r , whereas $s_{t,r}$ below is a witness slack; their indices separate the two uses. For a zero-test branch testing counter j_r , include

$$a_{t,r,j_r}; \quad (247)$$

for a positive-test branch, include

$$a_{t,r,j_r} - b_{t,r} - s_{t,r}. \quad (248)$$

There are Z branches of each of these two kinds.

For the electrical layer, set the omitted symbols $u_{-1} = u_{T+1} = 0$, without introducing them as variables, and include

$$m u_i - u_{i-1} - u_{i+1} - Q_c \mathbf{1}_{i=0} \quad (0 \leq i \leq T), \quad u_T - 1. \quad (249)$$

These are equations on a path of $T + 1$ configuration vertices. They will be lifted to the actual connected graph only after the machine layer has certified that the last vertex is a genuine halt.

77.2 Natural nonnegativity replaces squared routing gates

The selectors are automatically zero or one when they are natural and sum to one. Instead of squaring quadratic routing equalities, use the nonnegative polynomial

$$C_t = \sum_{j=0}^{d-1} \sum_{r \neq s} b_{t,s} a_{t,r,j}. \quad (250)$$

Let $\mathcal{L}_T$ be the complete list of linear residuals (241)–(249). Define

$$P_T(\mathbf{x}, \mathbf{w}) = \sum_{\ell \in \mathcal{L}_T} \ell^2 + \sum_{t=0}^{T-1} C_t. \quad (251)$$

This is a single integer polynomial of total degree at most two. On natural coordinates every summand is nonnegative. The construction does not assert nonnegativity on arbitrary signed real assignments.

Lemma 77.1 (Exact routing). *Suppose $\mathbf{b}, \mathbf{a}$ are natural, $\sum_r b_r = 1$, and $\sum_{j,r \neq s} b_s a_{r,j} = 0$. There is a unique selected branch r_* with $b_{r_*} = 1$, and $a_{r,j} = 0$ for every $r \neq r_*$. If additionally $c_j = \sum_r a_{r,j}$, then $a_{r_*,j} = c_j$.*

Proof. The sum-one equation on natural coordinates selects exactly one index. For each $r \neq r_*$, the nonnegative sum contains $b_{r_*} a_{r,j} = a_{r,j}$, so it forces all those entries to zero. The remaining assertion follows from the counter sum. $\square$

Theorem 77.2 (A quadratic, complete, unique finite certificate). *For every fixed program P , integer $m \geq 3$, horizon $T \geq 0$, and natural input vector $\mathbf{x}$, the following are equivalent:*

$$\exists \mathbf{w} \in \mathbb{N}^{N_T} : P_T(\mathbf{x}, \mathbf{w}) = 0 \iff P \text{ first halts after exactly } T \text{ transitions.} \quad (252)$$

Whenever a zero exists, its entire witness tuple is unique. It has

$$N_T = (d + 3)(T + 1) + 1 + T((d + 1)R + Z) \quad (253)$$

natural witnesses, excluding the d input parameters. The displayed compiler uses

$$d + 5 + T(5 + 2d + 2Z) \quad (254)$$

linear residuals and $dR(R - 1)T$ unsquared cross-term occurrences. Its electrical part is

$$u_i = D_{T-i}, \quad Q_c = D_{T+1}. \quad (255)$$

For $m \geq 5$ this tuple reconstructs the primitive connected-network certificate of Theorem 73.2.

Proof. At a natural zero every squared residual and every nonnegative cross term vanishes. At each step Lemma 77.1 selects exactly one branch and makes its routed counters the actual counters. The state residuals give that branch's correct source and target. A selected zero branch is permitted only when the tested counter vanishes. A selected positive branch has tested counter $1 + s_{t,r} \geq 1$. The update residuals implement its actual counter change. Inactive positive-branch slacks are forced to zero because their routed counter and selector are both zero.

Starting from (241), induction therefore recovers exactly the deterministic computation, including every state, counter, selector, routed counter, slack, and history value. If a halt occurred before T , there would be no branch with that source state to satisfy the next sum-one and state equations. The terminal equation forces a halt at T . Thus the halting time is exact, not merely bounded.

The final equation $u_T = 1$, followed by the circuit equations in reverse order, forces successively $u_{T-1} = m$, $u_{T-2} = m^2 - 1$, and in general $u_i = D_{T-i}$. The first row then forces $Q_c = D_{T+1}$. This also proves positivity of the charge without an extra positive slack and proves uniqueness of the remaining witnesses. For $T = 0$, the two circuit rows are simply $mu_0 - Q_c = 0$ and $u_0 - 1 = 0$.

Conversely, supply an actual first-halting trace. Give its selected branch value one, all other selectors zero, put the actual counters in its routed coordinates, and set selected positive slacks to the tested counter minus one and other slacks to zero. Append the prescribed history and the values (255). Every residual and cross term then vanishes. All witnesses are natural, proving existence and the stated bijection.

For the count, there are $(d + 2)(T + 1)$ configuration coordinates, $(d + 1)RT$ selectors and routed counters, ZT positive slacks, and $T + 2$ electrical coordinates. There are $d + 3$ initial/terminal rows, $4 + 2d + 2Z$ transition rows per step, and $T + 2$ electrical rows. These are exactly (253) and (254). Finally, place u_i on the positive rail of the i th recovered history vertex, $-u_i$ on its negative rail, and zero elsewhere. Its final vertex is a genuine halt; Lemma 73.1 verifies all graph equations, not just a finite principal block. $\square$

These counts describe the literal reference compiler, not an optimal quadratic representation. For example, the positive slacks and their rows can be projected away: on the selected positive branch, natural successor counters in (245) already force a positive tested counter. Their unique restoration is the tested counter minus one, or zero on an inactive branch. We retain them to expose the branch guards explicitly and keep the code, residual list, and witness count identical.

[write] The transition layer of this compiler (natural one-hot selectors, routed counters with the unsquared cross terms $b_{t,s}a_{t,r,j}$, guard slacks) is a variant of the degree-two counter-machine compilers with unsquared natural products in the report *liveness-beyond-halting* [41] (`lbh:qd:thm:compiler` and `lbh:ql:thm:compiler`, compared in `lbh:sec:threecompilers`) and of the quadratic bounded counter execution of [22] (`display cdc:eq:bigM`). It is printed with source 13's proof and without a novelty claim. New is the electrical layer (249), which forces $u_i = D_{T-i}$ and $Q_c = D_{T+1}$ and lifts to the connected network. The research programme's review [42] remarked that the routing cross sum could be factored by sharing the selector and routed-counter sums. The programme has since emitted complete paid straight-line evaluators of this unchanged polynomial [45] (commit 9bc875b6a): with $B = \sum_r b_{t,r}$ and $A_r^* = \sum_j a_{t,r,j}$, the identity $C_t = \sum_r (B - b_{t,r})A_r^*$ holds over all integer tuples, and the complete cost of evaluating P_T for a two-counter transfer program at horizon 3 falls from 245 to 206 operations. These are comparisons of explicitly emitted evaluators, not lower bounds or a fixed-arity universal equation. The slack projection above remains the author's observation and is not implemented.

77.3 Why this is not a finite-fold MRDP breakthrough

The parameter T in (252) chooses a different polynomial and witness dimension. It is not just another variable of one fixed polynomial. Taking the union over T gives the intended halting relation, but does not exhibit a fixed finite-dimensional polynomial whose complete natural witness is unique. Encoding all these finite tuples and applying MRDP may introduce further auxiliary witnesses with no preserved uniqueness theorem. We therefore make no claim about single-fold or finite-fold ordinary Diophantine representation.

Similarly, the total degree two here concerns a horizon-indexed family over natural coordinates. Replacing natural coordinates by unrestricted signed integers invalidates the nonnegativity argument for (250). One cannot infer an analogous signed quadratic theorem merely by reusing the same formal polynomial.

78 A complete numerical example and adversarial boundary checks

78.1 Countdown on one counter

Take states 0, 1, with 1 halting, and instruction

$$0 : \text{TEST}(0, 1, 0).$$

Tag its zero branch by 1 and its positive branch by 2, so $B = 3$. For input $x = 2$ the exact history is

t	state q_t	counter c_t	history h_t	selected branch
0	0	2	1	positive, tag 2
1	0	1	5	positive, tag 2
2	0	0	17	zero, tag 1
3	1	0	52	halt

There are $T = 3$ transitions and $L = 4$ configuration vertices. At $m = 5$,

$$D_0, D_1, D_2, D_3, D_4 = 1, 5, 24, 115, 551.$$

The primitive positive-rail potentials are $(115, 24, 5, 1)$, with charge 551; negative-rail potentials are their negatives. The circuit equations are visibly

$$5(115) - 24 = 551, \quad 5(24) - 115 - 5 = 0, \quad 5(5) - 24 - 1 = 0, \quad 5(1) - 5 = 0.$$

The unit-source effective resistance is

$$\mathcal{R} = \frac{230}{551}, \quad 551 = 19 \cdot 29.$$

This particular unit-source solution belongs to $\mathbb{Z}[1/19, 1/29]^{(W)}$, but not to $\mathbb{Z}[1/2, 1/3, 1/5]^{(W)}$. It nevertheless has an integral scaled certificate with charge 551, illustrating the fixed-amplitude qualification in Section 75.

For completeness the routed data are

t	$b_{t,1}$	$b_{t,2}$	$a_{t,1,0}$	$a_{t,2,0}$	$s_{t,2}$
0	0	1	0	2	1
1	0	1	0	1	0
2	1	0	0	0	0

These tables and the electrical values specify all 32 natural witnesses. There are 33 linear residuals and six cross terms. Expansion and collection produce 105 distinct nonzero monomials, of degree at most two, counting the input variable as part of the polynomial. The complete polynomial and its entire witness are exported as JSON; no term is omitted from that artifact.

Using the code's explicit vertex enumeration, the nonzero primitive network array is

positive vertex	value	negative vertex	value
24	115	25	-115
152	24	153	-24
1216	5	1217	-5
11020	1	11021	-1

Applying the full connected operator gives value 551 at vertex 24, value -551 at vertex 25, and zero at every other row, including all attachment hubs and every exterior neighbor.

78.2 The finite-principal-block trap

Suppose instead that the program increments forever. Truncate the first L configurations and assign the same formal path values D_{L-i-1} to them. Every equation inside that principal block is satisfied with charge D_L . But the actual next positive-rail vertex sees the residual -1 , because the last retained value is 1; its negative partner sees $+1$. The finite array is not a solution of the infinite system.

The verifier deliberately tests this failure for $1 \leq L \leq 19$. All four nonzero residuals are checked: the two source values and the two leakage values. A correct verifier must check exterior rows. A correct finite arithmetic compiler must also pay for the terminal-halting test; an invertible truncated circuit alone proves nothing about halting.

78.3 Other exact boundary checks

The tests include the initially halting case $T = 0$, both sides of zero branches, unused history digits, invalid inverse history candidates, multiple positive scalings of primitive certificates, and computations which never halt within the test horizon. They also check every single natural-coordinate perturbation by $+1$ and every permitted perturbation by -1 of the complete valid witnesses, including input parameters. These checks catch unbound input coordinates, free private witnesses, and several common boundary omissions. They are not substitutes for the inductive soundness and uniqueness proofs.

79 Reproducibility and a precise integration plan

79.1 What was executed

The implementation uses Python’s standard library, immutable program and configuration records, arbitrary-precision integers, exact rational arithmetic, and deterministic random seeds. Running

```
python verify.py
```

from the unpacked directory regenerates the examples and `verification.json`. The recorded run passed 38,524 assertions, distributed among 26 test categories. Important groups are shown below; the receipt contains the full list and exact counts.

Check group	Assertions
Connected neighbor symmetry, degree, and rail involution	7,500
Lucas rank divisibility over five odd parameters and 25 primes	15,000
Exact prime-valuation lifting	1,867
Continuant equations and coprimality	2,475
Single-coordinate natural witness mutations	3,698
Signed polynomial expansion identities	600
Exhaustive small natural routing gates	819
All categories combined	38,524

The polynomial identity tests allow signed assignments to compare two algebraic expressions; the zero-set and nonnegativity tests use natural assignments. This distinction is intentional. Determinant

tests use an independent rational Gaussian-elimination calculation, rather than merely repeating the continuant recurrence. Full network evaluations compute all neighbor rows, not just rows listed in the certificate.

The tests do not instantiate a particular published small universal machine. They verify the generic compiler on several finite programs, including one- and two-counter examples, initially halting programs, unconditional divergence, and a program with both halting and divergent inputs. The fixed-universal existence theorem is a mathematical composition with a standard universal finite counter program, not an empirical universality claim about these toy examples.

[write] *Review and shipped files.* The research programme’s review [42] (commit 899391bde) re-played this suite (38,524 checks) and found two defects in the reference API, not in the mathematics. The frozen `Certificate` keeps mutable row dictionaries and `export` returns the same objects, so mutating an export erases the live input-binding row; the review’s fixture changes a wrong-input energy from 9409 to 0 (delivered `src/substrate.py`, lines 343–348 and 373–377). Prime deduplication by `set(primes)` runs before the primality checks, so `localization_bound(5, [2, 2.0])` accepts an inexact scalar (lines 285 and 306). Its tested repair is `well_conditioned_certificate_snapshots.patch` (SHA-256 `cdd6307d...72f2a9`) in the research programme; it keeps every formula, passes the unchanged 38,524-check suite and reproduces all exports. The shipped `code/13-well-conditioned-substrate.py` is the unpatched original. In this report `verify.py` and `build.sh` are `code/13-well-conditioned-verify.py` and `code/13-well-conditioned-build.sh`, `src/substrate.py` is `code/13-well-conditioned-substrate.py`, the three examples and `verification.json` are `data/13-well-conditioned-*.json`, and `sources.md` is `13-well-conditioned-sources.md`; the PDF, the README and SHA256SUMS (11 of 11 verified at placement) are not shipped. `verify.py` imports `substrate` from a `src/` directory beside it and rewrites `examples/` and `verification.json` in its own directory, so it must run on a copy (README of this report).

79.2 What remains unformalized

No Lean or Rocq proof of this manuscript has been built. A suitable integration sequence would first formalize the branch-history bijection and finite-support module identities, then add the analytic and arithmetic layers. The following proposed modules describe interfaces, not existing completed source files:

Proposed module	Primary mathematical obligation
<code>ReversibleHistory</code>	Injective step map and finite-rooted-path theorem.
<code>PathResolvent</code>	Continuant recurrence, exact Green column, integral fiber.
<code>ConnectedDipole</code>	Neighbor presentation, involution, antisymmetric lifting.
<code>CoerciveSupport</code>	Norm bounds, inverse, approximation, support equivalence.
<code>LucasLocalization</code>	Ranks, valuation lifting including two, effective bound.
<code>QuadraticCertificate</code>	Literal compiler, soundness, completeness, uniqueness.

The first, second, third, and last layers are largely finitary and can be proved before analytic inversion. The analytic proof then shows why the same finite certificate coexists with a uniformly well-posed continuous problem. The existing MRDP interface can be applied to the final halting predicate, but it should not be used to bypass the structural proofs or claim preservation of primitive-witness uniqueness.

79.3 Source and trust boundaries

The archive includes the authored manuscript and code, not copies of the repository, third-party articles, or font files. Repository citations are pinned to the stated commit; the two earlier manuscript citations identify the supplied research documents by title and date. The executable checks and this manuscript's proofs have been reviewed within this work, but have not received independent peer review or a proof-assistant check. A literature search located classical antecedents but does not establish priority for the combined theorem. This is a substantive proposed research contribution with explicit claims and limitations, not a declaration that a recognized longstanding conjecture has been settled.

80 Further research questions

The following questions go beyond the proved statements. They are proposed research directions, not an assertion that every formulation already has a recognized open-problem status in the literature.

Research question 1 (Finite geometric presentation). Can the connected degree-three construction be realized by a fixed finite-range rule on a geometric lattice, with an explicitly paid ordinary input loader, while retaining exact compact-support equivalence and a uniform coercivity bound?

The current graph has an effective neighbor algorithm, but the vertex pairing and backbone are not a translation-invariant local geometric layout. An embedding must pay for wire routing, crossings, and source placement; simply renaming a computable graph as a lattice would not answer the question.

Research question 2 (Planarity with exact antisymmetric isolation). Can one retain a connected planar graph, bounded degree, a finite integer conductance alphabet, and the same two-terminal denominator law?

The two rails and their many backbone attachments do not come with a planar embedding theorem. A useful approach would be a planar symmetry-preserving crossing or isolation gadget whose Schur complement is exact and whose introduced denominators are controlled.

Research question 3 (Infinite recursive prime localizations). For which computable infinite prime sets S is $\{x : A_m^{-1}b_x \in \mathbb{Z}[S^{-1}]^{(W)}\}$ decidable, complete, or of intermediate computably enumerable degree?

The exact criterion is now concrete: it depends only on which continuants D_L have all prime factors in S . The finite-set proof supplies an obstruction and a benchmark. Extending it requires control of infinitely many rank constants, not merely a formal union of finite decision procedures.

Research question 4 (Prime-restricted charge without losing universality). Can another coercive connected compiler realize universal finite-support certificates while forcing every primitive charge to be a power of one fixed prime?

Our Lucas-denominator family cannot do so by Theorem 75.3. A positive answer would require a different recurrence, a different gadget, or another way to normalize a primitive certificate. The question concerns this structural design objective, not general Diophantine definability over a localization.

Research question 5 (Exact constants at the optimal row threshold). With four nonzeros per row and a fixed small integer coefficient alphabet, what are the best simultaneous coercivity, upper spectral, and denominator growth bounds for a connected universal compact-support compiler?

Increasing m makes the normalized operator nearly the identity but changes the alphabet and increases the charge growth. The tradeoff is not addressed by the row-sparsity theorem alone. It would be useful to optimize it within a rigorously specified gadget class before claiming a global optimum.

Research question 6 (Symmetry-breaking and robust replacements for exact support). Which quantified perturbations preserve a computable reduction, even when literal antisymmetry is lost? Can exact support be replaced by a precisely specified arithmetic invariant that survives a wider class of perturbations?

Our backbone cancellation is exact. Generic perturbation need not preserve it, and good analytic conditioning alone does not protect a zero-support predicate. A theorem here must state the permitted perturbations and the new exact decision problem rather than identify small numerical tails with zero.

Research question 7 (A finite-dimensional uniqueness-preserving compression). Can the canonical path charge and primitive finite-support certificate be compressed into a fixed finite tuple of integer witnesses without introducing uncontrolled multiplicities?

The horizon-indexed quadratic construction makes every finite witness explicit but does not answer this question. Any successful compression should distinguish the unique run, the unique primitive network array, and auxiliary arithmetic witnesses introduced by sequence coding. Its relation to finite-fold representation would have to be proved, not assumed.

Research question 8 (Other computable graphs and algebraic Green values). Can one obtain a prescribed finite menu of algebraic nonhalting resistance values, or higher arithmetical classifications of exact resistance properties, while retaining a single fixed connected coercive graph?

The present path resolvent produces one rational-versus-quadratic dichotomy. More elaborate computational subgraphs may produce richer algebraic behavior, but their exact integral fibers and localization properties should be analyzed alongside the real-valued observable.

Research question 9 (Effective versus nonuniform sparsity boundaries). What is the exact computational complexity of finite-support solvability when a degree-two connected operator is supplied as part of the input, rather than fixed in advance?

Theorem 76.1 deliberately allows an endpoint to be hardcoded for each fixed ray. A uniform classification must pay for discovering endpoints or certifying their absence. Separating these quantifier orders may lead to a useful hierarchy of sparse computational presentations.

Research question 10 (Kernel-checked end-to-end extraction). Can ProveIt extract the complete natural quadratic polynomial, its witness restoration map, and the connected-graph residual checker from a single verified program specification?

The reference implementation already gives a concrete target, including input zero, exact first-halting time, history inversion, prime two, and exterior rows. An initial formal milestone could be the full countdown example, followed by a parametric proof for the counter instruction type; only then should a universal table and operation ledger be instantiated.

81 Conclusion

The central separation is between analytic well-posedness and algebraic support. A single fixed connected grounded network can have a uniformly computable bounded inverse, an explicit spectral gap from zero, and a strongly convex energy, while its finite-support rational solutions encode halting. The input is only a two-terminal dipole. The graph's maximum degree is three, and the associated four-entry row bound is sharp in the fixed connected scalar class considered here.

The exact primitive certificate exposes more than undecidability: its charge is a specific Lucas continuant. That formula leads to a contrasting effective theorem over every finite-prime localization, and to the absence of any computable prime cutoff covering all halting certificates. A complete horizon-indexed quadratic polynomial connects these infinite-network statements to explicit finite natural arithmetic without disguising an external horizon as a fixed witness dimension.

These results give a concrete, proof-oriented extension of the repository's Diophantine-substrate program. Their next tests are independent mathematical review, stronger geometric realizations, and formal extraction of the compiler and its exact witness correspondences.

A A fully instantiated horizon-two polynomial schema

[write] Appendix A of source 02.

For the three-state alphabet (44), let $e_{t,0}, e_{t,R}, e_{t,L}$ be the natural selectors at time $t = 1, 2$. Its compressed information matrix is

$$A(e_t) = e_{t,0}H_0 + e_{t,R}H_R + e_{t,L}H_L = 3 \begin{pmatrix} e_{t,R} + e_{t,L} & e_{t,R} \\ e_{t,L} & e_{t,R} + e_{t,L} \end{pmatrix}.$$

Let $U = \mathbf{1}_2 \mathbf{1}_2^\top$, $Y_0 = I_2 + U$, $p_0 = 1$, $p_1 = 39$, and $p_2 = 1521$. The following is one explicit integer polynomial in fourteen natural variables (six selectors and eight entries of Y_1, Y_2):

$$\begin{aligned} \mathcal{I}_2 = & \sum_{t=1}^2 (e_{t,0} + e_{t,R} + e_{t,L} - 1)^2 \\ & + \sum_{t=1}^2 \sum_{a,b=1}^2 \left((Y_t)_{ab} - p_t - [A(e_t)(Y_{t-1} - p_{t-1}U)]_{ab} \right)^2 \\ & + \sum_{a,b=1}^2 ((Y_2)_{ab} - 1521)^2. \end{aligned}$$

All finite sums, coefficients, dimensions, and constant matrices are specified; expanding the squares is optional and does not change the polynomial. The word $(0, L)$ gives selectors $(1, 0, 0)$ and $(0, 0, 1)$, with $Y_1 = 39U$ and $Y_2 = 1521U$. Every displayed residual is then visibly zero. A word containing only R, L produces an invertible transverse product and cannot satisfy the terminal conditions.

This example illustrates the natural-domain uniqueness directly. A selector assignment such as $(2, -1, 0)$ is inadmissible, even though its entries sum to one; using arbitrary integer selectors without additional conditions would invalidate the one-hot proof. The checker enforces the natural domain exactly.

B Reproduction and source ledger

[write] Appendix B of source 02.

The archive contains the standalone `article.tex`, its compiled `article.pdf`, the Python implementation and independent evaluator, numerical example JSON files, a test receipt, and separate source/proof-status notes. From the package directory, run

```
python3 code/run_checks.py
python3 code/check_certificate.py \
    examples/seven_state_information_certificate.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first command rewrites the delivered examples and verification receipt. The checker returns exit status zero for an accepted certificate and nonzero for a rejected or ill-typed certificate. The deliberately incorrect uniform-target example is expected to fail, not to be treated as a regression.

[write] These commands use the delivered layout and names; the first one overwrites the recorded examples and receipt, and on Windows writes them with CRLF line endings. In this report the programs are `code/02-exact-erasure-*.py`, the examples and the three verification records are `data/02-exact-erasure-*.json`, and the delivered `article.tex` is printed as Part I; the duplicate `verification/run_output.txt` (byte-identical to the check results) and the source's PDF are not shipped. The README of this report gives a recipe that runs on a copy and leaves the shipped files untouched.

The numerical implementation uses only the Python standard library. The PDF build uses pdfLaTeX and standard TeX packages named in the preamble. The version used for the recorded Python checks was 3.13.5. The final PDF is rendered and checked separately; the PDF preflight record is included in the verification folder.

The external dependencies are narrow. The mortality bounds come from [1]; the halting/PCP interface is supported by [2]; the computably-enumerable/Diophantine correspondence is DPRM [3]. The MathOverflow discussion [4] supplies the precise research question, not a theorem used as an unproved premise. The repository sources [5, 6, 7] supply project context and claim boundaries, not automatic mathematical authority for the present results.

C Source audit and provenance

[write] Appendix A of source 10.

Repository inspection used the connected GitHub interface on October 2, 2026. The relevant reads were:

1. Computability/HilbertTenthProblem/README.md, first 210 lines, returned blob 880d2a13393ea55110b49e573180b51b011d9bc9. ([write] The link was the unpinned `/blob/main/`. Blob 880d2a13 is the README introduced by commit `db3b377f0` and replaced by `b7a9404d4`, both after the pin `297eb58e4` used for items 2 and 3, whose README blob is `138f234b`; both versions state the 75- and 87-operation figures quoted in Section 27.)
2. Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean, pinned at commit

297eb58e43157b7723cc28982083fecf8d2f29bf.

3. The probabilistic/quantum/continuous substrate collection’s README, pinned at the same commit; it identifies the strict-contraction and neural-precision reports as existing research contributions.
4. Targeted repository searches for Diophantine work, contraction, self-assembly, and Shapley operators. These are a scope check, not an exhaustive audit of every file in the repository.

The probabilistic collection is itself an unrefereed research collection. This article uses it for provenance and nonduplication, not as a substitute for proving the new simulation. The code in this archive was written for this manuscript and does not copy the repository’s imported compilers. No repository files were modified.

External sources were checked through primary publisher or author-preprint pages. In particular, the Bellman comparison uses arXiv revision 3 dated January 26, 2026, rather than only the February 2025 abstract. The Skolem status comparison includes the July 2026 preprint. No claim of exhaustive priority clearance follows from this targeted audit.

D A compact implementation contract

[write] Appendix B of source 10.

Program validates a finite source table and natural initial counters. **counter_circuit** emits the absorbing or erasing map. **layer_circuit** inserts shared delay copies and pads all outputs. **compile_program** emits paired-channel actions, normalization constants, and optional common reset and reward. **initialize** emits the complete dyadic payoff vector. ([write] The review [24] found that an explicitly supplied *integer* scale in **Program.encode** is divided in floating point, so that a counter value 1075 becomes 0.0 instead of 2^{-1075} ; the default rational scale, used by every shipped loader and export, is exact. Its patch **exact_convergence_integer_scale.patch** normalizes integer or rational scales and rejects floating-point and Boolean ones. The shipped **code/10-exact-convergence-bellman_diophantine.py** is the unpatched original.)

certificate accepts the game, initial numerator vector and positive denominator, a natural horizon, and either a scalar pair, consensus test, or rational constant full-vector target. It emits every squared linear form and complementary product, including all reward forcing coefficients. Its metadata reports the exact witness count, scaling convention, and endpoint-square count.

A JSON linear form is a list of integer pairs (j, a_j) ; index -1 denotes the constant monomial, and all other indices refer to the supplied variable label array. A complementarity term is an index pair (u, v) and means $W_u W_v$, not a residual that still needs squaring. The complete polynomial is the sum of the squares of all listed linear forms plus all listed complementarity terms. This specifies an ordinary integer polynomial without auxiliary syntax such as “run,” “minimum,” or “there exists a schedule.”

The independent checker evaluates that polynomial and, optionally, reconstructs the game trajectory from the serialized stochastic rows. Unsupported domains and malformed instructions are rejected. The implementation is intended as a transparent research reference, not as a hardened service for untrusted, resource-exhausting inputs.

E Reproduction and file inventory

[write] Appendix A of source 03.

The archive contains the following principal files:

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Source and compiled research article
<code>compiler.py</code>	Exact natural circuit and thermal routines
<code>verify.py</code>	Finite and symbolic verification suite
<code>verification.json</code>	Exact results and rational threshold enclosures
<code>examples.json</code>	Gate-level and expanded compiled examples
<code>README.md</code>	Usage, limitations, and reproduction instructions
<code>PROVENANCE.md</code>	Repository snapshot and primary-source links
<code>requirements.txt</code>	Symbolic verification dependency
<code>build.sh</code>	PDF build commands

Run the tests with Python 3.10 or later and SymPy installed:

```
python -m pip install -r requirements.txt
python verify.py
```

The arithmetic compiler and interval routines themselves use only the Python standard library. The manuscript can be rebuilt with a conventional LaTeX installation containing the packages listed in its preamble:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

[write] The delivered `build.sh` (shipped as `code/03-thermal-arithmetic-build.sh`) runs *three* `pdflatex` passes, not the two shown. File names in the table above are the delivered ones; their shipped names are listed in Section 38.5. `verify.py` imports `compiler` by its delivered name and rewrites `examples.json` and `verification.json` in its own directory, so it must be run on a copy (README of this report); `article.tex` is printed as Part III, and the source’s `README.md`, `article.pdf` and `SHA256SUMS.txt` are not shipped.

The verification receipt is generated by the delivered code. In particular, the optional accelerated ground-series path used for the near-boundary examples is identified in the implementation; it is not substituted for the general computability proof.

F Provenance of this report

[write] This appendix was written when the three batch-78 manuscripts were merged; Section F.3 records the batch-79 additions.

F.1 Sources

The report was built from three manuscripts of batch 78 of ProveIt’s incoming reports, cluster H2. Two arrived in the archives of commit 1977e6ea6 and one in those of commit 808b53ed8; all three were placed by commit 798b0c5d4, which staged their programs, data and audit notes byte-identical to the delivery.

Source	Archive and title	Arrival	Pin	Printed as
02 (base)	ProveIt_Exact_Erasure_Research.zip; <i>Exact Erasure without a Common Invariant Line</i> ; 29-page PDF	1977e6ea6	928ea9701	Part I (Sections 2–15 and conclusion), Appendices A–B
10	ProveIt_Exact_Convergence_Research.zip; <i>Exact Convergence as Computation</i> ; 24-page PDF	808b53ed8	297eb58e4 (README read unpinned; Appendix C)	Part II (Sections 17–29 and conclusion), Appendices C–D
03	Thermal_Arithmetic_Diophantine_Hamiltonians.zip; <i>Thermal Arithmetic</i> ; 25-page PDF	1977e6ea6	6914ccca6; blob 8bd07142 of the review it cites	Part III (Sections 31–39, ledger and conclusion), Appendix E

Contributions: source 02 the positive stochastic erasure compiler, its PCP front end, higher guards, decidable boundaries, zero–one law and natural quartics; source 10 the stochastic-game compiler, its boundaries and its quadratic certificates; source 03 the radially isolated Hamiltonian, its spectral and thermal classification and the critical fugacity. Every result, proof, example, remark, limitation, research question and appendix of the three manuscripts is printed. They share no theorem with one another, so no statement had to be printed once for two sources.

F.2 Where the merge had to choose

- *One report, three Parts.* The three manuscripts’ nearest neighbour is the report [21], which already had nine Parts; they were placed together as a sibling report because they share the spine of Section 1.2, not a construction. Placing sources 02 and 10 as a tenth Part of [21] and source 03 as an eleventh was the alternative considered at placement.
- *Base and order.* Source 02 is the base (the longest text, the earliest arrival, the widest set of results); its `article.tex` was staged unprefixed and has been replaced in place by this merged text. Source 10 follows because stochastic games extend stochastic matrices by minima and maxima; source 03 shares no lemma with either and comes last.
- *Re-proofs of collection results.* The core of Theorem 32.1, the fixed-arity quartic corollaries of Parts I and II, the no-cutoff clauses of all three Parts and the template of Theorem 34.4 are results of [22] (and partly of [21]). They are printed in place, with the sources’ own proofs, and marked by [write] pointers naming the host labels; no novelty is claimed for them. Remark 34.5 is an editorial consequence of two printed theorems.
- *Reviews and defects.* The research programme’s reviews of all three manuscripts [23, 24] are cited where their findings apply: the thermal compiler’s unchecked low-level descriptors (Section 38.5), the Bellman checker’s unbound initial vector (Section 26) and the Bellman loader’s floating-point integer scale (Appendix D). The delivered programs are shipped unpatched; the patches stay in the research programme.
- *Labels.* Every label carries the prefix `ste:.` The labels of source 02 are `ste:ee:` plus their delivered names, those of source 10 `ste:ec:` plus theirs, and those of source 03 `ste:ta:` plus theirs; no source label was dropped or renamed apart from the prefix. Added labels: 50 in all; 206 delivered labels give 256. Front section, Parts and provenance (12): `ste:sec:front`,

ste:sec:parts, ste:sec:spine, ste:tab:spine, ste:sec:notation, ste:tab:notation, ste:sec:relation, ste:sec:status, ste:part:ee, ste:part:ec, ste:part:ta, ste:app:provenance. Part I (9), the nine research questions: ste:ee:q:overhead, ste:ee:q:irreducible, ste:ee:q:lowdim, ste:ee:q:schedules, ste:ee:q:certificates, ste:ee:q:fixedalphabet, ste:ee:q:nonerasure, ste:ee:q:robust, ste:ee:q:formal. Part II (13), its four unlabelled statements, two appendices and seven research directions: ste:ec:cor:stopping, ste:ec:prop:onesided, ste:ec:prop:selectors, ste:ec:prop:threshold, ste:ec:q:instantiate, ste:ec:q:oneplayer, ste:ec:q:sparse, ste:ec:q:observations, ste:ec:q:economical, ste:ec:q:mrpd, ste:ec:q:formalize, ste:ec:app:audit, ste:ec:app:contract. Part III (16), four sections, its appendix, the ten research questions and the editorial remark: ste:ta:sec:intro, ste:ta:rem:universalcompact, ste:ta:sec:gap, ste:ta:sec:package, ste:ta:sec:questions, ste:ta:q:degree4, ste:ta:q:threemode, ste:ta:q:instantiate, ste:ta:q:height, ste:ta:q:compact, ste:ta:q:effective, ste:ta:q:singularities, ste:ta:q:couplings, ste:ta:q:cutoff, ste:ta:q:formalize, ste:ta:app:reproduction.

- *Numbering.* Source 02's Section n is Section $n + 1$ here (Sections 2–16) and its Appendices A–B keep their letters; source 10's Section n is Section $n + 16$ (Sections 17–30) and its Appendices A–B are Appendices C–D; source 03's Section n is Section $n + 30$ (Sections 31–41) and its Appendix A is Appendix E. Theorem numbers follow the sections.
- *Bibliography.* The three bibliographies are merged. Source 03's `matiyasevich` (the 2010 finite-fold paper) is `matiyasevichFF`, because source 02 uses that key for the 1993 book. The repository items read by several sources at different pins are merged into one entry each, recording every pin: `repoH10` (02's `repoH10`, 03's `reporeadme`, 10's `repo-h10`), `repoReview` (02's `repoReview`, 03's `repoaudit`) and `repoProb` (02's `repoProb`, 10's `repo-substrates`). Entries marked `[write]` were added.
- *Front matter and typography.* The three delivered title pages are replaced by one; each source's title, subtitle, author line, abstract and status statement open its Part verbatim. Source 02's remarks used the definition style; all remarks now use the remark style. No symbol was renamed (Section 1.3).
- *Files.* Delivered programs, data and audit notes are shipped byte-identical under prefixed names (the README maps them). Not shipped: the three PDFs, the delivered READMEs of all three and the manuscripts of sources 10 and 03 (printed as Parts II and III), the checksum ledgers of sources 02 and 03 (verified at placement), the two in-archive byte copies `verification/run_output.txt` (source 02) and `artifacts/test_run.txt` (source 10), and source 10's two regenerable certificates `countdown_certificate.json` and `fixedpoint_certificate.json`, which the README shows how to rebuild. All survive in the archives of their arrival commits.
- *Added text.* Every sentence written in the merge is marked `[write]`: Section 1; the openings of the three Parts; the notes in Part I on the MathOverflow claim, the repository relationship and review, the (3, 6) attribution, the nonnegative-mortality boundary, the radix-three encoding, geometric clocks, the circuit quadratization, the seven-state wording, the no-bound argument and shipped file names; the notes in Part II on the collection's contractions, the structural-subclasses question, the complementarity device, the quartic corollary, the no-cutoff and threshold arguments, the excluded certificates, the checker defect, the unpinned README read, the loader-scale defect and shipped file names; the notes in Part III on the fixed-quartic

route, the circuit quadratization, the Boolean energy landscape, the finite-fold template with Remark 34.5, the no-cutoff clause, the compiler defect, the degree floor, the build passes and shipped file names; the notes opening each appendix; and this appendix.

F.3 Batch 79: Parts IV and V

[write] This subsection was written when three manuscripts of batch 79 of ProveIt’s incoming reports, cluster J3, were added on 2 October 2026. All three arrived in the archives of commit ef2fc7990, pin the repository at commit 44983ed7e and were placed by commit bbaf322e5, which staged their programs, data and audit notes byte-identical to the delivery under the file prefixes 11-, 12- and 13-.

Source	Batch-79 manuscript, archive and title	Arrival	Pin	Printed as
11	12; <i>Mixing_Does_Not_Remove_Arithmetic.zip</i> ; <i>Mixing Does Not Remove Arithmetic</i> ; 24-page PDF	ef2fc7990	44983ed7e	Part IV (Sections 42–54), Appendices G–H
12 (base of Part V)	10; <i>Coercive_Green_Diophantine.zip</i> ; <i>Coercive Linear Algebra as a Universal Computational Substrate</i> ; 22-page PDF	ef2fc7990	44983ed7e	Part V.A (Sections 55–69), Appendices I–J
13	14; <i>Well_Conditioned_Diophantine_Computation.zip</i> ; <i>Well-Conditioned Diophantine Computation</i> ; 26-page PDF	ef2fc7990	44983ed7e	Part V.B (Sections 70–81), Appendices K–L

Contributions: source 11 the optimal positive doubly stochastic realization of polynomials, the quartic budget, the fixed-alphabet input curves, the degree–depth converse and the separated-target cutoff; source 12 the coercive path-forest operator with its canonical continuant certificate, Green dichotomy, Pell spectrum, decidable levels, fast approximation, polynomial-functional form, finite slices, network and killed-walk readings and the positive-characteristic obstruction; source 13 a second route to source 12’s main theorem together with the connected degree-three network, the finite-prime localization theorem, the row-three threshold, the positivity obstruction and the horizon quadratic. Every result, proof, example, remark, limitation, research question and appendix of the three manuscripts is printed. Statements that sources 12 and 13 share are printed in both, with bracketed pointers in source 13 and its proofs kept as second routes.

Where the batch-79 merge had to choose

- *This report, not a new one.* All three manuscripts have this report’s spine (Section 1.2): a uniform contraction or a uniformly bounded inverse with a computable rate, and an undecidable exact event. Placing them near Part V of [21], which has nine Parts, or as a new report for sources 12 and 13 were the alternatives considered at placement.
- *Order.* Part IV comes first because it re-uses Part I’s lift and answers part of Part I’s question of Section 15.6. New Parts were appended after Part III; no existing Part, section, statement or label was renumbered or renamed.
- *Base of Part V.* Source 12 covers every $\alpha \geq 3$ and more consequences of the common theorem; source 13 is printed after it in full. Its shared statements carry bracketed pointers, and source 12’s question on a connected network carries a dated note that source 13 answers it.

- *Re-proofs of collection results.* Part IV's lift is Part I's (Corollary 4.3); its gate lemma, binomial step and no-horizon corollary are `cdc:wf:lem:quadratzation`, `cdc:pt:prop:unipotent` and `cdc:bd:prop:nobound` of [22]; Part V's no-cutoff results are `cdc:bd:prop:nobound`; source 13's horizon quadratic is a variant of the counter compilers of [41]. They are printed in place with the sources' proofs and [write] pointers (Section 42.4 and the notes in Part V), with no novelty claim.
- *Reviews.* Sources 12 and 13 were reviewed by the research programme before placement [42]; its two findings on source 13's code are reported in Section 79, and the shipped programs are unpatched. Source 11 was reviewed shortly after placement [44], with no defect. All three reviews are cited where they apply; the three Parts were written after them.
- *Appendices.* The appendices of sources 11, 12 and 13 are Appendices G–L, placed after this provenance appendix so that the letters A–F of the earlier appendices are unchanged.
- *Labels.* The delivered labels of source 11 carry the prefix `ste:mx:`, those of source 12 `ste:cg:` and those of source 13 `ste:wc:`; none was dropped or renamed apart from the prefix (62, 85 and 75 labels, 222 in all). The batch-79 write added 45 labels, so the report has $256 + 222 + 45 = 523$. Front matter and provenance (5): `ste:part:mx`, `ste:part:cg`, `ste:tab:notation79`, `ste:cg:tab:conventions`, `ste:app:provenance79`. Part IV (17): `ste:mx:sec:intro`, `ste:mx:sec:pointers`, `ste:mx:sec:lift`, `ste:mx:sec:robust`, `ste:mx:sec:witnesses`, `ste:mx:sec:implementation`, `ste:mx:sec:questions`, `ste:mx:tab:examples`, `ste:mx:app:ledger`, `ste:mx:app:provenance` and the seven research subsections `ste:mx:q:instantiate`, `ste:mx:q:line`, `ste:mx:q:formal`, `ste:mx:q:zeroset`, `ste:mx:q:height`, `ste:mx:q:reversible`, `ste:mx:q:families`. Part V.A (12), the twelve research subsections: `ste:cg:q:frontend`, `ste:cg:q:history`, `ste:cg:q:local`, `ste:cg:q:connected`, `ste:cg:q:thresholds`, `ste:cg:q:offdiagonal`, `ste:cg:q:decidable`, `ste:cg:q:characteristic`, `ste:cg:q:coding`, `ste:cg:q:promise`, `ste:cg:q:killing`, `ste:cg:q:degrees`. Part V.B (11), the ten research questions and one appendix: `ste:wc:q:geometric`, `ste:wc:q:planar`, `ste:wc:q:infinite`, `ste:wc:q:primecharge`, `ste:wc:q:constants`, `ste:wc:q:robust`, `ste:wc:q:compression`, `ste:wc:q:algebraic`, `ste:wc:q:uniform`, `ste:wc:q:extraction`, `ste:wc:app:archive`.
- *Numbering.* Source 11's Section n is Section $n + 41$ (Sections 42–54; the [write] subsection 42.4 is added inside its first section), source 12's is $n + 54$ (55–69) and source 13's is $n + 69$ (70–81); source 13 is introduced by an unnumbered heading. Theorem numbers follow the sections; equations, tables, figures and source 13's research questions are numbered continuously through the report, as in Parts I–III.
- *Bibliography.* Merged entries: `repoH10` (adds 11's `repo-readme`, 12's `repo` and 13's `repoReadme`), `review1977` (11's `repo-review`), `repo-mrdp` (13's `repoMRDP`), `dudenhefner` (12's), `bennett` (12's `bennett` and 13's `Bennett`) and `linearreport` (12's `linearreport` and 13's `priorLinear`). Source 12's `matiyasevich` (the 1970 paper) is `matiyasevich1970`. Entries marked [write] were added.
- *Notation and typography.* No symbol was renamed and no normalization changed; Tables 8 and 10 record the collisions. Source 13's `\cref` references are written out; the three sources' `\ip` and `\norm` macros print with this report's definitions; source 12's New TX fonts are replaced by this report's Latin Modern. The three title pages are replaced by the Part openings, which print each source's title, author line and abstract, and source 11's status box.

- *Files.* Delivered programs, data and audit notes are shipped byte-identical under prefixed names (the README maps them), including all fourteen of source 11’s examples. Not shipped: the three PDFs, the three delivered READMEs and manuscripts (printed as Parts IV and V), the verified checksum ledgers of sources 11 (26 of 26) and 13 (11 of 11; source 12 ships none), and source 12’s `validation/test_output.txt`, a byte copy of `results.json`. All survive in the archives of commit `ef2fc7990`.
- *Added text.* Every sentence written for batch 79 is marked [write]: the changes to Section 1 (the source map, the Part summaries, the spine rows, Table 8 and its readings, the relation, review and status paragraphs), the title page and abstract, the openings of Parts IV and V and of Part V.B with Table 10, Section 42.4, the notes in Part IV on the README figures, the review of Part I’s source, the lift, the gate, the no-horizon corollary, the robust cutoff, the basis-dependent constants, the section numbers, shipped file names and the source’s provenance appendix; the notes in Part V on the README figures, batch-79 manuscript 01, the no-cutoff theorem, the killed walk, the answered connected-network question, the killing question, shipped file names and fonts, the MRDP interface, the antecedents, the shared statements of source 13, the counter compiler and the review findings; the dated notes in Part I after Corollary 4.3 and Theorem 8.2 and under its questions of Sections 15.6, 15.7 and 15.8; the appendix openings; this subsection; and the bibliography entries marked [write].

G Proof dependency and scope ledger

[write] Appendix A of source 11 (batch-79 manuscript 12).

Claim	Dependencies and qualification
Optimal $r(P) + 1$ lift	Finite differences, rational linear algebra, mass-zero lower bound; proved here. Optimal for the exact centered scalar series only.
Sharp quartic state budget	Derivative-order count and explicit radial quartic rank; proved here. The inverse-system dimension phenomenon is classical background.
Fixed universal quadratic curve	Classical MRDP plus the proved gate front end, joint-module lift, and rational loader. Universal numerical matrices not instantiated.
Fixed universal line segment	Same external MRDP dependency; positive-value filter and affine loader proved here. Degree at most five.
Degree–depth equivalence	Finite binomial expansion and the forward lift; no decision of the single-cubic natural problem.
Robust finite-search cutoff	Common entry minorization and total variation; requires strictly positive separation from the equilibrium target.
Exported examples	Symbolic identities and independent finite rational checks. Not a proof-assistant or historical-priority certification.

H Source provenance

[write] Appendix B of source 11.

The repository snapshot used for the selected-file comparison is

44983ed7ebfd545de55bfdb50e040c82f3d24295.

The two substantive repository sources are the H10 README and the six-report review listed in the bibliography. Their numerical circuit claims are reported as repository claims; this article does not re-audit those circuits. External primary papers and author-hosted manuscripts were consulted for the historical and theorem context. All sources were accessed during preparation of this note on 2 October 2026.

[write] The six-report review is [23] and the README is [5]; the source’s bibliography keys `repo-review` and `repo-readme` are merged into those entries.

The supplied proofs should be judged on their stated hypotheses. The literature search establishes substantial prior art but is not an exhaustive priority search. In particular, the classical rank theorem, positive stochastic embedding idea, inverse-system language, quartic normalization, and commutative-automaton undecidability must not be extracted from this note and advertised as first discoveries. The candidate contribution is the specific explicit and optimal-size combination, its input-geometry tradeoffs, its exact reverse translation, and the reproducible interface connecting them.

I Compiler and verifier specifications

[write] Appendix A of source 12 (batch-79 manuscript 10).

I.1 Local predecessor procedure

For a target (r, a, b, h) , compute $(h', d) = \text{divmod}(h, B)$. If d is not a branch tag, return “no predecessor.” Otherwise retrieve its unique branch. Reject a target-state mismatch. Invert the counter update and check its guard: an inverse increment requires a positive target counter, an inverse zero branch requires a zero counter, and an inverse decrement adds one. Return the resulting source only after its forward step is checked to equal the target. Every rejected case is a root candidate, not an error in the ambient vertex set.

I.2 Sparse verification

A verifier receives a validated program, α , a source s , an integer c , and a finite mapping u . It accumulates αu_v at each occupied vertex, subtracts u_v at each of its graph neighbors, and subtracts c at s . It accepts exactly when every accumulated residual is zero and the sum of coefficients at halting states is one. Because each occupied vertex has at most two neighbors, only finitely many rows are generated, including every row that can be nonzero. This procedure is polynomial in the complete sparse binary input size.

I.3 Exact interval procedure

```

Input: fixed program, root source s, integer alpha >= 3, bits b >= 0
n := max(1, ceil(b/2))
Simulate n-1 transitions, testing for a halt at the final configuration.
If a halt occurs with k configurations:
    compute D_0,...,D_k and return [D_(k-1)/D_k, D_(k-1)/D_k]
Otherwise:
    lo := 0; hi := 1/2
    repeat n times:
        lo := 1/(alpha-lo)
        hi := 1/(alpha-hi)
    return [lo, hi]

```

All arithmetic in the supplied implementation is integer or rational. The irrational endpoint is used only in the proof, never as a floating-point oracle in the algorithm.

J Artifact manifest and reproducibility

[write] Appendix B of source 12.

The archive contains the following original files:

File	Purpose
article.tex, article.pdf	Source and compiled manuscript.
README.md	Build, replay, and scope instructions.
code/green_machine.py	Exact history, algebra, and observable compiler.
code/verify.py	Reproducible exact validation suite.
examples/countdown.json	Full source, witness, and quadratic residuals.
examples/ modular_false_positive.json	Source-bound characteristic-two counterexample.
validation/results.json	Full 43-category check ledger.
validation/test_output.txt	Captured successful replay output.
SOURCE_AUDIT.md	Repository snapshot and external-source boundaries.

Run the following from the extracted directory:

```

python3 code/verify.py
python3 code/green_machine.py --export-example examples
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex

```

The Python code requires Python 3.10 or later and no third-party package. The document uses standard TeX Live packages, including New TX fonts and TikZ. External papers and font files are not redistributed in this archive.

[write] Shipped names: see the note at the end of Section 67. The delivered PDF used New TX fonts; this report is typeset in Latin Modern.

K The modular bound as executable pseudocode

[write] Appendix A of source 13 (batch-79 manuscript 14).

The proof of Corollary 75.4 is implemented by the following finite procedure. Here $U(m,n)$ is evaluated by the recurrence, not by floating-point exponentiation.

```

localization_bound(m, S):
    require m odd, m >= 3; require S a finite set of primes
    C = 1
    for p in S:
        find the first z >= 1 with U(m,z) = 0 mod p
        # An invertible pair recurrence guarantees z <= p*p - 1.
        c = max(0, valuation_p(U(m,z)) - valuation_p(z))
        C = C * p**c
    L = 1
    while (m-1)**L <= C*(L+1):
        L = L + 1
    return L

localized_unit_solution(P, input, m, S):
    B = localization_bound(m, S)
    if B <= 1: return false
    simulate P through transition number B-2
    if no halt occurred: return false
    L = number of configurations in that first-halting trace
    D = U(m,L+1)
    divide out every prime in S from D
    return (D == 1)

```

For the connected-network interpretation use odd $m \geq 5$. The bound itself is also valid for the disconnected $m = 3$ Fibonacci specialization.

L Archive contents and rebuilding

[write] Appendix B of source 13.

The archive is self-contained for the examples and verification. The principal files are:

File	Purpose
article.tex, article.pdf	This manuscript and its compiled version.
src/substrate.py	Counter compiler, connected graph, arithmetic and polynomial APIs.
verify.py	Deterministic exact verification and example regeneration.
verification.json	Actual test counts and worked-example metadata.
examples/	Full polynomial, full witness, and finite-prime bounds.
README.md	Commands, scope, and interpretation of the artifacts.
sources.md	Pinned source references and audit boundaries.
build.sh	Python verification followed by LaTeX compilation.

A standard pdfLaTeX installation with the packages listed in the preamble is sufficient. Run pdfLaTeX twice, or use `latexmk -pdf article.tex`. The verification code requires no downloaded

dependencies and makes no network requests. The PDF is rendered and visually inspected separately from mathematical verification.

[write] Shipped names: see the note in Section 79.

References

- [1] J. Cassaigne, V. Halava, T. Harju, and F. Nicolas, *Tighter Undecidability Bounds for Matrix Mortality, Zero-in-the-Corner Problems, and More*, arXiv:1404.0644v3, September 5, 2014. <https://arxiv.org/html/1404.0644v3>. The adopted bounds are $(d, m) = (3, 6), (5, 4), (9, 3), (15, 2)$; the paper’s general reducibility convention is oracle Turing reducibility.
- [2] Y. Forster, E. Heiter, and G. Smolka, *Verification of PCP-Related Computational Reductions in Coq*, arXiv:1711.07023v2, July 18, 2018. <https://arxiv.org/abs/1711.07023v2>. Related publication DOI: https://doi.org/10.1007/978-3-319-94821-8_15.
- [3] Y. V. Matiyasevich, *Hilbert’s Tenth Problem*, MIT Press, 1993. Author-hosted preface, including the statement of the computably-enumerable/Diophantine correspondence: <https://logic.pdmi.ras.ru/~yumat/H10Pbook/preface.htm>.
- [4] V. Miller, *Decidability of a matrix product being rank 1*, MathOverflow question 511960, June 2, 2026; subsequent discussion with B. Steinberg, June 2–4, 2026. The common-invariant-line restriction appears in the June 3 discussion. Accessed October 2, 2026. <https://mathoverflow.net/questions/511960/decidability-of-a-matrix-product-being-rank-1>.
- [5] V. Reshetnikov and contributors, *ProveIt: Hilbert’s tenth problem* (titled *Hilbert’s tenth problem: Jones’s articles on Diophantine representation* by sources 10 and 03), repository README Computability/HilbertTenthProblem/README.md, accessed October 2, 2026. Source 02 read it at commit 928ea97017a25ebe56d240c84f27d2275d818c75: <https://github.com/VladimirReshetnikov/ProveIt/blob/928ea97017a25ebe56d240c84f27d2275d818c75/Computability/HilbertTenthProblem/README.md>. Source 03 read it at the snapshot of [6], commit 6914ccca6685baf53b7a35f25efc89366c76ba74: <https://github.com/VladimirReshetnikov/ProveIt/blob/6914ccca6685baf53b7a35f25efc89366c76ba74/Computability/HilbertTenthProblem/README.md>. Source 10 read it through the unpinned link <https://github.com/VladimirReshetnikov/ProveIt/blob/main/Computability/HilbertTenthProblem/README.md>; the returned blob is recorded in Appendix C. [write] The three sources’ keys `repoH10` (02), `reporeadme` (03) and `repo-h10` (10) are merged into this entry. Sources 11, 12 and 13 read the README (source 12 also the directory) at commit 44983ed7ebfd545de55bfdb50e040c82f3d24295: <https://github.com/VladimirReshetnikov/ProveIt/blob/44983ed7ebfd545de55bfdb50e040c82f3d24295/Computability/HilbertTenthProblem/README.md>; their keys `repo-readme` (11), `repo` (12) and `repoReadme` (13) are merged here too.
- [6] *Review of the imported computational-substrate reports, 2 October 2026*, ProveIt research review, Computability/HilbertTenthProblem/Papers/research-h-wip/native-stream-queue/imported_substrate_review_20261002.md, blob 8bd07142c8b4298fb636e47b40d2d84e993a40ce. Source 02 read it at commit 928ea97017a25ebe56d240c84f27d2275d818c75: <https://github.com/VladimirReshetnikov/ProveIt/blob/928ea97017a25ebe56d240c84f27d2275d818c75/Computability/HilbertTenthProblem/README.md>.

- em/Papers/research-wip/native-stream-queue/imported_substrate_review_20261002.md. Source 03 read it at snapshot commit 6914ccca6685baf53b7a35f25efc89366c76ba74: https://github.com/VladimirReshetnikov/ProveIt/blob/6914ccca6685baf53b7a35f25efc89366c76ba74/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/imported_substrate_review_20261002.md. [write] The keys `repoReview` (02) and `repoaudit` (03) are merged; both pins carry the same blob.
- [7] *Diophantine Laws of Probabilistic, Quantum and Continuous Computation*, ProveIt combined report README, `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/probabilistic-quantum-and-continuous-computation/README.md`, an AI-assisted, unrefereed research collection; its source inventory includes *Strict Contractions with Diophantine-Complete Transients* and *Precision Is Memory*. Source 02 read it at commit 928ea97017a25ebe56d240c84f27d2275d818c75: <https://github.com/VladimirReshetnikov/ProveIt/blob/928ea97017a25ebe56d240c84f27d2275d818c75/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/probabilistic-quantum-and-continuous-computation/README.md>. Source 10 read it at commit 297eb58e43157b7723cc28982083fecf8d2f29bf, October 2, 2026: <https://github.com/VladimirReshetnikov/ProveIt/blob/297eb58e43157b7723cc28982083fecf8d2f29bf/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/probabilistic-quantum-and-continuous-computation/README.md>. [write] The keys `repoProb` (02) and `repo-substrates` (10) are merged. The report itself is [21].
- [8] L. S. Shapley. Stochastic games. *Proceedings of the National Academy of Sciences* 39(10) (1953), 1095–1100. doi:10.1073/pnas.39.10.1095.
- [9] A. Dudenhefner. Certified decision procedures for two-counter machines. *FSCD 2022*, LIPIcs 228, 16:1–16:18. doi:10.4230/LIPIcs.FSCD.2022.16. [write] Also cited by source 12 (same key), for the instruction-set qualifications and Theorem 6.
- [10] A. Varonka and K. Watanabe. On piecewise affine reachability with Bellman operators. [arXiv:2502.19923v3](https://arxiv.org/abs/2502.19923v3), January 26, 2026. Extended version of the MFCS 2025 paper, LIPIcs 345, 92:1–92:18.
- [11] M. Vahanwala. Skolem and positivity completeness of ergodic Markov chains. *Information Processing Letters* 186 (2024), 106481. doi:10.1016/j.ipl.2024.106481. Author preprint: [arXiv:2305.04881](https://arxiv.org/abs/2305.04881).
- [12] F. Luca, J. Ouaknine, and J. Worrell. Conjectural decidability of the Skolem problem. [arXiv:2607.15510](https://arxiv.org/abs/2607.15510), July 2026.
- [13] M. Grobelna, J. Křetínský, and M. Weininger. Stopping criteria for value iteration on concurrent stochastic reachability and safety games. [arXiv:2505.21087v2](https://arxiv.org/abs/2505.21087v2), September 10, 2025. Full version of the LICS 2025 paper.
- [14] D. Larchey-Wendling and Y. Forster. Hilbert’s tenth problem in Coq. *FSCD 2019*, LIPIcs 131, 27:1–27:20. doi:10.4230/LIPIcs.FSCD.2019.27.
- [15] V. Reshetnikov, ProveIt repository. *Diophantine/MRDP.lean*. Pinned repository source, commit 297eb58e43157b7723cc28982083fecf8d2f29bf, read October 2, 2026. Source 13 read it at commit 44983ed7ebfd545de55bfdb50e040c82f3d24295, citing the theorems `mrdp_dioph_iff`, `mrdp` and `mrdp_iff`; the blob is unchanged. [write] Source 13’s key `repoMRDP` is merged into this entry.

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- [19] D. Cantone, A. Casagrande, F. Fabris, and E. G. Omodeo, *The quest for Diophantine finiteness*, Le Matematiche **76** (2021), no. 1. <https://lematematiche.dmi.unict.it/index.php/lematematiche/article/view/2044>.
- [20] [write] T. Neary, *Undecidability in Binary Tag Systems and the Post Correspondence Problem for Five Pairs of Words*, in *32nd International Symposium on Theoretical Aspects of Computer Science (STACS 2015)*, LIPIcs **30** (2015), 649–661. A copy is kept at `Computability/HilbertTenthProblem/Lean/LIPIcs.STACS.2015.649.pdf`.
- [21] [write] ProveIt research report *Diophantine Laws of Probabilistic, Quantum and Continuous Computation*, collection `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/probabilistic-quantum-and-continuous-computation/`; Parts II (quantum substrates), IV (questions), V (exact probabilities), VII (exact observation in continuous-state dynamics), VIII (continuous time) and IX (exact stopping laws). Cited by label names, as at the write of this report.
- [22] [write] ProveIt research report *Canonical Diophantine Certificates*, collection `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/canonical-diophantine-certificates/`; its witness-faithful foundations, Part III (accelerators), Part IX (the boundary of fixed-arity compression) and Part XI (positivity-restricted degree threshold). Cited by label names, as at the write of this report.
- [23] [write] ProveIt research programme, *Six new substrate reports: review, constructor repairs and local reductions*, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/incoming_substrate_review_1977e6ea6.md` (commit 44b28c395), sections *Exact erasure* and *Thermal arithmetic and its repair*, with its receipt `incoming_substrate_review_1977e6ea6.json` (keys `erasure`, `thermal`, `thermal_repair`) and the patch `thermal_exact_dag_guardes.patch` in the same directory. Source 11 read the review at commit 44983ed7ebfd545de55bfdb50e040c82f3d24295 (its key `repo-review`, merged here): https://github.com/VladimirReshetnikov/ProveIt/blob/44983ed7ebfd545de55bfdb50e040c82f3d24295/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/incoming_substrate_review_1977e6ea6.md.
- [24] [write] ProveIt research programme, *Seven incoming computational-substrate reports and Exact Convergence and Total Quadratic Semantics: review of 808b53ed8*, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/incoming_substrate_review_808b53ed8.md` and `incoming_convergence_review_808b.md` (commit 126028588), with the receipt `incoming_substrate_review_808b53ed8.json` (keys `independent.convergence`, `independent.convergence_repairs`, `independent.convergence_scale`) and the patches `ex`

- `act_convergence_input_binding.patch` and `exact_convergence_integer_scale.patch` in the same directory.
- [25] [write] ProveIt research programme, research note `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/matrix_pcp_trace.md`, Section 1, as at the write of this report.
 - [26] Martin Davis. *Hilbert’s Tenth Problem is Unsolvable*. The American Mathematical Monthly 80(3), 233–269, 1973. [Primary article](#), [University of Maryland hosted copy](#).
 - [27] Paavo Turakainen. *Generalized Automata and Stochastic Languages*. Proceedings of the American Mathematical Society 21, 303–309, 1969. [Publisher archive record](#).
 - [28] Paul C. Bell. *Polynomially Ambiguous Probabilistic Automata on Restricted Languages*. [arXiv:1902.09407v3](#), 2020 (first submitted 2019).
 - [29] Paul C. Bell and Pavel Semukhin. *Decision Questions for Probabilistic Automata on Small Alphabets*. Logical Methods in Computer Science 19(4), article 36, 2023. [doi:10.46298/LMCS-19\(4:36\)2023](#). [Author manuscript](#), [arXiv:2105.10293v5](#).
 - [30] Borja Balle and Mehryar Mohri. *Learning Weighted Automata*. [Author-hosted manuscript](#). See Section 3.2 for the Hankel-rank characterization and minimality theorem.
 - [31] Joachim Jelisiejew. *Classifying Local Artinian Gorenstein Algebras*. [arXiv:1511.08007](#), first submitted 2015.
 - [32] S. K. Masuti and M. E. Rossi. *Artinian Level Algebras of Socle Degree 4*. [arXiv:1701.03180](#), first submitted 2017. See the inverse-system discussion and compressed-algebra context.
 - [33] Yu. V. Matiyasevich, *Enumerable Sets Are Diophantine*, Doklady Akademii Nauk SSSR **191**(2) (1970), 279–282 (Russian). Original-paper record and full-text link: <https://www.mathnet.ru/dan35274>. [write] Source 12’s key `matiyasevich`, renamed because source 02 uses that key for the 1993 book.
 - [34] C. H. Bennett, *Logical Reversibility of Computation*, IBM Journal of Research and Development **17**(6) (1973), 525–532. [doi:10.1147/rd.176.0525](#). History recording is the relevant antecedent; no physical energy claim from that paper is needed here (source 12). Source 13 inspected the original article through the UC San Diego-hosted scan, https://mathweb.ucsd.edu/~sbuss/CourseWeb/Math268_2013W/Bennett_Reversibiity.pdf. [write] Source 12’s key `bennett` and source 13’s key `Bennett` are merged into this entry.
 - [35] P. G. Doyle and J. L. Snell, *Random Walks and Electric Networks*, Carus Mathematical Monographs **22**, Mathematical Association of America, 1984. Author-posted version: [arXiv:math/0001057](#).
 - [36] Stephen Demko, William F. Moss, and Philip W. Smith, *Decay rates for inverses of band matrices*, Mathematics of Computation **43**(168) (1984), 491–499. DOI: [10.1090/S0025-5718-1984-0758197-9](#). This is an antecedent for inverse decay; all inverse bounds used here are proved directly in the text.
 - [37] *ProveIt*, `Diophantine/Paper1980/TuringPartrec100.lean`, commit `44983ed7ebfd545de55bfdb50e040c82f3d24295`. The inspected source exposes the finite-machine construction

- and the `re_tm0` interface; adjoining counter-simulation modules are listed in the project's Diophantine.lean facade. <https://github.com/VladimirReshetnikov/ProveIt/blob/44983ed7ebfd545de55bfdb50e040c82f3d24295/Computability/HilbertTenthProblem/Lean/Diophantine/Paper1980/TuringPartrec100.lean>. [write] Source 13's entry; the file is unchanged at this write.
- [38] *Linear Quasi-Diophantine Universality: Unique Polynomial Certificates, Boundary Tests, and an Exact Convex Integrality Gap*, research report prepared for Vladimir Reshetnikov, 2 October 2026. Source 12: earlier project report in the supplied research corpus, inspected in PDF/TeX form, cited for project provenance and comparison, not as a published priority claim or a machine-checked dependency of the present proofs. Source 13: supplied Library source `Linear_Quasi_Diophantine_Universality.tex`, cited as a prior research manuscript, not as an independently refereed result. [write] Source 12's key `linearreport` and source 13's key `priorLinear` are merged into this entry. The manuscript is batch-79 manuscript 01 (archive `Linear_Boundary_Transport_Research.zip`, arrival 060e08a07), placed in [40].
- [39] *Arithmetic of Rapidly Mixing Reversible Computation: Diophantine cuts, Liouville equilibria, and exact-symbolic barriers*. Research manuscript prepared for Vladimir Reshetnikov, 30 September 2026. Supplied Library source: `article(20260930-202816).tex`. Cited as a prior research manuscript, not as an independently refereed result. [write] Source 13's entry. The manuscript is source 17 of [21], printed in its Part V.
- [40] [write] ProveIt research report *polynomial-witness-histories*, collection `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/polynomial-witness-histories/`, placed by commit 224ca41df from batch-79 manuscripts 06, 15 and 01; manuscript 01 is to be printed there with the label prefix `pwh:bt:`. Not yet written at this write.
- [41] [write] ProveIt research report *Liveness beyond halting*, collection `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/liveness-beyond-halting/`; its degree-two counter-machine compilers `lbh:qd:thm:compiler` and `lbh:ql:thm:compiler` and their comparison `lbh:sec:threecompilers`. Cited by label names, as at this write.
- [42] [write] ProveIt research programme, *Review: coercive Green certificates and connected well-conditioned computation*, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_coercive_connected_aebfa386e.md` (commit 899391bde), with its checker `review_coercive_connected_aebfa386e.py`, its receipt `review_coercive_connected_aebfa386e.json` and the patch `well_conditioned_certificate_snapshots.patch` (SHA-256 `cdd6307d5659ae0c72e7715b72e22e8692b07362d0a0f008f89a9ac85372f2a9`) in the same directory.
- [43] [write] ProveIt research programme, byte-pinned intake `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/incoming_substrate_intake_aebfa386e.md` (commit b927b2290) and its continuing review table `incoming_substrate_review_aebfa386e.md` in the same directory, which lists the reviews of all three batch-79 manuscripts of this report as complete at commit 653349f6a.
- [44] [write] ProveIt research programme, *Review: Mixing Does Not Remove Arithmetic*, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_mixing_aebfa386e.md` (commit 5b633fcf9), with its checker `review_mixing_aebfa386e.py` and receipt `review_mixing_aebfa386e.json` in the same directory.

- [45] [write] ProveIt research programme, *Complete paid evaluators for connected counter certificates*, Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/connected_cross_routing_slp.md, with its compiler `connected_cross_routing_slp.py`, receipt `connected_cross_routing_slp.json` and independent review `review_connected_cross_routing_slp.md` in the same directory (commit 9bc875b6a).